

Monte Carlo methods and stochastic simulations: from integration to an approximation of SDEs¹

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Contents

1	Introduction	4
2	Deterministic quadrature rules for approximation of multidimensional Lebesgue integrals	5
2.1	Exercises	9
3	Crude Monte Carlo method	11
3.1	Monte Carlo simulation of π	11
3.2	Definition and basic properties of the crude Monte-Carlo method	12
3.2.1	Theoretical and empirical asymptotic confidence intervals	14
3.2.2	Hoeffding inequality, probability of large deviations and non-asymptotic confidence intervals	15
3.2.3	Almost sure convergence	16
3.3	Variance reduction	18
3.3.1	Control variates	18
3.3.2	Stratified sampling	21
3.4	There exists deterministic quadrature rule as good as crude Monte Carlo method	24
3.5	Some generalizations	25
3.5.1	Weighted integration	25
3.5.2	Abstract setting	26
3.6	There is always enough randomness	29
3.7	Implementation issues	29
3.8	Concluding remarks	30
3.9	Exercises	30
4	Deterministic vs randomized algorithms for ODEs	33
4.1	Deterministic Euler scheme	35
4.1.1	Optimality of the Euler algorithm in the Lipschitz class	37
4.2	Randomized Euler scheme	40
4.2.1	Coming back to integration: variance of the randomized Riemann sum vs variance of the crude Monte Carlo method	45
4.2.2	Convergence with probability one	47
4.3	Deterministic two-stage Runge-Kutta method	48
4.4	Randomized two-stage Runge-Kutta method	48
4.5	On connection between Euler/RK2 and gradient descent algorithms	48
4.6	Exercises	50
5	Canonical stochastic processes	52
5.1	Building blocks—Wiener and Poisson processes	52
5.2	Simulation of Poisson and Wiener processes	53
5.3	Brownian and Poisson bridges	54

5.4	Cauchy process	55
5.5	Optimal reconstruction of trajectories of the Wiener processes	55
5.6	Approximation of Lebesgue integral of Wiener process	58
5.7	Implementation issues	61
5.8	Concluding remarks	62
5.9	Exercises	62
6	Stochastic integration with respect to semimartingales	64
6.1	Exercises	69
7	Jump-diffusion stochastic differential equations	70
7.1	*SDEs driven by the Wiener process and compound Poisson process	73
7.2	On some relation between SDEs and ODEs	74
8	Euler scheme and its convergence	75
8.1	Optimality of the convergence rate 1/2 for the Euler-Maruyama algorithm	85
8.2	Implementation issues	87
8.3	Exercises	87
9	Randomized Euler scheme for jump-diffusion SDEs	89
9.0.1	Perturbed gradient descent method and Euler-Maruyama scheme	100
10	Introduction to optimal one-point approximation of solutions of SDEs	101
10.0.1	Some remarks on optimal reconstruction of trajectories of solutions of SDEs	110
10.1	Exercises	110
11	Weak approximation of solutions of SDEs with application to option pricing	111
11.1	Non-asymptotic (?) confidence intervals	116
11.2	Implementation issues	117
12	Parameter estimation and forecasting for SDEs-based models	120
12.1	Introductory example - parameter estimation in the geometric Brownian motion model	120
12.1.1	Quadratic variation based estimation	120
12.1.2	Linear regression approach	124
12.2	The maximum likelihood method (ML)	125
12.2.1	Geometric Brownian motion	125
12.2.2	Zero mean reverting OU-type process	127
12.2.3	Mean reverting OU-type process	128
12.2.4	Geometric mean reverting OU-type process	129
12.3	CLS-conditional least square method	129
12.4	Quasi maximum likelihood method (QML)	131
12.5	Monte Carlo forecasting in SDE-based models	132
12.5.1	Prediction intervals in the scalar case with $m = d = 1$	132
12.5.2	Ellipsoidal prediction regions in multidimensional case	134
12.5.3	Concluding remarks - why the geometry of prediction regions matters	137
12.6	Exercises	138
13	Auxiliary results—part 1	139

14 Auxiliary results–part 2	151
14.0.1 What does it mean that two processes are equivalent?	153
14.0.2 Three notions of measurability of stochastic processes	153
14.0.3 Stochastic process as a random element with values in functional space and vice versa	154
14.1 Exercises	155

Chapter 1

Introduction

In recent years, dynamical systems, both deterministic and stochastic, have grown increasingly complex, underpinning models across mathematics, physics, engineering, and finance. Classical numerical analysis, focused solely on deterministic algorithms, is often insufficient for such systems. For example, deterministic methods can be severely hampered by the curse of dimensionality. Consequently, Monte Carlo algorithms have gained popularity, together with closely related stochastic simulation techniques based on stochastic differential equations.

These lecture notes present the main ideas behind Monte Carlo methods and their theoretical properties. We apply them to problems such as numerical integration and the approximation of solutions to deterministic and stochastic differential equations. We also discuss Python implementations of representative algorithms and illustrate their use in option pricing.

Parts of these notes were used in lectures for PhD students at AGH University of Science and Technology in Kraków, Poland, during the summer semesters of 2020, 2021, 2023, and 2026.

Chapter 2

Deterministic quadrature rules for approximation of multidimensional Lebesgue integrals

Let us start with considering the following model problem in numerical analysis: the approximate value of the following multidimensional integral

$$I(f) = \int_{[0,1]^d} f(x) dx. \quad (2.1)$$

The integral above is understood in the Lebesgue sense concerning d -dimensional Lebesgue measure λ_d and $f : [0, 1]^d \rightarrow \mathbb{R}$ is at least Borel measurable and integrable.

In this chapter, we will assume that f is at least a continuous function. Then the Lebesgue integral $I(f)$ is equivalent to the Riemann integral of f by, for example, Theorem 4.7.3 in [61].

The simplest quadrature approximating (2.1) is the *rectangle quadrature*, defined as follows. Let $n \in \mathbb{N}$, $h = 1/n$, and we take

$$t_i = i \cdot h, \quad i = 0, 1, \dots, n, \quad (2.2)$$

$$A_{i_1, \dots, i_d} = [t_{i_1}, t_{i_1+1}] \times \dots \times [t_{i_d}, t_{i_d+1}], \quad (2.3)$$

for $i_k \in \{0, 1, \dots, n-1\}$, $k = 1, 2, \dots, d$. Then

$$[0, 1]^d = \bigcup_{i_1, \dots, i_d=0, 1, \dots, n-1} A_{i_1, \dots, i_d}. \quad (2.4)$$

Since for all $(i_1, \dots, i_d) \neq (j_1, \dots, j_d)$ we have that

$$\lambda_d(A_{i_1, \dots, i_d} \cap A_{j_1, \dots, j_d}) = 0, \quad (2.5)$$

we can write that

$$\begin{aligned} I(f) &= \sum_{i_1, \dots, i_d=0}^{n-1} \int_{A_{i_1, \dots, i_d}} f(x) dx \approx \sum_{i_1, \dots, i_d=0}^{n-1} \int_{A_{i_1, \dots, i_d}} f(\xi_{i_1, \dots, i_d}) dx \\ &= \sum_{i_1, \dots, i_d=0}^{n-1} f(\xi_{i_1, \dots, i_d}) \cdot \lambda_d(A_{i_1, \dots, i_d}) = h^d \cdot \sum_{i_1, \dots, i_d=0}^{n-1} f(\xi_{i_1, \dots, i_d}) \end{aligned} \quad (2.6)$$

with $\xi_{i_1, \dots, i_d} = (t_{i_1}, t_{i_2}, \dots, t_{i_d})$. Hence, the multidimensional rectangle rule is defined by

$$Q_n^d(f) = h^d \cdot \sum_{i_1, \dots, i_d=0}^{n-1} f(\xi_{i_1, \dots, i_d}). \quad (2.7)$$

Note that the cost of Q_n^d , measured by the number of function evaluations, is n^d .

We investigate the error of Q_n^d for integrands belonging to the following class

$$\mathcal{F}_L^d = \{f : [0, 1]^d \rightarrow \mathbb{R} \mid |f(x) - f(y)| \leq L \cdot \|x - y\|_\infty, \ x, y \in [0, 1]^d\}, \quad (2.8)$$

where $L \in (0, +\infty)$ is the class parameter and $\|\cdot\|_\infty$ is the maximum norm, i.e., $\|x\|_\infty = \max_{1 \leq i \leq d} |x_i|$ for $x = (x_1, \dots, x_d) \in \mathbb{R}^d$.

Theorem 2.0.1. *For any $f \in \mathcal{F}_L^d$ and for all $n \in \mathbb{N}$ we have that $Q_n^d(f)$ uses n^d values of f and*

$$|I(f) - Q_n^d(f)| \leq Ln^{-1}. \quad (2.9)$$

Proof. By Fubini's theorem and by using the substitution $u_j = x_j - t_{i_j}$ ($u_j \in [0, h]$ for $x_j \in [t_{i_j}, t_{i_j+1}]$), we have that

$$\begin{aligned} |I(f) - Q_n^d(f)| &\leq \sum_{i_1, \dots, i_d=0}^{n-1} \int_{A_{i_1, \dots, i_d}} |f(x_1, \dots, x_d) - f(t_{i_1}, \dots, t_{i_d})| dx_1 \dots dx_d \\ &\leq L \sum_{i_1, \dots, i_d=0}^{n-1} \int_{A_{i_1, \dots, i_d}} \|(x_1, \dots, x_d) - (t_{i_1}, \dots, t_{i_d})\|_\infty dx_1 \dots dx_d \\ &= L \sum_{i_1, \dots, i_d=0}^{n-1} \int_{t_{i_d}}^{t_{i_d+1}} \dots \int_{t_{i_1}}^{t_{i_1+1}} \max\{|x_1 - t_{i_1}|, \dots, |x_d - t_{i_d}|\} dx_1 \dots dx_d \\ &= Ln^d \int_0^h \dots \int_0^h \max\{u_1, \dots, u_d\} du_1 \dots du_d = Ln^d h^d \mathbb{E}(\max\{U_1, \dots, U_d\}) = \frac{Ld}{d+1} h, \end{aligned} \quad (2.10)$$

since for $U_1, \dots, U_d$ being iid random variables from $U([0, h])$, the random vector $(U_1, \dots, U_d)$ has uniform distribution over $[0, h]^d$. ■

In terms of cost, the above result can be formulated as follows: for all $f \in \mathcal{F}_L^d$ and $N \in \mathbb{N}$

$$|I(f) - Q_{N^{1/d}}^d(f)| \leq \frac{Ld}{d+1} N^{-1/d}, \quad (2.11)$$

where now $Q_{N^{1/d}}^d$ uses N evaluations of f . Hence, for a fixed error-level $\varepsilon \in (0, +\infty)$ we need to take

$$N \geq \left\lceil \left(\frac{L}{(1 + \frac{1}{d})\varepsilon} \right)^d \right\rceil = \Omega((1/\varepsilon)^d) \quad (2.12)$$

function values to ensure that the quadrature error is at most ε . For example, if one takes $\varepsilon = 1/2$, then the number of evaluations needed increases exponentially in d . One can ask whether this poor cost-error behavior is due to the use of this particular quadrature rule or to assumptions imposed on f . The answer is negative, following from the corresponding lower error bounds. Namely, we investigate the error of any deterministic quadrature $\mathcal{A}_n$ (that might be even nonlinear) in the Lipschitz class $\mathcal{F}_L^d$. We will use the Information-Based complexity framework (IBC), see [87], and we show a simplified proof of the lower error bound for the Lipschitz class $\mathcal{F}_L^d$, see [66] for more general results.

For a fixed $n \in \mathbb{N}$ we consider any algorithm of the form

$$\mathcal{A}_n(f) = \varphi_n(N_n(f)), \quad f \in \mathcal{F}_L^d, \quad (2.13)$$

where $\varphi_n : \mathbb{R}^n \rightarrow \mathbb{R}$ is, in general, a nonlinear function, and a vector $N_n(f) \in \mathbb{R}^n$ of the information used about f is

$$N_n(f) = [f(x_1), \dots, f(x_n)]. \quad (2.14)$$

Such information given by the values of f at some deterministic points x_k is called *standard information*. We assume that the information $N_n(f)$ about f might be adaptive, i.e., $x_1 \in [0, 1]^d$ is fixed for all $f \in \mathcal{F}_L^d$, and there exist functions $\psi_k : \mathbb{R}^{k-1} \rightarrow \mathbb{R}$, $k = 2, 3, \dots, n$, such that

$$x_k = \psi_k(f(x_1), f(x_2), \dots, f(x_{k-1})), \quad k = 2, 3, \dots, n. \quad (2.15)$$

The informational cost of the algorithm $\mathcal{A}_n(f)$ is equal to n for all $f \in \mathcal{F}_L^d$. The worst-case error of $\mathcal{A}_n$ in the class $\mathcal{F}_L^d$ is

$$e(\mathcal{A}_n, \mathcal{F}_L^d) = \sup_{f \in \mathcal{F}_L^d} |I(f) - \mathcal{A}_n(f)|. \quad (2.16)$$

To establish lower bounds on the worst-case error, we first state the following fact.

Proposition 2.0.2. *For any algorithm $\mathcal{A}_n$ of the form (2.13) we have*

$$e(\mathcal{A}_n, \mathcal{F}_L^d) \geq \frac{1}{2} \sup_{f_0, f_1 \in \mathcal{F}_L^d, N_n(f_0) = N_n(f_1)} |I(f_0) - I(f_1)|. \quad (2.17)$$

Proof. Let $f_0, f_1 \in \mathcal{F}_L^d$ be any functions such that $N_n(f_0) = N_n(f_1)$. Then $\mathcal{A}_n(f_0) = \mathcal{A}_n(f_1)$, and

$$e(\mathcal{A}_n, \mathcal{F}_L^d) \geq \sup_{f \in \{f_0, f_1\}} |I(f) - \mathcal{A}_n(f)| = \max_{i=0,1} |I(f_i) - \mathcal{A}_n(f_i)|, \quad (2.18)$$

since $\{f_0, f_1\} \subset \mathcal{F}_L^d$. Moreover, since $\mathcal{A}_n(f_0) = \mathcal{A}_n(f_1)$, by the triangle inequality we get

$$|I(f_0) - I(f_1)| \leq |I(f_0) - \mathcal{A}_n(f_0)| + |I(f_1) - \mathcal{A}_n(f_1)| \leq 2 \max_{i=0,1} |I(f_i) - \mathcal{A}_n(f_i)|, \quad (2.19)$$

which, together with (2.18), implies that

$$e(\mathcal{A}_n, \mathcal{F}_L^d) \geq \frac{1}{2} |I(f_0) - I(f_1)|. \quad (2.20)$$

Since $f_0, f_1 \in \mathcal{F}_L^d$ where arbitrary functions such that $N_n(f_0) = N_n(f_1)$, by taking supremum in (2.20) we get the thesis. $\square$

We now give the construction of two functions $f_0, f_1 \in \mathcal{F}_L^d$ (often called *fooling functions*), that have the same information and $|I(f_0) - I(f_1)| = \Omega(n^{-1/d})$. Let $P = \{p_1, \dots, p_n\}$ be an arbitrary subset of $[0, 1]^d$. We define the distant function (from the set P) in maximum norm as

$$\text{dist}_\infty(x, P) = \min_{p \in P} \|x - p\|_\infty, \quad x \in [0, 1]^d. \quad (2.21)$$

Fact 2.0.3. *For all $x, y \in [0, 1]^d$ it holds*

$$|\text{dist}_\infty(x, P) - \text{dist}_\infty(y, P)| \leq \|x - y\|_\infty. \quad (2.22)$$

Proof. For all $p \in P$, $x, y \in [0, 1]^d$ we have

$$\|x - p\|_\infty \leq \|x - y\|_\infty + \|y - p\|_\infty. \quad (2.23)$$

This implies that

$$\begin{aligned} \text{dist}_\infty(x, P) &= \min_{p \in P} \|x - p\|_\infty \leq \min_{p \in P} (\|x - y\|_\infty + \|y - p\|_\infty) \\ &= \|x - y\|_\infty + \min_{p \in P} \|y - p\|_\infty = \|x - y\|_\infty + \text{dist}_\infty(y, P), \end{aligned} \quad (2.24)$$

and

$$\text{dist}_\infty(x, P) - \text{dist}_\infty(y, P) \leq \|x - y\|_\infty. \quad (2.25)$$

Hence, we also have the following

$$\text{dist}_\infty(y, P) - \text{dist}_\infty(x, P) \leq \|y - x\|_\infty = \|x - y\|_\infty. \quad (2.26)$$

From (2.25) and (2.26), the thesis follows. $\square$

We now show the following useful estimate.

Lemma 2.0.4. *For any subset $P = \{p_1, p_2, \dots, p_n\} \subset [0, 1]^d$ we have*

$$\int_{[0,1]^d} \text{dist}_\infty(x, P) dx \geq \frac{d}{2(d+1)} n^{-1/d}. \quad (2.27)$$

Proof. From Lemma 13.0.18 we get

$$\int_{[0,1]^d} \text{dist}_\infty(x, P) dx = \int_0^{+\infty} \lambda_d(\{x \in [0, 1]^d \mid \text{dist}_\infty(x, P) > t\}) dt. \quad (2.28)$$

For any $t > 0$ define

$$P_t = \{x \in [0, 1]^d \mid \text{dist}_\infty(x, P) \leq t\}, \quad (2.29)$$

and then

$$P_t = \bigcup_{p \in P} \{x \in [0, 1]^d \mid \|x - p\|_\infty \leq t\} \subset \bigcup_{p \in P} \{x \in \mathbb{R}^d \mid \|x - p\|_\infty \leq t\}. \quad (2.30)$$

Therefore

$$\lambda_d(P_t) \leq \sum_{p \in P} \lambda(\{x \in \mathbb{R}^d \mid \|x - p\|_\infty \leq t\}). \quad (2.31)$$

Since for any $p = [p^1, \dots, p^d]$ and $t > 0$

$$\{x \in \mathbb{R}^d \mid \|x - p\|_\infty \leq t\} = \bigtimes_{j=1, \dots, d} [p^j - t, p^j + t], \quad (2.32)$$

we have

$$\lambda(\{x \in \mathbb{R}^d \mid \|x - p\|_\infty \leq t\}) = \prod_{j=1}^d \lambda_1([p^j - t, p^j + t]) = (2t)^d. \quad (2.33)$$

Hence,

$$\lambda_d(P_t) \leq n(2t)^d. \quad (2.34)$$

Since $[0, 1]^d = P_t^c \cup P_t$, we have

$$\lambda_d(\{x \in [0, 1]^d \mid \text{dist}_\infty(x, P) > t\}) = \lambda_d(P_t^c) = 1 - \lambda_d(P_t) \geq 1 - n(2t)^d. \quad (2.35)$$

Note that $1 - n(2t)^d \geq 0$ for all $t \in [0, t_{\max}]$, where $t_{\max} = \frac{1}{2} n^{-1/d}$. Therefore

$$\begin{aligned} \int_0^{+\infty} \lambda_d(\{x \in [0, 1]^d \mid \text{dist}_\infty(x, P) > t\}) dt &= \int_0^{+\infty} \lambda_d(P_t^c) dt \geq \int_0^{t_{\max}} \lambda_d(P_t^c) dt \\ &\geq \int_0^{t_{\max}} (1 - n(2t)^d) dt = t_{\max} - \frac{n2^d}{d+1} t_{\max}^{d+1} = \frac{d}{2(d+1)} n^{-1/d}, \end{aligned} \quad (2.36)$$

which ends the proof. $\square$

Let us take the function $\bar{f} \equiv 0$. Of course $\bar{f} \in \mathcal{F}_L^d$, and let us compute the adaptive information points x_j for $\bar{f}$. The first one, x_1 , is fixed, then

$$\begin{aligned} x_2 &= \psi_2(\bar{f}(x_1)) = \psi_2(0), \\ x_3 &= \psi_3(\bar{f}(x_1), \bar{f}(x_2)) = \psi_3(0, 0), \\ &\dots \\ x_n &= \psi_n(\bar{f}(x_1), \bar{f}(x_2), \dots, \bar{f}(x_{n-1})) = \psi_n(0, 0, \dots, 0), \end{aligned} \quad (2.37)$$

and $N_n(\bar{f}) = [0, \dots, 0] \in \mathbb{R}^n$. Take $P := \{x_1, x_2, \dots, x_n\} \subset [0, 1]^d$ and set

$$f_0(x) = L \cdot \text{dist}_\infty(x, P) = -f_1(x), \quad x \in [0, 1]^d. \quad (2.38)$$

From Fact 2.0.3 we have that $f_0, f_1 \in \mathcal{F}_L^d$. We now compute the adaptive information points x_k^i for f_i , $i = 0, 1$, $k = 1, 2, \dots, n$. The first one x_1 , is fixed, and we have $x_1^0 = x_1 = x_1^1$. Since $f_i(x_k) = 0$ for all $i = 0, 1$, $k = 1, 2, \dots, n$, we get for $i = 0, 1$ that

$$\begin{aligned} x_2^i &= \psi_2(f_i(x_1^i)) = \psi_2(f_i(x_1)) = \psi_2(0) = x_2, \\ x_3^i &= \psi_3(f_i(x_1^i), f_i(x_2^i)) = \psi_3(f_i(x_1), f_i(x_2)) = \psi_3(0, 0) = x_3, \\ &\dots \\ x_n^i &= \psi_n(f_i(x_1^i), f_i(x_2^i), \dots, f_i(x_{n-1}^i)) = \psi_n(f_i(x_1), f_i(x_2), \dots, f_i(x_{n-1})) = \psi_n(0, 0, \dots, 0) = x_n, \end{aligned}$$

and

$$N_n(f_i) = [f_i(x_1^i), f_i(x_2^i), \dots, f_i(x_n^i)] = [f_i(x_1), f_i(x_2), \dots, f_i(x_n)] = [0, 0, \dots, 0] = N_n(\bar{f}), \quad (2.39)$$

for $i = 0, 1$. From Proposition 2.0.2 and Lemma 2.0.4 we have

$$e(\mathcal{A}_n, \mathcal{F}_L^d) \geq \frac{1}{2} |I(f_0) - I(f_1)| = |I(f_0)| = L \int_{[0, 1]^d} \text{dist}_\infty(x, P) dx \geq \frac{Ld}{2(d+1)} n^{-1/d}. \quad (2.40)$$

Hence, the lower bound for the worst-case error of any deterministic algorithm $\mathcal{A}_n$ is $\Omega(n^{-1/d})$, where n is the number of evaluation points for f . Hence, if $e(\mathcal{A}_n, \mathcal{F}_L^d) \leq \varepsilon$ then $n = \Omega((1/\varepsilon)^d)$.

2.1 Exercises

1. Provide a version of Theorem 2.0.1 when we have non-uniform discretization of the cube $[0, 1]^d$.
2. Construct and investigate an error of the multidimensional version of the mid-point quadratic rule, approximating the integral (2.1), which is defined by (2.7), but this time with

$$\xi_{i_1, \dots, i_d} = \left((t_{i_1} + t_{i_1+1})/2, (t_{i_2} + t_{i_2+1})/2, \dots, (t_{i_d} + t_{i_d+1})/2 \right). \quad (2.41)$$

3. The Taylor-based composite quadrature rule for one-dimensional integrals is defined as follows: let $n \in \mathbb{N}$, $h = (b - a)/n$, $t_i = a + ih$, $i = 0, 1, \dots, n$, and

$$l_n(t) = \sum_{i=0}^{n-1} l_i(t) \cdot \mathbf{1}_{[t_i, t_{i+1})}(t), \quad t \in [a, b], \quad (2.42)$$

where

$$l_i(t) = \sum_{j=0}^{r-1} \frac{f^{(j)}(t_i)}{j!} (t - t_i)^j, \quad t \in [t_i, t_{i+1}], \quad i = 0, 1, \dots, n-1. \quad (2.43)$$

Then the quadrature is defined by taking

$$\mathcal{T}_n(f) := I(l_n) = \int_a^b l_n(t) dt = \sum_{i=0}^{n-1} \sum_{j=0}^{r-1} \frac{f^{(j)}(t_i)}{(j+1)!} (t_{i+1} - t_i)^j. \quad (2.44)$$

Show that for any $f \in C^r([a, b])$, $r \in \mathbb{N}$, it holds

$$\lim_{n \rightarrow +\infty} n^r \cdot (I(f) - \mathcal{T}_n(f)) = C_{\mathcal{T}}(f), \quad (2.45)$$

where

$$C_{\mathcal{T}}(f) = \frac{(b-a)^r}{(r+1)!} \left(f^{(r-1)}(b) - f^{(r-1)}(a) \right). \quad (2.46)$$

We call $C_{\mathcal{T}}(f)$ the *exact asymptotic constant*. Note that (2.45) gives the exact asymptotic behavior of error of $\mathcal{T}_n$.

We can define the following corrected Taylor-based quadrature by

$$\tilde{\mathcal{T}}_n(f) := \mathcal{T}_n(f) + \frac{C_{\mathcal{T}}(f)}{n^r} = \mathcal{T}_n(f) + \frac{h^r}{(r+1)!} \left(f^{(r-1)}(b) - f^{(r-1)}(a) \right). \quad (2.47)$$

Note that for any $n \in \mathbb{N}$ the quadrature $\tilde{\mathcal{T}}_n$ uses only values of $f^{(j)}(t_i)$ for $j = 0, 1, \dots, r-1$, $i = 0, 1, \dots, n-1$ (as $\mathcal{T}_n$) with one additional evaluation of $f^{(r-1)}$ at b .

Give a proof of the fact that for any $f \in C^r([a, b])$, $r \in \mathbb{N}$, it holds

$$\lim_{n \rightarrow +\infty} n^r (I(f) - \tilde{\mathcal{T}}_n(f)) = 0. \quad (2.48)$$

Hence, the convergence rate of the method $\tilde{\mathcal{T}}_n$ is greater than n^{-r} .

4. For $h > 0$ assume that $U_1, \dots, U_d$ are iid random variables from $U([0, h])$. By direct calculation show that the expected value of maximum of independent uniforms is

$$\mathbb{E}(\max\{U_1, \dots, U_d\}) = \frac{d}{d+1} h. \quad (2.49)$$

5. Show (2.30).

Chapter 3

Crude Monte Carlo method

Throughout this chapter, we denote a complete probability space by $(\Omega, \Sigma, \mathbb{P})$. We denote by $\|\cdot\|_2$ the Euclidean norm in $\mathbb{R}^d$.

We can roughly say that the Monte Carlo method is a direct consequence of the strong law of large numbers, which, in its simpler version, goes as follows (see, for example, [22], [14]).

Theorem 3.0.1. (*Strong Law of Large Numbers, SLLN*) *Let $(X_j)_{j \in \mathbb{N}}$ be a sequence of independent and identically distributed random variables with values in $\mathbb{R}^d$. If $\mathbb{E}\|X_1\|_2 < +\infty$, then*

$$\frac{1}{N} \sum_{j=1}^N X_j \rightarrow \mathbb{E}(X_1) \text{ as } N \rightarrow +\infty \text{ a.s.} \quad (3.1)$$

If $\mathbb{E}\|X_1\|_2 = +\infty$, then

$$\limsup_{N \rightarrow +\infty} \frac{1}{N} \sum_{j=1}^N X_j = +\infty \text{ a.s.} \quad (3.2)$$

In the next section, we start with the well-known example of the Monte Carlo simulation of π .

3.1 Monte Carlo simulation of π

Let us consider circle $O(C, r)$ with the centre $C = (c_1, c_2) \in \mathbb{R}^2$ and the radius $r \in (0, +\infty)$ inscribed in the square S with the side length $2r$. We randomly pick N points X_j from the square S and we count the number K of points X_j that lie in the circle $O(C, r)$, i.e., $(X_j)_{j=1,2,\dots,N}$ is the iid sequence of random variables with the common law $U([s_1 - r, s_1 + r] \times [s_2 - r, s_2 + r])$, and

$$K = \sum_{j=1}^N \mathbf{1}_{O(C,r)}(X_j). \quad (3.3)$$

Note that the ratio of the area of the circle O to the area of the square S is $(\pi r^2)/(2r)^2 = \pi/4$ and it is intuitively clear that we have

$$\frac{\pi}{4} \approx \frac{K}{N}, \quad (3.4)$$

as $N \rightarrow +\infty$. Hence, we can take

$$\hat{\pi}_N = 4K/N = \frac{4}{N} \sum_{j=1}^N \mathbf{1}_{O(C,r)}(X_j) \quad (3.5)$$

as the Monte Carlo approximation of π . Rigorously, let us now establish properties of the estimator $\hat{\pi}_N$.

Since the value of areas' ratio in (3.4) is independent of r and the circle's center C , we take for simplicity that $c_1 = c_2 = 0$ and $r = 1$. Moreover,

$$\begin{aligned}\mathbb{E}(\mathbf{1}_{O(C,r)}(X_j)) &= \mathbb{P}(X_j \in O(C,r)) = \int_{\mathbb{R}^2} \mathbf{1}_{O(C,r)}(x,y) \cdot \frac{1}{2} \mathbf{1}_{[-1,1]}(x) \cdot \frac{1}{2} \mathbf{1}_{[-1,1]}(y) dx dy \\ &= \frac{1}{4} \int_{O(C,r)} dx dy = \frac{1}{4} \lambda_2(O(C,r)) = \frac{\pi}{4},\end{aligned}\tag{3.6}$$

since $X_j = [X_j^1, X_j^2]$, and X_j^1, X_j^2 are independent and uniformly distributed random variables in $[-1, 1]$. Hence, by Theorem 3.0.1 we get that

$$\hat{\pi}_N \rightarrow 4 \cdot \mathbb{E}(\mathbf{1}_{O(C,r)}(X_1)) = \pi\tag{3.7}$$

almost surely as $N \rightarrow +\infty$. Furthermore, for all $N \in \mathbb{N}$

$$\mathbb{E}(\hat{\pi}_N) = \frac{4}{N} \sum_{j=1}^N \mathbb{P}(X_j \in O(C,r)) = 4 \cdot \mathbb{P}(X_1 \in O(C,r)) = \pi,\tag{3.8}$$

since $(X_j)_{j=1,2,\dots,N}$ are identically distributed. Hence, $\hat{\pi}_N$ is the unbiased estimator of π . We also compute the mean square error of the estimator $\hat{\pi}_N$,

$$(\mathbb{E}|\pi - \hat{\pi}_N|^2)^{1/2} = (\mathbb{E}(\hat{\pi}_N^2) - \pi^2)^{1/2},\tag{3.9}$$

where, by the independence of X_i and X_j for $i \neq j$, it holds that

$$\begin{aligned}\mathbb{E}(\hat{\pi}_N^2) &= \frac{16}{N^2} \mathbb{E}\left(\sum_{j=1}^N \mathbf{1}_{O(C,r)}(X_j)\right)^2 \\ &= \frac{16}{N^2} \left(\sum_{j=1}^N \mathbb{P}(X_j \in O(C,r)) + \sum_{i \neq j} \mathbb{P}(X_i \in O(C,r)) \cdot \mathbb{P}(X_j \in O(C,r))\right) \\ &= \frac{16}{N^2} \left(N \cdot \frac{\pi}{4} + (N^2 - N) \frac{\pi^2}{16}\right) = \frac{4\pi}{N} + \pi^2 \left(1 - \frac{1}{N}\right).\end{aligned}\tag{3.10}$$

Therefore, for all $N \in \mathbb{N}$

$$(\mathbb{E}|\pi - \hat{\pi}_N|^2)^{1/2} = (\pi(4 - \pi))^{1/2} \cdot N^{-1/2},\tag{3.11}$$

and the mean square error is inversely proportional to the square root of the number of samples N .

3.2 Definition and basic properties of the crude Monte-Carlo method

Again we consider the problem of approximating the integral (2.1) but now we assume even less than before for the integrand f . Namely, we assume that $f \in L^2([0, 1]^d, \mathcal{B}([0, 1]^d), \lambda_d)$. (Formally $L^2([0, 1]^d, \mathcal{B}([0, 1]^d), \lambda_d)$ consists of equivalence classes and by writing $f \in L^2([0, 1]^d, \mathcal{B}([0, 1]^d), \lambda_d)$ we mean that we take any representation of the equivalence class. As we will see later this does not cause any problem.) By the Schwarz inequality and the fact that $\lambda_d([0, 1]^d) = 1$ we have $I(|f|) < +\infty$. For $N \in \mathbb{N}$ we set

$$MC_N(f) = \frac{1}{N} \sum_{j=1}^N f(\xi_j),\tag{3.12}$$

where $(\xi_j)_{j \in \mathbb{N}}$ is a sequence of independent and identically distributed random variables on $(\Omega, \Sigma, \mathbb{P})$, and $\xi_j \sim U([0, 1]^d)$, $j \in \mathbb{N}$. Note that the cost of $MC_N(f)$ equals N evaluations of f .

By the SLLN we have that

$$MC_N(f) \rightarrow \mathbb{E}(f(\xi_1)) = I(f) \left(= \int_{[0,1]^d} f(x)dx \right) \text{ a.s.}, \quad (3.13)$$

and we can ask how good (in terms of suitably defined error) the approximation is.

Theorem 3.2.1. *For any $f \in L^2([0,1]^d, \mathcal{B}([0,1]^d), \lambda_d)$ the following holds:*

(i) *For all $N \in \mathbb{N}$*

$$\mathbb{E}(MC_N(f)) = I(f). \quad (3.14)$$

(ii) *For all $N \in \mathbb{N}$*

$$\left(\mathbb{E}|I(f) - MC_N(f)|^2 \right)^{1/2} = (Var(MC_N(f)))^{1/2} = \frac{\sigma(f)}{\sqrt{N}}, \quad (3.15)$$

where

$$\sigma^2(f) := I(f^2) - (I(f))^2. \quad (3.16)$$

(iii) *$MC_N(f)$ is independent of the choice of the representation of the equivalence class $f \in L^2([0,1]^d, \mathcal{B}([0,1]^d), \lambda_d)$ in the following sense: let f, g be two representations of the same equivalence class in $L^2([0,1]^d, \mathcal{B}([0,1]^d), \lambda_d)$. Then*

$$\mathbb{P}\left(\bigcap_{N \in \mathbb{N}} \{MC_N(f) = MC_N(g)\}\right) = 1. \quad (3.17)$$

$$(iv) \quad \forall_{f \in L^2([0,1]^d, \mathcal{B}([0,1]^d), \lambda_d)} \left[\sigma(f) = 0 \iff \exists_{c \in \mathbb{R}} \lambda_d(\{x \in [0,1]^d \mid f(x) = c\}) = 1 \right].$$

Proof. Note that $(f(\xi_j))_{j \in \mathbb{N}}$ is the sequence of identically distributed and independent random variables. Hence

$$\mathbb{E}(MC_N(f)) = \frac{1}{N} \sum_{j=1}^N \mathbb{E}(f(\xi_j)) = \frac{1}{N} \cdot N \cdot \mathbb{E}(f(\xi_1)) = \frac{1}{\lambda_d([0,1]^d)} \int_{[0,1]^d} f(x)dx = I(f). \quad (3.18)$$

Therefore, by the properties of variance, we get

$$\begin{aligned} \mathbb{E}|I(f) - MC_N(f)|^2 &= \mathbb{E}\left(|\mathbb{E}(MC_N(f)) - MC_N(f)|^2\right) \\ &= Var(MC_N(f)) = \frac{1}{N^2} \sum_{j=1}^N Var(f(\xi_j)) = \frac{1}{N^2} \cdot N \cdot Var(f(\xi_1)) = \frac{\sigma^2(f)}{N}, \end{aligned}$$

since

$$Var(f(\xi_1)) = \sigma^2(f). \quad (3.19)$$

Let f, g be two representations of the same equivalence class in $L^2([0,1]^d, \mathcal{B}([0,1]^d), \lambda_d)$ and set

$$A = \{x \in [0,1]^d \mid f(x) \neq g(x)\}. \quad (3.20)$$

Of course $\lambda_d(A) = 0$. Hence, for all $N \in \mathbb{N}$ we get by the triangle inequality

$$\begin{aligned} \|MC_N(f) - MC_N(g)\|_{L^1(\Omega; \mathbb{R})} &\leq \frac{1}{N} \sum_{j=1}^N \|f(\xi_j) - g(\xi_j)\|_{L^1(\Omega; \mathbb{R})} \\ &= \|(f - g)(\xi_1)\|_{L^1(\Omega; \mathbb{R})} = I(|f - g|) \\ &= \int_{[0,1]^d \setminus A} |f(x) - g(x)|dx + \int_A |f(x) - g(x)|dx = 0. \end{aligned} \quad (3.21)$$

Thereby, for all $N \in \mathbb{N}$

$$\mathbb{P}\left(\{MC_N(f) = MC_N(g)\}\right) = 1, \quad (3.22)$$

which implies (3.17). The proof of (iv) is left as an exercise. ■

Remark 3.2.2. Note that if we assume that $f \in L^2([0, 1]^d, \mathcal{B}([0, 1]^d), \lambda_d)$, then any deterministic quadrature rule based on function evaluations is not well-defined. This is because the evaluation of an (in fact) equivalence class of a function at a point $x \in [0, 1]^d$ is mathematically senseless - in particular, the value of such quadrature will depend on the choice of representation of the equivalence class. Having in mind Theorem 3.2.1 (iii), this is another issue where Monte Carlo quadratures overcome the deterministic ones. □

Remark 3.2.3. Formally in Theorem 3.2.1 we have shown that for any equivalence class $[f] \in L^2([0, 1]^d, \mathcal{B}([0, 1]^d), \lambda_d)$ it is mathematically correct to take

$$MC_N([f]) := MC_N(f). \quad (3.23)$$

□

Remark 3.2.4. From (iv) in the Theorem above we see that the method MC_N is exact only in the trivial case when f is (almost surely) a constant function. Therefore, contrary to deterministic quadrature rules, it does not make sense to define the degree of the quadrature MC_N . □

The theorem above tells that the mean-square error of $MC_N(f)$ is $O(N^{-1/2})$ with the cost of N evaluations of f and independently of the dimension d . But how good is our approximation for fixed $\omega \in \Omega$, when we compute $MC_N(f)(\omega)$ based on the realizations $\xi_1(\omega), \dots, \xi_N(\omega)$? The question is very important and natural since such a situation occurs in practice on a computer. To answer this question we will investigate empirical confidence intervals and the error of the Monte-Carlo method in the almost sure sense.

3.2.1 Theoretical and empirical asymptotic confidence intervals

Recall that by the Central Limit Theorem 13.0.19 and (13.50) it holds

$$\lim_{N \rightarrow +\infty} \mathbb{P}\left(\left\{\sqrt{N} \cdot \frac{MC_N(f) - I(f)}{\sigma(f)} \in [a, b]\right\}\right) = \Phi(b) - \Phi(a), \quad (3.24)$$

for all $-\infty < a < b < +\infty$, where $\Phi(t) = \int_{-\infty}^t \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$. Let us fix confidence level $\alpha \in (0, 1)$ and let q_α be the two-sided α -quantile defined as the (unique) solution of the equation

$$\mathbb{P}(\{|Z| \leq q_\alpha\}) = \alpha, \quad (3.25)$$

where $Z \sim N(0, 1)$. We also have that $\mathbb{P}(\{|Z| \leq q_\alpha\}) = \Phi(q_\alpha) - \Phi(-q_\alpha) = 2\Phi(q_\alpha) - 1$. For example $q_\alpha = 2$ corresponds to $\alpha \simeq 0, 95$. We set $a := -q_\alpha$, $b = q_\alpha$ and we define the following confidence interval

$$\mathcal{J}_N^\alpha(f) = \left[MC_N(f) - q_\alpha \frac{\sigma(f)}{\sqrt{N}}, MC_N(f) + q_\alpha \frac{\sigma(f)}{\sqrt{N}}\right]. \quad (3.26)$$

By (3.24) we have that

$$\mathbb{P}(\{I(f) \in \mathcal{J}_N^\alpha(f)\}) = \mathbb{P}\left(\left\{\sqrt{N} \cdot \frac{|MC_N(f) - I(f)|}{\sigma(f)} \leq q_\alpha\right\}\right) \rightarrow \Phi(q_\alpha) - \Phi(-q_\alpha) = \alpha, \quad (3.27)$$

as $N \rightarrow +\infty$.

We call $\mathcal{J}_N^\alpha$ the *theoretical confidence interval*, since it uses the value of $\sigma(f)$ which, in practice, is usually not known. And now comes the *magic trick*: the variance $\sigma^2(f)$ can be simulated in parallel with $MC_N(f)$. Namely, we can use the following estimator of $\sigma^2(f)$

$$\widehat{\sigma}_N^2(f) = \frac{1}{N-1} \sum_{j=1}^N \left(f(\xi_j) - MC_N(f) \right)^2 \quad (3.28)$$

$$= \frac{1}{N-1} \left(\sum_{j=1}^N \left(f(\xi_j) \right)^2 - N \cdot \left(MC_N(f) \right)^2 \right). \quad (3.29)$$

(Although the single-loop implementation of the equation (3.29) is straightforward, it should not be used in practice because of the danger of cancellations. The numerically stable recursive one-pass formula will be presented in the sequel.) Again by the SLLN, we have that

$$\widehat{\sigma}_N^2(f) \rightarrow \sigma^2(f) \quad (3.30)$$

almost surely as $N \rightarrow +\infty$, and for all $N \in \mathbb{N}$

$$\mathbb{E}(\widehat{\sigma}_N^2(f)) = \sigma^2(f). \quad (3.31)$$

By using the Central Limit Theorem together with Slutsky's lemma we get that

$$\lim_{N \rightarrow +\infty} \mathbb{P} \left(\left\{ \sqrt{N} \cdot \frac{MC_N(f) - I(f)}{\sqrt{\widehat{\sigma}_N^2(f)}} \in [a, b] \right\} \right) = \Phi(b) - \Phi(a) \quad (3.32)$$

for all $-\infty < a < b < +\infty$. The *empirical confidence interval* is thus given by

$$\hat{\mathcal{J}}_N^\alpha(f) = \left[MC_N(f) - q_\alpha \sqrt{\widehat{\sigma}_N^2(f)/N}, MC_N(f) + q_\alpha \sqrt{\widehat{\sigma}_N^2(f)/N} \right], \quad (3.33)$$

and by (3.32) we have that

$$\mathbb{P}(\{I(f) \in \hat{\mathcal{J}}_N^\alpha(f)\}) \rightarrow \alpha, \text{ as } N \rightarrow +\infty. \quad (3.34)$$

3.2.2 Hoeffding inequality, probability of large deviations and non-asymptotic confidence intervals

When f is a bounded Borel function it is possible to show an exponential decay of the probability of the so-called *exceptional set*. Namely, we have the following result that follows from the Hoeffding inequality (see Theorems 13.0.21, 13.0.22).

Proposition 3.2.5. *Let $(\xi_j)_{j \in \mathbb{N}}$ be an i.i.d sequence of random vectors and $f : \mathbb{R}^d \rightarrow \mathbb{R}$ a bounded Borel function. Then for all $\varepsilon \in (0, +\infty)$ and $N \in \mathbb{N}$*

$$\mathbb{P} \left(\left| \frac{1}{N} \sum_{j=1}^N f(\xi_j) - \mathbb{E}f(\xi_1) \right| > \varepsilon \right) \leq 2 \exp \left(- \frac{\varepsilon^2 N}{8 \|f\|_\infty^2} \right), \quad (3.35)$$

where $\|f\|_\infty = \sup_{x \in \mathbb{R}^d} |f(x)|$.

(We left proof of Proposition 3.2.5 as an exercise.)

For (3.12) we directly get what follows.

Corollary 3.2.6. *Let $f : [0, 1]^d \rightarrow \mathbb{R}$ be a bounded Borel function. Then for all $C \in (0, +\infty)$, $N \in \mathbb{N}$*

$$\mathbb{P}\left(\left\{|MC_N(f) - I(f)| > C \cdot N^{-1/2}\right\}\right) \leq 2 \exp\left(-\frac{C^2}{8\|f\|_\infty^2}\right), \quad (3.36)$$

where $\|f\|_\infty = \sup_{x \in [0, 1]^d} |f(x)|$.

We can see that the probability of the exceptional set tends to zero exponentially fast with $C \rightarrow +\infty$.

The second nice thing that we can get from the Hoeffding inequality is the *non-asymptotic confidence intervals*. Namely, for any $\delta \in (0, 1)$ let us take

$$C(\delta) = \sqrt{8 \ln(2/\delta)} \cdot \|f\|_\infty, \quad (3.37)$$

provided that $\|f\|_\infty$ (or its 'good' approximation) is known. Then, by Corollary 3.2.6 we obtain for all $\delta \in (0, 1)$, $N \in \mathbb{N}$

$$\mathbb{P}\left(\left\{|MC_N(f) - I(f)| > C(\delta) \cdot N^{-1/2}\right\}\right) \leq \delta. \quad (3.38)$$

Hence, for all $\delta \in (0, 1)$, $N \in \mathbb{N}$

$$I(f) \in \left[MC_N(f) - \frac{C(\delta)}{\sqrt{N}}, MC_N(f) + \frac{C(\delta)}{\sqrt{N}}\right] \quad (3.39)$$

with probability at least $1 - \delta$. This probability of success does not depend on N , which means that we can design confidence intervals for Monte Carlo simulation uniformly in the number of samples N .

Let us now fix $\varepsilon, \delta \in (0, 1)$. From (3.38) we have that if we take

$$N = \Theta\left(\left(\frac{1}{\varepsilon}\right)^2 \ln\left(\frac{2}{\delta}\right)\right) \quad (3.40)$$

then the probability that the error $|MC_N(f) - I(f)|$ is $O(\varepsilon)$ is at least $1 - \delta$. Moreover, this error is achieved by $O\left(\left(\frac{1}{\varepsilon}\right)^2 \ln\left(\frac{2}{\delta}\right)\right)$ evaluations of f .

3.2.3 Almost sure convergence

In this section, we will investigate the convergence of the crude Monte Carlo method in the almost sure sense. For this, we need to prove the following auxiliary result.

Theorem 3.2.7. *Let $f \in L^p([0, 1]^d, \mathcal{B}([0, 1]^d), \lambda_d)$ for some $p \in [2, +\infty)$. Then there exists $C_p(f) \in (0, +\infty)$ such that for all $N \in \mathbb{N}$ it holds*

$$\left(\mathbb{E}|I(f) - MC_N(f)|^p\right)^{1/p} \leq C_p(f) \cdot N^{-1/2}, \quad (3.41)$$

and

$$\left(\mathbb{E}|I(f) - MC_N(f)|^q\right)^{1/q} \leq C_p(f) \cdot N^{-1/2} \quad (3.42)$$

for all $q \in (0, p]$.

Proof. Let

$$\mathcal{G}_j = \sigma(\xi_1, \dots, \xi_j), \quad j \geq 1, \quad (3.43)$$

with $\mathcal{G}_0 = \{\emptyset, \Omega\}$. Moreover, we set

$$Z_j = \sum_{k=1}^j \left(f(\xi_k) - I(f)\right), \quad 1 \leq j \leq N, \quad (3.44)$$

and $Z_0 := 0$. The sequence $(Z_j, \mathcal{G}_j)_{j=0,1,\dots,N}$ is a discrete-time martingale such that $\max_{1 \leq j \leq N} \mathbb{E}|Z_j|^p < +\infty$ (see exercise 11.). Moreover, its quadratic variation is given by

$$[Z]_j = \sum_{k=1}^j |Z_k - Z_{k-1}|^2 = \sum_{k=1}^j \delta_k^2, \quad 1 \leq j \leq N. \quad (3.45)$$

where $\delta_k := f(\xi_k) - I(f)$. Therefore, by the Burkholder-Davis-Gundy inequality (Theorem 13.0.23) we get

$$\mathbb{E}\left(\max_{0 \leq j \leq N} |Z_j|^p\right) \leq C_p^p \cdot \mathbb{E}([Z]_N^{p/2}) = C_p^p \cdot \mathbb{E}\left(\sum_{k=1}^N \delta_k^2\right)^{p/2}. \quad (3.46)$$

Since $p \in [2, +\infty)$, the function $[0, +\infty) \ni x \rightarrow x^{p/2}$ is convex and by the Jensen inequality we get

$$\left(\sum_{k=1}^N \delta_k^2\right)^{p/2} \leq N^{\frac{p}{2}-1} \sum_{k=1}^N |\delta_k|^p. \quad (3.47)$$

Hence,

$$\mathbb{E}([Z]_N^{p/2}) \leq N^{\frac{p}{2}-1} \sum_{k=1}^N \mathbb{E}|\delta_k|^p = N^{\frac{p}{2}} \cdot \int_{[0,1]^d} |f(x) - I(f)|^p dx, \quad (3.48)$$

and

$$\mathbb{E}|I(f) - MC_N(f)|^p = \frac{1}{N^p} \cdot \mathbb{E}|Z_N|^p \leq C_p^p \cdot N^{-\frac{p}{2}} \cdot \int_{[0,1]^d} |f(x) - I(f)|^p dx, \quad (3.49)$$

which ends the proof (3.41) with $C_p(f) = C_p \cdot \left(\int_{[0,1]^d} |f(x) - I(f)|^p dx\right)^{1/p} < +\infty$. For $q \in (0, p]$ we have from the Lyapunov inequality that

$$\left(\mathbb{E}|I(f) - MC_N(f)|^q\right)^{1/q} \leq \left(\mathbb{E}|I(f) - MC_N(f)|^p\right)^{1/p} \leq C_p(f) \cdot N^{-1/2}. \quad (3.50)$$

This ends the proof of (3.42). $\blacksquare$

As a corollary, we obtain the following main result of the section.

Theorem 3.2.8. *Let $f \in \bigcap_{p \in [2, +\infty)} L^p([0, 1]^d, \mathcal{B}([0, 1]^d), \lambda_d)$. Then for all $\varepsilon \in (0, 1/2)$ there exists $\eta_\varepsilon(f) \in \bigcap_{p \in [1, +\infty)} L^p(\Omega)$ such that*

$$\mathbb{P}\left(\bigcap_{N \in \mathbb{N}} \left\{|I(f) - MC_N(f)| \leq \eta_\varepsilon(f) \cdot N^{-\frac{1}{2} + \varepsilon}\right\}\right) = 1. \quad (3.51)$$

Proof. From Theorem 3.2.7 we have that for all $p \in [1, +\infty)$ and all $N \in \mathbb{N}$

$$\left(\mathbb{E}|I(f) - MC_N(f)|^p\right)^{1/p} \leq C_{\max\{2, p\}}(f) \cdot N^{-1/2}. \quad (3.52)$$

Applying Theorem 13.0.20 we get the thesis. $\blacksquare$

Under the assumptions of Theorem 3.2.8 we only know that $\mathbb{E}|\eta_\varepsilon(f)|^p < +\infty$ and currently there are no further (more accurate) estimates available in the literature.

3.3 Variance reduction

In this section, we discuss possibilities of variance reduction by two (seemingly the most common) approaches:

- the use of control variates,
- stratified sampling.

3.3.1 Control variates

Let us assume that we are able to decompose function $f \in L^2([0, 1]^d, \mathcal{B}([0, 1]^d), \lambda_d)$ in the following way

$$f = g + (f - g), \quad (3.53)$$

where g is such that we can compute the integral $I(g)$ explicitly. (For example, g can be a piecewise polynomial that interpolates f . However, in general, it can be any function that is in $L^2([0, 1]^d, \mathcal{B}([0, 1]^d), \lambda_d)$.) We now take

$$\widetilde{MC}_N(f) := I(g) + MC_N(f - g). \quad (3.54)$$

Applying Theorem 3.2.1 we get

$$\left(\mathbb{E} |I(f) - \widetilde{MC}_N(f)|^2 \right)^{1/2} = (\text{Var}(MC_N(f - g)))^{1/2} = \sigma(f - g)/\sqrt{N}. \quad (3.55)$$

Hence, we look for g , satisfying (3.53), such that $I(g)$ is known and $\sigma(f - g) < \sigma(f)$. Then g is called the *control variate* for f .

We now present a good choice of g in the scalar case (i.e., $d = 1$) for $f \in C^r([0, 1])$, $r \in \mathbb{N} = \{1, 2, \dots\}$, with Hölder continuous r th derivative. Firstly, we prove the following useful lemma, see [15] for an alternative method of proof.

Lemma 3.3.1. *Let $f \in C^r([a, b])$ with $r \in \mathbb{N} = \{1, 2, \dots\}$, $-\infty < a < b < +\infty$, and let us assume that there exist $H \in (0, +\infty)$, $\varrho \in (0, 1]$ such that for all $x, y \in [a, b]$*

$$|f^{(r)}(x) - f^{(r)}(y)| \leq H|x - y|^\varrho. \quad (3.56)$$

Let w_r be the Lagrange interpolation polynomial of degree at most r , that interpolates f at $r + 1$ points $x_0 < x_1 < \dots < x_r$ such that $x_i \in [a, b]$ for $i = 0, 1, \dots, r$. Then for all $x \in [a, b]$

$$|f(x) - w_r(x)| \leq \frac{H}{r!} (b - a)^{r+\varrho}. \quad (3.57)$$

Proof. If $x = x_j$ for some j then $|f(x) - w_r(x)| = 0$. So let us assume that $x \in [a, b] \setminus \{x_0, x_1, \dots, x_r\}$. Then by the classical error formula for the Lagrange interpolation (see, for example, [35]) we get for all $x \in [a, b]$

$$f(x) - w_r(x) = f[x_0, \dots, x_r, x] \cdot \prod_{k=0}^r (x - x_k), \quad (3.58)$$

with

$$w_r(x) = \sum_{j=0}^r f[x_0, x_1, \dots, x_j] \cdot \prod_{k=0}^{j-1} (x - x_k) = \sum_{j=0}^r f(x_j) \cdot \prod_{i=0, i \neq j}^r \frac{x - x_i}{x_j - x_i}, \quad (3.59)$$

where $f[x_0, \dots, x_k, x]$ is the divided difference based on the nodes $x_0, \dots, x_k, x$. Recall that for pairwise distinct points $x_0, \dots, x_r, x$ we also have

$$f[x_0, \dots, x_r, x] = \frac{f[x_1, \dots, x_r, x] - f[x_0, \dots, x_{r-1}, x_r]}{x - x_0}. \quad (3.60)$$

From (3.58) and (3.60) we obtain

$$f(x) - w_r(x) = \left(f[x_1, \dots, x_r, x] - f[x_0, \dots, x_{r-1}, x_r] \right) \cdot \prod_{j=1}^r (x - x_j). \quad (3.61)$$

Since $f^{(r)}$ is continuous on $[a, b]$ we have that for any $x \in [a, b]$ there exist $\xi_x, \eta \in [a, b]$ such that

$$\begin{aligned} f[x_1, \dots, x_r, x] &= \frac{f^{(r)}(\xi_x)}{r!}, \\ f[x_0, \dots, x_{r-1}, x_r] &= \frac{f^{(r)}(\eta)}{r!}. \end{aligned} \quad (3.62)$$

Hence, by (3.61) and (3.62) we get for any $x \in [a, b]$

$$\begin{aligned} |f(x) - w_r(x)| &= \frac{1}{r!} \cdot |f^{(r)}(\xi_x) - f^{(r)}(\eta)| \cdot \prod_{j=1}^r |x - x_j| \\ &\leq \frac{H}{r!} |\xi_x - \eta|^\varrho \cdot (b - a)^r \leq \frac{H}{r!} \cdot (b - a)^{r+\varrho}, \end{aligned} \quad (3.63)$$

and the proof is finished. $\blacksquare$

We now take $a = 0, b = 1$, $x_i = ih$, $i = 0, 1, \dots, n$, where $h = 1/n$, $n \in \mathbb{N}$. On each subinterval $[x_i, x_{i+1}]$, $i = 0, 1, \dots, n-1$ we interpolate f at the equidistant points $x_i^{(j)} = x_i + \frac{h}{r}j$, $j = 0, 1, \dots, r$, by the Lagrange polynomial $w_{i,n,r}$. The control variate $g_n : [0, 1] \rightarrow \mathbb{R}$ is defined as follows :

$$g_n(x) = w_{i,n,r}(x) = \sum_{j=0}^r f(x_i^{(j)}) \cdot \prod_{k=0, k \neq j}^r \frac{x - x_i^{(k)}}{x_i^{(j)} - x_i^{(k)}}, \quad x \in [x_i, x_{i+1}], \quad i = 0, 1, \dots, n-1. \quad (3.64)$$

Of course, $g_n \in C([0, 1])$, $I(g_n)$ can be computed explicitly (since it is a piecewise polynomial function), and by Lemma 3.3.1 we have that

$$\|f - g_n\|_\infty = \sup_{x \in [0, 1]} |f(x) - g_n(x)| = \max_{0 \leq i \leq n-1} \sup_{x_i \leq x \leq x_{i+1}} |f(x) - w_{i,n,r}(x)| \leq \frac{H}{r!} \cdot n^{-(r+\varrho)}. \quad (3.65)$$

In addition

$$\sigma^2(f - g_n) \leq I(|f - g_n|^2) \leq \|f - g_n\|_\infty^2 \leq \left(\frac{H}{r!} \right)^2 \cdot n^{-2(r+\varrho)}, \quad (3.66)$$

and $\sigma(f - g_n) < \sigma(f)$ for sufficiently large n . Hence

$$\left(\mathbb{E} |I(f) - \widetilde{MC}_{N,n}(f)|^2 \right)^{1/2} = \sigma(f - g_n) / \sqrt{N} \leq \frac{H}{r!} \cdot n^{-(r+\varrho)} \cdot N^{-1/2}, \quad (3.67)$$

where

$$\widetilde{MC}_{N,n}(f) = I(g_n) + \frac{1}{N} \sum_{j=1}^N (f - g_n)(\xi_j) \quad (3.68)$$

and $(\xi_j)_{j \in \mathbb{N}}$ is an i.i.d sequence of random variables, uniformly distributed on $[0, 1]$. The informational cost of $\widetilde{MC}_{N,n}(f)$ is $O(n + N)$ evaluations of f . For $n = N$ we arrive at the method $\widetilde{MC}_{N,N}$ that has the error

$$\left(\mathbb{E} |I(f) - \widetilde{MC}_{N,N}(f)|^2 \right)^{1/2} \leq \frac{H}{r!} \cdot N^{-(r+\varrho+\frac{1}{2})}, \quad (3.69)$$

with the cost of $O(N)$ evaluations of f . Note that $\widetilde{MC}_{N,N}$ evaluates f at points chosen both in deterministic and randomized ways. As we can see, in this case, we can significantly decrease the error from $O(N^{-1/2})$ to $O(N^{-(r+\varrho+\frac{1}{2})})$, and, essentially, the cost $O(N)$ remains the same.

Remark 3.3.2. For a definition of the control variate g_n in the case when $f \in C([0, 1])$ (i.e., $r = 0$) and f is Hölder continuous with the exponent $\varrho \in (0, 1]$ see (3.167) in Exercise 15.

Remark 3.3.3. The rate $O(N^{-(r+\varrho+\frac{1}{2})})$ is optimal, see Proposition 1.3.9, page 34 in [66].

Having the results above we now investigate the probability of the exceptional set for the method $\widetilde{MC}_{N,n}$. Note that for $h_n = f - g_n$

$$I(f) - \widetilde{MC}_{N,n}(f) = I(h_n) - MC_N(h_n) = -\frac{1}{N}Z_N^n, \quad (3.70)$$

where

$$Z_j^n = \sum_{k=1}^j \left(h_n(\xi_k) - I(h_n) \right), \quad 1 \leq j \leq N, \quad (3.71)$$

$$Z_0^n = 0. \quad (3.72)$$

Let $\mathcal{G}_0 = \{\emptyset, \Omega\}$ and $\mathcal{G}_j = \sigma(\xi_1, \dots, \xi_j)$ for $j \geq 1$. Since

$$\max_{1 \leq j \leq N} |Z_j^n| \leq 2N \|f - g_n\|_\infty < +\infty, \quad (3.73)$$

and for $j = 0, 1, \dots, N-1$

$$\mathbb{E}(Z_{j+1}^n - Z_j^n \mid \mathcal{G}_j) = \mathbb{E}(h_n(\xi_{j+1})) - I(h_n) = 0, \quad (3.74)$$

we get that for all $n \in \mathbb{N}$ the process $(Z_j^n, \mathcal{G}_j)_{1 \leq j \leq N}$ is a discrete-time martingale. Moreover, for all $j = 1, 2, \dots, N$ we have

$$|Z_j^n - Z_{j-1}^n| \leq |h_n(\xi_j)| + |I(h_n)| \leq 2\|f - g_n\|_\infty, \quad (3.75)$$

hence from the Hoeffding inequality (see Theorem 13.0.22) we get for any $\varepsilon > 0$, $N, n \in \mathbb{N}$ that

$$\begin{aligned} \mathbb{P}(|I(f) - \widetilde{MC}_{N,n}(f)| > \varepsilon) &= \mathbb{P}(|I(h_n) - MC_N(h_n)| > \varepsilon) \\ &= \mathbb{P}(|Z_N^n| > N \cdot \varepsilon) \leq 2 \exp\left(-\frac{\varepsilon^2 N}{8\|f - g_n\|_\infty^2}\right), \end{aligned} \quad (3.76)$$

and by (3.66) we obtain

$$\mathbb{P}(|I(f) - \widetilde{MC}_{N,n}(f)| > \varepsilon) \leq 2 \exp\left(-\frac{\varepsilon^2 \cdot N \cdot n^{2(r+\varrho)}}{8C_1}\right), \quad (3.77)$$

where $C_1 = (H/r!)^2$. Therefore, for any $C \in (0, +\infty)$, $N, n \in \mathbb{N}$

$$\mathbb{P}(|I(f) - \widetilde{MC}_{N,n}(f)| > C \cdot n^{-(r+\varrho)} \cdot N^{-1/2}) \leq 2 \exp\left(-\frac{C^2}{8C_1}\right). \quad (3.78)$$

For $\delta \in (0, 1)$ set

$$C(\delta) = \sqrt{8 \ln(2/\delta)} \cdot (H/r!). \quad (3.79)$$

Then for any $\delta \in (0, 1)$, $N, n \in \mathbb{N}$ we have

$$\mathbb{P}(|I(f) - \widetilde{MC}_{N,n}(f)| > C(\delta) \cdot n^{-(r+\varrho)} \cdot N^{-1/2}) \leq \delta \quad (3.80)$$

and

$$I(f) \in \left[\widetilde{MC}_{N,n}(f) - \frac{C(\delta)}{n^{r+\varrho} \cdot N^{1/2}}, \widetilde{MC}_{N,n}(f) + \frac{C(\delta)}{n^{r+\varrho} \cdot N^{1/2}} \right] \quad (3.81)$$

with the probability at least $1 - \delta$.

Let us now fix $\varepsilon, \delta \in (0, 1)$ and take $n = N$. From (3.80) we have that if we set

$$N = \Theta \left(\left(\frac{1}{\varepsilon} \right)^{\frac{1}{r+\varrho+1/2}} \left(\ln \left(\frac{2}{\delta} \right) \right)^{\frac{1}{2(r+\varrho)+1}} \right) \quad (3.82)$$

then the probability that the error $|\widetilde{MC}_{N,N}(f) - I(f)|$ is $O(\varepsilon)$ is at least $1 - \delta$. Moreover, this error is achieved by $O \left(\left(\frac{1}{\varepsilon} \right)^{\frac{1}{r+\varrho+1/2}} \left(\ln \left(\frac{2}{\delta} \right) \right)^{\frac{1}{2(r+\varrho)+1}} \right)$ evaluations of f .

We have shown the possible definition of control variate g for scalar integration of functions from $C^r([0, 1])$ with Hölder r th derivative $f^{(r)}$. We now show how to decrease the variance by choosing suitable g for multidimensional integration of functions from the Hölder class

$$\mathcal{F}_L^{\varrho,d} = \{f : [0, 1]^d \rightarrow \mathbb{R} \mid |f(x) - f(y)| \leq L \cdot \|x - y\|_\infty^\varrho, \ x, y \in [0, 1]^d\}, \quad (3.83)$$

where $L \in (0, +\infty)$, $\varrho \in (0, 1]$ are the class parameters and $\|\cdot\|_\infty$ is the maximum norm. For a fixed $N \in \mathbb{N}$, and assuming (without the loss of generality) that $N^{1/d} \in \mathbb{N}$, we take

$$g(x) = \sum_{i_1, \dots, i_d=0}^{N^{1/d}-1} f(\xi_{i_1, \dots, i_d}) \cdot \mathbf{1}_{A_{i_1, \dots, i_d}}(x), \quad (3.84)$$

where $\xi_{i_1, \dots, i_d}$, $A_{i_1, \dots, i_d}$ are defined as in (2.6) and (2.3), respectively. Note that g uses N values of f and, in fact, we have divided $[0, 1]^d$ into N sub-boxes $A_{i_1, \dots, i_d}$ such that (2.4) and (2.5) hold. Then, proceeding analogously as in the proof of Theorem 2.0.1, we have

$$\sigma^2(f - g) \leq I[(f - g)^2] = \sum_{i_1, \dots, i_d=0}^{N^{1/d}-1} \int_{A_{i_1, \dots, i_d}} |f(x) - f(\xi_{i_1, \dots, i_d})|^2 dx \leq L^2 \cdot N^{-2\varrho/d}. \quad (3.85)$$

Moreover

$$I(g) = \frac{1}{N} \sum_{i_1, \dots, i_d=0}^{N^{1/d}-1} f(\xi_{i_1, \dots, i_d}) = Q_{N^{1/d}}^d(f), \quad (3.86)$$

with $Q_{N^{1/d}}^d(f)$ defined by (2.7). Hence, for

$$\widetilde{MC}_N(f) = Q_{N^{1/d}}^d(f) + \frac{1}{N} \sum_{j=1}^N (f - g)(\xi_j), \quad (3.87)$$

where $(\xi_j)_{j=1, \dots, N}$ is an iid sequence sampled from $U([0, 1]^d)$, we have that

$$\left(\mathbb{E} |I(f) - \widetilde{MC}_N(f)|^2 \right)^{1/2} = \sigma(f - g)/\sqrt{N} \leq L \cdot N^{-(\frac{\varrho}{d} + \frac{1}{2})}. \quad (3.88)$$

Note also that $\widetilde{MC}_N(f)$ uses $2N$ values of f .

3.3.2 Stratified sampling

In this approach, we divide the box $D = [0, 1]^d$ (or in the more general setting, the integration domain) into K sub-boxes D_i , $i = 1, 2, \dots, K$, in such a way that

$$\bigcup_{i=1}^K D_i = [0, 1]^d, \quad (3.89)$$

and for all $i \neq j$

$$\lambda_d(D_i \cap D_j) = 0. \quad (3.90)$$

We then have

$$I(f) = \sum_{j=1}^K I_i(f), \quad (3.91)$$

where $I_i(f) = I(\mathbf{1}_{D_i} f) = \int_{D_i} f(x) dx$. Let us take

$$\overline{MC}_N(f) = \sum_{i=1}^K \overline{MC}_{N_i}^{(i)}(f), \quad (3.92)$$

with $N = \sum_{i=1}^K N_i$ and

$$\overline{MC}_{N_i}^{(i)}(f) = \frac{\text{vol}(D_i)}{N_i} \sum_{j=1}^{N_i} f(\xi_j^i), \quad (3.93)$$

and $(\xi_j^i)_{j=1, \dots, N_i, i=1, \dots, K}$ are independent random vectors, where for each i the random vectors $(\xi_j^i)_{j=1, \dots, N_i}$ are uniformly distributed in D_i . Then for all $i = 1, 2, \dots, K$

$$\mathbb{E}(\overline{MC}_{N_i}^{(i)}(f)) = \frac{\text{vol}(D_i)}{N_i} \sum_{j=1}^{N_i} \mathbb{E}(f(\xi_j^i)) = I_i(f), \quad (3.94)$$

and hence

$$\mathbb{E}(\overline{MC}_N(f)) = \sum_{i=1}^K I_i(f) = I(f). \quad (3.95)$$

Since $\{\overline{MC}_{N_1}^{(1)}(f), \dots, \overline{MC}_{N_K}^{(K)}(f)\}$ is a family of independent random variables, we get from (3.94) and the properties of variance

$$\mathbb{E}|I(f) - \overline{MC}_N(f)|^2 = \text{Var}(\overline{MC}_N(f)) = \sum_{i=1}^K \text{Var}(\overline{MC}_{N_i}^{(i)}(f)), \quad (3.96)$$

where, by the independence of $\{f(\xi_1^i), \dots, f(\xi_{N_i}^i)\}$ for given i , we have

$$\text{Var}(\overline{MC}_{N_i}^{(i)}(f)) = \frac{\text{vol}^2(D_i)}{N_i^2} \sum_{j=1}^{N_i} \text{Var}(f(\xi_j^i)). \quad (3.97)$$

Note that

$$\text{Var}(f(\xi_j^i)) = \mathbb{E}(f(\xi_j^i))^2 - (\mathbb{E}(f(\xi_j^i)))^2 = \frac{1}{\text{vol}(D_i)} I_i(f^2) - \frac{1}{\text{vol}^2(D_i)} (I_i(f))^2 \quad (3.98)$$

for all $j = 1, \dots, N_i$. Therefore

$$\text{Var}(\overline{MC}_{N_i}^{(i)}(f)) = \frac{1}{N_i} \left(\text{vol}(D_i) \cdot I_i(f^2) - (I_i(f))^2 \right), \quad (3.99)$$

and

$$\mathbb{E}|I(f) - \overline{MC}_N(f)|^2 = \sum_{i=1}^K \frac{1}{N_i} \left(\text{vol}(D_i) \cdot I_i(f^2) - (I_i(f))^2 \right). \quad (3.100)$$

Assuming for $i = 1, \dots, K$ that

$$N_i = \text{vol}(D_i) \cdot N \in \mathbb{N}, \quad (3.101)$$

we obtain

$$\mathbb{E}|I(f) - \overline{MC}_N(f)|^2 = \frac{1}{N} \sum_{i=1}^K \left(I_i(f^2) - \frac{(I_i(f))^2}{\text{vol}(D_i)} \right) = \frac{1}{N} \left(I(f^2) - \sum_{i=1}^K \frac{(I_i(f))^2}{\text{vol}(D_i)} \right). \quad (3.102)$$

We compare the error (3.102) of the stratified sampling method (3.92) with the error of the crude Monte Carlo method

$$MC_N(f) = \frac{1}{N} \sum_{j=1}^N f(\xi_j), \quad (3.103)$$

where $(\xi_j)_{j=1, \dots, N}$ is an iid sequence sampled from $U([0, 1]^d)$, and $N = \sum_{i=1}^K N_i$. Note that each algorithm $\overline{MC}_N(f)$, $MC_N(f)$ uses N (random) evaluations of f . From the Cauchy-Schwarz inequality

$$(I(f))^2 = \left(\sum_{i=1}^K (\text{vol}(D_i))^{1/2} \cdot \frac{I_i(f)}{(\text{vol}(D_i))^{1/2}} \right)^2 \leq \sum_{i=1}^K \text{vol}(D_i) \cdot \sum_{i=1}^K \frac{(I_i(f))^2}{\text{vol}(D_i)} = \sum_{i=1}^K \frac{(I_i(f))^2}{\text{vol}(D_i)}, \quad (3.104)$$

since $\sum_{i=1}^K \text{vol}(D_i) = \text{vol}(D) = 1$. Hence

$$\|I(f) - MC_N(f)\|_{L^2(\Omega)} = \frac{(I(f^2) - (I(f))^2)^{1/2}}{\sqrt{N}} \geq \frac{\left(I(f^2) - \sum_{i=1}^K \frac{(I_i(f))^2}{\text{vol}(D_i)} \right)^{1/2}}{\sqrt{N}} = \|I(f) - \overline{MC}_N(f)\|_{L^2(\Omega)}, \quad (3.105)$$

and the error of $\overline{MC}_N(f)$ is not bigger than the error of $MC_N(f)$. Moreover, by the properties of Cauchy-Schwarz inequality, the equality in (3.105) holds if and only if $I_i(f)/\text{vol}(D_i) = \text{const}$ for $i = 1, \dots, K$.

We now consider the following (exemplary) class of Hölder continuous functions $f : [0, 1]^d \rightarrow \mathbb{R}$ in order to see how much we can gain by using stratified sampling. Namely, let

$$\mathcal{F}_L^{\varrho, d} = \{f : [0, 1]^d \rightarrow \mathbb{R} \mid |f(x) - f(y)| \leq L \cdot \|x - y\|_{\infty}^{\varrho}, \ x, y \in [0, 1]^d\}, \quad (3.106)$$

where $L \in (0, +\infty)$, $\varrho \in (0, 1]$ are the class parameters and $\|\cdot\|_{\infty}$ is the maximum norm.

We have

$$\text{Var}(f(\xi_j^i)) = \mathbb{E} \left(f(\xi_j^i) - \mathbb{E}(f(\xi_j^i)) \right)^2 = \mathbb{E} \left(f(\xi_j^i) - \frac{I_i(f)}{\text{vol}(D_i)} \right)^2 = \frac{1}{\text{vol}(D_i)} I_i \left[(f - A_i)^2 \right], \quad (3.107)$$

where $A_i = \frac{I_i(f)}{\text{vol}(D_i)}$. By the mean value theorem 13.0.8 we have that for all $i = 1, 2, \dots, K$ there exist $\bar{x}_i \in D_i$ such that $A_i = f(\bar{x}_i)$. Hence

$$I_i \left[(f - A_i)^2 \right] = \int_{D_i} |f(x) - f(\bar{x}_i)|^2 dx \leq L^2 \int_{D_i} \|x - \bar{x}_i\|_{\infty}^{2\varrho} dx \leq L^2 \cdot \text{vol}(D_i) \cdot \sup_{x \in D_i} \|x - \bar{x}_i\|_{\infty}^{2\varrho}, \quad (3.108)$$

which in turn implies that

$$\text{Var}(f(\xi_j^i)) \leq L^2 \cdot \sup_{x \in D_i} \|x - \bar{x}_i\|_{\infty}^{2\varrho}, \quad (3.109)$$

$$\text{Var}(\overline{MC}_{N_i}^{(i)}(f)) = \frac{\text{vol}^2(D_i)}{N_i^2} \sum_{j=1}^{N_i} \text{Var}(f(\xi_j^i)) \leq \frac{L^2 \text{vol}^2(D_i)}{N_i} \cdot \sup_{x \in D_i} \|x - \bar{x}_i\|_{\infty}^{2\varrho}, \quad (3.110)$$

and finally, since $N_i = \text{vol}(D_i) \cdot N$,

$$\mathbb{E}|I(f) - \overline{MC}_N(f)|^2 = \sum_{i=1}^K \text{Var}(\overline{MC}_{N_i}^{(i)}(f)) \leq \frac{L^2}{N} \sum_{i=1}^K \text{vol}(D_i) \cdot \sup_{x \in D_i} \|x - \bar{x}_i\|_{\infty}^{2\varrho}. \quad (3.111)$$

Let us divide D into $K = N$ sub-boxes D_i , each of the edge length equal to $N^{-1/d}$, and where we assume that $N^{1/d} \in \mathbb{N}$. Hence $\text{vol}(D_i) = (N^{-1/d})^d = 1/N$ and we have that $N_i = 1$ for $i = 1, 2, \dots, N$. It means that we are using only one random sample on each D_i . Thus in this case the stratified sampling Monte Carlo method (3.92) has the form

$$\overline{MC}_N(f) = \frac{1}{N} \sum_{i=1}^N f(\xi_1^i) \quad (3.112)$$

where $\xi_1^1, \dots, \xi_1^N$ are independent random vectors and $\xi_1^i \sim U(D_i)$. Moreover for $i = 1, 2, \dots, N$

$$\sup_{x \in D_i} \|x - \bar{x}_i\|_\infty^{2\varrho} \leq N^{-2\varrho/d}, \quad (3.113)$$

and

$$\left(\mathbb{E} |I(f) - \overline{MC}_N(f)|^2 \right)^{1/2} \leq LN^{-(\frac{\varrho}{d} + \frac{1}{2})}, \quad (3.114)$$

where $\overline{MC}_N(f)$ uses N values of f .

Remark 3.3.4. *The most visible gain of using the stratified sampling Monte Carlo algorithm is for $d = 1$, since in the class $\mathcal{F}_L^{\varrho, d}$ its $L^2(\Omega)$ -error is $O(N^{-(\varrho+1/2)})$, the error of the crude Monte Carlo method is $O(N^{-1/2})$ while the error of the deterministic rectangle quadrature rule is $O(N^{-\varrho})$.*

Remark 3.3.5. *In the Hölder class $\mathcal{F}_L^{\varrho, d}$ the upper bound on the $L^2(\Omega)$ -error for the both methods $\overline{MC}_N(f)$ and $\widetilde{MC}_N(f)$ is the same, however, the algorithm $\widetilde{MC}_N(f)$ uses two times more function evaluations than $\overline{MC}_N(f)$.*

3.4 There exists deterministic quadrature rule as good as crude Monte Carlo method

There is an interesting consequence of Theorem 3.2.1 and mean-value theorem for integrals. Namely, combining those two we can show that for every $f \in C([0, 1]^d)$ and $N \in \mathbb{N}$ there exist nodes

$$x_1^*, x_2^*, \dots, x_N^* \in [0, 1]^d \quad (3.115)$$

such that

$$\left| I(f) - \frac{1}{N} \sum_{j=1}^N f(x_j^*) \right| = \sigma(f)/\sqrt{N}. \quad (3.116)$$

This means that the deterministic quadrature rule $Q_N^{\text{det}}(f) = \frac{1}{N} \sum_{j=1}^N f(x_j^*)$ achieves the error $O(N^{-1/2})$, the same as the crude Monte Carlo method MC_N , but by using only deterministic evaluations of f .

To show (3.116) note the joint law $\mu_{(\xi_1, \dots, \xi_N)}$ of independent identically distributed random variables $\xi_1, \dots, \xi_N$ ($\xi_i \sim U([0, 1]^d)$) is the product measure $\mu_{(\xi_1, \dots, \xi_N)} = \mu_{\xi_1} \times \dots \times \mu_{\xi_N} = \lambda_d \times \dots \times \lambda_d$. Hence, by Theorems 3.2.1, 13.0.7 we get

$$\sigma^2(f)/N = \mathbb{E} |I(f) - MC_N(f)|^2 = \int_{\substack{[0,1]^d \times \dots \times [0,1]^d \\ N\text{-times}}} \left| I(f) - \frac{1}{N} \sum_{j=1}^N f(x_j) \right|^2 (\lambda_d \times \dots \times \lambda_d)(dx_1, \dots, dx_N). \quad (3.117)$$

Since the function

$$[0, 1]^d \times \dots \times [0, 1]^d \ni (x_1, \dots, x_N) \rightarrow \left| I(f) - \frac{1}{N} \sum_{j=1}^N f(x_j) \right|^2 \in [0, +\infty) \quad (3.118)$$

is continuous, we get by Theorem 13.0.8 that there exist points (3.115) such that

$$\int_{\substack{[0,1]^d \times \dots \times [0,1]^d \\ N\text{-times}}} \left| I(f) - \frac{1}{N} \sum_{j=1}^N f(x_j) \right|^2 (\lambda_d \times \dots \times \lambda_d)(dx_1, \dots, dx_N) = \left| I(f) - \frac{1}{N} \sum_{j=1}^N f(x_j^*) \right|^2. \quad (3.119)$$

(Note that $(\lambda_d \times \dots \times \lambda_d)([0, 1]^d \times \dots \times [0, 1]^d) = 1$.)

The error of Q_N^{det} is the same as for MC_N . However, the nodes (3.115) (that depend on f) are not known (we have only established the existence of such points), and the method Q_N^{det} cannot be used in practice. The result (3.116) is of a purely theoretical nature. Nevertheless, the technique described above is often used for nonconstructive proving of the existence of deterministic quadrature rules and it is called *probabilistic method*.

3.5 Some generalizations

We have discussed in detail the crude Monte Carlo method for (2.1). Of course, this is not the most general problem that we can address.

3.5.1 Weighted integration

We consider the problem of approximating the following weighted integral

$$I_w(f) = \int_{\mathbb{R}^d} w(x) f(x) dx, \quad (3.120)$$

where we assume that $f : \mathbb{R}^d \rightarrow \mathbb{R}$, $w : \mathbb{R}^d \rightarrow [0, +\infty)$ are Borel measurable and

$$W := \int_{\mathbb{R}^d} w(x) dx < +\infty, \quad (3.121)$$

$$\max \left\{ \int_{\mathbb{R}^d} w(x) |f(x)| dx, \int_{\mathbb{R}^d} w(x) |f(x)|^2 dx \right\} < +\infty. \quad (3.122)$$

In this case, the function $\mathbb{R}^d \ni x \rightarrow g(x) := w(x)/W$ is a probability density function, that is absolutely continuous with respect to the d -dimensional Lebesgue measure λ_d . Such integrals often appear in mathematical finances and option pricing.

Assuming that we can sample from g we define

$$MC_N(f) = \frac{W}{N} \sum_{j=1}^N f(\xi_j), \quad (3.123)$$

where $(\xi_j)_{j \in \mathbb{N}}$ is an iid sequence of random vectors with the common density g . It is easy to see that

$$\mathbb{E}(MC_N(f)) = I_w(f), \quad (3.124)$$

and

$$Var(MC_N(f)) = \frac{1}{N} \sigma_w^2(f), \quad (3.125)$$

where

$$\sigma_w^2(f) = W \cdot I_w(f^2) - (I_w(f))^2. \quad (3.126)$$

This implies that

$$\|I_w(f) - MC_N(f)\|_{L^2(\Omega)} = \frac{\sigma_w(f)}{\sqrt{N}}, \quad (3.127)$$

for all $N \in \mathbb{N}$.

Example 3.5.1. For the weighted integral

$$I_w(f) = \int_{\mathbb{R}} e^{-x^2/2} f(x) dx, \quad (3.128)$$

the crude Monte Carlo method has the form

$$MC_N(f) = \frac{\sqrt{2\pi}}{N} \sum_{j=1}^N f(\xi_j), \quad (3.129)$$

where $(\xi_j)_{j \in \mathbb{N}}$ are independently sampled from the standard normal distribution $N(0, 1)$.

Example 3.5.2. We apply weighted integration to the estimation of the Lebesgue measure of compact subsets in $\mathbb{R}^d$. For a compact subset D of $\mathbb{R}^d$ let M be a d -dimensional cube such that $D \subset M$. We assume that the volume $\text{vol}(M) = \lambda_d(M)$ is known. We take $f = \mathbf{1}_D$ and $w = \mathbf{1}_M$. Then

$$I_w(f) = \lambda_d(D). \quad (3.130)$$

Hence, if we take $(\xi_j)_{j \in \mathbb{N}}$ as an iid sequence of random vectors that are uniformly distributed on the cube M , then we have with probability one

$$MC_N(f) = \frac{\text{vol}(M)}{N} \sum_{j=1}^N \mathbf{1}_D(\xi_j) \rightarrow \lambda_d(D), \quad (3.131)$$

as $N \rightarrow +\infty$.

3.5.2 Abstract setting

Let μ be a probabilistic measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, i.e., $\mu(\mathbb{R}^d) = 1$. For $f \in L^2(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mu)$ we consider the problem of Monte Carlo approximation of the following Lebesgue integral

$$I(f) = \int_{\mathbb{R}^d} f(x) \mu(dx). \quad (3.132)$$

The crude Monte Carlo estimator of (3.132) is analogous to (3.12), namely

$$MC_N(f) = \frac{1}{N} \sum_{j=1}^N f(\xi_j), \quad (3.133)$$

where this time $(\xi_j)_{j \in \mathbb{N}}$ is an i.i.d sequence of random variables with distribution μ . We stress that we can use (3.133) only in the case when we can sample from the distribution μ . Note that the weighted integration problem, described in the previous section, is a special case of (3.132) with $\mu(dx) = \frac{1}{W} w(x) \lambda_d(dx)$, since

$$\int_{\mathbb{R}^d} f(x) \mu(dx) = \int_{\mathbb{R}^d} \frac{w(x)}{W} f(x) dx. \quad (3.134)$$

We can go even further. Let us switch from $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ into arbitrary measurable space $(X, \mathcal{A})$ and let us consider the random variable ξ with values in X , i.e., Σ -to- $\mathcal{A}$ measurable mapping

$$\xi : \Omega \rightarrow X. \quad (3.135)$$

The law of ξ is given by

$$\mu(A) = \mathbb{P}(\{\xi \in A\}) = \mathbb{P}(\xi^{-1}(A)), \quad (3.136)$$

for all $A \in \mathcal{A}$. Then the space $(X, \mathcal{A}, \mu)$ is a measure space equipped with the probability measure μ ($\mu(X) = 1$). Let $f : X \rightarrow \mathbb{R}$ be a function from $L^2(X, \mathcal{A}, \mu)$. Note that by Theorem 13.0.7 we get

$$\mathbb{E}(f(\xi)) = \int_{\Omega} (f \circ \xi)(\omega) \mathbb{P}(d\omega) = I(f), \quad (3.137)$$

where this time

$$I(f) = \int_X f(x) \mu(dx). \quad (3.138)$$

(Since $\mu(X) = 1$, we get by the Hölder inequality $I(|f|) \leq (I(f^2))^{1/2} < +\infty$.) The Monte Carlo method approximating (3.138) is again defined analogously as before, i.e.,

$$MC_N(f) = \frac{1}{N} \sum_{j=1}^N f(\xi_j), \quad (3.139)$$

where $(\xi_j)_{j \in \mathbb{N}}$ is an i.i.d sequence of X -valued random variables with distribution μ . Again $(f(\xi_j))_{j \in \mathbb{N}}$ are identically distributed and independent real random variables. Hence, by Theorem 3.0.1, $MC_N(f) \rightarrow I(f)$ almost surely as $N \rightarrow +\infty$, and by using the same proof technique as for Theorem 3.2.1 we get for all $N \in \mathbb{N}$

$$\left(\mathbb{E} |I(f) - MC_N(f)|^2 \right)^{1/2} = \sigma(f)/\sqrt{N}, \quad (3.140)$$

where $\sigma^2(f) = I(f^2) - (I(f))^2$. Moreover, it is easy to see that the following generalizations of Theorems 3.2.7, 3.2.8 are true.

Theorem 3.5.3. *Let $f \in L^p(X, \mathcal{A}, \mu)$ for some $p \in [2, +\infty)$. Then for*

$$K_p(f) = C_p \cdot \left(\int_X |f(x) - I(f)|^p \mu(dx) \right)^{1/p} < +\infty, \quad (3.141)$$

where C_p is the constant from the BDG inequality, and for all $N \in \mathbb{N}$ it holds

$$\left(\mathbb{E} |I(f) - MC_N(f)|^p \right)^{1/p} \leq K_p(f) \cdot N^{-1/2}, \quad (3.142)$$

and

$$\left(\mathbb{E} |I(f) - MC_N(f)|^q \right)^{1/q} \leq K_p(f) \cdot N^{-1/2} \quad (3.143)$$

for all $q \in (0, p]$.

Theorem 3.5.4. *Let $f \in \bigcap_{p \in [2, +\infty)} L^p(X, \mathcal{A}, \mu)$. Then for all $\varepsilon \in (0, 1/2)$ there exists $\eta_\varepsilon(f) \in \bigcap_{p \in [1, +\infty)} L^p(\Omega)$ such that*

$$\mathbb{P} \left(\bigcap_{N \in \mathbb{N}} \left\{ |I(f) - MC_N(f)| \leq \eta_\varepsilon(f) \cdot N^{-\frac{1}{2} + \varepsilon} \right\} \right) = 1. \quad (3.144)$$

However, to use (3.139) we have to be able to generate realizations $\xi(\omega)$ of the random variable ξ with values in the abstract set X . This is a much harder task than sampling from the finite-dimensional distribution as it was for (3.133). One possible solution is to construct a random variable

$$\hat{\xi} : \Omega \rightarrow X, \quad (3.145)$$

with a simpler structure than ξ , that approximates ξ in a suitable way, and (what is the most important) for which we have access to the values (samples) $\hat{\xi}(\omega) \in X$. Such a construction might be highly nontrivial

and involve additional computational costs. We will face such a situation in future lectures, where ξ will be a solution of the underlying stochastic differential equation, while $\hat{\xi}$ will be a suitable approximation of ξ . (For example, $\hat{\xi}$ might be the output of the Euler or Milstein method.) Then approximate computation of $\mathbb{E}(f(\xi))$ corresponds to the weak approximation of a solution of SDE.

Now, let

$$\widehat{MC}_N(f) = \frac{1}{N} \sum_{j=1}^N f(\hat{\xi}_j), \quad (3.146)$$

where $(\hat{\xi}_j)_{j \in \mathbb{N}}$ is an i.i.d sequence of X -valued random variables with distribution $\hat{\mu}$, where $\hat{\mu}$ is a law of $\hat{\xi}$. Let us now additionally assume that

$$\mathbb{E}|f(\hat{\xi})|^2 = \int_X |f(x)|^2 \hat{\mu}(dx) < +\infty. \quad (3.147)$$

Then (3.146) is a crude Monte Carlo method that approximates

$$\mathbb{E}(f(\hat{\xi})) = \int_X f(x) \hat{\mu}(dx), \quad (3.148)$$

and

$$\left(\mathbb{E}|\mathbb{E}(f(\hat{\xi})) - \widehat{MC}_N(f)|^2 \right)^{1/2} = \hat{\sigma}(f)/\sqrt{N}, \quad (3.149)$$

where

$$\hat{\sigma}^2(f) = \int_X |f(x)|^2 \hat{\mu}(dx) - \left(\int_X f(x) \hat{\mu}(dx) \right)^2. \quad (3.150)$$

Furthermore

$$\begin{aligned} \left(I(f) - \widehat{MC}_N(f) \right)^2 &= \left(\left(\mathbb{E}(f(\xi)) - \mathbb{E}(f(\hat{\xi})) \right) + \left(\mathbb{E}(f(\hat{\xi})) - \widehat{MC}_N(f) \right) \right)^2 \\ &= \left(\mathbb{E}(f(\xi)) - \mathbb{E}(f(\hat{\xi})) \right)^2 + \left(\mathbb{E}(f(\hat{\xi})) - \widehat{MC}_N(f) \right)^2 \\ &\quad + 2 \left(\mathbb{E}(f(\xi)) - \mathbb{E}(f(\hat{\xi})) \right) \cdot \left(\mathbb{E}(f(\hat{\xi})) - \widehat{MC}_N(f) \right). \end{aligned} \quad (3.151)$$

Since $\mathbb{E}(\widehat{MC}_N(f)) = \mathbb{E}(f(\hat{\xi}))$, by (3.149) we arrive at

$$\mathbb{E} \left(I(f) - \widehat{MC}_N(f) \right)^2 = |\mathbb{E}(f(\xi)) - \mathbb{E}(f(\hat{\xi}))|^2 + \frac{\hat{\sigma}^2(f)}{N}. \quad (3.152)$$

The term

$$|\mathbb{E}(f(\xi)) - \mathbb{E}(f(\hat{\xi}))| = \left| \int_X f(x) \mu(dx) - \int_X f(x) \hat{\mu}(dx) \right| \quad (3.153)$$

is the so-called *weak error* and, roughly speaking, it measures how accurate the law $\hat{\mu}$ approximates μ . Hence, the mean square error (3.152) consists of two types of errors: the weak error and the error of Monte Carlo simulation.

We always have

$$|\mathbb{E}(f(\xi)) - \mathbb{E}(f(\hat{\xi}))| \leq \mathbb{E}|f(\xi) - f(\hat{\xi})|, \quad (3.154)$$

and sometimes this inequality might be used to provide a further upper bound on the weak error. For example, in the case when (X, d) is a separable metric space (what will be the most common case) equipped with the

Borel σ -field $\mathcal{A} = \mathcal{B}(X)$, let us assume that f is Lipschitz functional, i.e., there exists $L \in (0, +\infty)$ such that for all $x, y \in X$

$$|f(x) - f(y)| \leq Ld(x, y). \quad (3.155)$$

(For instance, f might be the payoff of the European option.) Then we have that

$$\mathbb{E}|f(\xi) - f(\hat{\xi})| \leq L\mathbb{E}(d(\xi, \hat{\xi})). \quad (3.156)$$

(Note that $d(\xi, \hat{\xi})$ is Σ -to- $\mathcal{B}([0, +\infty))$ measurable non-negative function, see Exercises 3.9 at the end of this chapter.) The value of $\mathbb{E}(d(\xi, \hat{\xi}))$ is the average distance between trajectories of ξ and $\hat{\xi}$, and it is called *strong error*. (Compared to the weak error, it relies on the joint law $\mu_{(\xi, \hat{\xi})}$ of $(\xi, \hat{\xi})$.) Hence in this case we get

$$\left(\mathbb{E}|I(f) - \widehat{MC}_N(f)|^2\right)^{1/2} \leq L\mathbb{E}(d(\xi, \hat{\xi})) + \hat{\sigma}(f)/\sqrt{N}. \quad (3.157)$$

Note, however, that for more smooth functions f it is possible to get better upper bounds not by using (3.154) but by estimating directly the weak error.

The setting described above for Monte Carlo simulations is general and might look abstract. However, in future lectures, it will turn out that this setting is a natural and convenient framework for a problem of weak approximation of solutions of SDEs, and, as a result, for option pricing.

3.6 There is always enough randomness

For the Monte Carlo methods, we always use sequences of independent random variables or random elements with values in metric spaces. Then the natural question arises if a probability space is large enough to carry such infinite sequences.

For the proof of these fundamental results see [46] or page 243. in [51].

Theorem 3.6.1. (*Ulam - Lomnicki theorem*) *Let T be an arbitrary set. Let for every $t \in T$ a measurable space $(E_t, \mathcal{E}_t)$ and probability measure $\mathbb{P}_t$ on it be given. Then there exists a probability space $(\Omega, \Sigma, \mathbb{P})$ and independent random elements $(X_t)_{t \in T}$ such that each $X_t : \Omega \rightarrow E_t$ is $\Sigma/\mathcal{E}_t$ -measurable and $\mathbb{P}(\{X_t \in B\}) = \mathbb{P}_t(B)$ for all $B \in \mathcal{E}_t$.*

3.7 Implementation issues

We present a simple implementation of the crude Monte Carlo method for the following integral

$$\int_a^b f(x)dx. \quad (3.158)$$

First, we will make a suitable change of variable and we get that

$$\int_a^b f(x)dx = \int_0^1 g(x)dx, \quad (3.159)$$

with $g(x) = (b-a)f(a + (b-a)x)$. In the Python implementation, we use the NumPy module and, the so-called, *vectorization*.

```
import numpy as np

def quad_MC(f, a, b, N):
    random_samples_0 = np.random.uniform(0, 1, N).astype('float32')
```

```

random_samples_0 = a + (b - a) * random_samples_0
g_values = (b-a) * f(random_samples_0)

MC_g = np.mean(g_values)

random_samples_1 = np.power(g_values - MC_g, 2)
Var_g = (1.0 / (N - 1)) * np.sum(random_samples_1)

L_CI = MC_g - 2.0 * np.sqrt(Var_g / N)
R_CI = MC_g + 2.0 * np.sqrt(Var_g / N)

return MC_g, Var_g, L_CI, R_CI

```

3.8 Concluding remarks

As summary, we recall after [67] three mandatory steps while performing Monte Carlo simulation to approximate $\mathbb{E}(f(\xi))$ for a given f :

1. Specification of a confidence level $\alpha \in (0, 1)$, that should be close to 1.
2. Simulation of an N -sample $\xi_1, \dots, \xi_N$ of i.i.d random vectors having the same distribution μ as ξ and (possibly recursive) computation of both its empirical mean $MC_N(f)$ and its empirical variance $\widehat{\sigma}_N^2(f)$.
3. Computation of the resulting empirical confidence interval $\hat{J}_N^\alpha(f)$ at the confidence level α . This is the only trustworthy output of the performed Monte Carlo simulation.

Remark 3.8.1. *References for MC method in literature...*

3.9 Exercises

1. Show that $(f(\xi_j))_{j \in \mathbb{N}}$ is the sequence of identically distributed and independent random variables.
2. Show that $Var(f(\xi_1)) = \sigma^2(f)$.
3. Let $(A_j)_{j \in \mathbb{N}} \subset \Sigma$ be a sequence of events such that $\mathbb{P}(A_j) = 1$ for all $j \in \mathbb{N}$. Show that $\mathbb{P}\left(\bigcap_{j \in \mathbb{N}} A_j\right) = 1$.
4. Construct a crude Monte-Carlo method that approximates the integral

$$I(f) = \int_{\mathbb{R}^d} f(x) e^{-\|x\|_2^2/2} dx, \quad (3.160)$$

for a Borel measurable functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that belong to the weighted $L^2(\mathbb{R}^d; e^{-\|x\|_2^2/2})$ space. (Such integrals often appear in mathematical finance.)

Hint. Note that

$$I(f) = (2\pi)^{d/2} \cdot \int_{\mathbb{R}^d} f(x) \mu(dx), \quad (3.161)$$

where $\mu(A) = \int_A \frac{1}{(2\pi)^{d/2}} e^{-\|x\|_2^2/2} \lambda_d(dx)$, $A \in \mathcal{B}(\mathbb{R}^d)$, is the Gaussian measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

5. (*Option pricing in the Black-Scholes model*) Construct Monte Carlo estimator of the following expected value

$$\mathbb{E}((S(T) - K)^+) \quad (3.162)$$

for $K \in [0, +\infty)$ and $S(t) = S(0) \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W(t)\right)$, $t \in [0, T]$, with $S(0) > 0$, $\mu \in \mathbb{R}$, $\sigma > 0$. Recall that $(W(t))_{t \in [0, +\infty)}$ is a one-dimensional Wiener process, see Chapter 5.

6. Give a proof of Theorem 3.2.1 (iv).
7. Show (3.30), (3.31), and (3.32) in full details.
8. Show that the value of (3.139) is independent of the choice of the representation of the equivalence class $f \in L^2(X, \mathcal{A}, \mu)$ in the following sense: let f, g be two representations of the same equivalence class in $L^2(X, \mathcal{A}, \mu)$. Then

$$\mathbb{P}\left(\bigcap_{N \in \mathbb{N}} \{MC_N(f) = MC_N(g)\}\right) = 1. \quad (3.163)$$

9. Justify that the mapping

$$\Omega \ni \omega \rightarrow d(\xi(\omega), \hat{\xi}(\omega)) \in [0, +\infty) \quad (3.164)$$

is Σ -to- $\mathcal{B}(\mathbb{R})$ measurable.

Hint. Firstly show that the function $X \times X \ni (x, y) \rightarrow d(x, y) \in [0, +\infty)$ is continuous, and therefore $\mathcal{B}(X \times X)/\mathcal{B}(\mathbb{R})$ -measurable. Then use the fact that for a separable metric space X we have $\mathcal{B}(X \times X) = \mathcal{B}(X) \otimes \mathcal{B}(X)$.

10. Give a proof of the following lemma (see Lemma 5.1 in [27]).

Lemma 3.9.1. *Given a sequence of real-valued, independent and identically distributed random variables $\{Z_n\}_{n \in \mathbb{N}_0}$ with $\mathbb{P}(Z_1 > 0) = 1$, consider the sequence of random variables $\{Y_n\}_{n \in \mathbb{N}}$ defined by*

$$Y_n = \left(\prod_{i=0}^{n-1} Z_i\right) Y_0, \quad (3.165)$$

where $\mathbb{P}(Y_0 > 0) = 1$. If $\ln(Z_1)$ is integrable, then

$$\mathbb{E}(\ln(Z_1)) < 0 \Rightarrow \lim_{n \rightarrow +\infty} Y_n = 0, \text{ with probability } 1 \Rightarrow \mathbb{E}(\ln(Z_1)) \leq 0. \quad (3.166)$$

Hint. Use the fact that $\ln(Y_n) = \ln(Y_0) + S_n$ with $S_n = \sum_{i=0}^{n-1} \ln Z_i$.

11. Show that $(Z_j, \mathcal{G}_j)_{j=0,1,\dots,N}$, defined in (3.44), is a discrete-time martingale such that $\max_{1 \leq j \leq N} \mathbb{E}|Z_j|^p < +\infty$.
12. Investigate the $L^p(\Omega)$ -error, $p \in [2, +\infty)$, for $\widetilde{MC}_{N,N}$ defined in (3.68).
13. Construct empirical confidence intervals for $\widetilde{MC}_{N,N}$.
14. Construct empirical confidence intervals for $MC_N(f)$ defined in (3.139).
15. $f \in C^r([0, 1])$ with ϱ -Hölder continuous r th derivative $f^{(r)}$, $r \in \mathbb{N}_0$, $\varrho \in (0, 1]$. Investigate the mean square error of $\widetilde{MC}_{N,n}$ when the control variate $g_n : [0, 1] \rightarrow \mathbb{R}$ is a piecewise Taylor polynomial

$$g_n(x) = \sum_{i=0}^{n-1} \mathbf{1}_{[x_i, x_{i+1})}(x) \cdot \sum_{j=0}^r \frac{f^{(j)}(x_i)}{j!} (x - x_i)^j, \quad (3.167)$$

for $x_i = i/n$, $i = 0, 1, \dots, n$.

16. Let $f \in C^r([0, 1])$, $r \in \mathbb{N}$. As a control variate, we take the following piecewise Taylor polynomial

$$g_n(x) = \sum_{i=0}^{n-1} \mathbf{1}_{[x_i, x_{i+1})}(x) \cdot \sum_{j=0}^{r-1} \frac{f^{(j)}(x_i)}{j!} (x - x_i)^j, \quad x \in [a, b] \quad (3.168)$$

for $x_i = i/n$, $i = 0, 1, \dots, n$. For $\widetilde{MC}_{n,n}$ show that

$$\lim_{n \rightarrow +\infty} n^{r+\frac{1}{2}} \cdot \|I(f) - \widetilde{MC}_{n,n}(f)\|_{L^2(\Omega)} = C_{MC}(f), \quad (3.169)$$

where

$$C_{MC}(f) = \left[\frac{1}{(r!)^2 \cdot (2r+1)} \int_0^1 (f^{(r)}(t))^2 dt - \frac{1}{((r+1)!)^2} (f^{(r-1)}(1) - f^{(r-1)}(0))^2 \right]^{1/2}. \quad (3.170)$$

Chapter 4

Deterministic vs randomized algorithms for ODEs

This chapter shows how to define randomized algorithms for approximate solving of ordinary differential equations. We consider initial-value problems of the following form

$$\begin{cases} z'(t) = f(t, z(t)), & t \in [a, b], \\ z(a) = \xi, \end{cases} \quad (4.1)$$

where $d \in \mathbb{N}$, $\xi \in \mathbb{R}^d$, $-\infty < a < b < +\infty$. In $\mathbb{R}^d$ we consider the Euclidean norm $\|\cdot\|$. We assume that $f \in C([a, b] \times \mathbb{R}^d; \mathbb{R}^d)$ is of at most linear growth and satisfies the local Lipschitz condition. We consider explicit randomized Euler and Runge-Kutta schemes.

We impose the following assumptions on the right-hand side function f :

(A1) $f \in C([a, b] \times \mathbb{R}^d; \mathbb{R}^d)$,

(A2) there exists $K \in [0, +\infty)$ such that for all $t \in [a, b]$, $y \in \mathbb{R}^d$

$$\|f(t, y)\| \leq K(1 + \|y\|), \quad (4.2)$$

(A3) for every $R > 0$ there exists $L_R \in [0, +\infty)$ such that for all $t \in [a, b]$, and all $x, y \in \mathbb{R}^d$ satisfying $\|x\| \leq R$, $\|y\| \leq R$ it holds

$$\|f(t, x) - f(t, y)\| \leq L_R \|x - y\|, \quad (4.3)$$

(A4) there exists $\varrho \in (0, 1]$ such that for every $H > 0$ there exists $L_H \in [0, +\infty)$ such that for all $t, s \in [a, b]$, and all $x \in \mathbb{R}^d$ satisfying $\|x\| \leq H$ it holds

$$\|f(t, x) - f(s, x)\| \leq L_H |t - s|^\varrho. \quad (4.4)$$

Therefore $f = f(t, x)$ is globally continuous, of at most linear growth, it satisfies local Lipschitz assumption with respect to x , and it is globally Hölder continuous with respect to t .

Example 4.0.1. The exemplary function satisfying (A1)-(A4) is

$$f(t, x) = x \cdot \sin(x^2 \cdot |t|^\varrho), \quad (t, x) \in [a, b] \times \mathbb{R}. \quad (4.5)$$

In this section, we will use the following version of Peano's theorem, see, for example, Theorem 70.4, page 292, in [23] for its proof. (For further comments on the existence of solutions of (4.1) check Remarks 4.0.5, 4.0.6).

Theorem 4.0.2. ([23]) *If the function f satisfies Assumptions (A1) and (A2), then the initial value problem (4.1) has at least one solution on $[a, b]$.*

Lemma 4.0.3. *Let $\xi \in \mathbb{R}^d$.*

(i) *Let us assume that f satisfies the assumptions (A1)-(A3). Then there exists a unique solution $z : [a, b] \rightarrow \mathbb{R}^d$, $z \in C^1([a, b] \times \mathbb{R}^d; \mathbb{R}^d)$, of (4.1). Moreover*

$$\sup_{a \leq t \leq b} \|z(t)\| \leq \max\{1, K(b-a)\} e^{K(b-a)} (1 + \|\xi\|), \quad (4.6)$$

and for all $s, t \in [a, b]$

$$\|z(t) - z(s)\| \leq K \left(1 + \max\{1, K(b-a)\} e^{K(b-a)} \right) (1 + \|\xi\|) |t - s|. \quad (4.7)$$

(ii) *Let us assume that f satisfies the assumptions (A1)-(A4). Then there exists $L_1 = L_1(a, b, K, \xi) \in [0, +\infty)$ such that for all $t, s \in [a, b]$*

$$\|z'(t) - z'(s)\| \leq L_1 |t - s|^\varrho. \quad (4.8)$$

Proof. Let us assume that f satisfies the assumptions (A1)-(A3). According to Theorem 4.0.2 under the conditions (A1), (A2) the equation (4.1) has at least one solution $z \in C^1([a, b] \times \mathbb{R}^d; \mathbb{R}^d)$. For all $t \in [a, b]$ we have

$$\|z(t)\| \leq \|\xi\| + K(b-a) + \int_a^t \|z(s)\| ds, \quad (4.9)$$

and by Gronwall's lemma 13.0.11 we get (4.6). Moreover, for all $t, s \in [a, b]$

$$\|z(t) - z(s)\| = \|z(t \vee s) - z(t \wedge s)\| \leq \int_{t \wedge s}^{t \vee s} \|f(u, z(u))\| du \leq K \left(1 + \sup_{a \leq t \leq b} \|z(t)\| \right) |t - s|, \quad (4.10)$$

and from (4.6) we get (4.7).

Let z and $\tilde{z}$ be two solutions of (4.1). From (4.6) we have that $\max\{\|z(t)\|, \|\tilde{z}(t)\|\} \leq R$ for all $t \in [a, b]$, where $R = \max\{1, K(b-a)\} e^{K(b-a)} (1 + \|\xi\|)$. Then, by the assumption (A3) we have that there exists $L_R \in [0, +\infty)$ such that for all $t \in [a, b]$

$$\|z(t) - \tilde{z}(t)\| \leq \int_a^t \|f(s, z(s)) - f(s, \tilde{z}(s))\| ds \leq L_R \int_a^t \|z(s) - \tilde{z}(s)\| ds. \quad (4.11)$$

Again, by Gronwall's lemma 13.0.11 for all $t \in [a, b]$ we have $z(t) = \tilde{z}(t)$, and the uniqueness of the solution z follows. Finally, let us assume that f satisfies the assumptions (A1)-(A4). From (4.6) we have that $\|z(t)\| \leq R$ for all $t \in [a, b]$, where $R = H = \max\{1, K(b-a)\} e^{K(b-a)} (1 + \|\xi\|)$. Hence, by (A3), (A4), (4.6), (4.7) we have that there exist $L_R, L_H \in [0, +\infty)$ such that for all $t, s \in [a, b]$

$$\begin{aligned} \|z'(t) - z'(s)\| &= \|f(t, z(t)) - f(s, z(s))\| \leq \|f(t, z(t)) - f(t, z(s))\| + \|f(t, z(s)) - f(s, z(s))\| \\ &\leq L_R \|z(t) - z(s)\| + L_H |t - s|^\varrho. \end{aligned} \quad (4.12)$$

This and the fact that $|t - s|^{1-\varrho} \leq 1 + |t - s| \leq 1 + b - a$ implies (4.8). $\square$

The following result states that the solution z depends on the initial value ξ in a continuous-like way. Note that the L_2 in the proposition below depends on the initial value, which is due to the local Lipschitz condition for f . This is not the case when f satisfies the Lipschitz condition globally.

Proposition 4.0.4. *Let us assume that f satisfies (A1)-(A3). Let $z = z(\xi, f)$ denotes the solution of (4.1) under the initial value $\xi \in \mathbb{R}^d$, while $y = y(\eta, f)$ the solution of (4.1) with the initial-value $\eta \in \mathbb{R}^d$. Then there exists $L_2 = L_2(a, b, K, \xi, \eta) \in [0, +\infty)$ such that for all $t \in [a, b]$*

$$\|z(t) - y(t)\| \leq e^{L_2(t-a)} \|\eta - \xi\|. \quad (4.13)$$

Proof. By Lemma 4.0.3 (i) we have that

$$\sup_{a \leq t \leq b} \|z(t)\| \leq R_1 := \max\{1, K(b-a)\} e^{K(b-a)} (1 + \|\xi\|), \quad (4.14)$$

and

$$\sup_{a \leq t \leq b} \|y(t)\| \leq R_2 := \max\{1, K(b-a)\} e^{K(b-a)} (1 + \|\eta\|). \quad (4.15)$$

Hence, for $R = R_1 \vee R_2$ there exists $L_R \in [0, +\infty)$ such that for all $s \in [a, b]$

$$\|f(s, z(s)) - f(s, y(s))\| \leq L_R \|z(s) - y(s)\|. \quad (4.16)$$

Therefore, for all $t \in [a, b]$

$$\|z(t) - y(t)\| \leq \|\xi - \eta\| + L_R \int_a^t \|z(s) - y(s)\| ds \quad (4.17)$$

and by Gronwall's lemma 13.0.11 we get the thesis. $\square$

Remark 4.0.5. *We refer to Theorem (2.12) on page 252 in [1] where an even more general version of Peano's theorem was established than Theorem 4.0.2.*

Remark 4.0.6. *The existence and uniqueness of the solution to (4.1) under the assumptions (A1)-(A3) can also be derived from Theorem 2.2 on pages 104-105 in [18].*

4.1 Deterministic Euler scheme

Let $n \in \mathbb{N}$, $h = (b-a)/n$, $t_k = a + kh$ for $k = 0, 1, \dots, n$. We define the Euler scheme as follows. Take

$$y_0 = \xi, \quad (4.18)$$

and

$$y_{k+1} = y_k + hf(t_k, y_k), \quad (4.19)$$

for $k = 0, 1, \dots, n-1$. Moreover, as a time-continuous approximation of z in the whole interval $[a, b]$ we set

$$l_n(t) := l_{k,n}(t) \text{ for } t \in [t_k, t_{k+1}], \quad k = 0, 1, \dots, n-1, \quad (4.20)$$

where

$$l_{k,n}(t) = y_k + (t - t_k)f(t_k, y_k). \quad (4.21)$$

Of course $l_n \in C([a, b]; \mathbb{R}^d)$ and it is a piecewise linear function. In order to construct the function l_n we use n evaluations of f .

Theorem 4.1.1. *Let $\xi \in \mathbb{R}^d$ and let f satisfy the assumptions (A1)-(A4). Then there exists $C \in [0, +\infty)$ such that for all $n \in \mathbb{N}$ the following holds*

$$\sup_{a \leq t \leq b} \|z(t) - l_n(t)\| \leq Ch^q. \quad (4.22)$$

Proof. Note that for all $t \in [a, b]$ we can write

$$l_n(t) = \xi + \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) f(t_k, y_k) ds, \quad (4.23)$$

and

$$z(t) = \xi + \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) f(s, z(s)) ds. \quad (4.24)$$

Hence, for all $t \in [a, b]$

$$\begin{aligned} \|z(t) - l_n(t)\| &\leq \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \|f(s, z(s)) - f(t_k, y_k)\| ds \\ &\leq \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \|f(s, z(s)) - f(s, y_k)\| ds + \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \|f(s, y_k) - f(t_k, y_k)\| ds. \end{aligned} \quad (4.25)$$

Moreover, from Lemma 4.0.3 (i) we have that

$$\sup_{a \leq t \leq b} \|z(t)\| \leq R_1 := \max\{1, K(b-a)\} e^{K(b-a)} (1 + \|\xi\|), \quad (4.26)$$

and for $k = 0, 1, \dots, n-1$

$$\|y_{k+1}\| \leq (1 + Kh)\|y_k\| + Kh \quad (4.27)$$

so by the discrete version of Gronwall's lemma (Lemma 13.0.12), we have for all $n \in \mathbb{N}$

$$\max_{0 \leq k \leq n} \|y_k\| \leq R_2 := e^{K(b-a)} (1 + \|\xi\|). \quad (4.28)$$

By (A3), (A4) for $H = R = R_1 \vee R_2$ there exist $L_R, L_H \in [0, +\infty)$ such that for all $s \in [t_k, t_{k+1}]$, $k = 0, 1, \dots, n$, and all $n \in \mathbb{N}$

$$\begin{aligned} \|f(s, z(s)) - f(s, y_k)\| &\leq L_R \|z(s) - y_k\| \leq L_R \|z(s) - z(t_k)\| + L_R \|z(t_k) - l_n(t_k)\| \\ &\leq L_R C_1 h + L_R \|z(t_k) - l_n(t_k)\|, \end{aligned} \quad (4.29)$$

where $C_1 = K(1 + \max\{1, K(b-a)\} e^{K(b-a)})(1 + \|\xi\|)$ and

$$\|f(s, y_k) - f(t_k, y_k)\| \leq L_H |s - t_k|^\varrho \leq L_H h^\varrho. \quad (4.30)$$

Hence, by (4.25), (4.29), (4.30), (4.150) we get for all $t \in [a, b]$

$$\begin{aligned} \|z(t) - l_n(t)\| &\leq (b-a)L_R C_1 h + L_R \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \|z(t_k) - l_n(t_k)\| ds \\ &\quad + (b-a)L_H h^\varrho \\ &\leq (b-a) \left(L_R C_1 (1 + b-a) + L_H \right) h^\varrho + L_R \int_a^t \sup_{a \leq u \leq s} \|z(u) - l_n(u)\| ds, \end{aligned} \quad (4.31)$$

which in turn implies that for all $t \in [a, b]$

$$\sup_{a \leq u \leq t} \|z(u) - l_n(u)\| \leq (b-a) \left(L_R C_1 (1 + b-a) + L_H \right) h^\varrho + L_R \int_a^t \sup_{a \leq u \leq s} \|z(u) - l_n(u)\| ds. \quad (4.32)$$

The function $[a, b] \ni t \rightarrow \sup_{a \leq u \leq t} \|z(u) - l_n(u)\|$ is Borel measurable (as a nondecreasing function, see Proposition 13.0.5) and bounded, since $z, l_n \in C([a, b]; \mathbb{R}^d)$ and therefore

$$\sup_{a \leq t \leq b} \sup_{a \leq u \leq t} \|z(u) - l_n(u)\| \leq \sup_{a \leq u \leq b} \|z(u) - l_n(u)\| \leq \sup_{a \leq u \leq b} \|z(u)\| + \sup_{a \leq u \leq b} \|l_n(u)\| < +\infty. \quad (4.33)$$

Hence, by Gronwall's lemma 13.0.11 we get

$$\sup_{a \leq t \leq b} \|z(t) - l_n(t)\| \leq e^{L_R(b-a)}(b-a) \left(L_R C_1(1+b-a) + L_H \right) h^e, \quad (4.34)$$

from which the thesis follows. $\square$

4.1.1 Optimality of the Euler algorithm in the Lipschitz class

In this section, we investigate a lower error bound of any deterministic algorithm $\mathcal{A}_n$, that approximates the value of $z(b)$, in the Lipschitz class $\mathcal{F}_L$ of scalar right-hand side functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Namely,

$$\mathcal{F}_L = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid \forall x, y \in \mathbb{R} |f(x) - f(y)| \leq L|x - y|\}, \quad (4.35)$$

and we also consider the following subclass of $\mathcal{F}_L$

$$\mathcal{F}_{L,d_0}^+ = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid \forall x, y \in \mathbb{R} |f(x) - f(y)| \leq L|x - y|, f(x) \geq d_0\}, \quad (4.36)$$

where $d_0 > 0$, $L \in [0, +\infty)$ are the class parameters. We also consider the following class of admissible initial-values ξ

$$\mathcal{I}_L = \{\xi \in \mathbb{R} \mid |\xi| \leq K\}, \quad (4.37)$$

where $K \in [0, +\infty)$ is the parameter of the class. Hence, we consider the following class of the pairs (ξ, f)

$$\mathcal{F}_{K,L,d_0}^+ = \mathcal{I}_K \times \mathcal{F}_{L,d_0}^+. \quad (4.38)$$

As in Chapter 1, we will use the Information-Based complexity framework (IBC), see [87], and we show a simplified proof of the lower error bound for the Lipschitz class $\mathcal{F}_L$. For more general results, see [38], [40].

For $n \in \mathbb{N}$ and $(\xi, f) \in \mathcal{F}_{K,L,d_0}^+$ we consider any algorithm $\mathcal{A}_n(\xi, f)$, that approximates $z(\xi, f)(b)$, of the form

$$\mathcal{A}_n(\xi, f) = \varphi_n(N_n(\xi, f)), \quad (4.39)$$

where $\varphi_n : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ is a function, and a vector $N_n(\xi, f) \in \mathbb{R}^{n+1}$ of the information used about f is as follows

$$N_n(\xi, f) = [\xi, f(x_1), \dots, f(x_n)]. \quad (4.40)$$

Such information given by the values of f at some deterministic points x_k is called *standard information*. We assume that the information $N_n(\xi, f)$ about (ξ, f) might be adaptive, i.e., there exist functions ψ_k , $k = 1, 2, \dots, n$, such that

$$x_1 = \psi_1(\xi) \quad (4.41)$$

and

$$x_k = \psi_k(x_1, \dots, x_{k-1}, f(x_1), f(x_2), \dots, f(x_{k-1})), \quad k = 2, 3, \dots, n. \quad (4.42)$$

The informational cost of the algorithm $\mathcal{A}_n(\xi, f)$ is equal to n for all $(\xi, f) \in \mathcal{F}_{K,L,d_0}^+$. The worst-case error of $\mathcal{A}_n$ in the class $\mathcal{F}_{K,L,d_0}^+$ is

$$e(\mathcal{A}_n, \mathcal{F}_{K,L,d_0}^+) = \sup_{(\xi, f) \in \mathcal{F}_{K,L,d_0}^+} |z(\xi, f)(b) - \mathcal{A}_n(\xi, f)|. \quad (4.43)$$

To establish lower bounds on the worst-case error, we first state the following fact.

Fact 4.1.2. For all $x \geq 0$

$$\frac{x}{x+1} \geq \frac{1}{2} \min\{x, 1\}. \quad (4.44)$$

Proof. Consider the case when $x \geq 1$. Then $\min\{x, 1\} = 1$, $1/(x+1) \leq 1/2$, and

$$\frac{x}{x+1} = 1 - \frac{1}{x+1} \geq \frac{1}{2} = \frac{1}{2} \min\{x, 1\}. \quad (4.45)$$

When $0 \leq x \leq 1$ then $\min\{x, 1\} = x$, and $1/(x+1) \geq 1/2$. Hence,

$$\frac{x}{x+1} \geq \frac{1}{2}x = \frac{1}{2} \min\{x, 1\}. \quad (4.46)$$

This ends the proof. $\square$

For any $(\xi, f) \in \mathcal{F}_{K,L,d_0}^+$ there exists a unique solution $z(\xi, f) : [a, b] \rightarrow \mathbb{R}$ of the scalar autonomous ODE

$$\begin{cases} z'(t) = f(z(t)), & t \in [a, b], \\ z(a) = \xi. \end{cases} \quad (4.47)$$

Moreover, since $f(y) \geq d_0 > 0$, the solution $z(\xi, f)$ is strictly increasing on $[a, b]$. Therefore, by (4.47) and change of variables formula we get for all $t \in [a, b]$ that

$$t - a = \int_a^t 1 ds = \int_a^t \frac{z'(s)}{f(z(s))} ds = \int_{\xi}^{z(t)} \frac{du}{f(u)}. \quad (4.48)$$

The proof of the following estimate proceeds analogously to that of Proposition 2.0.2.

Proposition 4.1.3. For any algorithm $\mathcal{A}_n$ of the form (4.39) we have

$$e(\mathcal{A}_n, \mathcal{F}_{K,L,d_0}^+) \geq \frac{1}{2} \sup_{(\xi_0, f_0), (\xi_1, f_1) \in \mathcal{F}_{K,L,d_0}^+, N_n(\xi_0, f_0) = N_n(\xi_1, f_1)} |z(\xi_0, f_0)(b) - z(\xi_1, f_1)(b)|. \quad (4.49)$$

Take $\xi_0 = \xi_1 = 0$, $f_0(x) = d_0$. Of course $(\xi_0, f_0) \in \mathcal{F}_{K,L,d_0}^+$ and we compute the adaptive evaluation points x_k for f_0 as follows

$$\begin{aligned} x_1 &= \psi_1(\xi_0) = \psi_1(0), \\ x_2 &= \psi_2(x_1, f_0(x_1)) = \psi_2(x_1, d_0), \\ &\dots, \\ x_n &= \psi_n(x_1, \dots, x_{n-1}, f_0(x_1), \dots, f_0(x_{n-1})) = \psi_n(x_1, \dots, x_{n-1}, d_0, \dots, d_0). \end{aligned} \quad (4.50)$$

Moreover, $N_n(\xi_0, f_0) = [0, d_0, \dots, d_0] \in \mathbb{R}^{n+1}$. Take $P = \{x_1, x_2, \dots, x_n\}$ and set

$$f_1(x) = f_0(x) + L \cdot \text{dist}(x, P) = d_0 + L \cdot \text{dist}(x, P), \quad (4.51)$$

where $\text{dist}(x, P) = \min_{1 \leq j \leq n} |x - x_j| \geq 0$. Hence, for all $x, y \in \mathbb{R}$, $f_1(x) \geq d_0$ and by Fact 2.0.3

$$|f_1(x) - f_1(y)| = L |\text{dist}(x, P) - \text{dist}(y, P)| \leq L|x - y|. \quad (4.52)$$

Hence, $(\xi_1, f_1) \in \mathcal{F}_{K,L,d_0}^+$ and $f_1(x_k) = d_0$ for all $k = 1, 2, \dots, n$. We now compute the adaptive evaluation points x_k^1 for f_1 as follows

$$\begin{aligned} x_1^1 &= \psi_1(\xi_1) = \psi_1(0) = x_1, \\ x_2^1 &= \psi_2(x_1^1, f_1(x_1^1)) = \psi_2(x_1, f_1(x_1)) = \psi_2(x_1, d_0) = x_2, \\ &\dots, \\ x_n^1 &= \psi_n(x_1^1, \dots, x_{n-1}^1, f_1(x_1^1), \dots, f_1(x_{n-1}^1)) = \psi_n(x_1, \dots, x_{n-1}, f_1(x_1), \dots, f_1(x_{n-1})) \\ &= \psi_n(x_1, \dots, x_{n-1}, d_0, \dots, d_0) = x_n, \end{aligned} \quad (4.53)$$

and therefore $N_n(\xi_1, f_1) = [0, f_1(x_1^1), \dots, f_1(x_n^1)] = [0, f_1(x_1), \dots, f_1(x_n)] = [0, d_0, \dots, d_0] = N(\xi_0, f_0)$. By Proposition 4.1.3 and for the constructed pairs (ξ_0, f_0) , (ξ_1, f_1) , in what follows we estimate the lower bound for $|z(\xi_0, f_0)(b) - z(\xi_1, f_1)(b)|$. Note that

$$z(\xi_0, f_0)(t) = d_0(t - a), \quad t \in [a, b], \quad (4.54)$$

and by (4.48)

$$b - a = \int_0^{z(\xi_1, f_1)(b)} \frac{du}{f_1(u)} = \int_0^{z(\xi_1, f_1)(b)} \frac{du}{d_0 + L \cdot \text{dist}(u, P)} \leq \frac{1}{d_0} \int_0^{z(\xi_1, f_1)(b)} du = \frac{z(\xi_1, f_1)(b)}{d_0}. \quad (4.55)$$

Hence,

$$z(\xi_0, f_0)(b) = d_0(b - a) \leq z(\xi_1, f_1)(b), \quad (4.56)$$

and

$$\begin{aligned} (0 \leq) z(\xi_1, f_1)(b) - z(\xi_0, f_0)(b) &= \int_0^{z(\xi_1, f_1)(b)} 1 du - d_0 \cdot \int_0^{z(\xi_1, f_1)(b)} \frac{du}{d_0 + L \cdot \text{dist}(u, P)} \\ &= \int_0^{z(\xi_1, f_1)(b)} \frac{L \cdot \text{dist}(u, P)}{d_0 + L \cdot \text{dist}(u, P)} du \geq \int_0^{d_0(b-a)} \frac{L \cdot \text{dist}(u, P)}{d_0 + L \cdot \text{dist}(u, P)} du. \end{aligned} \quad (4.57)$$

From Fact 4.1.2 we have that for all $u \geq 0$

$$\begin{aligned} \frac{L \cdot \text{dist}(u, P)}{d_0 + L \cdot \text{dist}(u, P)} &= \frac{(L/d_0) \cdot \text{dist}(u, P)}{1 + (L/d_0) \cdot \text{dist}(u, P)} \geq \frac{1}{2} \min\{(L/d_0) \cdot \text{dist}(u, P), 1\} \\ &= \frac{L}{2d_0} \min\{\text{dist}(u, P), d_0/L\}. \end{aligned} \quad (4.58)$$

Therefore, from Lemma 13.0.18 we have

$$\begin{aligned} z(\xi_1, f_1)(b) - z(\xi_0, f_0)(b) &\geq \frac{L}{2d_0} \int_0^{d_0(b-a)} \min\{\text{dist}(u, P), d_0/L\} du \\ &= \frac{L}{2d_0} \int_0^{d_0/L} \lambda_1(\{u \in [0, d_0(b-a)] \mid \text{dist}(u, P) > t\}) dt. \end{aligned} \quad (4.59)$$

Proceeding analogously as in (2.34) we have that for all $t \geq 0$

$$\lambda_1(\{u \in [0, d_0(b-a)] \mid \text{dist}(u, P) \leq t\}) \leq 2nt, \quad (4.60)$$

and for $t \in [0, d_0(b-a)/(2n)]$

$$\lambda_1(\{u \in [0, d_0(b-a)] \mid \text{dist}(u, P) > t\}) \geq d_0(b-a) - 2nt \geq 0. \quad (4.61)$$

Hence, for all $n \geq \lceil L(b-a)/2 \rceil$ we have $\min\{d_0/L, d_0(b-a)/(2n)\} = d_0(b-a)/(2n)$ and

$$\begin{aligned}
z(\xi_1, f_1)(b) - z(\xi_0, f_0)(b) &\geq \frac{L}{2d_0} \int_0^{d_0/L} \lambda_1(\{u \in [0, d_0(b-a)] \mid \text{dist}(u, P) > t\}) dt \\
&\geq \frac{L}{2d_0} \int_0^{\min\{d_0/L, d_0(b-a)/(2n)\}} \lambda_1(\{u \in [0, d_0(b-a)] \mid \text{dist}(u, P) > t\}) dt \\
&= \frac{L}{2d_0} \int_0^{d_0(b-a)/(2n)} \lambda_1(\{u \in [0, d_0(b-a)] \mid \text{dist}(u, P) > t\}) dt \\
&\geq \frac{L}{2d_0} \int_0^{d_0(b-a)/(2n)} (d_0(b-a) - 2nt) dt = \frac{L}{2d_0} \cdot \frac{d_0^2(b-a)^2}{4} n^{-1} = \frac{Ld_0(b-a)^2}{8} n^{-1}. \tag{4.62}
\end{aligned}$$

By Proposition 4.1.3, we have that for all $n \geq \lceil L(b-a)/2 \rceil$ and for any algorithm $\mathcal{A}_n$ of the form (4.39) we have

$$e(\mathcal{A}_n, \mathcal{F}_L) \geq e(\mathcal{A}_n, \mathcal{F}_{K,L,d_0}^+) \geq \frac{Ld_0(b-a)^2}{16} n^{-1}. \tag{4.63}$$

On the other hand for the Euler algorithm $\mathcal{A}_n^E := l_n(b)$, defined by (4.20), that uses n evaluations of f we get by Theorem 4.1.1

$$e(\mathcal{A}_n^E, \mathcal{F}_L) = O(n^{-1}). \tag{4.64}$$

Hence, the Euler algorithm $\mathcal{A}_n^E$ is optimal in the Lipschitz class $\mathcal{F}_L$.

The results presented in this section can be extended to the multidimensional case and also other classes of right-hand side functions f can be considered, see [38], [40].

4.2 Randomized Euler scheme

Let $(\Omega, \Sigma, \mathbb{P})$ be a complete probability space. Let $(\tau_j)_{j \in \mathbb{N}_0}$ be a sequence of independent random variables on $(\Omega, \Sigma, \mathbb{P})$ that are identically uniformly distributed on $[0, 1]$. Let $n \in \mathbb{N}$, $h = (b-a)/n$, $t_k = a + kh$ for $k = 0, 1, \dots, n$. We define the randomized Euler scheme as follows. Take

$$y_0 = \xi, \tag{4.65}$$

and

$$y_{k+1} = y_k + hf(\theta_k, y_k), \tag{4.66}$$

for $k = 0, 1, \dots, n-1$, where $\theta_k = t_k + h\tau_k$ is uniformly distributed in $[t_k, t_{k+1}]$. Moreover, as a time-continuous approximation of z in the whole interval $[a, b]$ we set

$$l_n(t) := l_{k,n}(t) \text{ for } t \in [t_k, t_{k+1}], \quad k = 0, 1, \dots, n-1, \tag{4.67}$$

where

$$l_{k,n}(t) = y_k + (t - t_k)f(\theta_k, y_k). \tag{4.68}$$

Note that this time $\{l_n(t)\}_{t \in [a,b]}$ is a stochastic process with continuous trajectories. To construct the process l_n we use n (random) evaluations of f .

Lemma 4.2.1. *Let $p \in [2, +\infty)$, $\xi \in \mathbb{R}^d$ and let f satisfy the assumptions (A1)-(A4). Then for all $n \in \mathbb{N}$ the function*

$$[a, b] \ni t \rightarrow \mathbb{E} \left(\sup_{a \leq u \leq t} \|z(u) - l_n(u)\|^p \right) \tag{4.69}$$

is Borel measurable and bounded.

Proof. Note that for all $u \in [a, b]$ the mapping $\Omega \ni \omega \rightarrow \|z(u) - l_n(u, \omega)\|^p$ is a nonnegative random variable. Moreover, trajectories of the stochastic process $\{\|z(u) - l_n(u, \omega)\|^p\}_{u \in [a, b]}$ are continuous and therefore for all $t \in [a, b]$, $c \in \mathbb{R}$

$$\begin{aligned} & \left\{ \omega \in \Omega \mid \sup_{a \leq u \leq t} \|z(u) - l_n(u, \omega)\|^p \leq c \right\} = \left\{ \omega \in \Omega \mid \sup_{u \in [a, t] \cap \mathbb{Q}} \|z(u) - l_n(u, \omega)\|^p \leq c \right\} \\ & = \bigcap_{u \in [a, t] \cap \mathbb{Q}} \left\{ \omega \in \Omega \mid \|z(u) - l_n(u, \omega)\|^p \leq c \right\} \in \Sigma. \end{aligned} \quad (4.70)$$

This means that for every $t \in [a, b]$ the function $\Omega \ni \omega \rightarrow \sup_{a \leq u \leq t} \|z(u) - l_n(u, \omega)\|^p$ is a nonnegative random variable. Hence the function (4.69) is well-defined. The function (4.69) is Borel measurable as a nondecreasing function. Boundedness is left as an exercise. $\square$

Theorem 4.2.2. *Let $p \in [2, +\infty)$, $\xi \in \mathbb{R}^d$ and let f satisfy the assumptions (A1)-(A4). Then there exists $C \in [0, +\infty)$ such that for all $n \in \mathbb{N}$ the following holds*

$$\left\| \sup_{a \leq t \leq b} \|z(t) - l_n(t)\| \right\|_p \leq Ch^{\min\{\varrho + \frac{1}{2}, 1\}}. \quad (4.71)$$

Proof. We have that for all $t \in [a, b]$

$$l_n(t) = \xi + \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) f(\theta_k, y_k) ds, \quad (4.72)$$

and, as before,

$$z(t) = \xi + \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) f(s, z(s)) ds. \quad (4.73)$$

Hence, for all $t \in [a, b]$

$$\begin{aligned} \|z(t) - l_n(t)\| & \leq \left\| \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \left(f(s, z(s)) - f(s, z(t_k)) \right) ds \right\| \\ & + \left\| \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \left(f(s, z(t_k)) - f(\theta_k, z(t_k)) \right) ds \right\| \\ & + \left\| \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \left(f(\theta_k, z(t_k)) - f(\theta_k, y_k) \right) ds \right\|. \end{aligned} \quad (4.74)$$

Again from Lemma 4.0.3 (i) we have that

$$\sup_{a \leq t \leq b} \|z(t)\| \leq R_1 := \max\{1, K(b-a)\} e^{K(b-a)} (1 + \|\xi\|), \quad (4.75)$$

and for $k = 0, 1, \dots, n-1$

$$\|y_{k+1}\| \leq (1 + Kh) \|y_k\| + Kh, \quad (4.76)$$

so by the discrete version of Gronwall's lemma

$$\max_{0 \leq k \leq n} \|y_k\| \leq R_2 := e^{K(b-a)} (1 + \|\xi\|) \quad (4.77)$$

for all $n \in \mathbb{N}$ with probability one. By (A3), (A4) for $H = R = R_1 \vee R_2$ there exist $L_R, L_H \in [0, +\infty)$ such that for all $s \in [t_k, t_{k+1}]$, $k = 0, 1, \dots, n$, and all $n \in \mathbb{N}$

$$\|f(s, z(s)) - f(s, z(t_k))\| \leq L_R \|z(s) - z(t_k)\| \leq L_R C_1 h, \quad (4.78)$$

where $C_1 = K \left(1 + \max\{1, K(b-a)\} e^{K(b-a)}\right) (1 + \|\xi\|)$,

$$\|f(s, z(t_k)) - f(\theta_k, z(t_k))\| \leq L_H |s - \theta_k|^\varrho \leq L_H h^\varrho, \quad (4.79)$$

$$\|f(\theta_k, z(t_k)) - f(\theta_k, y_k)\| \leq L_R \|z(t_k) - l_n(t_k)\|, \quad (4.80)$$

with probability one. Hence, by the estimates above we get for all $t \in [a, b]$

$$\left\| \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \left(f(s, z(s)) - f(s, z(t_k)) \right) ds \right\| \leq (b-a) L_R C_1 h, \quad (4.81)$$

$$\left\| \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \left(f(\theta_k, z(t_k)) - f(\theta_k, y_k) \right) ds \right\| \leq L_R \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \|z(t_k) - l_n(t_k)\| ds. \quad (4.82)$$

We define

$$i(t) = \sup\{i = 0, 1, \dots, n \mid t_i \leq t\}, \quad (4.83)$$

$$\zeta(t) = t_{i(t)}, \quad (4.84)$$

for $t \in [a, b]$. Then

$$\left\| \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \left(f(s, z(t_k)) - f(\theta_k, z(t_k)) \right) ds \right\| \leq A_{1,n}^{RE}(t) + A_{2,n}^{RE}(t), \quad (4.85)$$

with

$$A_{1,n}^{RE}(t) = \left\| \sum_{k=0}^{i(t)-1} \int_{t_k}^{t_{k+1}} \left(f(s, z(t_k)) - f(\theta_k, z(t_k)) \right) ds \right\| \quad (4.86)$$

and

$$A_{2,n}^{RE}(t) = \left\| \int_{\zeta(t)}^t \left(f(s, z(\zeta(t))) - f(\theta_{i(t)}, z(\zeta(t))) \right) ds \right\|, \quad (4.87)$$

where $A_{2,n}^{RE}(b) = 0$. By (A4) and (4.75) we have for all $t \in [a, b]$ that $0 \leq t - \zeta(t) \leq h$ and hence

$$A_{2,n}^{RE}(t) \leq L_H \int_{\zeta(t)}^t |s - \theta_{i(t)}|^\varrho ds \leq L_H h^{\varrho+1}. \quad (4.88)$$

Let for $k = 0, 1, \dots, n-1$

$$Y_k = \int_{t_k}^{t_{k+1}} \left(f(s, z(t_k)) - f(\theta_k, z(t_k)) \right) ds, \quad (4.89)$$

then by (4.79)

$$\|Y_k\| \leq L_H h^{\varrho+1} \quad (4.90)$$

with probability one. Moreover, let

$$Z_j = \sum_{k=0}^j Y_k, \quad j = 0, 1, \dots, n-1, \quad (4.91)$$

and $Z_{-1} := 0$. Then we can write that for $t \in [a, b]$

$$A_{1,n}^{RE}(t) = \|Z_{i(t)-1}\| = \left\| \sum_{k=0}^{n-1} Z_{k-1} \cdot \mathbf{1}_{[t_k, t_{k+1})}(t) + Z_{n-1} \cdot \mathbf{1}_{\{b\}}(t) \right\|. \quad (4.92)$$

Therefore, we have for all $t \in [a, b]$ that

$$\mathbb{E} \left(\sup_{t \in [a, b]} (A_{1,n}^{RE}(t))^p \right) = \mathbb{E} \left(\max_{0 \leq j \leq n-1} \|Z_j\|^p \right). \quad (4.93)$$

Note that for all $n \in \mathbb{N}$ the process

$$(Z_j, \sigma(\theta_0, \theta_1, \dots, \theta_j))_{j=0,1,\dots,n-1} \quad (4.94)$$

is a discrete-time martingale, since for $j = 0, 1, \dots, n-1$

$$\|Z_j\| \leq (b-a)L_H h^\varrho < +\infty, \quad (4.95)$$

and for $j = 0, 1, \dots, n-2$

$$\mathbb{E}(Z_{j+1} - Z_j \mid \sigma(\theta_0, \theta_1, \dots, \theta_j)) = \mathbb{E}(Y_{j+1} \mid \sigma(\theta_0, \theta_1, \dots, \theta_j)) = \mathbb{E}(Y_{j+1}) = 0. \quad (4.96)$$

In (4.96) we used the following facts:

- $\sigma(Y_{j+1})$ and $\sigma(\theta_0, \theta_1, \dots, \theta_j)$ are independent σ -fields,
- since $\theta_{j+1} \sim U[t_{j+1}, t_{j+2}]$, it holds

$$\mathbb{E}(f(\theta_{j+1}, z(t_{j+1}))) = \frac{1}{h} \int_{t_{j+1}}^{t_{j+2}} f(s, z(t_{j+1})) ds. \quad (4.97)$$

By using the Burkholder inequality (see Theorem 13.0.23) and the estimate (13.58) we obtain

$$\mathbb{E} \left(\max_{0 \leq j \leq n-1} \|Z_j\|^p \right) \leq C_p^p \mathbb{E} \left[\left(\sum_{k=0}^{n-1} \|Y_k\|^2 \right)^{p/2} \right] \leq C_p^p n^{\frac{p}{2}-1} \sum_{k=0}^{n-1} \mathbb{E} \|Y_k\|^p \leq C_p^p L_H^p (b-a)^{p/2} h^{p(\varrho+\frac{1}{2})}. \quad (4.98)$$

Combining (4.85), (4.88), (4.98) we get

$$\mathbb{E} \left(\sup_{a \leq t \leq b} \left\| \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) (f(s, z(t_k)) - f(\theta_k, z(t_k))) ds \right\|^p \right) \leq ch^{p(\varrho+\frac{1}{2})}, \quad (4.99)$$

where $c > 0$ does not depend on n . From (4.74), (4.81), (4.82), by Hölder and Jensen inequalities we get for all $t \in [a, b]$

$$\begin{aligned}
& \sup_{a \leq u \leq t} \|z(u) - l_n(u)\|^p \leq c_1 h^p + c_2 \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \|z(t_k) - l_n(t_k)\|^p ds \\
& + c_3 \sup_{a \leq t \leq b} \left\| \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \left(f(s, z(t_k)) - f(\theta_k, z(t_k)) \right) ds \right\|^p \\
& \leq c_1 h^p + c_2 \int_a^t \sup_{a \leq u \leq s} \|z(u) - l_n(u)\|^p ds \\
& + c_3 \sup_{a \leq t \leq b} \left\| \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) \left(f(s, z(t_k)) - f(\theta_k, z(t_k)) \right) ds \right\|^p
\end{aligned} \tag{4.100}$$

hence by Fubini's theorem (see exercise 5.) and (4.99)

$$\mathbb{E} \left(\sup_{a \leq u \leq t} \|z(u) - l_n(u)\|^p \right) \leq c_4 h^{p \min\{\varrho + \frac{1}{2}, 1\}} + c_2 \int_a^t \mathbb{E} \left(\sup_{a \leq u \leq s} \|z(u) - l_n(u)\|^p \right) ds. \tag{4.101}$$

By applying Gronwall's lemma we get the thesis. $\square$

Remark 4.2.3. When $\varrho = 1$ then both algorithms have the same rate of convergence equal to 1. However, for $\varrho \in (0, 1)$ the randomized Euler algorithm outperforms its deterministic counterpart. Moreover, in the limiting case when $\varrho = 0$ and f is only continuous (or even only Borel measurable) with respect to the time variable t then the randomized Euler scheme has the error $O(h^{1/2})$, while the error of its deterministic version is $O(1)$ and there is lack of convergence, see [49], [64].

Note that in the special case, when $f = f(t)$, does not depend on the space variable y , we get the following result.

Proposition 4.2.4. Let us assume that $\xi = 0$ and let there exist $L_H \in [0, +\infty)$, $\varrho \in (0, 1]$ such that for all $t, s \in [a, b]$

$$\|f(t) - f(s)\| \leq L_H |t - s|^\varrho. \tag{4.102}$$

Then there exists $C \in [0, +\infty)$ such that for all $n \in \mathbb{N}$ the following holds

$$\left\| \sup_{a \leq t \leq b} \|z(t) - l_n(t)\| \right\|_p \leq C h^{\varrho + \frac{1}{2}}, \tag{4.103}$$

the functions $[a, b] \ni t \rightarrow \mathbb{E}(l_n(t)) \in \mathbb{R}$, $[a, b] \ni t \rightarrow \text{Var}(l_n(t)) \in [0, +\infty)$ are continuous, and for all $t \in [a, b]$ we have

$$\mathbb{E}(l_n(t)) = \sum_{k=0}^{n-1} \left(z(t_k) + \frac{t - t_k}{h} (z(t_{k+1}) - z(t_k)) \right) \cdot \mathbf{1}_{[t_k, t_{k+1})}(t) + z(b) \cdot \mathbf{1}_{\{b\}}(t), \tag{4.104}$$

while for $t \in [t_k, t_{k+1}]$, $k = 0, 1, \dots, n-1$

$$\begin{aligned}
\text{Var}(l_n(t)) &= h \cdot \left[\int_a^{t_k} (f(s))^2 ds + \left(\frac{t - t_k}{h} \right)^2 \int_{t_k}^{t_{k+1}} (f(s))^2 ds \right] \\
&\quad - \sum_{j=0}^{k-1} \left(\int_{t_j}^{t_{j+1}} f(s) ds \right)^2 - \left(\frac{t - t_k}{h} \right)^2 \cdot \left(\int_{t_k}^{t_{k+1}} f(s) ds \right)^2.
\end{aligned} \tag{4.105}$$

Proof. We have that for all $t \in [a, b]$

$$l_n(t) = \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) f(\theta_k) ds, \quad (4.106)$$

and, as before,

$$z(t) = \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) f(s) ds = \int_a^t f(s) ds. \quad (4.107)$$

Hence, by (4.99) we get for all $t \in [a, b]$ that

$$\mathbb{E} \left(\sup_{a \leq t \leq b} \|z(t) - l_n(t)\|^p \right) = \mathbb{E} \left(\sup_{a \leq t \leq b} \left\| \int_a^t \sum_{k=0}^{n-1} \mathbf{1}_{[t_k, t_{k+1})}(s) (f(s) - f(\theta_k)) ds \right\|^p \right) \leq ch^{p(\varrho + \frac{1}{2})}. \quad (4.108)$$

The equalities (4.104), (4.105) follow from direct computations, and their proofs are left as an exercise. $\square$

In this case, l_n (which can be seen as a randomized Riemann sum) approximates in $[a, b]$ the function z , which is the anti-derivative of f . Moreover, $l_n(t)$ for $t \in \{t_0, t_1, \dots, t_n\}$ is the unbiased estimator of $z(t)$. We can see that when $f = f(t)$ does not depend on y , the order of convergence is $\varrho + 1/2$ which might be faster than $\min\{\varrho + 1/2, 1\}$.

We will look closer at the variance of the randomized Riemann sum in the next subsection.

4.2.1 Coming back to integration: variance of the randomized Riemann sum vs variance of the crude Monte Carlo method

We assume that $f \in L^2[a, b]$ and let us consider the problem of approximation of the Lebesgue integral

$$I(f) = \int_a^b f(t) dt (= z(b)). \quad (4.109)$$

The crude Monte Carlo method is defined as

$$MC_n(f) = h \sum_{k=0}^{n-1} f(\xi_k), \quad (4.110)$$

where $(\xi_k)_{k=0,1,\dots,n-1}$ is the iid sequence of random variables with the law $U([a, b])$, while the randomized Riemann sum is defined as

$$\mathcal{R}_n(f) = h \sum_{k=0}^{n-1} f(\theta_k), \quad (4.111)$$

where $(\theta_k)_{k=0,1,\dots,n-1}$ is the sequence of independent random variables such that each θ_k has the law $U[t_k, t_{k+1}]$, $t_k = a + kh$, $k = 0, 1, \dots, n$, $h = (b - a)/n$. Note that both quadratures use n random evaluations of f .

It holds that $\mathbb{E}(MC_n(f)) = I(f)$ and

$$\mathbb{E}|I(f) - MC_n(f)|^2 = \text{Var}(MC_n(f)) = (b - a) \cdot s^2(f) \cdot n^{-1}, \quad (4.112)$$

where

$$s^2(f) = I(f^2) - \frac{1}{b - a} (I(f))^2.$$

For $\mathcal{R}_n$ we have that $\mathbb{E}(\mathcal{R}_n(f)) = I(f)$ and

$$\mathbb{E}|I(f) - \mathcal{R}_n(f)|^2 = \text{Var}(\mathcal{R}_n(f)) = (b - a) \cdot s_n^2(f) \cdot n^{-1}, \quad (4.113)$$

with

$$s_n^2(f) = I(f^2) - \frac{1}{h} \sum_{k=0}^{n-1} \left(\int_{t_k}^{t_{k+1}} f(s) ds \right)^2, \quad (4.114)$$

where $0 \leq s_n^2(f) \leq s^2(f)$. Moreover, an even stronger result holds, see the thesis of the following lemma.

Lemma 4.2.5. *For any $f \in L^2[a, b]$ we have*

$$\lim_{n \rightarrow +\infty} s_n^2(f) = 0. \quad (4.115)$$

Proof. First, assume $f \in C[a, b]$. Then by the mean value theorem for integrals, we get that there exists $\xi_k \in [t_k, t_{k+1}]$ for $k = 0, 1, \dots, n-1$ and therefore

$$\frac{1}{h} \sum_{k=0}^{n-1} \left(\int_{t_k}^{t_{k+1}} f(s) ds \right)^2 = \sum_{k=0}^{n-1} (f(\xi_k))^2 \cdot h \rightarrow \int_a^b (f(s))^2 ds \quad (4.116)$$

as $n \rightarrow +\infty$. Hence, (4.115) holds for any $f \in C[a, b]$.

Recall that $C[a, b]$ is a dense subset of $L^2[a, b]$, see, for example, Proposition 3.4.4., page 101 in [14]. Hence, for $f \in L^2[a, b]$ and all $\varepsilon > 0$ there exists $f_\varepsilon \in C[a, b]$ such that

$$\|f - f_\varepsilon\|_{L^2[a, b]} < \varepsilon. \quad (4.117)$$

In particular, by the first part of the proof we have for all $\varepsilon > 0$ that

$$\lim_{n \rightarrow +\infty} s_n^2(f_\varepsilon) = 0. \quad (4.118)$$

Moreover, we can write that

$$s_n^2(f) = \|f\|_{L^2[a, b]}^2 - \frac{1}{h} \sum_{k=0}^{n-1} \left(\int_{t_k}^{t_{k+1}} f(s) ds \right)^2, \quad (4.119)$$

and

$$|s_n^2(f) - s_n^2(f_\varepsilon)| \leq E_{1, n, \varepsilon}(f) + \frac{1}{h} \cdot E_{2, n, \varepsilon}(f), \quad (4.120)$$

where

$$E_{1, n, \varepsilon}(f) = \left| \|f\|_{L^2[a, b]}^2 - \|f_\varepsilon\|_{L^2[a, b]}^2 \right| \quad (4.121)$$

$$E_{2, n, \varepsilon}(f) = \left| \sum_{k=0}^{n-1} \left(\int_{t_k}^{t_{k+1}} f(s) ds \right)^2 - \sum_{k=0}^{n-1} \left(\int_{t_k}^{t_{k+1}} f_\varepsilon(s) ds \right)^2 \right|. \quad (4.122)$$

By (4.117) we have for all $n \in \mathbb{N}$ and all $\varepsilon > 0$ that

$$\begin{aligned} E_{1, n, \varepsilon}(f) &= \left| \|f\|_{L^2[a, b]} - \|f_\varepsilon\|_{L^2[a, b]} \right| \cdot \left(\|f\|_{L^2[a, b]} + \|f_\varepsilon\|_{L^2[a, b]} \right) \\ &\leq \|f - f_\varepsilon\|_{L^2[a, b]} \cdot \left(2\|f\|_{L^2[a, b]} + \|f - f_\varepsilon\|_{L^2[a, b]} \right) \leq \varepsilon \cdot (2\|f\|_{L^2[a, b]} + \varepsilon), \end{aligned} \quad (4.123)$$

and, by the Schwarz inequality,

$$\begin{aligned}
E_{2,n,\varepsilon}(f) &\leq \sum_{k=0}^{n-1} \left| \left(\int_{t_k}^{t_{k+1}} f(s) ds \right)^2 - \left(\int_{t_k}^{t_{k+1}} f_\varepsilon(s) ds \right)^2 \right| \\
&= \sum_{k=0}^{n-1} \left| \int_{t_k}^{t_{k+1}} (f(s) - f_\varepsilon(s)) ds \right| \cdot \left| \int_{t_k}^{t_{k+1}} (f(s) + f_\varepsilon(s)) ds \right| \\
&\leq \left(\sum_{k=0}^{n-1} \left| \int_{t_k}^{t_{k+1}} (f(s) - f_\varepsilon(s)) ds \right|^2 \right)^{1/2} \times \left(\sum_{k=0}^{n-1} \left| \int_{t_k}^{t_{k+1}} (f(s) + f_\varepsilon(s)) ds \right|^2 \right)^{1/2} \\
&\leq \left(\sum_{k=0}^{n-1} h \cdot \int_{t_k}^{t_{k+1}} |f(s) - f_\varepsilon(s)|^2 ds \right)^{1/2} \times \left(\sum_{k=0}^{n-1} h \cdot \int_{t_k}^{t_{k+1}} |f(s) + f_\varepsilon(s)|^2 ds \right)^{1/2} \\
&= h \cdot \|f - f_\varepsilon\|_{L^2[a,b]} \cdot \|f + f_\varepsilon\|_{L^2[a,b]} \leq h \cdot \|f - f_\varepsilon\|_{L^2[a,b]} \cdot \left(\|f\|_{L^2[a,b]} + \|f_\varepsilon\|_{L^2[a,b]} \right) \\
&\leq h \cdot \|f - f_\varepsilon\|_{L^2[a,b]} \cdot \left(2\|f\|_{L^2[a,b]} + \|f - f_\varepsilon\|_{L^2[a,b]} \right) \leq h \cdot \varepsilon \cdot (2\|f\|_{L^2[a,b]} + \varepsilon). \tag{4.124}
\end{aligned}$$

Hence, by (4.120), (4.123), (4.124) we get for all $n \in \mathbb{N}$ and all $\varepsilon > 0$

$$|s_n^2(f) - s_n^2(f_\varepsilon)| \leq 2\varepsilon \cdot (2\|f\|_{L^2[a,b]} + \varepsilon), \tag{4.125}$$

which, by (4.118) and the fact that $s_n^2(f) \geq 0$, gives for all $\varepsilon > 0$

$$0 \leq \liminf_{n \rightarrow +\infty} s_n^2(f) \leq \limsup_{n \rightarrow +\infty} s_n^2(f) \leq 2\varepsilon \cdot (2\|f\|_{L^2[a,b]} + \varepsilon), \tag{4.126}$$

and this implies the thesis. $\square$

By Lemma 4.2.5 we have that $s_n^2(f) < s^2(f)$ for sufficiently large n . Hence, the randomized Riemann quadrature $\mathcal{R}_n$ has asymptotically smaller variance than the classical Monte Carlo method MC_n . Moreover, due to the low smoothness of $f \in L^2[a,b]$, for approximation of $I(f)$ there are not too many other options than the randomized Riemann sum $\mathcal{R}_n$.

Note that we cannot use CLT in order to construct empirical confidence intervals for $\mathcal{R}_n$, as it was for MC_n , since the random variables $f(\theta_k)$ do not have the same distribution. However, in the case when the Borel measurable function $f : [a,b] \rightarrow \mathbb{R}$ is bounded (i.e., $\|f\|_\infty < +\infty$) we can use the Hoeffding-Azuma inequality to design non-asymptotic confidence intervals, see Theorem 13.0.22.

4.2.2 Convergence with probability one

By using Lemma 13.0.20 and Theorem 4.2.2 we get the following result on convergence with probability one of the randomized Euler scheme.

Corollary 4.2.6. *Let $\xi \in \mathbb{R}^d$ and let f satisfy the assumptions (A1)-(A4) for some $\varrho \in (0,1]$. Then for all $\varepsilon \in (0, \min\{\varrho + \frac{1}{2}, 1\})$ there exists a random variable η_ε such that*

$$\sup_{a \leq u \leq b} \|z(u) - l_n(u)\| \leq \eta_\varepsilon \cdot n^{-\min\{\varrho + \frac{1}{2}, 1\} + \varepsilon} \text{ almost surely} \tag{4.127}$$

for all $n \in \mathbb{N}$.

4.3 Deterministic two-stage Runge-Kutta method

The definition of the deterministic two-stage Runge-Kutta method (also known as an *explicit mid-point method*) goes as follows

$$y^0 = \xi, \quad (4.128)$$

and

$$y_{1/2}^j = y^{j-1} + \frac{h}{2} f(t_{j-1}, y^{j-1}), \quad (4.129)$$

$$y^j = y^{j-1} + h f(t_{j-1} + \frac{h}{2}, y_{1/2}^j), \quad (4.130)$$

for $j = 1, 2, \dots, n$. As a time-continuous approximation of z in the whole interval $[a, b]$ we set

$$l_n(t) := l_{j,n}(t) \text{ for } t \in [t_j, t_{j+1}], \quad j = 0, 1, \dots, n-1, \quad (4.131)$$

where

$$l_{k,n}(t) = \frac{y^{j+1} - y^j}{t_{j+1} - t_j} (t - t_j) + y^j = y^j + (t - t_j) f(t_j + \frac{h}{2}, y_{1/2}^{j+1}). \quad (4.132)$$

4.4 Randomized two-stage Runge-Kutta method

As in the case of randomized Euler scheme, let $(\tau_j)_{j \in \mathbb{N}_0}$ be a sequence of independent random variables on $(\Omega, \Sigma, \mathbb{P})$ that are identically uniformly distributed on $[0, 1]$. Let $n \in \mathbb{N}$, $h = (b - a)/n$, $t_j = a + jh$ for $j = 0, 1, \dots, n$. We define the randomized two-stage Runge-Kutta method as follows

$$y^0 = \xi, \quad (4.133)$$

and

$$y_\tau^j = y^{j-1} + h \tau_j f(t_{j-1}, y^{j-1}), \quad (4.134)$$

$$y^j = y^{j-1} + h f(\theta_j, y_\tau^j), \quad (4.135)$$

for $j = 1, 2, \dots, n$, where $\theta_j = t_{j-1} + h \tau_j$ is uniformly distributed in $[t_{j-1}, t_j]$. Moreover, as a time-continuous approximation of z in the whole interval $[a, b]$ we set

$$l_n(t) := l_{j,n}(t) \text{ for } t \in [t_j, t_{j+1}], \quad j = 0, 1, \dots, n-1, \quad (4.136)$$

where

$$l_{k,n}(t) = \frac{y^{j+1} - y^j}{t_{j+1} - t_j} (t - t_j) + y^j = y^j + (t - t_j) f(\theta_{j+1}, y_\tau^{j+1}). \quad (4.137)$$

In particular, if we take $\tau_j = 1/2$ for all j then we recover the deterministic explicit mid-point method, see the previous section.

Remark 4.4.1. *Randomized algorithms for ODEs have been considered, for example, in [?], [?]*

4.5 On connection between Euler/RK2 and gradient descent algorithms

Let us consider the following optimization problem

$$\min_{x \in \mathbb{R}^d} f(x) \quad (4.138)$$

for a given $f \in C^1(\mathbb{R}^d; \mathbb{R})$. We associate with (4.138) the following ODE

$$\begin{cases} x'(t) = -\nabla f(x(t)), & t \in [0, +\infty), \\ x(0) = x_0. \end{cases} \quad (4.139)$$

Such systems are called *gradient systems*. We assume that the gradient $\mathbb{R}^d \ni x \rightarrow \nabla f(x) \in \mathbb{R}^d$ satisfies the assumptions (A1), (A2), (A3). Then the equation (4.139) has a unique solution x that is defined on the whole $[0, +\infty)$, see Remark 4.5.2. Note that we have

$$\frac{d}{dt}f(x(t)) = \langle \nabla f(x(t)), x'(t) \rangle = -\|\nabla f(x(t))\|^2 \leq 0. \quad (4.140)$$

Hence, f is non-increasing along the trajectory of x . Furthermore, we have the following result about the convergence of x to the local minimum of f . The proof can be found in [4], page 142.

Theorem 4.5.1. *If x^* is an isolated critical point of f which is also a local minimum, then the stationary solution $x = x^*$ is an asymptotically stable solution of (4.139).*

Theorem above implies that if the initial-value x_0 is sufficiently close to x^* then $\lim_{t \rightarrow +\infty} x(t) = x^*$. Approaching the local minimum x^* of f by observing the trajectory $x = x(t)$ while $t \rightarrow +\infty$ is sometimes called *continuous learning*.

For $h > 0$ and $t_k = kh$, $k = 0, 1, \dots$, the Euler scheme for (4.139) takes the form

$$\begin{aligned} y_0 &= x_0, \\ y_{k+1} &= y_k - h \cdot \nabla f(y_k), \quad k = 0, 1, \dots \end{aligned} \quad (4.141)$$

This is the well-known gradient descent (GD) algorithm for searching a (local) minimum of f when starting from x_0 . In this context, the stepsize h can be interpreted as the learning rate hyperparameter. Using the (GD) method in order to minimize f is also called *discrete learning*. For discrete learning, when h is sufficiently small and ∇f is globally Lipschitz, we can only be sure that either $f(x_k) \rightarrow -\infty$ either $\nabla f(x_k) \rightarrow 0$ when $k \rightarrow +\infty$, see.... In that sense, we can say that discrete learning is harder than its continuous counterpart. However, since in general we do not know the exact solution to (4.139), we can provide on computer only the discrete one.

We discussed the origins of the gradient descent method (GD) from the perspective of gradient ODEs (9.93). We now show how to obtain its stochastic counterpart called *stochastic gradient descent* (SGD). In ML many loss functions f that we need to minimize have the following special form

$$f(x) = \frac{1}{n} \sum_{j=1}^n f_j(x). \quad (4.142)$$

For such a function the (full) GD method has the form

$$\begin{aligned} y_0 &= x_0, \\ y_{k+1} &= y_k - h \cdot \frac{1}{n} \sum_{j=1}^n \nabla f_j(y_k), \quad k = 0, 1, \dots \end{aligned} \quad (4.143)$$

For large n (which is mostly the size of the training data) computation of the next y_{k+1} may be very computationally demanding and, therefore, time-consuming. Note that we have

$$f(x) = \frac{1}{n} \sum_{j=1}^n f_j(x) = \mathbb{E}(f_\xi(x)) \quad (4.144)$$

where $\xi \sim U(\{1, 2, \dots, n\})$. The idea of constructing the SGD is now to use Monte Carlo approximation to $\mathbb{E}(f_\xi(x))$. This can be done in the following way:

- (i) we approximate $\mathbb{E}(f_\xi(x))$ by using only one sample per step, i.e., we take $(\xi_j)_{j \in \mathbb{N}}$ an iid sequence from $U(\{1, 2, \dots, n\})$ and

$$\begin{aligned} y_0 &= x_0, \\ y_{k+1} &= y_k - h \nabla f_{\xi_{k+1}}(y_k), \quad k = 0, 1, \dots \end{aligned} \quad (4.145)$$

- (ii) we fix M , the size of the mini-batch, at each $(k+1)$ th-step we sample M ($M \ll n$) independent random numbers $\xi_{k+1}^{(1)}, \xi_{k+1}^{(2)}, \dots, \xi_{k+1}^{(M)}$ from $U(\{1, 2, \dots, n\})$ and then we set

$$\begin{aligned} y_0 &= x_0, \\ y_{k+1} &= y_k - h \cdot \frac{1}{M} \cdot \sum_{j=1}^M \nabla f_{\xi_{k+1}^{(j)}}(y_k), \quad k = 0, 1, \dots \end{aligned} \quad (4.146)$$

(We also assume that all random variables $(\xi_k^{(j)})_{j=1,2,\dots,M,k \in \mathbb{N}}$ are independent.)

Hence, in the SGD version, we do not compute the full gradient but, instead, we randomly pick an index j and function f_j for which we are computing gradient and then using it to establish the next step y_{k+1} .

There are of course other options for discretizing (4.139). For example, we can also apply randomized RK2 method to (4.139) and obtain the approximation sequence:

$$\begin{aligned} y^0 &= x_0, \\ y_\tau^{k+1} &= y^k - h \tau_{k+1} \cdot \nabla f(y^k) \\ y^{k+1} &= y^k - h \cdot \nabla f(y_\tau^{k+1}), \quad k = 0, 1, \dots, \end{aligned} \quad (4.147)$$

where $(\tau_k)_{k \in \mathbb{N}}$ is an iid sequence sampled from $U(0, 1)$. Analogously, we can define the following *randomized two-stage stochastic gradient descent method* for the functions (4.142):

$$\begin{aligned} \bar{y}^0 &= x_0, \\ \bar{y}_\tau^j &= \bar{y}^{j-1} - h \tau_j \nabla f_{\xi_j}(\bar{y}^{j-1}), \\ \bar{y}^j &= \bar{y}^{j-1} - h \nabla f_{\zeta_j}(\bar{y}_\tau^j), \quad j = 1, 2, \dots, \end{aligned} \quad (4.148)$$

where $(\tau_k)_{k \in \mathbb{N}}$ is an iid sequence sampled from $U(0, 1)$, $(\xi_j)_{j \geq 1}$ and $(\zeta_j)_{j \geq 1}$ are iid sequences sampled from $U(\{1, 2, \dots, n\})$, and finally $(\tau_j)_{j \geq 1}$, $(\xi_j)_{j \geq 1}$, $(\zeta_j)_{j \geq 1}$ are independent.

We were unable to find any references or analysis of this method in the literature, however, we conjecture that this method possesses better convergence properties than the gradient descent method (9.93).

Remark 4.5.2. *Since the gradient ∇f satisfies (A1), (A2), (A3), we have the existence and uniqueness of solution x for (4.139) on the interval $[0, T]$ for any $T > 0$. Moreover, this solution is bounded on $[0, T]$ (which follows from the fact that ∇f is of at most linear growth), and therefore x can be continued on the whole $[0, +\infty)$, see, for example, Theorem 2.11 and Example 2.1, pages 46-47 in [4].*

4.6 Exercises

1. Show that $x \vee y - x \wedge y = |x - y|$ for all $x, y \in \mathbb{R}$.
2. Show that the function (4.5) satisfies the assumptions (A1)-(A4).
3. Justify the boundedness of the function (4.69).
4. Implement deterministic and randomized Euler scheme in Python and perform numerical experiments for the function (4.5).

5. Prove that the stochastic process $\{\sup_{a \leq u \leq t} \|z(u) - l_n(u)\|^p\}_{t \in [a, b]}$ is product measurable.
6. Give a proof of (4.104), (4.105).
7. Show that $0 \leq s_n^2(f) \leq s^2(f)$ for all $n \in \mathbb{N}$, $f \in L^2[a, b]$.
8. Let us assume that f is Borel measurable and bounded. Show that for any $n \in \mathbb{N}$, $\delta \in (0, 1)$ we have that

$$I(f) \in \left[\mathcal{R}_n(f) - \frac{C(\delta)}{\sqrt{n}}, \mathcal{R}_n(f) + \frac{C(\delta)}{\sqrt{n}} \right] \quad (4.149)$$

with the probability at least $1 - \delta$, where $C(\delta) = (8 \ln(2/\delta))^{1/2} \cdot (b - a) \cdot \|f\|_\infty$.

9. Show that for any discretization $a = t_0 < t_1 < \dots < t_n = b$ and any function $\phi : [a, b] \rightarrow [0, +\infty)$ the following holds

$$\sum_{k=0}^{n-1} \phi(t_k) \cdot \mathbf{1}_{[t_k, t_{k+1})}(s) \leq \sup_{a \leq u \leq s} \phi(u) \quad (4.150)$$

for any $s \in [a, b]$.

Chapter 5

Canonical stochastic processes

In this chapter we assume that $(\Omega, \Sigma, (\Sigma_t)_{t \in [0, +\infty)}, \mathbb{P})$ is a filtered probability space. We start with recalling definitions of Wiener and Poisson processes (non-homogeneous and compound).

5.1 Building blocks—Wiener and Poisson processes

Definition 5.1.1. An $\mathbb{R}^m$ -valued stochastic process $W = (W(t))_{t \in [0, +\infty)}$ is called an *m-dimensional $(\Sigma_t)_{t \in [0, +\infty)}$ -Wiener process* on $(\Omega, \Sigma, \mathbb{P})$ (or a *m-dimensional Wiener process with respect to the filtration $(\Sigma_t)_{t \in [0, +\infty)}$*) if

- (i) $\mathbb{P}(W(0) = 0) = 1$,
- (ii) $\sigma(W(t)) \subset \Sigma_t$ for all $t \in [0, +\infty)$,
- (iii) almost all trajectories of W are continuous,
- (iv) $\sigma(W(t) - W(s))$ is independent of Σ_s for all $0 \leq s < t$,
- (v) for $0 \leq s < t$ the increment $W(t) - W(s)$ is normally distributed with mean zero and covariance matrix equal to $(t - s)I_m$, where I_m is the $\mathbb{R}^{m \times m}$ -identity matrix.

Definition 5.1.2. An $\mathbb{R}^m$ -valued stochastic process $N = (N(t))_{t \in [0, +\infty)}$ is called an *m-dimensional non-homogeneous $(\Sigma_t)_{t \in [0, +\infty)}$ -Poisson process* on $(\Omega, \Sigma, \mathbb{P})$ (or an *m-dimensional non-homogeneous Poisson process with respect to the filtration $(\Sigma_t)_{t \in [0, +\infty)}$*) if

- (i) $\mathbb{P}(N(0) = 0) = 1$,
- (ii) $\sigma(N(t)) \subset \Sigma_t$ for all $t \in [0, +\infty)$,
- (iii) almost all trajectories of N are càdlàg,
- (iv) $\sigma(N(t) - N(s))$ is independent of Σ_s for all $0 \leq s < t$,
- (v) for $j \in \{1, 2, \dots, m\}$, $0 \leq s < t$ the increment $N^j(t) - N^j(s)$ has Poisson law with the intensity parameter $\int_s^t \lambda_j(u) du$, where the intensity function $\lambda_j : [0, +\infty) \rightarrow (0, +\infty)$ is Borel and locally integrable,
- (vi) $\{\Sigma_\infty^{N^1}, \dots, \Sigma_\infty^{N^m}\}$ is a family of independent σ -fields.

We also denote by $m_j(t) = \int_0^t \lambda_j(s) ds$ and $\Lambda(t, s) = m(t) - m(s)$ for $s, t \in [0, T]$ and $j = 1, 2, \dots, m_N$. If λ_j is a constant function then we say that N^j is a homogeneous Poisson process.

Definition 5.1.3. Let $N = (N_t)_{t \in [0, +\infty)}$ be a one-dimensional homogeneous Poisson process, with respect to a given filtration $(\Sigma_t)_{t \in [0, +\infty)}$, and let $(\xi_j)_{j \in \mathbb{N}}$ be an i.i.d sequence of real random variables with distribution μ , such that the σ -fields Σ_∞^N and $\sigma\left(\bigcup_{j \in \mathbb{N}} \sigma(\xi_j)\right)$ are independent. The process

$$J(t) = \sum_{j=1}^{N(t)} \xi_j, \quad t \in [0, +\infty), \quad (5.1)$$

is called the *compound Poisson process*.

Analogously we can define the non-homogeneous compound Poisson process.

Due to Definition 13.9 and Theorem 13.11 in [89] the condition (iv) in the above definitions implies that $Z \in \{N, W\}$ is a process with *independent increments*, i.e., for all $n \in \mathbb{N}$, $0 \leq t_1 < \dots < t_n$ the family

$$\{\sigma(Z(t_1)), \sigma(Z(t_2) - Z(t_1)), \dots, \sigma(Z(t_n) - Z(t_{n-1}))\} \quad (5.2)$$

is a collection of independent σ -fields. The compound Poisson process Y has also independent increments in the sense that the collection (5.2) for $Z = J$ consists of independent σ -fields. Hence, by Theorem 13.10 in [89] for all $0 \leq s < t$ the σ -fields $\sigma(J(t) - J(s))$ and Σ_s^J are independent.

The independence-of-increments property is crucial both from theoretical and practical point of view. In the next section we provide, based on this property, simple methods of simulating the processes above.

5.2 Simulation of Poisson and Wiener processes

We will use the fact that processes N and W have independent increments.

Firstly, we describe how to generating values of $Z \in \{N, W\}$ at fixed grid points

$$0 = t_0 < t_1 < \dots < t_n = T, \quad n \in \mathbb{N}, \quad T \in (0, +\infty). \quad (5.3)$$

We have that

$$Z(t_i) = \sum_{j=0}^{i-1} \Delta Z_j, \quad i = 0, 1, \dots, n, \quad (5.4)$$

where $\Delta Z_j = Z(t_{j+1}) - Z(t_j)$ and the random variables $(\Delta Z_j)_{j=0}^{n-1}$ are independent. Hence, let us take $(V_j)_{j=0}^{n-1}$, an i.i.d sequence of random variables, where $V_1 \sim N(0, 1)$, and let $(U_j)_{j=0}^{n-1}$ be a sequence of in-

dependent random variables, such that $U_j \sim \text{Pois}\left(\int_{t_j}^{t_{j+1}} \lambda(s) ds\right)$. (We also assume that $(V_j)_{j=0}^{n-1}$ and $(U_j)_{j=0}^{n-1}$

are independent, i.e., the σ -fields $\sigma\left(\bigcup_{j=0}^{n-1} \sigma(U_j)\right)$ and $\sigma\left(\bigcup_{j=0}^{n-1} \sigma(V_j)\right)$ are independent.) Then

- if $Z = W$, then $\Delta Z_j \sim N(0, t_{j+1} - t_j)$ has the same law as $(t_{j+1} - t_j)^{1/2} \cdot V_j$, and we set

$$Z(t_i) = \sum_{j=0}^{i-1} (t_{j+1} - t_j)^{1/2} \cdot V_j, \quad (5.5)$$

- if $Z = N$, then ΔZ_j and U_j has the same law $\text{Pois}\left(\int_{t_j}^{t_{j+1}} \lambda(s) ds\right)$, and

$$Z(t_i) = \sum_{j=0}^{i-1} U_j. \quad (5.6)$$

- for $Z = J$ let us assume that the values of the Poisson process $N(t_1), \dots, N(t_n)$ has already been generated by the method described above. Then sample an i.i.d sequence $\xi_1, \dots, \xi_{N(t_n)}$ according to the distribution μ and take

$$Z(t_i) = \sum_{k=1}^{N(t_i)} \xi_k \quad (5.7)$$

for $i = 1, \dots, n$.

5.3 Brownian and Poisson bridges

Recall that conditional variance of a square integrable random variable X , given sub- σ -field $\mathcal{G}$, is defined as

$$\text{Var}(X \mid \mathcal{G}) = \mathbb{E}\left((X - \mathbb{E}(X \mid \mathcal{G}))^2 \mid \mathcal{G}\right). \quad (5.8)$$

Lemma 5.3.1. (*[26], [63], [67], [74]*) *Let us fix $n \in \mathbb{N}$ and $0 = t_0 < t_1 < \dots < t_n = T$. Then for $t \in (t_i, t_{i+1})$ and $i = 0, 1, \dots, n-1$*

(i) *conditioned on $(W(t_1), \dots, W(t_n))$ the random variable $W(t)$ has the law $N(\mu_W(t), \sigma_W^2(t))$ with*

$$\mu_W(t) = \mathbb{E}(W(t) \mid W(t_1), \dots, W(t_n)) = \frac{(t - t_i) \cdot W(t_{i+1}) + (t_{i+1} - t) \cdot W(t_i)}{t_{i+1} - t_i}, \quad (5.9)$$

$$\sigma_W^2(t) = \text{Var}(W(t) \mid W(t_1), \dots, W(t_n)) = \frac{(t_{i+1} - t) \cdot (t - t_i)}{t_{i+1} - t_i}. \quad (5.10)$$

(ii) *conditioned on $(N(t_1), \dots, N(t_n))$ the increment $N(t) - N(t_i)$ has a binomial distribution with the number of trials $N(t_{i+1}) - N(t_i)$ and with the probability of success in each trial equal to $\frac{\Lambda(t, t_i)}{\Lambda(t_{i+1}, t_i)}$.*

Hence

$$\mu_N(t) = \mathbb{E}(N(t) \mid N(t_1), \dots, N(t_n)) = \frac{\Lambda(t, t_i) \cdot N(t_{i+1}) + \Lambda(t_{i+1}, t) \cdot N(t_i)}{\Lambda(t_{i+1}, t_i)}, \quad (5.11)$$

$$\sigma_N^2(t) = \text{Var}(N(t) \mid N(t_1), \dots, N(t_n)) = (N(t_{i+1}) - N(t_i)) \cdot \frac{\Lambda(t_{i+1}, t) \cdot \Lambda(t, t_i)}{(\Lambda(t_{i+1}, t_i))^2}, \quad (5.12)$$

for $t \in [t_i, t_{i+1}]$, $i = 0, 1, \dots, n-1$.

In particular, since W, N have independent increments we have that for all $t \in [t_i, t_{i+1}]$, $i = 0, 1, \dots, n-1$, $A \in \mathcal{B}(\mathbb{R})$, $k = 0, 1, \dots$

$$\mathbb{P}(W(t) \in A \mid W(t_1), \dots, W(t_n)) = \mathbb{P}(W(t) \in A \mid W(t_i), W(t_{i+1})), \quad (5.13)$$

and

$$\mathbb{P}(N(t) - N(t_i) = k \mid N(t_1), \dots, N(t_n)) = \mathbb{P}(N(t) - N(t_i) = k \mid N(t_i), N(t_{i+1})). \quad (5.14)$$

This, together with Lemma 5.3.1, allows us to generate in easy way additional values of W and N in between previously generated values $Z(t_1), \dots, Z(t_n)$ of respective process $Z \in \{N, W\}$.

For $Z \in \{N, W\}$ let

$$\bar{Z}_n = (\bar{Z}_n(t))_{t \in [0, T]} = \left(\mathbb{E}(Z(t) \mid Z(t_1), \dots, Z(t_n)) \right)_{t \in [0, T]} = \left(\sum_{i=0}^{n-1} \mu_Z(t) \cdot \mathbf{1}_{[t_i, t_{i+1}]}(t) \right)_{t \in [0, T]}, \quad (5.15)$$

be the piecewise linear interpolation of $\{(t_i, Z(t_i))\}_{i=0,1,\dots,n}$ and define the stochastic process

$$\hat{Z}_n = (\hat{Z}_n(t))_{t \in [0, T]} = \left(Z(t) - \bar{Z}_n(t) \right)_{t \in [0, T]}. \quad (5.16)$$

Then the process $\hat{Z}_n$ consists of independent Brownian bridges (for $Z = W$) or Poisson bridges (when $Z = N$) corresponding to the respective (disjoint) subintervals. Moreover, $\hat{Z}_n(t_j) = 0$ for all $j = 0, 1, \dots, n$, and $\mathbb{E}(\hat{Z}_n(t)) = 0$ for all $t \in [0, T]$. This process often appears in the theory of optimal approximation of SDEs. It is easy to see that $\sigma(\hat{Z}_n(t)) \subset \sigma(\mathcal{N}_n(Z))$ for all $t \in [0, T]$, however, $\hat{Z}_n$ is not adapted to $(\Sigma_t)_{t \in [0, T]}$.

5.4 Cauchy process

In this section, we take a look at the scalar symmetric Cauchy process, which is another important example of the Lévy process. It is frequently used to model extreme events due to its infinite variance and lack of a finite value of expectation at any time $t > 0$.

Definition 5.4.1. An $\mathbb{R}$ -valued stochastic process $C = (C(t))_{t \in [0, +\infty)}$ is called a $(\Sigma_t)_{t \in [0, +\infty)}$ -Cauchy process on $(\Omega, \Sigma, \mathbb{P})$ (or a Cauchy process with respect to the filtration $(\Sigma_t)_{t \in [0, +\infty)}$) if

- (i) $\mathbb{P}(C(0) = 0) = 1$,
- (ii) $\sigma(C(t)) \subset \Sigma_t$ for all $t \in [0, +\infty)$,
- (iii) almost all trajectories of C are càdlàg,
- (iv) $\sigma(C(t) - C(s))$ is independent of Σ_s for all $0 \leq s < t$,
- (v) for $0 \leq s < t$ the increment $C(t) - C(s)$ has the (symmetric) Cauchy law with location 0 and scale $t - s$, i.e.,

$$\mathbb{E}\left(\exp(iu(C(t) - C(s)))\right) = \exp(-(t - s)|u|), \quad u \in \mathbb{R}. \quad (5.17)$$

Remark 5.4.2. From (5.17) we have for any $t \geq 0$

$$\mathbb{E}\left(\exp(iuC(t))\right) = \exp(-t|u|), \quad u \in \mathbb{R}, \quad (5.18)$$

hence $C(t) \sim \text{Cauchy}(0, t)$ with density

$$\mathbb{P}(C(t) \in dx) = \frac{1}{\pi} \frac{t}{t^2 + x^2} dx. \quad (5.19)$$

In particular, $\mathbb{E}|C(t)| = +\infty$ for $t > 0$ (and $\mathbb{E}C(t)$ is not well-defined).

Remark 5.4.3. The Cauchy process is a Lévy process (since it has independent and stationary increments and it is continuous in probability), and it is 1-stable, i.e., for any $c > 0$,

$$C(ct) \stackrel{d}{=} cC(t). \quad (5.20)$$

Moreover, it has no Brownian part, and it is a pure jump process (but it is not a compound Poisson process).

5.5 Optimal reconstruction of trajectories of the Wiener processes

In this section, we show how to approximate optimally (in the Information-Based Complexity sense) trajectories of the Wiener process. Firstly, we rigorously define what we mean by an optimal algorithm.

We consider the vector space $C([0, T])$ of all continuous functions $f : [0, T] \rightarrow \mathbb{R}$, equipped with the supremum norm $\|\cdot\|_\infty$. Then $(C([0, T]), \|\cdot\|_\infty)$ is a separable Banach space. In $C([0, T])$ we consider the Borel σ -field $\mathcal{B}(C([0, T]))$.

Any approximation method $\bar{X} = (\bar{X}_n)_{n \in \mathbb{N}}$ is defined by two sequences $\bar{\varphi} = (\varphi_n)_{n \in \mathbb{N}}$, $\bar{\Delta} = (\Delta_n)_{n \in \mathbb{N}}$, where every

$$\varphi_n : \mathbb{R}^n \rightarrow C([0, T]) \quad (5.21)$$

is Borel function (i.e., $\mathcal{B}(\mathbb{R}^n)$ -to- $\mathcal{B}(C([0, T]))$ measurable) and

$$\Delta_n = \{t_{0,n}, t_{1,n}, \dots, t_{n,n}\} \quad (5.22)$$

is a (possibly) non-uniform partition of $[0, T]$ such that

$$0 = t_{0,n} < t_{1,n} < \dots < t_{n,n} = T. \quad (5.23)$$

By

$$\bar{\mathcal{N}}(W) = (\mathcal{N}_n(W))_{n \in \mathbb{N}} \quad (5.24)$$

we mean a sequence of vectors $\mathcal{N}_n(W)$ which provides *standard information* with n evaluations about the Wiener process W at the discrete points from Δ_n , i.e.,

$$\mathcal{N}_n(W) = [W(t_{1,n}), W(t_{2,n}), \dots, W(t_{n,n})]. \quad (5.25)$$

(Recall that $W(0) = 0$ almost surely.) After computing the information $\mathcal{N}_n(W)$, we apply the mapping φ_n in order to obtain the n th approximation $\bar{X}_n = (\bar{X}_n(t))_{t \in [0, T]}$ in the following way

$$\bar{X}_n = \varphi_n(\mathcal{N}_n(W)). \quad (5.26)$$

The informational cost of each $\bar{X}_n$ is equal to n evaluations of W . The set of all methods $\bar{X} = (\bar{X}_n)_{n \in \mathbb{N}}$, defined as above, is denoted by χ^{noneq} . We also consider the following subclass of χ^{noneq}

$$\chi^{\text{eq}} = \{\bar{X} \in \chi^{\text{noneq}} \mid \forall n \in \mathbb{N} \Delta_n = \Delta_n^{\text{eq}} := \{iT/n \mid i = 0, 1, \dots, n\}\}. \quad (5.27)$$

Before we give a definition of error of $\bar{X}$, we discuss some analytical properties of $\bar{X}_n$. Note that for each $n \in \mathbb{N}$ the function

$$\bar{X}_n : \Omega \rightarrow C([0, T]) \quad (5.28)$$

is $\sigma(\mathcal{N}_n(W))/\mathcal{B}(C([0, T]))$ -measurable as composition of measurable mappings. Hence, it is a random element in $C([0, T])$. By taking

$$\tilde{X}_n(\omega, t) = (\bar{X}_n(\omega))(t), \quad (\omega, t) \in \Omega \times [0, T], \quad (5.29)$$

we obtain $\sigma(\mathcal{N}_n(W)) \otimes \mathcal{B}([0, T])$ -to- $\mathcal{B}(\mathbb{R})$ measurable stochastic process, see Theorem 14.0.6. Hence, $\bar{X}_n$ and $\tilde{X}_n$ are two representations of the same n th approximation. In the rest of this chapter, we shall abuse notation and write $\bar{X}_n$ instead of $\tilde{X}_n$.

The n th error of a method $\bar{X}$ is defined as

$$e_n(\bar{X}) = \|W - \bar{X}_n\|_{L^2(\Omega \times [0, T])} = \left(\int_{\Omega \times [0, T]} |W(\omega, t) - (\bar{X}_n(\omega))(t)|^2 \mathbb{P} \times \lambda_1(d\omega, dt) \right)^{1/2}. \quad (5.30)$$

We implicitly assume that for all $n \in \mathbb{N}$

$$\mathbb{E} \int_0^T |\bar{X}_n(t)|^2 dt < +\infty, \quad (5.31)$$

since otherwise (by the Minkowski inequality) the error $e_n(\bar{X})$ would be infinite for some n . Note that by Tonelli's Theorem, the error $e_n(\bar{X})$ can be equivalently written as

$$e_n(\bar{X}) = \left(\mathbb{E} \|W - \bar{X}_n\|_{L^2([0, T])}^2 \right)^{1/2} = \left(\mathbb{E} \int_0^T |W(t) - \bar{X}_n(t)|^2 dt \right)^{1/2}. \quad (5.32)$$

Finally, the n th minimal error, in the respective class of algorithms, is given by

$$e^\diamond(n) = \inf_{\bar{X} \in \chi^\diamond} e_n(\bar{X}), \quad \diamond \in \{\text{eq}, \text{noneq}\}. \quad (5.33)$$

We will investigate (possibly) sharp bounds on n th minimal errors. Moreover, we determine the optimal methods $\bar{X}^\diamond, \diamond \in \{\text{eq}, \text{noneq}\}$ that attain the infimum in (5.33).

At first, we establish a lower bound for the error of an arbitrary method from χ^{noneq} .

Lemma 5.5.1. *For any $\bar{X} \in \chi^{\text{noneq}}$ and for all $n \in \mathbb{N}$ we have that*

$$e_n(\bar{X}) \geq \frac{T}{\sqrt{6}} \cdot n^{-1/2}. \quad (5.34)$$

Proof. Let us fix any method $\bar{X} \in \chi^{\text{noneq}}$ based on a sequence of discretizations $(\Delta_n)_{n \in \mathbb{N}}$. Hence, by (5.31), Tonelli's theorem, and basic properties of Lebesgue integral we get that

$$\mathbb{E}|\bar{X}_n(t)|^2 < +\infty \quad (5.35)$$

for all $n \in \mathbb{N}$ and almost all $t \in [0, T]$. Then by the projection property for conditional expectation (see Theorem 13.0.29) we have for all $n \in \mathbb{N}$ and almost all $t \in [0, T]$ that

$$\mathbb{E}|W(t) - \bar{X}_n(t)|^2 \geq \mathbb{E}|W(t) - \mathbb{E}(W(t) \mid \mathcal{N}_n(W))|^2, \quad (5.36)$$

since $\mathbb{E}|W(t)|^2 < +\infty$, for all $t \in [0, T]$, and $\sigma(\bar{X}_n(t)) \subset \sigma(\mathcal{N}_n(W))$. Due to Tonelli's theorem and product measurability of the underlying stochastic processes, both functions $[0, T] \ni t \rightarrow \mathbb{E}|W(t) - \bar{X}_n(t)|^2$ and $[0, T] \ni t \rightarrow \mathbb{E}|W(t) - \mathbb{E}(W(t) \mid \mathcal{N}_n(W))|^2$ are Borel. Therefore, (5.36) and Lemma 5.3.1 imply for all $n \in \mathbb{N}$ that

$$\begin{aligned} (e_n(\bar{X}))^2 &= \int_0^T \mathbb{E}|W(t) - \bar{X}_n(t)|^2 dt \geq \int_0^T \mathbb{E}|W(t) - \mathbb{E}(W(t) \mid \mathcal{N}_n(W))|^2 dt \\ &= \int_0^T \mathbb{E}\left(\text{Var}(W(t) \mid \mathcal{N}_n(W))\right) dt = \sum_{i=0}^{n-1} \int_{t_i}^{t_{i+1}} \mathbb{E}|\hat{W}_n(t)|^2 dt = \frac{1}{6} \sum_{i=0}^{n-1} (t_{i+1} - t_i)^2, \end{aligned} \quad (5.37)$$

and by using the Jensen inequality we arrive at

$$(e_n(\bar{X}))^2 \geq \frac{1}{6} \sum_{i=0}^{n-1} (t_{i+1} - t_i)^2 \geq \frac{1}{6n} \left(\sum_{i=0}^{n-1} (t_{i+1} - t_i) \right)^2 = \frac{T^2}{6n}. \quad (5.38)$$

This implies (5.34). $\blacksquare$

Now we show that the method $\bar{W}^{\text{eq}}$ based on the sequence of equidistant discretizations $\bar{\Delta}^{\text{eq}}$ achieves the error (5.34).

Theorem 5.5.2. *For the method*

$$\bar{W}_n^{\text{eq}} = (\bar{W}_n^{\text{eq}}(t))_{t \in [0, T]} = \left(\sum_{i=0}^{n-1} \frac{(t - t_i) \cdot W(t_{i+1}) + (t_{i+1} - t) \cdot W(t_i)}{t_{i+1} - t_i} \cdot \mathbf{1}_{[t_i, t_{i+1}]}(t) \right)_{t \in [0, T]} \quad (5.39)$$

where $t_i = iT/n$, $i = 0, 1, \dots, n$, we have

$$e_n(\bar{W}^{\text{eq}}) = \frac{T}{\sqrt{6}} \cdot n^{-1/2}. \quad (5.40)$$

Therefore, the method $\bar{W}^{\text{eq}} = (\bar{W}_n^{\text{eq}})_{n \in \mathbb{N}}$ is optimal in the class χ^{noneq} , i.e., for all $n \in \mathbb{N}$

$$e_n(\bar{W}^{\text{eq}}) = \inf_{\bar{X} \in \chi^{\text{noneq}}} e_n(\bar{X}) = \frac{T}{\sqrt{6}} \cdot n^{-1/2}. \quad (5.41)$$

Proof. By (5.34) we have that

$$(e_n(\bar{W}^{\text{eq}}))^2 = \sum_{i=0}^{n-1} \int_{t_i}^{t_{i+1}} \mathbb{E}|W(t) - \bar{W}^{\text{eq}}(t)|^2 dt = \frac{1}{6} \sum_{i=0}^{n-1} (t_{i+1} - t_i)^2 = \frac{T^2}{6n} \leq \left(\inf_{\bar{X} \in \mathcal{X}^{\text{noneq}}} e_n(\bar{X}) \right)^2, \quad (5.42)$$

hence we get (5.41). $\blacksquare$

Remark 5.5.3. *With the help of Theorems 14.0.5 (iii), 14.0.6 (iii) it turns out that the thesis of Theorem 5.5.2 holds true even if we consider the mappings (5.21) as $\mathcal{B}(\mathbb{R}^n)$ -to- $\mathcal{B}(L^2([0, T]))$ measurable functions*

$$\varphi_n : \mathbb{R}^n \rightarrow L^2([0, T]). \quad (5.43)$$

5.6 Approximation of Lebesgue integral of Wiener process

In this section, we investigate the minimal error for the problem of approximation of the following integral

$$\mathcal{I}(W) = \int_0^T W(t) dt. \quad (5.44)$$

The integrals of the form (5.44) appear, for example, in the Wagner-Platen scheme that is used for the approximation of solutions of SDEs. See also Chapter 10 where a more general problem of weighted integration of the Wiener process is considered.

Remark 5.6.1. *Due to Fubini's Theorem, the function $\Omega \ni \omega \rightarrow \mathcal{I}(W)(\omega) = \int_0^T W(\omega, t) dt \in \mathbb{R}$ is Σ -to- $\mathcal{B}(\mathbb{R})$ measurable. Moreover, it is a Gaussian random variable.*

An approximation method $\bar{X} = (\bar{X}_n)_{n \in \mathbb{N}}$ is defined by two sequences $\bar{\varphi} = (\varphi_n)_{n \in \mathbb{N}}$, $\bar{\Delta} = (\Delta_n)_{n \in \mathbb{N}}$, where every

$$\varphi_n : \mathbb{R}^n \rightarrow \mathbb{R} \quad (5.45)$$

is a Borel function and

$$\Delta_n = \{t_{0,n}, t_{1,n}, \dots, t_{n,n}\} \quad (5.46)$$

is a (possibly) non-uniform partition of $[0, T]$ such that

$$0 = t_{0,n} < t_{1,n} < \dots < t_{n,n} = T. \quad (5.47)$$

By

$$\bar{\mathcal{N}}(W) = (\mathcal{N}_n(W))_{n \in \mathbb{N}} \quad (5.48)$$

we mean a sequence of vectors $\mathcal{N}_n(W)$ which provides *standard information* with n evaluations about the Wiener process W at the discrete points from Δ_n , i.e.,

$$\mathcal{N}_n(W) = [W(t_{1,n}), W(t_{2,n}), \dots, W(t_{n,n})]. \quad (5.49)$$

(Recall that $W(0) = 0$ almost surely.) After computing the information $\mathcal{N}_n(W)$, we apply the mapping φ_n in order to obtain the n th approximation $\bar{X}_n$ in the following way

$$\bar{X}_n = \varphi_n(\mathcal{N}_n(W)). \quad (5.50)$$

The informational cost of each $\bar{X}_n$ is equal to n evaluations of W . The set of all methods $\bar{X} = (\bar{X}_n)_{n \in \mathbb{N}}$, defined as above, is denoted by χ^{noneq} . We also consider the following subclass of χ^{noneq}

$$\chi^{\text{eq}} = \{\bar{X} \in \chi^{\text{noneq}} \mid \forall_{n \in \mathbb{N}} \Delta_n = \Delta_n^{\text{eq}} := \{iT/n \mid i = 0, 1, \dots, n\}\}. \quad (5.51)$$

Note that for each $n \in \mathbb{N}$ the function

$$\bar{X}_n : \Omega \rightarrow \mathbb{R} \quad (5.52)$$

is $\sigma(\mathcal{N}_n(W))/\mathcal{B}(\mathbb{R})$ -measurable as composition of measurable mappings. The n th error of a method $\bar{X}$ is defined as

$$e_n(\bar{X}) = \|\mathcal{I}(W) - \bar{X}_n\|_{L^2(\Omega)} = \left(\mathbb{E}|\mathcal{I}(W) - \bar{X}_n|^2 \right)^{1/2}. \quad (5.53)$$

We implicitly assume that for all $n \in \mathbb{N}$

$$\mathbb{E}|\bar{X}_n|^2 < +\infty, \quad (5.54)$$

since otherwise the error $e_n(\bar{X})$ would be infinite for some n .

Finally, the n th minimal error, in the respective class of algorithms, is given by

$$e^\diamond(n) = \inf_{\bar{X} \in \chi^\diamond} e_n(\bar{X}), \quad \diamond \in \{\text{eq}, \text{noneq}\}. \quad (5.55)$$

We will investigate (possibly) sharp bounds on n th minimal errors. Moreover, we determine the optimal methods $\bar{X}^\diamond, \diamond \in \{\text{eq}, \text{noneq}\}$ that attain the infimum in (5.55).

The following useful technical fact can be shown by direct calculations, so we left it as an exercise.

Lemma 5.6.2. ([26], [67]) *For any $s, t \in [t_i, t_{i+1}]$, $i = 0, 1, \dots, n-1$ and $n \in \mathbb{N}$ we have that*

$$\text{Cov}(\hat{W}_n(t), \hat{W}_n(s)) = \frac{(t_{i+1} - \max\{t, s\})(\min\{t, s\} - t_i)}{t_{i+1} - t_i}, \quad (5.56)$$

and

$$\int_{t_i}^{t_{i+1}} \int_{t_i}^{t_{i+1}} \text{Cov}(\hat{W}_n(t), \hat{W}_n(s)) ds dt = \frac{(t_{i+1} - t_i)^3}{12}. \quad (5.57)$$

Moreover, for any $s \in [t_j, t_{j+1}]$, $t \in [t_i, t_{i+1}]$, $i, j = 0, 1, \dots, n-1$, $i \neq j$

$$\text{Cov}(\hat{W}_n(t), \hat{W}_n(s)) = 0. \quad (5.58)$$

We now prove the following lower bound.

Lemma 5.6.3. *For any $\bar{X} \in \chi^{\text{noneq}}$ and for all $n \in \mathbb{N}$ we have that*

$$e_n(\bar{X}) \geq \frac{T^{3/2}}{\sqrt{12}} \cdot n^{-1}. \quad (5.59)$$

Proof. Let us fix any method $\bar{X} \in \chi^{\text{noneq}}$ based on a sequence of discretizations $(\Delta_n)_{n \in \mathbb{N}}$. Then by the projection property for conditional expectation (Theorem 13.0.29) we have that for all $n \in \mathbb{N}$ that

$$\mathbb{E}|\mathcal{I}(W) - \bar{X}_n|^2 \geq \mathbb{E}|\mathcal{I}(W) - \mathbb{E}(\mathcal{I}(W) \mid \mathcal{N}_n(W))|^2, \quad (5.60)$$

since $\mathbb{E}|\mathcal{I}(W)|^2 < +\infty$ and $\sigma(\bar{X}_n) \subset \sigma(\mathcal{N}_n(W))$. By the conditional version of Fubini's Theorem (Theorem 13.0.30) we get

$$\mathbb{E}(\mathcal{I}(W) \mid \mathcal{N}_n(W)) = \int_0^T \mathbb{E}(W(t) \mid \mathcal{N}_n(W)) dt \text{ a.s.} \quad (5.61)$$

Hence

$$(e_n(\bar{X}))^2 = \mathbb{E}|\mathcal{I}(W) - \bar{X}_n|^2 \geq \mathbb{E} \left| \sum_{j=0}^{n-1} \int_{t_j}^{t_{j+1}} \hat{W}_n(t) dt \right|^2 = \sum_{j=0}^{n-1} \mathbb{E}(S_j^2) + \sum_{i \neq j} \mathbb{E}(S_i S_j), \quad (5.62)$$

where $S_i = \int_{t_i}^{t_{i+1}} \hat{W}_n(t) dt$. From Lemma 5.6.2 we obtain

$$\mathbb{E}(S_j^2) = \int_{t_j}^{t_{j+1}} \int_{t_j}^{t_{j+1}} \mathbb{E}(\hat{W}_n(t) \cdot \hat{W}_n(s)) ds dt = \frac{(t_{j+1} - t_j)^3}{12}, \quad (5.63)$$

$$\mathbb{E}(S_i S_j) = \int_{t_i}^{t_{i+1}} \int_{t_j}^{t_{j+1}} \mathbb{E}(\hat{W}_n(t) \cdot \hat{W}_n(s)) ds dt = 0. \quad (5.64)$$

Therefore, by the Jensen inequality, we get

$$(e_n(\bar{X}))^2 \geq \frac{1}{12} \sum_{i=0}^{n-1} (t_{i+1} - t_i)^3 \geq \frac{T^3}{12n^2}. \quad (5.65)$$

This implies (5.59). $\blacksquare$

We now show that the trapezoidal quadrature rule is the optimal one.

Theorem 5.6.4. *For the trapezoidal quadrature rule*

$$\bar{X}_n^{\text{eq}} = \frac{T}{2n} \sum_{i=0}^{n-1} (W(t_i) + W(t_{i+1})) = \frac{T}{n} \sum_{i=0}^{n-1} W(t_i) + \frac{T}{2n} W(t_n), \quad (5.66)$$

with $t_i = iT/n$, $i = 0, 1, \dots, n$, we have

$$e_n(\bar{X}^{\text{eq}}) = \frac{T^{3/2}}{\sqrt{12}} \cdot n^{-1}. \quad (5.67)$$

Therefore, the method $\bar{X}^{\text{eq}} = (\bar{X}_n^{\text{eq}})_{n \in \mathbb{N}}$ is optimal in the class χ^{noneq} , i.e., for all $n \in \mathbb{N}$

$$e_n(\bar{X}^{\text{eq}}) = \inf_{\bar{X} \in \chi^{\text{noneq}}} e_n(\bar{X}) = \frac{T^{3/2}}{\sqrt{12}} \cdot n^{-1}. \quad (5.68)$$

Proof. The following holds

$$\bar{X}_n^{\text{eq}} = \mathbb{E}(\mathcal{I}(W) \mid W(iT/n), i = 1, \dots, n) = \int_0^T \bar{W}_n^{\text{eq}}(t) dt. \quad (5.69)$$

Thereby

$$(e_n(\bar{X}^{\text{eq}}))^2 = \mathbb{E} \left| \sum_{j=0}^{n-1} \int_{t_j}^{t_{j+1}} (W(t) - \bar{W}_n^{\text{eq}}(t)) dt \right|^2 = \frac{1}{12} \sum_{j=0}^{n-1} (t_{j+1} - t_j)^3 = \frac{T^3}{12n^2}. \quad (5.70)$$

This and (5.59) imply (5.68). $\blacksquare$

5.7 Implementation issues

Drawing trajectories of the Wiener process.

```
import numpy as np

def Wiener_path( T, n ):
    h = float( T / n )
    sq_h = np.sqrt( h )

    W = 0.0
    x = [0.0]
    y = [W]

    Normals = sq_h * np.random.normal(0, 1, n)

    for i in range(1, n+1):
        W += Normals[i-1]

        x.append(i * h)
        y.append(W)

    return x, y
```

Drawing trajectories of the Poisson process.

```
import numpy as np

def Poisson_path(lam, T, n):
    h = float(T / n)
    lh = lam * h

    P = 0.0
    x = [0.0]
    y = [P]

    Poiss = np.random.poisson(lh, n)

    for i in range(1, n+1):
        P += Poiss[i - 1]

        x.append(i*h)
        y.append(P)

    return x, y
```

Drawing trajectories of compound Poisson process.

```
def Compound_Poisson(T, n, lam):
    h = float(T / n)
    lh = lam * h

    dN = np.random.poisson(lh, n)
    N = np.cumsum(dN)

    N = np.insert(N, 0, 0.0)

    t = [0.0]
    Y = [0.0]
    num_T = len(N) - 1

    xi = np.random.normal(0.0, 3.0, N[num_T])
    for i in range(1, num_T+1):
        p = 0.0
```

```

for k in range(1, N[i]):
    p += xi[k]
t.append(i * h)
Y.append(p)

return t, Y

```

5.8 Concluding remarks

Due to Theorem 5.6.4 we have for all $n \in \mathbb{N}$

$$\begin{aligned}
& \inf_{\varphi_n: \mathbb{R}^n \rightarrow \mathbb{R} - \text{Borel}, 0 < t_1 < t_2 < \dots < t_n = T} \mathbb{E} \left| \int_0^T W(t) dt - \varphi_n(\mathcal{N}_n(W)) \right|^2 \\
&= \inf_{\varphi_n: \mathbb{R}^n \rightarrow \mathbb{R} - \text{Borel}, 0 < t_1 < t_2 < \dots < t_n = T} \int_{C([0, T])} \left| \int_0^T f(t) dt - \varphi_n(f(t_1), \dots, f(t_n)) \right|^2 w(df) = \frac{T^{3/2}}{\sqrt{12}} n^{-1}, \quad (5.71)
\end{aligned}$$

where w is the Wiener measure on $(C([0, T]), \mathcal{B}(C([0, T])))$. Hence, Theorem 5.6.4 answers the question: *in average, what is the error when approximating the Lebesgue integral of continuous function via deterministic quadrature rule based on finite number of function evaluations?* From the point of view of Information-Based Complexity such a model of computation is called the *average case setting*, see [79], [87]. See also [26], where a more general problem of nonlinear Lebesgue integration in the average-case setting was considered. Finally, note that the problem of (nonlinear) Lebesgue integration is connected with the complexity of stochastic integration, see [86], [26].

5.9 Exercises

- Let X be a square-integrable random variable and let $\mathcal{G}$ be a sub- σ -sigma field of Σ . Moreover, let Y be a $\mathcal{G}$ -measurable random variable.

(i) In addition, let $\mathbb{E}Y^2 < +\infty$. Show that

$$\text{Var}(X + Y \mid \mathcal{G}) = \text{Var}(X \mid \mathcal{G}). \quad (5.72)$$

(ii) Additionally assume that $\mathbb{E}|XY|^2 < +\infty$. Justify that

$$\text{Var}(XY \mid \mathcal{G}) = Y^2 \cdot \text{Var}(X \mid \mathcal{G}). \quad (5.73)$$

- Let X, Y be square integrable random variables and let $\mathcal{F}$ be a sub- σ -field of Σ . We define the conditional covariance of the random variables X, Y as follows

$$\text{Cov}(X, Y \mid \mathcal{F}) = \mathbb{E} \left((X - \mathbb{E}(X \mid \mathcal{F})) \cdot (Y - \mathbb{E}(Y \mid \mathcal{F})) \mid \mathcal{F} \right). \quad (5.74)$$

- Prove that the conditional covariance (5.74) is well-defined, i.e., give a proof that the conditional expectation

$$\mathbb{E} \left((X - \mathbb{E}(X \mid \mathcal{F})) \cdot (Y - \mathbb{E}(Y \mid \mathcal{F})) \mid \mathcal{F} \right) \quad (5.75)$$

exists.

- Prove that the following equality holds

$$\text{Cov}(X, Y \mid \mathcal{F}) = \mathbb{E}(XY \mid \mathcal{F}) - \mathbb{E}(X \mid \mathcal{F}) \cdot \mathbb{E}(Y \mid \mathcal{F}). \quad (5.76)$$

(c) Prove the following

$$Cov(X, Y) = \mathbb{E}(Cov(X, Y \mid \mathcal{F})) + Cov(\mathbb{E}(X \mid \mathcal{F}), \mathbb{E}(Y \mid \mathcal{F})). \quad (5.77)$$

3. Let $(\mathcal{G}_t)_{t \in [0, +\infty)}$ be an arbitrary filtration on $(\Omega, \Sigma, \mathbb{P})$. Show that

$$\bigcap_{n \in \mathbb{N}} \mathcal{G}_{t + \frac{1}{n}} = \bigcap_{s > t} \mathcal{G}_s. \quad (5.78)$$

4. Let $(\Sigma_t)_{t \in [0, +\infty)}$ be a filtration on $(\Omega, \Sigma, \mathbb{P})$, satisfying the usual conditions, and let $\mathcal{G}$ be a sub- σ -field of Σ that is independent of Σ_∞ . Show that $(\mathcal{H}_t)_{t \in [0, +\infty)}$, where

$$\mathcal{H}_t = \sigma(\Sigma_t \cup \mathcal{G}), \quad (5.79)$$

is a filtration that also satisfies the usual conditions.

Hint. Chamont, Yor, "Exercices...."

5. Let $W = (W_t)_{t \in [0, +\infty)}$ be a $(\Sigma_t)_{t \in [0, +\infty)}$ -Wiener process on $(\Omega, \Sigma, \mathbb{P})$, and let $\mathcal{G}$ be a sub- σ -field of Σ that is independent of Σ_∞ . Show that $W = (W_t)_{t \in [0, +\infty)}$ is a Wiener process with respect to the filtration $(\mathcal{H}_t)_{t \in [0, +\infty)}$, defined as in (5.79).

6. What is the law of $\mathcal{I}(W)$?

7. Show that $\hat{W}_n$ is a Gaussian process.

8. Provide a more efficient implementation of the procedure that generates trajectories of the compound Poisson process.

9. Write a function that simulates trajectories of the ruin process with perturbations which is defined as follows

$$R(t) = x_0 + ct + \sigma W(t) + J(t), t \in [0, T], \quad (5.80)$$

where $x_0, c, \sigma \in (0, +\infty)$ and W, J are Wiener and compound Poisson process, respectively. Approximate, by using a suitably constructed Monte Carlo method, the probability of ruin

$$\mathbb{P}(R(T) < 0). \quad (5.81)$$

Do not forget to compute empirical confidence intervals.

10. Give a proof of Lemma 5.6.2.

11. Show that the processes W, N, C are continuous in probability.

Hint. For the Cauchy process C , use the characteristic function of its law.

Chapter 6

Stochastic integration with respect to semimartingales

Let $(\Omega, \Sigma, \mathbb{P})$ be a complete probability space equipped with filtration $(\Sigma_t)_{t \in [0, +\infty)}$ that satisfies the usual conditions. In this chapter, we mainly based on the results from [2], [55], [72], [82], and [88]. We focus only on stochastic integration of processes from the space $\mathbb{L}$ (see (14.22)) which, as we will see, is sufficient in our case.

A process X is a *semimartingale* if it admits a decomposition

$$X = M + A, \quad (6.1)$$

where M is a right-continuous local L^2 -martingale and A is right-continuous, adapted, and of locally finite variation in the sense that almost surely the paths of A are of finite variation on each compact subinterval of $[0, +\infty)$. This decomposition is not unique unless we restrict it to the case when A is continuous, see, for example, page 150. in [88]. Below we provide main examples of semimartingales.

- The Wiener process W is a (continuous) semimartingale since it is a martingale, so $A = 0$,
- The process $(W^2(t))_{t \geq 0}$ is a semimartingale with the following decomposition

$$W^2(t) = (W^2(t) - t) + t, \quad (6.2)$$

$(W^2(t) - t, \bar{\Sigma}_t^W)_{t \geq 0}$ is a càdlàg L^2 -martingale.

- The non-homogeneous Poisson process is a semimartingale due to the fact that

$$N(t) = (N(t) - m(t)) + m(t), \quad (6.3)$$

and the process $(N(t) - m(t), \bar{\Sigma}_t^N)_{t \geq 0}$ is a càdlàg L^2 -martingale. It is denoted by $\tilde{N}$ and called the *compensated Poisson process*.

- The process $(\tilde{N}^2(t))_{t \geq 0}$ is a semimartingale with the decomposition

$$\tilde{N}^2(t) = (\tilde{N}^2(t) - m(t)) + m(t), \quad (6.4)$$

where $(\tilde{N}^2(t) - m(t), \bar{\Sigma}_t^{\tilde{N}^2})_{t \geq 0}$ is a càdlàg L^2 -martingale.

- Provided that $\mathbb{E}(\xi_1^2) < +\infty$, the compound Poisson process is a semimartingale since

$$J(t) = (J(t) - t\lambda\mathbb{E}(\xi_1)) + t\lambda\mathbb{E}(\xi_1) \quad (6.5)$$

and $(J(t) - t\lambda\mathbb{E}(\xi_1), \bar{\Sigma}_t^J)_{t \geq 0}$ is a càdlàg L^2 -martingale called *compensated compound Poisson process*. We denote it by $\tilde{J}$.

- If $\mathbb{E}(\xi_1^2) < +\infty$, then the process $(\tilde{J}^2(t))_{t \geq 0}$ is a semimartingale with the decomposition

$$\tilde{J}^2(t) = (\tilde{J}^2(t) - t\lambda\mathbb{E}(\xi_1^2)) + t\lambda\mathbb{E}(\xi_1^2) \quad (6.6)$$

and $(\tilde{J}^2(t) - t\lambda\mathbb{E}(\xi_1^2), \bar{\Sigma}_t^J)_{t \geq 0}$ is a càdlàg L^2 -martingale.

Note that the processes N , $\tilde{N}^2$, J , $\tilde{J}^2$ are itself processes with locally finite variation.

Suppose that $Y \in \mathbb{L}$ and X is a semimartingale with the decomposition (6.1). Then the stochastic integral of Y with respect to X is defined as

$$\int_0^t Y(s) dX(s) = \int_0^t Y(s) dM(s) + \int_0^t Y(s) dA(s). \quad (6.7)$$

Note that the value of the stochastic integral is independent of the semimartingale decomposition (6.1), see [88]. In (6.7) the integral

- $\int_0^t Y(s) dA(s)$ is a Lebesgue-Stieltjes integral, computed path-by-path,
- $\int_0^t Y(s) dM(s)$ is a stochastic integral with respect to the càdlàg L^2 -martingale M .

The precise construction of stochastic integrals with respect to martingales is given, for example, in [82]. Here we only state the main properties of such integrals that we will use in the context of approximation of solutions of jump-diffusion stochastic differential equations.

The fundamental theorem that allows the definition of stochastic integral with respect to càdlàg L^2 -martingales is the following theorem, called the Doob-Meyer decomposition.

Theorem 6.0.1. ([82]) *Let $(M(t), \Sigma_t)_{t \geq 0}$ be a square integrable càdlàg martingale. Then there is a unique decomposition*

$$M^2(t) = Y(t) + A(t), \quad t \geq 0, \quad (6.8)$$

where $(Y(t), \Sigma_t)_{t \geq 0}$ is càdlàg martingale and $A = (A(t))_{t \geq 0}$ is a predictable, right continuous, and increasing process such that $A(0) = 0$ and $\mathbb{E}(A(t)) < +\infty$ for all $t \geq 0$. From now on the process $A = (A(t))_{t \geq 0}$ is denoted by $\langle M \rangle = (\langle M \rangle_t)_{t \geq 0}$ and it is called the **predictable quadratic variation** of M .

It is easy to see that (6.2) and (6.4) are two (very important) examples of Doob-Meyer decomposition.

From Theorem (6.0.1) we have what follows.

Corollary 6.0.2. *Let $(M_i(t), \Sigma_t)_{t \geq 0}$, $i = 1, 2$, be square integrable càdlàg martingales. Then the process $(M_1(t) \cdot M_2(t))_{t \geq 0}$ has a unique decomposition*

$$M_1(t) \cdot M_2(t) = Y(t) + A(t), \quad t \geq 0, \quad (6.9)$$

where $(Y(t), \Sigma_t)_{t \geq 0}$ is càdlàg martingale and $A = (A(t))_{t \geq 0}$ is a difference of two processes $A_i = (A_i(t))_{t \geq 0}$ where A_i is a predictable, right continuous, and increasing process such that $A_i(0) = 0$ and $\mathbb{E}(A_i(t)) < +\infty$ for all $t \geq 0$. From now on the process $A = (A(t))_{t \geq 0}$ is denoted by $\langle M_1, M_2 \rangle = (\langle M_1, M_2 \rangle_t)_{t \geq 0}$ and it is called the **predictable cross quadratic variation** of $(M_1(t) \cdot M_2(t))_{t \geq 0}$.

The following result is a direct consequence of Corollary 6.0.2.

Corollary 6.0.3. *Suppose that $(M_i(t), \Sigma_t)_{t \geq 0}$, $i = 1, 2$, are square integrable càdlàg martingales such that the process $(M_1(t) \cdot M_2(t), \Sigma_t)_{t \geq 0}$ is a martingale. Then $\langle M_1, M_2 \rangle_t = 0$ for all $t \geq 0$.*

Proposition 6.0.4. Let $W = (W(t))_{t \in [0, +\infty)}$ be a m_W -dimensional $(\Sigma_t)_{t \in [0, +\infty)}$ -Wiener process, while $N = (N(t))_{t \in [0, +\infty)}$ is a m_N -dimensional non-homogeneous $(\Sigma_t)_{t \in [0, +\infty)}$ -Poisson process, both defined on $(\Omega, \Sigma, \mathbb{P})$, and such that the σ -fields $\Sigma_\infty^W, \Sigma_\infty^N$ are independent. (i.e. W and N are independent). We assume that each intensity $\lambda_j : [0, T] \rightarrow (0, +\infty)$ of $(N^j(t))_{t \in [0, +\infty)}$ is Borel and integrable, $j = 1, 2, \dots, m_N$. If $i \neq j$ then for any $t \geq 0$

$$\langle W^i, W^j \rangle_t = 0, \quad (6.10)$$

$$\langle \tilde{N}^i, \tilde{N}^j \rangle_t = 0, \quad (6.11)$$

while for $i = j$

$$\langle W^i \rangle_t = \langle W^i, W^i \rangle_t = t, \quad (6.12)$$

$$\langle \tilde{N}^i \rangle_t = \langle \tilde{N}^i, \tilde{N}^i \rangle_t = m_i(t). \quad (6.13)$$

If $\sigma([W(t) - W(s), N(t) - N(s)])$ is independent of Σ_s for any $0 \leq s < t$ then for any $i \in \{1, \dots, m_W\}$, $j \in \{1, \dots, m_N\}$ and $t \geq 0$

$$\langle W^i, \tilde{N}^j \rangle_t = 0, \quad (6.14)$$

Proof. Equalities (6.12) are direct consequences of (6.2) and (6.4).

We only show (6.14), since the proofs of (6.10), (6.11) are analogous.

For any i, j and $0 \leq s < t$ we have

$$\begin{aligned} & \mathbb{E}(W^i(t) \cdot \tilde{N}^j(t) \mid \Sigma_s) = \\ &= \mathbb{E}\left((W^i(t) - W^i(s) + W^i(s)) \cdot (\tilde{N}^j(t) - \tilde{N}^j(s) + \tilde{N}^j(s)) \mid \Sigma_s\right) \\ &= \mathbb{E}\left((W^i(t) - W^i(s)) \cdot (\tilde{N}^j(t) - \tilde{N}^j(s)) \mid \Sigma_s\right) \\ &\quad + \tilde{N}^j(s) \cdot \mathbb{E}\left(W^i(t) - W^i(s) \mid \Sigma_s\right) + W^i(s) \cdot \mathbb{E}\left(\tilde{N}^j(t) - \tilde{N}^j(s) \mid \Sigma_s\right) \\ &\quad + \mathbb{E}\left(W^i(s) \cdot \tilde{N}^j(s) \mid \Sigma_s\right) \\ &= \mathbb{E}\left((W^i(t) - W^i(s)) \cdot (\tilde{N}^j(t) - \tilde{N}^j(s))\right) + W^i(s) \cdot \tilde{N}^j(s) \\ &= \mathbb{E}\left(W^i(t) - W^i(s)\right) \cdot \mathbb{E}\left(\tilde{N}^j(t) - \tilde{N}^j(s)\right) + W^i(s) \cdot \tilde{N}^j(s) \\ &= W^i(s) \cdot \tilde{N}^j(s). \end{aligned} \quad (6.15)$$

Therefore, the process $(W^i(t) \cdot \tilde{N}^j(t), \Sigma_t)_{t \geq 0}$ is a martingale and by Corollary 6.0.3 we get (6.14). $\blacksquare$

We now recall the basic properties of stochastic integrals. Let X be a semimartingale with respect to $(\Sigma_t)_{t \in [0, +\infty)}$. Moreover, let $f, f_1, f_2 \in \mathbb{L}$ and $\alpha, \beta \in \mathbb{R}$ and suppose that ρ, τ are $(\Sigma_t)_{t \in [0, +\infty)}$ -stopping times, such that $\rho \leq \tau$ a.s. Recall that the stochastic interval $(\rho, \tau]$ is defined as follows

$$(\rho, \tau] = \{(\omega, t) \in \Omega \times [0, +\infty) \mid \rho(\omega) < t \leq \tau(\omega)\}. \quad (6.16)$$

Moreover the process $\mathbf{1}_{(\rho, \tau]}$ belongs to $\mathbb{L}$. The following holds.

- For all $t \in [0, +\infty)$

$$\int_0^t (\alpha f_1(s) + \beta f_2(s)) dX(s) = \alpha \int_0^t f_1(s) dX(s) + \beta \int_0^t f_2(s) dX(s) \text{ a.s.} \quad (6.17)$$

- The stochastic integral process $Y = (Y(t))_{t \geq 0} = \left(\int_0^t f_1(s) dX(s) \right)_{t \in [0, +\infty)}$ is itself a semimartingale, and

$$\int_0^t f_2(s) dY(s) = \int_0^t f_2(s) f_1(s) dX(s). \quad (6.18)$$

- For all $t \in [0, +\infty)$

$$\begin{aligned} \int_0^t \mathbf{1}_{(\rho, \tau]}(s) f(s) dX(s) &= \int_0^t \mathbf{1}_{(0, \tau]}(s) f(s) dX(s) - \int_0^t \mathbf{1}_{(0, \rho]}(s) f(s) dX(s) \\ &= \int_0^{t \wedge \tau} f(s) dX(s) - \int_0^{t \wedge \rho} f(s) dX(s) = \int_{t \wedge \rho}^{t \wedge \tau} f(s) dX(s) \end{aligned} \quad (6.19)$$

- If Z is a Σ_ρ -measurable random variable then

$$\int_\rho^\tau Z f(s) dX(s) = Z \int_\rho^\tau f(s) dX(s) \text{ a.s.} \quad (6.20)$$

- Let $f = (f(t))_{t \geq 0}$ be a **predictable simple process**, i.e., there is a sequence of stopping times

$$0 = \tau_0 < \tau_1 < \dots < \tau_n < \dots \quad (6.21)$$

such that for all $t \geq 0$

$$f(t) = \xi_0 \mathbf{1}_{\{0\}}(t) + \sum_{i=0}^{+\infty} \xi_i \mathbf{1}_{(\tau_i, \tau_{i+1}]}(t), \quad (6.22)$$

where ξ_0 is Σ_0 -measurable and ξ_i are Σ_{τ_i} -measurable random variables. Then for all $t \geq 0$

$$\begin{aligned} \int_0^t f(s) dX(s) &= \sum_{i=0}^{+\infty} \xi_i \cdot \left(X(\tau_{i+1} \wedge t) - X(\tau_i \wedge t) \right) \\ &= \sum_{i=0}^{+\infty} \xi_i \cdot \left(M(\tau_{i+1} \wedge t) - M(\tau_i \wedge t) \right) + \sum_{i=0}^{+\infty} \xi_i \cdot \left(A(\tau_{i+1} \wedge t) - A(\tau_i \wedge t) \right) \text{ a.s.} \end{aligned} \quad (6.23)$$

- If the semimartingale X has paths of finite variation on compacts, then the stochastic integral $\int_0^t f(s) dX(s)$ is indistinguishable from the Lebesgue-Stieltjes integral, computed path-by-path.

If $Y \in \mathbb{L}$ then

$$\int_0^t Y(s) dN(s) = \int_0^t Y(s) d\tilde{N}(s) + \int_0^t Y(s) dm(s) = \int_0^t Y(s) d\tilde{N}(s) + \int_0^t Y(s) \lambda(s) ds, \quad (6.24)$$

but, on the other hand, the stochastic integrals $\int_0^t Y(s) dN(s)$, $\int_0^t Y(s) d\tilde{N}(s)$ are equivalent to the respective Lebesgue-Stieltjes integrals. Moreover, since N is a pure jump process, we get

$$\int_0^t Y(s) dN(s) = \sum_{0 < s \leq t} Y(s) \Delta N(s) = \sum_{k=1}^{N(t)} Y(\tau_k), \quad (6.25)$$

while

$$\int_0^t Y(s) dJ(s) = \sum_{0 < s \leq t} Y(s) \Delta J(s) = \sum_{k=1}^{N(t)} Y(\tau_k) \xi_k, \quad (6.26)$$

where $(\tau_k)_{k \in \mathbb{N}}$ are the jump times of N .

Below we present the main properties of stochastic integrals with respect to càdlàg L^2 -martingales.

For a given L^2 -martingale M let us consider the following subset of $\mathbb{L}$

$$\mathbb{L}^2(M) = \left\{ X \in \mathbb{L} \mid \forall T > 0 \quad \mathbb{E} \int_0^T |X(s)|^2 d\langle M \rangle_s < +\infty \right\}. \quad (6.27)$$

Suppose that $(M(t), \Sigma_t)_{t \in [0, +\infty)}$, $(M_1(t), \Sigma_t)_{t \in [0, +\infty)}$, $(M_2(t), \Sigma_t)_{t \in [0, +\infty)}$ are square integrable càdlàg martingales. Let $f \in \mathbb{L}^2(M)$.

- For all $0 \leq s < t$ we have

$$\mathbb{E} \left(\int_0^t f(u) dM(u) \mid \Sigma_s \right) = \int_0^s f(u) dM(u) \text{ a.s.}, \quad (6.28)$$

and, in particular,

$$\mathbb{E} \left(\int_0^t f(u) dM(u) \right) = 0. \quad (6.29)$$

•

$$\mathbb{E} \left(\left(\int_s^t f(u) dM(u) \right)^2 \mid \Sigma_s \right) = \mathbb{E} \left(\int_s^t |f(u)|^2 d\langle M \rangle_u \mid \Sigma_s \right) \text{ a.s.} \quad (6.30)$$

- If $f_i \in \mathbb{L}^2(M_i)$, $i = 1, 2$, then for all $0 \leq s < t$

$$\mathbb{E} \left(\int_s^t f_1(u) dM_1(u) \cdot \int_s^t f_2(u) dM_2(u) \mid \Sigma_s \right) = \mathbb{E} \left(\int_s^t f_1(u) f_2(u) d\langle M_1, M_2 \rangle_u \mid \Sigma_s \right) \text{ a.s.} \quad (6.31)$$

If X is an $\mathbb{R}^d$ -valued process such that each coordinate X^i , $i = 1, 2, \dots, d$, is an $\mathbb{R}$ -valued semimartingale, then we call X the vector valued semimartingale. For an $\mathbb{R}^{d \times m}$ -valued stochastic process $Y \in \mathbb{L}$ we define

$$\int_0^t Y(s) dX(s) = \left(\sum_{j=1}^m \int_0^t Y^{ij}(s) dX^j(s) \right)_{i=1,2,\dots,d}. \quad (6.32)$$

For example, in the case of multidimensional non-homogeneous Poisson process N we have that

$$\int_0^t Y(s) dN(s) = \left(\sum_{j=1}^{m_N} \int_0^t Y^{ij}(s) dN^j(s) \right)_{i=1,2,\dots,d} = \int_0^t Y(s) d\tilde{N}(s) + \int_0^t Y(s) dm(s), \quad (6.33)$$

where

$$\int_0^t Y(s) d\tilde{N}(s) = \left(\sum_{j=1}^{m_N} \int_0^t Y^{ij}(s) d\tilde{N}^j(s) \right)_{i=1,2,\dots,d}, \quad (6.34)$$

$$\int_0^t Y(s) dm(s) = \left(\sum_{j=1}^{m_N} \int_0^t Y^{ij}(s) dm_j(s) \right)_{i=1,2,\dots,d} = \left(\sum_{j=1}^{m_N} \int_0^t Y^{ij}(s) \lambda_j(s) ds \right)_{i=1,2,\dots,d}. \quad (6.35)$$

Let us now assume that $Y^{ij} \in \mathbb{L}^2(\tilde{N}^j)$ for $i = 1, 2, \dots, d$, $j = 1, 2, \dots, m_N$. From the definition above, Proposition 6.0.4 and (6.31) we arrive at the following version of the multidimensional Itô isometry

$$\begin{aligned} \mathbb{E} \left\| \int_0^t Y(s) d\tilde{N}(s) \right\|^2 &= \mathbb{E} \sum_{k=1}^d \left| \sum_{j=1}^{m_N} \int_0^t Y^{kj}(s) d\tilde{N}^j(s) \right|^2 \\ &= \sum_{k=1}^d \sum_{j_1, j_2=1}^{m_N} \mathbb{E} \left(\int_0^t Y^{kj_1}(s) d\tilde{N}^{j_1}(s) \cdot \int_0^t Y^{kj_2}(s) d\tilde{N}^{j_2}(s) \right) \\ &= \sum_{k=1}^d \sum_{j_1, j_2=1}^{m_N} \mathbb{E} \left(\int_0^t Y^{kj_1}(s) Y^{kj_2}(s) d\langle \tilde{N}^{j_1}, \tilde{N}^{j_2} \rangle_s \right) \\ &= \sum_{k=1}^d \sum_{j=1}^{m_N} \mathbb{E} \int_0^t |Y^{kj}(s)|^2 \lambda_j(s) ds = \sum_{j=1}^{m_N} \mathbb{E} \int_0^t \|Y^{(j)}(s)\|^2 \lambda_j(s) ds, \end{aligned} \quad (6.36)$$

where $Y^{(j)}$ is the j th column of the matrix $Y = [Y^{kj}]_{1 \leq k \leq d, 1 \leq j \leq m_N}$. In similar way, if $Y^{ij} \in \mathbb{L}^2(W^j)$ for $i = 1, 2, \dots, d$, $j = 1, 2, \dots, m_W$ we get

$$\mathbb{E} \left\| \int_0^t Y(s) dW(s) \right\|^2 = \mathbb{E} \int_0^t \|Y(s)\|^2 ds. \quad (6.37)$$

TODO: Itô formula for semimartingales....

6.1 Exercises

1. Let $-\infty < \alpha < \beta < +\infty$ and suppose $Z : \Omega \rightarrow \mathbb{R}^{d \times m_W}$ to be Σ_α -measurable random vector. Show that

$$\int_\alpha^\beta Z dW(t) = Z \cdot (W(\beta) - W(\alpha)). \quad (6.38)$$

2. Let $-\infty < \alpha < \beta < +\infty$ and suppose $Z : \Omega \rightarrow \mathbb{R}^{d \times m_N}$ to be Σ_α -measurable random vector. Show that

$$\int_\alpha^\beta Z dN(t) = Z \cdot (N(\beta) - N(\alpha)). \quad (6.39)$$

Chapter 7

Jump-diffusion stochastic differential equations

In this section, by $\|\cdot\|$ we denote either the Euclidean or the Frobenius norm, and the meaning is clear from the context.

Let $(\Omega, \Sigma, \mathbb{P})$ be a complete probability space equipped with filtration $(\Sigma_t)_{t \in [0, +\infty)}$ that satisfies the usual conditions. Let $W = (W(t))_{t \in [0, +\infty)}$ be a m_W -dimensional $(\Sigma_t)_{t \in [0, +\infty)}$ -Wiener process, while $N = (N(t))_{t \in [0, +\infty)}$ is a m_N -dimensional non-homogeneous $(\Sigma_t)_{t \in [0, +\infty)}$ -Poisson process, both defined on $(\Omega, \Sigma, \mathbb{P})$, and such that the σ -fields $\Sigma_\infty^W, \Sigma_\infty^N$ are independent. (i.e. W and N are independent). We assume that each intensity $\lambda_j : [0, T] \rightarrow (0, +\infty)$ of $(N^j(t))_{t \in [0, +\infty)}$ is Borel and integrable, $j = 1, 2, \dots, m_N$.

(A0) Let $\xi : \Omega \rightarrow \mathbb{R}^d$ be a Σ_0 -measurable random vector, such that $\mathbb{E}\|\xi\|^2 < +\infty$.

Let us fix functions

$$a : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad (7.1)$$

$$b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m_W}, \quad (7.2)$$

$$c : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m_N}, \quad (7.3)$$

such that

(A1) $a, b, c \in C([0, T] \times \mathbb{R}^d)$

(A2) there exists $K \in (0, +\infty)$ such that for $f \in \{a, b, c\}$ and for all $t \in [0, T], x, y \in \mathbb{R}^d$

$$\|f(t, x) - f(t, y)\| \leq K\|x - y\|. \quad (7.4)$$

We consider the following *jump-diffusion stochastic differential equation*

$$\begin{cases} dX(t) = a(t, X(t-))dt + b(t, X(t-))dW(t) + c(t, X(t-))dN(t), & t \in [0, T], \\ X(0) = \xi, \end{cases} \quad (7.5)$$

where we denote $X(t-) = \lim_{s \rightarrow t-} X(s)$, we take $X(0-) := X(0)$ (no jump at $t = 0$) and $X(T+) := X(T)$. The SDE (7.5) can also be rewritten as an SDE driven by the multidimensional semimartingale $Y = [t, W, N] = [t, W^1, \dots, W^{m_W}, N^1, \dots, N^{m_N}]$

$$\begin{cases} dX(t) = F(t, X(t-))dY(t), & t \in [0, T], \\ X(0) = \xi, \end{cases} \quad (7.6)$$

where

$$F(t, y) = [F_{ij}(t, y)]_{1 \leq i \leq d, 1 \leq j \leq 1+m_W+m_N} = \begin{bmatrix} a_1 & b_{12} & \cdots & b_{1m_W} & c_{11} & \cdots & c_{1m_N} \\ a_2 & b_{22} & \cdots & b_{2m_W} & c_{21} & \cdots & c_{2m_N} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_d & b_{d,1} & \cdots & b_{dm_W} & c_{d1} & \cdots & c_{dm_N} \end{bmatrix} (t, y). \quad (7.7)$$

Under the assumptions (A1), (A2) we have that $F \in C([0, T] \times \mathbb{R}^d)$ and for all $t \in [0, T], x, y \in \mathbb{R}^d$

$$\|F(t, x) - F(t, y)\| \leq K_1 \|x - y\|. \quad (7.8)$$

Of course the stochastic differential equation (7.6) must be understood as an integral equation

$$X(t) = \xi + \int_0^t F(s, X(s-)) dY(s), \quad t \in [0, T]. \quad (7.9)$$

It follows from Theorem 12.8 at page 390 in [56] that under (A0) and (7.8) the SDE (7.6) has a unique strong solution $X = (X(t))_{t \in [0, T]} \in \mathbb{D}$.

A classical example is the following linear jump-diffusion SDEs

$$\begin{cases} dX(t) = \mu X(t)dt + \sigma X(t)dW(t) + cX(t-)dN(t), & t \in [0, T], \\ X(0) = x_0, \end{cases} \quad (7.10)$$

that models the stock price in Merton's model, see [71]. We assume that $\mu \in \mathbb{R}, \sigma > 0, x_0 > 0$ and $c > -1$. The solution of (7.10) is

$$X(t) = x_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W(t)\right) \cdot (1 + c)^{N(t)}. \quad (7.11)$$

The results below are known for general SDEs driven by semimartingales.

Lemma 7.0.1. *Under the assumptions (A0), (A1), (A2) there exists $C \in (0, +\infty)$ such that*

$$\mathbb{E}\left(\sup_{0 \leq t \leq T} \|X(t)\|^2\right) \leq C(1 + \mathbb{E}\|\xi\|^2), \quad (7.12)$$

and for all $t, s \in [0, T]$

$$\mathbb{E}\|X(t) - X(s)\|^2 \leq C \cdot (1 + \mathbb{E}\|\xi\|^2) \cdot |t - s|. \quad (7.13)$$

Moreover, if there exists $p \in [2, +\infty)$ such that $\mathbb{E}\|\xi\|^p < +\infty$ then there exists $C_p \in (0, +\infty)$ such that

$$\mathbb{E}\left(\sup_{0 \leq t \leq T} \|X(t)\|^p\right) \leq C_p(1 + \mathbb{E}\|\xi\|^p), \quad (7.14)$$

and for all $t, s \in [0, T]$

$$\left(\mathbb{E}\|X(t) - X(s)\|^p\right)^{1/p} \leq C_p \cdot (1 + (\mathbb{E}\|\xi\|^p)^{1/p}) \cdot |t - s|^{1/2}, \quad (7.15)$$

Remark 7.0.2. *Note that if $c = 0$ then for any $p \in [2, +\infty)$ there exists $C_p \in (0, +\infty)$ such that for all $t, s \in [0, T]$*

$$\left(\mathbb{E}\|X(t) - X(s)\|^p\right)^{1/p} \leq C_p \cdot (1 + (\mathbb{E}\|\xi\|^p)^{1/p}) \cdot |t - s|^{1/2}, \quad (7.16)$$

provided that $\mathbb{E}\|\xi\|^p < +\infty$. This is no longer true if $c \neq 0$. Due to the properties of the Poisson process, we have in that case

$$\left(\mathbb{E}\|X(t) - X(s)\|^p\right)^{1/p} \leq C_p \cdot (1 + (\mathbb{E}\|\xi\|^p)^{1/p}) \cdot |t - s|^{1/p}. \quad (7.17)$$

So in the jump-diffusion case the $L^p(\Omega)$ -regularity of X is the same as in the pure Gaussian case only for $p = 2$. This, rather low for large $p > 2$, L^p -regularity of X in the jump case affects the rates of convergence for approximating algorithms.

We also prove to following useful fact on properties of the left continuous modification of X , i.e. $(X(t-))_{t \in [0, T]}$.

Proposition 7.0.3. *Let us assume (A0), (A1), (A2) and let there exists $p \in [2, +\infty)$ such that $\mathbb{E}\|\xi\|^p < +\infty$. Then we have that*

(i) for all $t \in [0, T]$

$$\mathbb{P}(X(t) = X(t-)) = 1, \quad (7.18)$$

(ii) for all $t, s \in [0, T]$, $\diamond, * \in \{-, +\}$

$$\|X(t^\diamond) - X(s^*)\|_{L^p(\Omega)} = \|X(t) - X(s)\|_{L^p(\Omega)}, \quad (7.19)$$

(iii) the function

$$[0, T] \ni t \rightarrow \|X(t)\|_{L^p(\Omega)} \in [0, +\infty) \quad (7.20)$$

is continuous.

Proof. We start by showing (i). By (7.15) we have for all $t \in [0, T]$ that

$$\lim_{s \rightarrow t-} \mathbb{E}\|X(t) - X(s)\|^p = 0. \quad (7.21)$$

Since $\mathbb{R}^d \ni x \rightarrow \|x\|^p$ is a continuous function, we get for all $t \in [0, T]$

$$\lim_{s \rightarrow t-} \|X(t) - X(s)\|^p = \|\Delta X(t)\|^p \quad (7.22)$$

and, for all $s \in [0, T]$,

$$\|X(t) - X(s)\|^p \leq 2c_p \cdot \sup_{0 \leq v \leq T} \|X(v)\|^p. \quad (7.23)$$

where, by (7.14), we have that $\mathbb{E}\left(\sup_{0 \leq v \leq T} \|X(v)\|^p\right) < +\infty$. Hence, by the Lebesgue's dominated convergence theorem, (7.21), and (7.22) we get for all $t \in [0, T]$

$$0 = \lim_{s \rightarrow t-} \mathbb{E}\|X(t) - X(s)\|^p = \mathbb{E}\|\Delta X(t)\|^p, \quad (7.24)$$

which implies (7.18).

In order to proof (ii) note that for any $t \in [0, T]$, $* \in \{-, +\}$ we have

$$\|X(t^*) - X(t)\|_{L^p(\Omega)} \in \{0, \|\Delta X(t)\|_{L^p(\Omega)}\}, \quad (7.25)$$

hence, by (7.24), we have

$$\|X(t^*) - X(t)\|_{L^p(\Omega)} = 0. \quad (7.26)$$

Applying (7.26), triangle inequality, and reverse triangle inequality we get

$$\begin{aligned} \|X(t^\diamond) - X(s^*)\|_{L^p(\Omega)} &\leq \|X(t^\diamond) - X(t)\|_{L^p(\Omega)} + \|X(t) - X(s)\|_{L^p(\Omega)} + \|X(s) - X(s^*)\|_{L^p(\Omega)} \\ &= \|X(t) - X(s)\|_{L^p(\Omega)}, \end{aligned} \quad (7.27)$$

and

$$\begin{aligned} \|X(t^\diamond) - X(s^*)\|_{L^p(\Omega)} &\geq \|X(t) - X(s)\|_{L^p(\Omega)} \\ -\|(X(t^\diamond) - X(t)) + (X(s) - X(s^*))\|_{L^p(\Omega)} &= \|X(t) - X(s)\|_{L^p(\Omega)}, \end{aligned} \quad (7.28)$$

since, again by (7.26),

$$0 \leq \|(X(t^\diamond) - X(t)) + (X(s) - X(s^*))\|_{L^p(\Omega)} \leq \|X(t^\diamond) - X(t)\|_{L^p(\Omega)} + \|X(s) - X(s^*)\|_{L^p(\Omega)} = 0. \quad (7.29)$$

Combining (7.27) and (7.28) we get (7.19).

Let us denote by $g(t) = \|X(t)\|_{L^p(\Omega)}$. We have for all $t \in [0, T]$ that

$$\lim_{s \rightarrow t-} \|X(s)\|^p = \|X(t-)\|^p, \quad (7.30)$$

and for all $s \in [0, T]$

$$\|X(s)\|^p \leq \sup_{0 \leq v \leq T} \|X(v)\|^p, \quad (7.31)$$

where $\mathbb{E} \left(\sup_{0 \leq v \leq T} \|X(v)\|^p \right) < +\infty$. Hence, Lebesgue's dominated convergence theorem gives

$$g(t-) = \lim_{s \rightarrow t-} g(s) = \|X(t-)\|_{L^p(\Omega)}. \quad (7.32)$$

From (7.24), (7.32) we have for all $t \in [0, T]$

$$0 \leq |g(t) - g(t-)| \leq \|X(t) - X(t-)\|_{L^p(\Omega)} = \|\Delta X(t)\|_{L^p(\Omega)} = 0, \quad (7.33)$$

and hence

$$g(t-) = g(t). \quad (7.34)$$

Moreover, since X is càdlàg, we have for all $t \in [0, T]$

$$\lim_{s \rightarrow t+} \|X(s)\|^p = \|X(t+)\|^p = \|X(t)\|^p, \quad (7.35)$$

and by using again the Lebesgue's dominated convergence theorem we get for all $t \in [0, T]$ that

$$g(t+) = \lim_{s \rightarrow t+} g(s) = g(t). \quad (7.36)$$

From (7.34), (7.36) we have that the function g is continuous on $[0, T]$. □

Remark 7.0.4. *Note that Proposition 7.0.3 holds for any process X such that...*

7.1 *SDEs driven by the Wiener process and compound Poisson process

In the scalar case, we briefly comment on the case when the jump part of the underlying SDE is driven by the compound Poisson process Y , i.e.

$$\begin{cases} dX(t) = a(t, X(t-))dt + b(t, X(t-))dW(t) + c(t, X(t-))dJ(t), & t \in [0, T], \\ X(0) = \xi. \end{cases} \quad (7.37)$$

In this case

$$\int_0^t c(s, X(s-))dJ(s) = \sum_{0 < s \leq t} c(s, X(s-))\Delta J(s) = \sum_{k=1}^{N(t)} c(\tau_k, X(\tau_k-))\xi_k. \quad (7.38)$$

Note that more general equations are considered, for example, when the noise comes from Levy processes, Poisson random measures, etc.

Merton model, driven by the compound Poisson process, has the following form

$$\begin{cases} dX(t) = \mu X(t)dt + \sigma X(t)dW(t) + X(t-)dJ(t), & t \in [0, T], \\ X(0) = x_0, \end{cases} \quad (7.39)$$

with the explicit solution

$$X(t) = x_0 \exp\left((\mu - \frac{1}{2}\sigma^2)t + \sigma W(t)\right) \prod_{k=1}^{N(t)} (\xi_k + 1). \quad (7.40)$$

TODO: Give examples of other SDEs, OU processes etc. and how to solve them via the Itô formula

7.2 On some relation between SDEs and ODEs

Let us consider the following SDE with additive diffusion and jump terms

$$\begin{cases} dX(t) = a(t, X(t-))dt + b(t)dW(t) + c(t)dJ(t), & t \in [0, T], \\ X(0) = \xi. \end{cases} \quad (7.41)$$

Let us take $Y(t) = X(t) - Z(t)$, $Z(t) = \int_0^t b(s)dW(s) + \int_0^t c(s)dJ(s)$. The process Y , with differentiable trajectories, satisfies (in the pathwise sense) the following ODE

$$\begin{cases} Y'(t) = G(t, Z(t), Y(t)), & t \in [0, T], \\ Y(0) = \xi, \end{cases} \quad (7.42)$$

with $G(t, z, y) = a(t, y + z)$.

Chapter 8

Euler scheme and its convergence

One of the most basic, but still fundamental, algorithms for approximation of solutions of SDEs is the Euler-Maruyama algorithm.

We now describe heuristically its construction...

For the functions

$$a : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad (8.1)$$

$$b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m_W}, \quad (8.2)$$

$$c : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m_N}, \quad (8.3)$$

we assume that

(B) there exists $K \in (0, +\infty)$ such that for $f \in \{a, b, c\}$ and for all $s, t \in [0, T]$, $x, y \in \mathbb{R}^d$

$$\|f(t, x) - f(s, y)\| \leq K(|t - s| + \|x - y\|), \quad (8.4)$$

(C) the intensity functions $\lambda_j : [0, T] \rightarrow (0, +\infty)$, $j = 1, 2, \dots, m_N$, are Borel measurable and bounded.

For $f = a$ the norm $\|\cdot\|$ is the euclidean norm $\|\cdot\|_2$, while for $f \in \{b, c\}$ the norm $\|\cdot\|$ is the Frobenius norm $\|\cdot\|_F$. From now we will denote both norms by $\|\cdot\|$ and the notion will be clear from the context. (Note that if f satisfies (B) then it also satisfies (A1) and (A2).)

The following facts are left as simple exercises.

Fact 8.0.1. *If f satisfies (B) then there exists $K_1 \in (0, +\infty)$ such that for all $(t, x) \in [0, T] \times \mathbb{R}^d$*

$$\|f(t, x)\| \leq K_1(1 + \|x\|). \quad (8.5)$$

Fact 8.0.2. *Suppose that the random vectors $X : \Omega \rightarrow \mathbb{R}^{d \times m}$ and $Y : \Omega \rightarrow \mathbb{R}^m$ are independent. Then*

$$\mathbb{E}\|X \cdot Y\|^2 \leq \mathbb{E}\|X\|^2 \cdot \mathbb{E}\|Y\|^2. \quad (8.6)$$

The classical Euler scheme is defined as follows.

Let $n \in \mathbb{N}$ and $0 = t_0 < t_1 < \dots < t_n = T$ be (possibly) non-uniform discretization of $[0, T]$. The classical Euler scheme for the jump-diffusion SDE (7.5) is defined as follows:

$$X_n^E(0) = \xi, \quad (8.7)$$

and for $k = 0, 1, \dots, n - 1$

$$X_n^E(t_{k+1}) = X_n^E(t_k) + a(U_{k,n}^E) \cdot \Delta t_k + b(U_{k,n}^E) \cdot \Delta W_k + c(U_{k,n}^E) \cdot \Delta N_k, \quad (8.8)$$

where

$$U_{k,n}^E = (t_k, X_n^E(t_k)), \quad (8.9)$$

$$\begin{aligned} \Delta t_k &= t_{k+1} - t_k, \\ \Delta W_k &= [\Delta W_k^1, \dots, \Delta W_k^{m_W}]^T, \\ \Delta N_k &= [\Delta N_k^1, \dots, \Delta N_k^{m_N}]^T, \end{aligned} \quad (8.10)$$

and

$$\Delta Z_k^j = Z^j(t_{k+1}) - Z^j(t_k), \quad Z \in \{N, W\}. \quad (8.11)$$

Since

$$b(U_{k,n}^E) \cdot \Delta W_k = \left(\sum_{j=1}^{m_W} b_{ij}(U_{k,n}^E) \cdot \Delta W_k^j \right)_{i=1,2,\dots,d}, \quad (8.12)$$

$$c(U_{k,n}^E) \cdot \Delta N_k = \left(\sum_{j=1}^{m_N} c_{ij}(U_{k,n}^E) \cdot \Delta N_k^j \right)_{i=1,2,\dots,d}, \quad (8.13)$$

we can write each component of the Euler scheme as follows

$$X_{n,i}^E(0) = \xi_i, \quad (8.14)$$

$$\begin{aligned} X_{n,i}^E(t_{k+1}) &= X_{n,i}^E(t_k) + a_i(U_{k,n}^E) \cdot \Delta t_k \\ &\quad + \sum_{j=1}^{m_W} b_{ij}(U_{k,n}^E) \cdot \Delta W_k^j \\ &\quad + \sum_{j=1}^{m_N} c_{ij}(U_{k,n}^E) \cdot \Delta N_k^j, \end{aligned} \quad (8.15)$$

for $k = 0, 1, \dots, n-1$, $i = 1, 2, \dots, d$.

Example 8.0.3. For the scalar equation (7.10) we have that

$$X_n^E(t_{k+1}) = X_n^E(t_k)(1 + \mu \Delta t_k + \sigma \Delta W_k + c \Delta N_k) = x_0 \cdot \prod_{j=0}^k (1 + \mu \Delta t_j + \sigma \Delta W_j + c \Delta N_j), \quad (8.16)$$

for $k = 0, 1, \dots, n-1$.

The aim of this chapter is to prove the following result.

Theorem 8.0.4. Let us assume that a, b, c satisfy (B), (C) and ξ satisfies (A0). Then there exists $C \in (0, +\infty)$ such that for all $n \in \mathbb{N}$ and $0 = t_0 < t_1 < \dots < t_n = T$ we have

$$\max_{0 \leq k \leq n} \left(\mathbb{E} \|X(t_k) - X_n^E(t_k)\|^2 \right)^{1/2} \leq C \cdot \left(1 + (\mathbb{E} \|\xi\|^2)^{1/2} \right) \cdot \max_{0 \leq i \leq n-1} (t_{i+1} - t_i)^{1/2}. \quad (8.17)$$

Before we prove this fact we need several auxiliary lemmas concerning the so-called *time-continuous Euler approximation* $\tilde{X}_n^E = (\tilde{X}_n^E(t))_{t \in [0, T]}$. It is defined as follows:

$$\tilde{X}_n^E(0) = \xi, \quad (8.18)$$

and for $t \in (t_k, t_{k+1}]$, $k = 0, 1, \dots, n-1$ we set

$$\tilde{X}_n^E(t) = X_n^E(t_k) + a(U_{k,n}^E) \cdot (t - t_k) + b(U_{k,n}^E) \cdot (W(t) - W(t_k)) + c(U_{k,n}^E) \cdot (N(t) - N(t_k)). \quad (8.19)$$

The process $\tilde{X}_n^E$ can be seen as a specific interpolation between the points $X^E(t_k)$, $k = 0, 1, \dots, n$, with the coefficients a, b, c 'frozen' at $U_{k,n}^E$. The approximation $\tilde{X}_n^E$ is not implementable, since it requires the complete knowledge of trajectories of N and W .

By induction, we have that

$$\tilde{X}_n^E(t_k) = X_n^E(t_k), \quad k = 0, 1, \dots, n. \quad (8.20)$$

Since the trajectories of $[N, W]$ are right-continuous we get

$$\lim_{t \rightarrow t_k+} \tilde{X}_n^E(t) = X_n^E(t_k) = \tilde{X}_n^E(t_k), \quad (8.21)$$

and

$$\begin{aligned} \lim_{t \rightarrow t_{k+1}-} \tilde{X}_n^E(t) &= X_n^E(t_k) + a(U_{k,n}^E) \cdot (t_{k+1} - t_k) + b(U_{k,n}^E) \cdot (W(t_{k+1}) - W(t_k)) \\ &\quad + c(U_{k,n}^E) \cdot (N(t_{k+1}-) - N(t_k)) \neq \tilde{X}_n^E(t_{k+1}) (= X_n^E(t_{k+1})) \end{aligned} \quad (8.22)$$

for $k = 0, 1, \dots, n-1$. Moreover, it is easy to see that $\tilde{X}_n^E = (\tilde{X}_n^E(t))_{t \in [0, T]}$ is adapted to $(\Sigma_t)_{t \in [0, +\infty)}$ and has càdlàg paths. Hence, $\tilde{X}_n^E \in \mathbb{D}$.

Lemma 8.0.5. *Let us assume that a, b, c satisfy (B), (C) and ξ satisfies (A0). Then there exists $C \in (0, +\infty)$ such that for all $n \in \mathbb{N}$ and $0 = t_0 < t_1 < \dots < t_n = T$ we have*

$$\sup_{t \in [0, T]} \mathbb{E} \|\tilde{X}_n^E(t)\|^2 \leq C(1 + \mathbb{E} \|\xi\|^2). \quad (8.23)$$

Proof. Firstly we show that

$$\max_{0 \leq k \leq n} \mathbb{E} \|X_n^E(t_k)\|^2 < +\infty. \quad (8.24)$$

We proceed by induction and let us assume that there exists $s \in \{0, 1, \dots, n-1\}$ with the property that

$$\max_{0 \leq k \leq s} \mathbb{E} \|X_n^E(t_k)\|^2 < +\infty. \quad (8.25)$$

(This is of course satisfied for $s = 0$.) Since $\{\sigma(b(U_{s,n}^E)), \sigma(\Delta W_s)\}$ and $\{\sigma(c(U_{s,n}^E)), \sigma(\Delta N_s)\}$ are two families of independent σ -fields, we get by the Fact 8.0.2

$$\begin{aligned} \mathbb{E} \|X_n^E(t_{s+1})\|^2 &\leq C_1 \left(\mathbb{E} \|X_n^E(t_s)\|^2 + (\Delta t_s)^2 \cdot \mathbb{E} \|a(U_{s,n}^E)\|^2 \right. \\ &\quad \left. + \mathbb{E} \|b(U_{s,n}^E)\|^2 \cdot \mathbb{E} \|\Delta W_s\|^2 + \mathbb{E} \|c(U_{s,n}^E)\|^2 \cdot \mathbb{E} \|\Delta N_s\|^2 \right). \end{aligned} \quad (8.26)$$

Since there exist $c_W, c_N \in (0, +\infty)$ such that for all $t \in [t_k, t_{k+1}]$, $k = 0, 1, \dots, n-1$ it holds

$$\begin{aligned} \mathbb{E} \|W(t) - W(t_k)\|^2 &\leq c_W(t - t_k), \\ \mathbb{E} \|N(t) - N(t_k)\|^2 &\leq c_N(t - t_k), \end{aligned} \quad (8.27)$$

we get by Fact 8.0.1 that

$$\begin{aligned} \mathbb{E} \|X_n^E(t_{s+1})\|^2 &\leq C_2 \left(\mathbb{E} \|X_n^E(t_s)\|^2 + (\Delta t_s)^2 \cdot (1 + \mathbb{E} \|X_n^E(t_s)\|^2) \right. \\ &\quad \left. + (1 + \mathbb{E} \|X_n^E(t_s)\|^2) \cdot \Delta t_s + (1 + \mathbb{E} \|X_n^E(t_s)\|^2) \cdot \Delta t_s \right) < +\infty. \end{aligned} \quad (8.28)$$

Hence,

$$\max_{0 \leq k \leq s+1} \mathbb{E} \|X_n^E(t_k)\|^2 < +\infty. \quad (8.29)$$

By the rules of induction, this completes the proof of (8.24).

From (8.19) it follows that for all $t \in [t_k, t_{k+1}]$, $k = 0, 1, \dots, n-1$

$$\begin{aligned} \mathbb{E}\|\tilde{X}_n^E(t)\|^2 &\leq C_2 \left(\mathbb{E}\|X_n^E(t_k)\|^2 + (t - t_k)^2 \cdot (1 + \mathbb{E}\|X_n^E(t_k)\|^2) \right. \\ &\quad \left. + (1 + \mathbb{E}\|X_n^E(t_k)\|^2) \cdot (t - t_k) + (1 + \mathbb{E}\|X_n^E(t_k)\|^2) \cdot (t - t_k) \right) \\ &\leq C_3(1 + \mathbb{E}\|X_n^E(t_k)\|^2), \end{aligned} \quad (8.30)$$

and by (8.24)

$$\sup_{0 \leq t \leq T} \mathbb{E}\|\tilde{X}_n^E(t)\|^2 = \max_{0 \leq k \leq n-1} \sup_{t_k \leq t \leq t_{k+1}} \mathbb{E}\|\tilde{X}_n^E(t)\|^2 \leq C_3(1 + \max_{0 \leq k \leq n-1} \mathbb{E}\|X_n^E(t_k)\|^2) < +\infty. \quad (8.31)$$

Note that for now the bound in (8.31) depends on n . In the second part of the proof, we will show, with the help of Gronwall's lemma, that we can obtain the bound (8.23), with C that is independent of n .

We can write for all $t \in [0, T]$ that

$$\tilde{X}_n^E(t) = \xi + \int_0^t \tilde{a}_n(s) ds + \int_0^t \tilde{b}_n(s) dW(s) + \int_0^t \tilde{c}_n(s) dN(s), \quad (8.32)$$

where for $f \in \{a, b, c\}$

$$\tilde{f}_n(s) = \sum_{k=0}^{n-1} f(U_{k,n}^E) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s), \quad (8.33)$$

and recall that $U_{k,n}^E = (t_k, X_n^E(t_k)) = (t_k, \tilde{X}_n^E(t_k))$. Note that $(\tilde{f}_n(t))_{t \in [0, T]} \in \mathbb{L}$ for $f \in \{a, b, c\}$. Furthermore, for $f \in \{a, b, c\}$ and all $t \in [0, T]$ it holds

$$\begin{aligned} \|\tilde{f}_n(t)\|^2 &\leq \left(\sum_{k=0}^{n-1} \|f(U_{k,n}^E)\| \cdot \mathbf{1}_{(t_k, t_{k+1}]}(t) \right)^2 \\ &= \sum_{k=0}^{n-1} \|f(U_{k,n}^E)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(t), \end{aligned} \quad (8.34)$$

which, together with Fact 8.0.1, implies

$$\begin{aligned} \int_0^t \|\tilde{f}_n(s)\|^2 ds &\leq \int_0^t \sum_{k=0}^{n-1} \|f(U_{k,n}^E)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &\leq K_1^2 \int_0^t \sum_{k=0}^{n-1} (1 + \|\tilde{X}_n^E(t_k)\|^2) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &\leq 2K_1^2 \int_0^t \sum_{k=0}^{n-1} (1 + \|\tilde{X}_n^E(t_k)\|^2) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &\leq 2K_1^2 \int_0^T \sum_{k=0}^{n-1} \mathbf{1}_{(t_k, t_{k+1}]}(s) ds + 2K_1^2 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &= 2K_1^2 T + 2K_1^2 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds. \end{aligned} \quad (8.35)$$

By the fact that $N(t) = \tilde{N}(t) + m(t)$, we have

$$\begin{aligned} \mathbb{E} \|\tilde{X}_n^E(t)\|^2 &\leq C_1 \left(\mathbb{E} \|\xi\|^2 + \mathbb{E} \left\| \int_0^t \tilde{a}_n(s) ds \right\|^2 + \mathbb{E} \left\| \int_0^t \tilde{b}_n(s) dW(s) \right\|^2 \right. \\ &\quad \left. + \mathbb{E} \left\| \int_0^t \tilde{c}_n(s) d\tilde{N}(s) \right\|^2 + \mathbb{E} \left\| \int_0^t \tilde{c}_n(s) dm(s) \right\|^2 \right). \end{aligned} \quad (8.36)$$

From the Hölder inequality, Itô isometry for stochastic integrals driven by martingales, and (8.35) we get for $t \in [0, T]$

$$\begin{aligned} \mathbb{E} \left\| \int_0^t \tilde{a}_n(s) ds \right\|^2 &\leq \mathbb{E} \left(\int_0^t \|\tilde{a}_n(s)\| ds \right)^2 \leq T \mathbb{E} \int_0^t \|\tilde{a}_n(s)\|^2 ds \\ &\leq K_2 + K_3 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds, \end{aligned} \quad (8.37)$$

$$\begin{aligned} \mathbb{E} \left\| \int_0^t \tilde{b}_n(s) dW(s) \right\|^2 &= \mathbb{E} \int_0^t \|\tilde{b}_n(s)\|^2 ds \\ &\leq K_4 + K_5 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds, \end{aligned} \quad (8.38)$$

$$\begin{aligned} \mathbb{E} \left\| \int_0^t \tilde{c}_n(s) d\tilde{N}(s) \right\|^2 &= \sum_{j=1}^{m_N} \mathbb{E} \int_0^t \|\tilde{c}_n^{(j)}(s)\|^2 \cdot \lambda_j(s) ds \\ &\leq \max_{1 \leq j \leq m_N} \|\lambda_j\|_\infty \cdot \mathbb{E} \int_0^t \|\tilde{c}_n(s)\|^2 ds \\ &\leq K_6 + K_7 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds, \end{aligned} \quad (8.39)$$

and

$$\begin{aligned}
& \mathbb{E} \left\| \int_0^t \tilde{c}_n(s) dm(s) \right\|^2 = \sum_{i=1}^d \mathbb{E} \left| \sum_{j=1}^{m_N} \int_0^t \tilde{c}_{n,ij}(s) \cdot \lambda_j(s) ds \right|^2 \\
& \leq m_N \cdot \sum_{i=1}^d \sum_{j=1}^{m_N} \mathbb{E} \left| \int_0^t \tilde{c}_{n,ij}(s) \cdot \lambda_j(s) ds \right|^2 \\
& \leq m_N T \cdot \sum_{i=1}^d \sum_{j=1}^{m_N} \int_0^t \mathbb{E} |\tilde{c}_{n,ij}(s)|^2 \cdot \lambda_j^2(s) ds \\
& \leq m_N T \cdot \max_{1 \leq j \leq m_N} \|\lambda_j\|_\infty^2 \cdot \mathbb{E} \int_0^t \|\tilde{c}_n(s)\|^2 ds \\
& \leq K_8 + K_9 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds.
\end{aligned} \tag{8.40}$$

Hence

$$\begin{aligned}
\mathbb{E} \|\tilde{X}_n^E(t)\|^2 & \leq K_{10}(1 + \mathbb{E}\|\xi\|^2) + K_{11} \int_0^t \sum_{k=0}^{n-1} \mathbb{E} \|\tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\
& \leq K_{10}(1 + \mathbb{E}\|\xi\|^2) + K_{11} \int_0^t \sup_{0 \leq u \leq z} \mathbb{E} \|\tilde{X}_n^E(u)\|^2 dz
\end{aligned} \tag{8.41}$$

where K_{10}, K_{11} depend only on $K_1, T, m_N, \max_{1 \leq j \leq m_N} \|\lambda_j\|_\infty$. Therefore, for all $t \in [0, T]$

$$\sup_{0 \leq s \leq t} \mathbb{E} \|\tilde{X}_n^E(s)\|^2 \leq K_{10}(1 + \mathbb{E}\|\xi\|^2) + K_{11} \int_0^t \sup_{0 \leq u \leq z} \mathbb{E} \|\tilde{X}_n^E(u)\|^2 dz. \tag{8.42}$$

Note that the function

$$[0, T] \ni t \rightarrow \sup_{0 \leq s \leq t} \mathbb{E} \|\tilde{X}_n^E(s)\|^2 \tag{8.43}$$

is Borel measurable (as a nondecreasing function) and bounded, which follows from (8.31). By using Lemma 13.0.11 we get the thesis. ■

Proof of Theorem 8.0.4. We have that for all $t \in [0, T]$

$$X(t) = \xi + \int_0^t \hat{a}(s) ds + \int_0^t \hat{b}(s) dW(s) + \int_0^t \hat{c}(s) dN(s), \tag{8.44}$$

where for $f \in \{a, b, c\}$

$$\hat{f}(s) = \sum_{k=0}^{n-1} f(s, X(s-)) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s). \tag{8.45}$$

(Note that for $f \in \{a, b, c\}$ the process $(\hat{f}(t))_{t \in [0, T]}$ is in $\mathbb{L}$ so all Lebesgue and stochastic integrals appearing in (8.44) are well-defined.) From the representations above we thus get

$$\mathbb{E} \|X(t) - \tilde{X}_n^E(t)\|^2 \leq 3 \left(A_{1,n}(t) + A_{2,n}(t) + A_{3,n}(t) \right) \tag{8.46}$$

where

$$\begin{aligned}
A_{1,n}(t) &= \mathbb{E} \left\| \int_0^t (\hat{a}(s) - \tilde{a}_n(s)) ds \right\|^2, \\
A_{2,n}(t) &= \mathbb{E} \left\| \int_0^t (\hat{b}(s) - \tilde{b}_n(s)) dW(s) \right\|^2, \\
A_{3,n}(t) &= \mathbb{E} \left\| \int_0^t (\hat{c}(s) - \tilde{c}_n(s)) dN(s) \right\|^2.
\end{aligned} \tag{8.47}$$

By the Schwarz inequality, we get

$$A_{1,n}(t) \leq T \mathbb{E} \int_0^t \|\hat{a}(s) - \tilde{a}_n(s)\|^2 ds, \tag{8.48}$$

while from the Itô isometry

$$A_{2,n}(t) = \mathbb{E} \int_0^t \|\hat{b}(s) - \tilde{b}_n(s)\|^2 ds. \tag{8.49}$$

Furthermore

$$A_{3,n}(t) \leq 2 \left(B_{1,n}(t) + B_{2,n}(t) \right), \tag{8.50}$$

where

$$\begin{aligned}
B_{1,n}(t) &= \mathbb{E} \left\| \int_0^t (\hat{c}(s) - \tilde{c}_n(s)) d\tilde{N}(s) \right\|^2, \\
B_{2,n}(t) &= \mathbb{E} \left\| \int_0^t (\hat{c}(s) - \tilde{c}_n(s)) dm(s) \right\|^2.
\end{aligned} \tag{8.51}$$

From the isometry for martingale-driven stochastic integrals, we get

$$\begin{aligned}
B_{1,n}(t) &= \sum_{j=1}^{m_N} \mathbb{E} \int_0^t \|\hat{c}^{(j)}(s) - \tilde{c}_n^{(j)}(s)\|^2 \cdot \lambda_j(s) ds \\
&\leq \max_{1 \leq j \leq m_N} \|\lambda_j\|_\infty \cdot \mathbb{E} \int_0^t \|\hat{c}(s) - \tilde{c}_n(s)\|^2 ds
\end{aligned} \tag{8.52}$$

and by the Schwarz inequality

$$\begin{aligned}
B_{2,n}(t) &= \sum_{i=1}^d \mathbb{E} \left| \sum_{j=1}^{m_N} \int_0^t \left(\hat{c}_{ij}(s) - \tilde{c}_{n,ij}(s) \right) \cdot \lambda_j(s) ds \right|^2 \\
&\leq m_N \cdot \sum_{i=1}^d \sum_{j=1}^{m_N} \mathbb{E} \left| \int_0^t \left(\hat{c}_{ij}(s) - \tilde{c}_{n,ij}(s) \right) \cdot \lambda_j(s) ds \right|^2 \\
&\leq m_N T \cdot \sum_{i=1}^d \sum_{j=1}^{m_N} \int_0^t \mathbb{E} |\hat{c}_{ij}(s) - \tilde{c}_{n,ij}(s)|^2 \cdot \lambda_j^2(s) ds \\
&\leq m_N T \cdot \max_{1 \leq j \leq m_N} \|\lambda_j\|_\infty^2 \cdot \mathbb{E} \int_0^t \|\hat{c}(s) - \tilde{c}_n(s)\|^2 ds.
\end{aligned} \tag{8.53}$$

Hence, by (8.50), (8.52), and (8.53) we get

$$A_{3,n}(t) \leq C \cdot \mathbb{E} \int_0^t \|\hat{c}(s) - \tilde{c}_n(s)\|^2 ds. \tag{8.54}$$

Combining (8.46), (8.48), (8.49), and (8.54) we arrive at

$$\mathbb{E} \|X(t) - \tilde{X}_n^E(t)\|^2 \leq C \sum_{f \in \{a,b,c\}} \mathbb{E} \int_0^t \|\hat{f}(s) - \tilde{f}_n(s)\|^2 ds. \tag{8.55}$$

Now, for $f \in \{a, b, c\}$ and $s \in [0, T]$ we get

$$\begin{aligned}
\|\hat{f}(s) - \tilde{f}_n(s)\| &= \left\| \sum_{k=0}^{n-1} \left(f(s, X(s-)) - f(U_{k,n}^E) \right) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \right\| \\
&\leq \sum_{k=0}^{n-1} \|f(s, X(s-)) - f(U_{k,n}^E)\| \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \\
&\leq K \sum_{k=0}^{n-1} \left((s - t_k) + \|X(s-) - \tilde{X}_n^E(t_k)\| \right) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \\
&\leq K \max_{0 \leq k \leq n-1} \Delta t_k + K \sum_{k=0}^{n-1} \|X(s-) - X(t_k)\| \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \\
&\quad + K \sum_{k=0}^{n-1} \|X(t_k) - \tilde{X}_n^E(t_k)\| \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s)
\end{aligned} \tag{8.56}$$

and hence

$$\begin{aligned}
\|\hat{f}(s) - \tilde{f}_n(s)\|^2 &\leq 3K^2 \max_{0 \leq k \leq n-1} (\Delta t_k)^2 + 3K^2 \sum_{k=0}^{n-1} \|X(s-) - X(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \\
&\quad + 3K^2 \sum_{k=0}^{n-1} \|X(t_k) - \tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s).
\end{aligned} \tag{8.57}$$

Since any trajectory of X has only finite number of (jump) discontinuities in $[0, T]$, the functions

$$[0, T] \ni s \rightarrow \sum_{k=0}^{n-1} \|X(s-) - X(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s), \quad (8.58)$$

$$[0, T] \ni s \rightarrow \sum_{k=0}^{n-1} \|X(s) - X(t_k)\|^2 \cdot \mathbf{1}_{[t_k, t_{k+1})}(s) \quad (8.59)$$

differs only on a finite subset of $[0, T]$ (the first is a càglàd modification of the second one). This implies for $f \in \{a, b, c\}$ and $t \in [0, T]$ that

$$\begin{aligned} \int_0^t \|\hat{f}(s) - \tilde{f}_n(s)\|^2 ds &\leq 3K^2 T \max_{0 \leq k \leq n-1} (\Delta t_k)^2 + 3K^2 \int_0^t \sum_{k=0}^{n-1} \|X(s) - X(t_k)\|^2 \cdot \mathbf{1}_{[t_k, t_{k+1})}(s) ds \\ &\quad + 3K^2 \int_0^t \sum_{k=0}^{n-1} \|X(t_k) - \tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds. \end{aligned} \quad (8.60)$$

By Lemma 7.0.1 we have that

$$\begin{aligned} \int_0^t \sum_{k=0}^{n-1} \mathbb{E} \|X(s) - X(t_k)\|^2 \cdot \mathbf{1}_{[t_k, t_{k+1})}(s) ds &\leq C(1 + \mathbb{E}\|\xi\|^2) \int_0^T \sum_{k=0}^{n-1} (s - t_k) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &\leq C_1(1 + \mathbb{E}\|\xi\|^2) \cdot \max_{0 \leq k \leq n-1} \Delta t_k. \end{aligned} \quad (8.61)$$

Since $\max_{0 \leq k \leq n-1} (\Delta t_k)^2 \leq T \max_{0 \leq k \leq n-1} \Delta t_k$, we get for all $t \in [0, T]$

$$\begin{aligned} \mathbb{E} \int_0^t \|\hat{f}(s) - \tilde{f}_n(s)\|^2 ds &\leq K_1 \cdot (1 + \mathbb{E}\|\xi\|^2) \cdot \max_{0 \leq k \leq n-1} \Delta t_k \\ &\quad + K_2 \int_0^t \sum_{k=0}^{n-1} \mathbb{E} \|X(t_k) - \tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds. \end{aligned} \quad (8.62)$$

and hence

$$\begin{aligned} \mathbb{E} \|X(t) - \tilde{X}_n^E(t)\|^2 &\leq K_3 \cdot (1 + \mathbb{E}\|\xi\|^2) \cdot \max_{0 \leq k \leq n-1} \Delta t_k \\ &\quad + K_4 \int_0^t \sum_{k=0}^{n-1} \mathbb{E} \|X(t_k) - \tilde{X}_n^E(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &\leq K_3 \cdot (1 + \mathbb{E}\|\xi\|^2) \cdot \max_{0 \leq k \leq n-1} \Delta t_k \\ &\quad + K_4 \int_0^t \sup_{0 \leq u \leq z} \mathbb{E} \|X(u) - \tilde{X}_n^E(u)\|^2 dz. \end{aligned} \quad (8.63)$$

Therefore, for all $t \in [0, T]$

$$\begin{aligned} \sup_{0 \leq s \leq t} \mathbb{E} \|X(s) - \tilde{X}_n^E(s)\|^2 &\leq K_3 \cdot (1 + \mathbb{E}\|\xi\|^2) \cdot \max_{0 \leq k \leq n-1} \Delta t_k \\ &\quad + K_4 \int_0^t \sup_{0 \leq u \leq z} \mathbb{E} \|X(u) - \tilde{X}_n^E(u)\|^2 dz. \end{aligned} \quad (8.64)$$

Note that the function

$$[0, T] \ni t \rightarrow \sup_{0 \leq s \leq t} \mathbb{E} \|X(s) - \tilde{X}_n^E(s)\|^2 \quad (8.65)$$

is bounded (due to Lemmas 7.0.1, 8.0.5) and Borel measurable (as a nondecreasing function). Hence, by Gronwall's lemma 13.0.11 we get

$$\sup_{0 \leq t \leq T} \mathbb{E} \|X(t) - \tilde{X}_n^E(t)\|^2 \leq K_5 \cdot (1 + \mathbb{E} \|\xi\|^2) \cdot \max_{0 \leq k \leq n-1} \Delta t_k, \quad (8.66)$$

which completes the proof. $\blacksquare$

In order to have approximation of X in between discretization points $(t_k)_{k=0,1,\dots,n}$ we consider the following piecewise linear interpolation of the data $(t_k, X_n^E(t_k))_{k=0,1,\dots}$

$$\bar{X}_n^E(t) = \sum_{k=0}^{n-1} \frac{(t - t_k) \cdot X_n^E(t_{k+1}) + (t_{k+1} - t) \cdot X_n^E(t_k)}{t_{k+1} - t_k} \cdot \mathbf{1}_{[t_k, t_{k+1}]}(t), \quad t \in [0, T]. \quad (8.67)$$

The process $\bar{X}_n^E = (\bar{X}_n^E(t))_{t \in [0, T]}$ has continuous trajectories, however, it is not adapted to the underlying filtration $(\Sigma_t)_{t \in [0, +\infty)}$. If needed, one can obtain an adapted approximation of X by considering the following step process

$$\hat{X}_n^E(t) = \sum_{k=0}^{n-1} X_n^E(t_k) \cdot \mathbf{1}_{[t_k, t_{k+1})}(t), \quad t \in [0, T], \quad (8.68)$$

and $\hat{X}_n^E(T) = X_n^E(T)$. Of course $\hat{X}_n^E \in \mathbb{D}$. Such approximations, defined in the whole $[0, T]$, are useful when considering path-dependent option pricing, see Chapter 9.

Theorem 8.0.6. *Let us assume that a, b, c satisfy (B), (C) and ξ satisfies (A0). Then there exists $C \in (0, +\infty)$ such that for all $n \in \mathbb{N}$ and $0 = t_0 < t_1 < \dots < t_n = T$ we have*

$$\left(\mathbb{E} \int_0^T \|X(t) - \bar{X}_n^E(t)\|^2 dt \right)^{1/2} \leq C(1 + (\mathbb{E} \|\xi\|^2)^{1/2}) \max_{0 \leq k \leq n-1} (\Delta t_k)^{1/2}. \quad (8.69)$$

and

$$\left(\mathbb{E} \int_0^T \|X(t) - \hat{X}_n^E(t)\|^2 dt \right)^{1/2} \leq C(1 + (\mathbb{E} \|\xi\|^2)^{1/2}) \max_{0 \leq k \leq n-1} (\Delta t_k)^{1/2}. \quad (8.70)$$

Proof. We have for $t \in [t_k, t_{k+1}]$, $k = 0, 1, \dots, n-1$, that

$$X(t) - \bar{X}_n^E(t) = \frac{t_{k+1} - t}{t_{k+1} - t_k} (X(t) - X_n^E(t_k)) - \frac{t - t_k}{t_{k+1} - t_k} (X_n^E(t_{k+1}) - X(t)), \quad (8.71)$$

and hence

$$\begin{aligned} \|X(t) - \bar{X}_n^E(t)\| &\leq \|X(t) - X(t_k)\| + \|X(t_k) - X_n^E(t_k)\| \\ &\quad + \|X_n^E(t_{k+1}) - X(t_{k+1})\| + \|X(t_{k+1}) - X(t)\|. \end{aligned} \quad (8.72)$$

Therefore, by Lemma 7.0.1, (8.66), and the fact that $(a + b + c + d)^2 \leq 4(a^2 + b^2 + c^2 + d^2)$ for all $a, b, c, d \geq 0$, we get

$$\begin{aligned} \mathbb{E} \int_0^T \|X(t) - \bar{X}_n^E(t)\|^2 dt &= \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} \mathbb{E} \|X(t) - \bar{X}_n^E(t)\|^2 dt \\ &\leq 4 \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} \mathbb{E} \|X(t) - X(t_k)\|^2 dt + 4 \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} \mathbb{E} \|X(t_{k+1}) - X(t)\|^2 dt \\ &\quad + 8T \sup_{0 \leq t \leq T} \mathbb{E} \|X(t) - \tilde{X}_n^E(t)\|^2 \leq K(1 + \mathbb{E} \|\xi\|^2) \max_{0 \leq k \leq n-1} \Delta t_k. \end{aligned} \quad (8.73)$$

By (8.66) we have that

$$\begin{aligned}
\mathbb{E} \int_0^T \|X(t) - \hat{X}_n^E(t)\|^2 dt &= \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} \mathbb{E} \|X(t) - X_n^E(t_k)\|^2 dt \\
&\leq 2 \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} \mathbb{E} \|X(t) - X(t_k)\|^2 dt + 2 \sum_{k=0}^{n-1} \Delta t_k \cdot \mathbb{E} \|X(t_k) - X_n^E(t_k)\|^2 \\
&\leq C_1(1 + \mathbb{E}\|\xi\|^2) \max_{0 \leq k \leq n-1} \Delta t_k + C_2 \sup_{0 \leq t \leq T} \mathbb{E} \|X(t) - \tilde{X}_n^E(t)\|^2 \\
&\leq C_3(1 + \mathbb{E}\|\xi\|^2) \max_{0 \leq k \leq n-1} \Delta t_k,
\end{aligned} \tag{8.74}$$

and this gives (8.70). $\blacksquare$

8.1 Optimality of the convergence rate $1/2$ for the Euler-Maruyama algorithm

Let us consider the Euler-Maruyama algorithm X_n^E . In the case of equidistant discretization, we show that the error bound $O(n^{-1/2})$ is sharp in a class of coefficients a, b, c, ξ satisfying the (Lipschitz) assumptions (B), (C), and (A0).

Consider the geometric Brownian motion given by the SDE

$$dX(t) = \sigma X(t) dW(t), \quad t \in [0, T], \tag{8.75}$$

and with the initial-value $X(0) = 1$, $\sigma > 0$. The coefficients $a = c = 0$, $b(t, x) = \sigma x$, $\xi = 1$ satisfy the assumptions (B), (C) and (A0). Moreover, $X(t) = \exp\left(-\frac{1}{2}\sigma^2 t + \sigma W(t)\right)$ and $\mathbb{E}(X(t))^2 = \exp(\sigma^2 t) \geq 1$ for all $t \in [0, T]$. By (8.32), (8.44) and Itô isometry we have for all $n \in \mathbb{N}$ and any discretization $0 = t_0 < t_1 < \dots < t_n = T$ that

$$\begin{aligned}
\mathbb{E}|X(T) - X_n^E(T)|^2 &= \sigma^2 \cdot \mathbb{E} \left| \int_0^T \left[\sum_{k=0}^{n-1} (X(s) - X_n^E(t_k)) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \right] dW(s) \right|^2 \\
&= \sigma^2 \cdot \mathbb{E} \int_0^T \left[\sum_{k=0}^{n-1} (X(s) - X_n^E(t_k)) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \right]^2 ds \\
&= \sigma^2 \cdot \mathbb{E} \int_0^T \sum_{k=0}^{n-1} (X(s) - X_n^E(t_k))^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds = \sigma^2 \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} \mathbb{E}(X(s) - X_n^E(t_k))^2 ds.
\end{aligned} \tag{8.76}$$

Since $X(s) = X(t_k) + \sigma \int_{t_k}^s X(u) dW(u)$ for $s \in [t_k, t_{k+1}]$, we have that

$$\begin{aligned}
\mathbb{E}(X(s) - X_n^E(t_k))^2 &= \mathbb{E}\left[(X(t_k) - X_n^E(t_k)) + \sigma \int_{t_k}^s X(u) dW(u)\right]^2 \\
&= \mathbb{E}(X(t_k) - X_n^E(t_k))^2 + \sigma^2 \mathbb{E}\left(\int_{t_k}^s X(u) dW(u)\right)^2 \\
&\quad + 2\sigma \mathbb{E}\left((X(t_k) - X_n^E(t_k)) \cdot \int_{t_k}^s X(u) dW(u)\right) \\
&\geq \sigma^2 \mathbb{E}\left(\int_{t_k}^s X(u) dW(u)\right)^2 + 2\sigma \mathbb{E}\left((X(t_k) - X_n^E(t_k)) \cdot \int_{t_k}^s X(u) dW(u)\right),
\end{aligned} \tag{8.77}$$

where

$$\mathbb{E}\left(\int_{t_k}^s X(u) dW(u)\right)^2 = \int_{t_k}^s \mathbb{E}(X(u))^2 du \geq s - t_k, \tag{8.78}$$

$$\begin{aligned}
&\mathbb{E}\left((X(t_k) - X_n^E(t_k)) \cdot \int_{t_k}^s X(u) dW(u)\right) \\
&= \mathbb{E}\left[(X(t_k) - X_n^E(t_k)) \cdot \mathbb{E}\left(\int_{t_k}^s X(u) dW(u) \middle| \Sigma_{t_k}\right)\right] = 0.
\end{aligned} \tag{8.79}$$

Therefore, by (8.76)-(8.79) and Jensen inequality we get that

$$\mathbb{E}|X(T) - X_n^E(T)|^2 \geq \frac{\sigma^4}{2} \sum_{k=0}^{n-1} (\Delta t_k)^2 \geq \frac{\sigma^4 T^2}{2} n^{-1}, \tag{8.80}$$

which in turn implies the lower error bound $\Omega(n^{-1/2})$. Hence, under the assumptions of Theorem 8.0.4 the $L^2(\Omega)$ -error $O(n^{-1/2})$ of the Euler-Maruyama algorithm, based on equidistant mesh, is sharp.

If one considers the scalar case ($d = m = 1$) and assumes only global Lipschitz continuity of a and σ (without additional differentiability assumptions), the sharp minimal rate for *arbitrary* algorithms based on standard information about Brownian motion appears to be less settled in the literature. The example presented above only shows that the Euler-Maruyama method itself may have a sharp rate $n^{-1/2}$ on some equations (e.g., geometric Brownian motion).

Consider now the case when $d, m \geq 2$. Recall that Clark and Cameron (1980) exhibited a simple two-dimensional example

$$\begin{aligned}
dX^1(t) &= dW^1(t), \\
dX^2(t) &= X^1(t) dW^2(t), \quad t \in [0, T],
\end{aligned} \tag{8.81}$$

where $X^1(0) = X^2(0) = 0$, for which the solution $X^2(T)$ at the final time T is the Lévy area $\mathcal{I}(W^1, W^2)$. For that example, they proved that, given only discrete values of $W = [W^1, W^2]$, no approximation method can achieve a better strong convergence order than $1/2$. Consequently, within the class of methods driven by

standard information about W and in the case of the class of Lipschitz drift and diffusion coefficients, the Euler-Maruyama scheme (with the error order $1/2$) is the optimal one in the worst-case model of error.

For scalar SDEs ($d = m = 1$) the Lévy-area issue disappears. Under additional smoothness assumptions on the coefficients (e.g. existence and Lipschitz continuity of suitable derivatives), the information complexity of one-point strong approximation is different: the minimal achievable L_p -error at the final time might be of order n^{-1} , and Milstein-type (or adaptive) schemes are optimal up to multiplicative constants, see [63].

8.2 Implementation issues

Draw $\bar{X}_n^E(0)$ accordingly to the distribution of ξ and for $k = 0, 1, \dots, n-1$

$$\bar{X}_n^E(t_{k+1}) = \bar{X}_n^E(t_k) + a(\bar{U}_{k,n}^E) \cdot \Delta t_k + (\Delta t_k)^{1/2} \cdot b(\bar{U}_{k,n}^E) \cdot Z_k + c(\bar{U}_{k,n}^E) \cdot U_k, \quad (8.82)$$

where $\bar{U}_{k,n}^E = (t_k, \bar{X}_n^E(t_k))$ and

$$Z_k = [Z_k^1, \dots, Z_k^{m_W}]^T, \quad (8.83)$$

$$U_k = [U_k^1, \dots, U_k^{m_N}]^T, \quad (8.84)$$

where

- $(Z_k^j)_{k=0, \dots, n-1, j=1, \dots, m_W}$ is an i.i.d sequence of random variables with $Z_1^1 \sim N(0, 1)$,
- $(U_k^j)_{k=0, \dots, n-1, j=1, \dots, m_N}$ is a sequence of independent random variables, where $U_k^j \sim \text{Pois}\left(\int_{t_k}^{t_{k+1}} \lambda_j(s) ds\right)$,
- the collection $\{\xi, (Z_k^j)_{k=0, \dots, n-1, j=1, \dots, m_W}, (U_k^j)_{k=0, \dots, n-1, j=1, \dots, m_N}\}$ consists of independent random variables.

Then

$$\bar{X}_{n,i}^E(0) = \xi_i \text{ (in law)}, \quad (8.85)$$

$$\begin{aligned} \bar{X}_{n,i}^E(t_{k+1}) = \bar{X}_{n,i}^E(t_k) &+ a_i(\bar{U}_{k,n}^E) \cdot \Delta t_k \\ &+ (\Delta t_k)^{1/2} \cdot \sum_{j=1}^{m_W} b_{ij}(\bar{U}_{k,n}^E) \cdot Z_k^j \\ &+ \sum_{j=1}^{m_N} c_{ij}(\bar{U}_{k,n}^E) \cdot U_k^j, \end{aligned} \quad (8.86)$$

for $k = 0, 1, \dots, n-1$, $i = 1, 2, \dots, d$.

TODO: Dodac kod z implementacja oraz przykładowe rysunki...

8.3 Exercises

1. Let $U : \Omega \rightarrow \mathbb{R}^{d \times m}$ be Σ_α -measurable where $0 \leq \alpha < \beta$. Moreover, let W and N be m -dimensional Wiener process and m -dimensional non-homogeneous Poisson process, respectively. Show that for $Z \in \{N, W\}$

$$\int_{\alpha}^{\beta} U dZ(t) = U \cdot (Z(\beta) - Z(\alpha)). \quad (8.87)$$

2. Show that there exist c_W, c_N such that for all $0 \leq \alpha \leq \beta$ we have

$$\begin{aligned}\mathbb{E}\|W(\beta) - W(\alpha)\|^2 &\leq c_W \cdot (\beta - \alpha), \\ \mathbb{E}\|N(\beta) - N(\alpha)\|^2 &\leq c_N \cdot (\beta - \alpha).\end{aligned}\tag{8.88}$$

3. For the Euler process

(i) show (8.20).

(ii) give a proof that $\tilde{X}_n^E = (\tilde{X}_n^E(t))_{t \in [0, T]}$ is adapted to $(\Sigma_t)_{t \in [0, +\infty)}$.

(iii) show that

$$X_n^E(t_k) = \eta + \sum_{j=0}^{k-1} a(U_{j,n}^E) \cdot (t_{j+1} - t_j) + \sum_{j=0}^{k-1} b(U_{j,n}^E) \cdot \Delta W_j + \sum_{j=0}^{k-1} c(U_{j,n}^E) \cdot \Delta N_j \tag{8.89}$$

for $k = 0, 1, \dots, n$. (We use the convention that $\sum_{j=0}^{-1} c_j := 0$.)

(iv) give a proof that for all $t \in [0, T]$ we have

$$\tilde{X}_n^E(t) = \eta + \int_0^t \tilde{a}_n(s) ds + \int_0^t \tilde{b}_n(s) dW(s) + \int_0^t \tilde{c}_n(s) dN(s), \tag{8.90}$$

where for $f \in \{a, b, c\}$

$$\tilde{f}_n(t) = \sum_{k=0}^{n-1} f(U_{k,n}^E) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(t). \tag{8.91}$$

4. Give a proof of Fact 8.0.2.

Chapter 9

Randomized Euler scheme for jump-diffusion SDEs

For the functions

$$a : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad (9.1)$$

$$b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m_W}, \quad (9.2)$$

$$c : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m_N}, \quad (9.3)$$

we assume that

(B) there exists $K \in (0, +\infty)$ such that for $f \in \{b, c\}$ and for all $s, t \in [0, T]$, $x, y \in \mathbb{R}^d$

$$\|f(t, x) - f(s, y)\| \leq K(|t - s| + \|x - y\|), \quad (9.4)$$

(C) the function a is Borel measurable and there exists $L \in (0, +\infty)$ such that for all $t \in [0, T]$, $x, y \in \mathbb{R}^d$

$$\|a(t, x) - a(t, y)\| \leq L\|x - y\|, \quad (9.5)$$

(D) there exists $D \in (0, +\infty)$ such that for all $t \in [0, T]$

$$\|a(t, 0)\| \leq D. \quad (9.6)$$

(E) the intensity functions $\lambda_j : [0, T] \rightarrow (0, +\infty)$, $j = 1, 2, \dots, m_N$, are Borel measurable and bounded.

For $f = a$ the norm $\|\cdot\|$ is the euclidean norm $\|\cdot\|_2$, while for $f \in \{b, c\}$ the norm $\|\cdot\|$ is the Frobenius norm $\|\cdot\|_F$. From now we will denote both norms by $\|\cdot\|$ and the notion will be clear from the context.

Note that (C) and (D) imply that for all $(t, y) \in [0, T] \times \mathbb{R}^d$

$$\|a(t, y)\| \leq \bar{D}(1 + \|y\|), \quad (9.7)$$

where $\bar{D} = \max\{L, D\}$.

The randomized Euler scheme is defined as follows.

Let $n \in \mathbb{N}$ and $t_k = kT/n$, $k = 0, 1, \dots, n$, be a uniform discretization of $[0, T]$. Let $(\tau_j)_{j \in \mathbb{N}_0}$ be a sequence of independent random variables on $(\Omega, \Sigma, \mathbb{P})$ that are identically uniformly distributed on $[0, 1]$, and such that the σ -fields Σ_∞ and $\sigma\left(\bigcup_{j \geq 0} \sigma(\tau_j)\right)$ are independent. The classical Euler scheme for the jump-diffusion SDE (7.5) is defined as follows:

$$X_n^{RE}(0) = \xi, \quad (9.8)$$

and for $k = 0, 1, \dots, n-1$

$$X_n^{RE}(t_{k+1}) = X_n^{RE}(t_k) + a(\theta_k, X_n^{RE}(t_k)) \cdot \Delta t_k + b(U_{k,n}^{RE}) \cdot \Delta W_k + c(U_{k,n}^{RE}) \cdot \Delta N_k, \quad (9.9)$$

where

$$\theta_k = t_k + \tau_k \cdot \Delta t_k, \quad (9.10)$$

$$U_{k,n}^{RE} = (t_k, X_n^{RE}(t_k)), \quad (9.11)$$

$$\begin{aligned} \Delta t_k &= t_{k+1} - t_k = \frac{T}{n}, \\ \Delta W_k &= [\Delta W_k^1, \dots, \Delta W_k^{m_W}]^T, \\ \Delta N_k &= [\Delta N_k^1, \dots, \Delta N_k^{m_N}]^T, \end{aligned} \quad (9.12)$$

and

$$\Delta Z_k^j = Z^j(t_{k+1}) - Z^j(t_k), \quad Z \in \{N, W\}. \quad (9.13)$$

Since

$$b(U_{k,n}^{RE}) \cdot \Delta W_k = \left(\sum_{j=1}^{m_W} b_{ij}(U_{k,n}^{RE}) \cdot \Delta W_k^j \right)_{i=1,2,\dots,d}, \quad (9.14)$$

$$c(U_{k,n}^{RE}) \cdot \Delta N_k = \left(\sum_{j=1}^{m_N} c_{ij}(U_{k,n}^{RE}) \cdot \Delta N_k^j \right)_{i=1,2,\dots,d}, \quad (9.15)$$

we can write each component of the randomized Euler scheme as follows

$$X_{n,i}^{RE}(0) = \xi_i, \quad (9.16)$$

$$\begin{aligned} X_{n,i}^{RE}(t_{k+1}) &= X_{n,i}^{RE}(t_k) + a_i(\theta_k, X_n^{RE}(t_k)) \cdot \Delta t_k \\ &\quad + \sum_{j=1}^{m_W} b_{ij}(U_{k,n}^{RE}) \cdot \Delta W_k^j \\ &\quad + \sum_{j=1}^{m_N} c_{ij}(U_{k,n}^{RE}) \cdot \Delta N_k^j, \end{aligned} \quad (9.17)$$

for $k = 0, 1, \dots, n-1$, $i = 1, 2, \dots, d$. The aim of this chapter is to prove the following result.

Theorem 9.0.1. *Let us assume that $a, b, c, \{\lambda_j\}_{j=1,\dots,m_N}$ satisfy (B), (C), (D), (E) and ξ satisfies (A0). Then there exists $C \in (0, +\infty)$ such that for all $n \in \mathbb{N}$ we have*

$$\max_{1 \leq k \leq n} \left(\mathbb{E} \|X(t_k) - X_n^{RE}(t_k)\|^2 \right)^{1/2} \leq C \cdot \left(1 + (\mathbb{E} \|\xi\|^2)^{1/2} \right) \cdot n^{-1/2}. \quad (9.18)$$

For the error analysis of the randomized Euler scheme we need to consider the following extended filtration $(\tilde{\Sigma}_t^n)_{t \geq 0}$, defined as

$$\tilde{\Sigma}_t^n = \Sigma_t \vee \sigma(\tau_0, \tau_1, \dots, \tau_{n-1}), \quad t \geq 0. \quad (9.19)$$

Note that W is also a Wiener process with respect to $(\tilde{\Sigma}_t^n)_{t \geq 0}$, while N is again Poisson process also with respect to $(\tilde{\Sigma}_t^n)_{t \geq 0}$. Hence, we can consider stochastic integrals of processes from $\mathbb{L}$ that are adapted to $(\tilde{\Sigma}_t^n)_{t \geq 0}$. Before we prove this fact we need several auxiliary lemmas concerning the so-called *time-continuous Euler approximation* $\tilde{X}_n^{RE} = (\tilde{X}_n^{RE}(t))_{t \in [0, T]}$. It is defined as follows:

$$\tilde{X}_n^{RE}(0) = \xi, \quad (9.20)$$

and for $t \in (t_k, t_{k+1}]$, $k = 0, 1, \dots, n-1$ we set

$$\tilde{X}_n^{RE}(t) = X_n^{RE}(t_k) + a(\theta_k, X_n^{RE}(t_k)) \cdot (t - t_k) + b(U_{k,n}^{RE}) \cdot (W(t) - W(t_k)) + c(U_{k,n}^{RE}) \cdot (N(t) - N(t_k)). \quad (9.21)$$

The approximation $\tilde{X}_n^{RE}$ is not implementable, since it requires the complete knowledge of trajectories of N and W .

By induction, we have that

$$\tilde{X}_n^{RE}(t_k) = X_n^{RE}(t_k), \quad k = 0, 1, \dots, n. \quad (9.22)$$

Since the trajectories of $[N, W]$ are right-continuous we get

$$\lim_{t \rightarrow t_k+} \tilde{X}_n^{RE}(t) = X_n^{RE}(t_k) = \tilde{X}_n^{RE}(t_k), \quad (9.23)$$

and

$$\begin{aligned} \lim_{t \rightarrow t_{k+1}-} \tilde{X}_n^{RE}(t) &= X_n^{RE}(t_k) + a(\theta_k, X_n^{RE}(t_k)) \cdot (t_{k+1} - t_k) + b(U_{k,n}^{RE}) \cdot (W(t_{k+1}) - W(t_k)) \\ &\quad + c(U_{k,n}^{RE}) \cdot (N(t_{k+1}-) - N(t_k)) \neq \tilde{X}_n^{RE}(t_{k+1}) (= X_n^{RE}(t_{k+1})) \end{aligned} \quad (9.24)$$

for $k = 0, 1, \dots, n-1$. Moreover, it is easy to see that $\tilde{X}_n^{RE} = (\tilde{X}_n^{RE}(t))_{t \in [0, T]}$ is adapted to $(\tilde{\Sigma}_t^n)_{t \in [0, +\infty)}$ and has càdlàg paths.

Lemma 9.0.2. *Let us assume that $a, b, c, \{\lambda_j\}_{j=1, \dots, m_N}$ satisfy (B), (C), (D), (E) and ξ satisfies (A0). Then there exists $C \in (0, +\infty)$ such that for all $n \in \mathbb{N}$ we have*

$$\sup_{t \in [0, T]} \mathbb{E} \|\tilde{X}_n^{RE}(t)\|^2 \leq C(1 + \mathbb{E} \|\xi\|^2). \quad (9.25)$$

Proof. Firstly we show that

$$\max_{0 \leq k \leq n} \mathbb{E} \|X_n^{RE}(t_k)\|^2 < +\infty. \quad (9.26)$$

We proceed by induction and let us assume that there exists $s \in \{0, 1, \dots, n-1\}$ with the property that

$$\max_{0 \leq k \leq s} \mathbb{E} \|X_n^{RE}(t_k)\|^2 < +\infty. \quad (9.27)$$

(This is of course satisfied for $s = 0$.) Since $\{\sigma(b(U_{s,n}^{RE})), \sigma(\Delta W_s)\}$ and $\{\sigma(c(U_{s,n}^{RE})), \sigma(\Delta N_s)\}$ are two families of independent σ -fields, we get by the Fact 8.0.2

$$\begin{aligned} \mathbb{E} \|X_n^{RE}(t_{s+1})\|^2 &\leq C_1 \left(\mathbb{E} \|X_n^{RE}(t_s)\|^2 + (\Delta t_s)^2 \cdot \mathbb{E} \|a(\theta_s, X_n^{RE}(t_s))\|^2 \right. \\ &\quad \left. + \mathbb{E} \|b(U_{s,n}^{RE})\|^2 \cdot \mathbb{E} \|\Delta W_s\|^2 + \mathbb{E} \|c(U_{s,n}^{RE})\|^2 \cdot \mathbb{E} \|\Delta N_s\|^2 \right). \end{aligned} \quad (9.28)$$

Since there exist $c_W, c_N \in (0, +\infty)$ such that for all $t \in [t_k, t_{k+1}]$, $k = 0, 1, \dots, n-1$ it holds

$$\begin{aligned} \mathbb{E} \|W(t) - W(t_k)\|^2 &\leq c_W(t - t_k), \\ \mathbb{E} \|N(t) - N(t_k)\|^2 &\leq c_N(t - t_k), \end{aligned} \quad (9.29)$$

we get by Fact 8.0.1 that

$$\begin{aligned} \mathbb{E} \|X_n^{RE}(t_{s+1})\|^2 &\leq C_2 \left(\mathbb{E} \|X_n^{RE}(t_s)\|^2 + (\Delta t_s)^2 \cdot (1 + \mathbb{E} \|X_n^{RE}(t_s)\|^2) \right. \\ &\quad \left. + (1 + \mathbb{E} \|X_n^{RE}(t_s)\|^2) \cdot \Delta t_s + (1 + \mathbb{E} \|X_n^{RE}(t_s)\|^2) \cdot \Delta t_s \right) < +\infty. \end{aligned} \quad (9.30)$$

Hence,

$$\max_{0 \leq k \leq s+1} \mathbb{E} \|X_n^{RE}(t_k)\|^2 < +\infty. \quad (9.31)$$

By the rules of induction, this completes the proof of (9.26).

From (9.21) it follows that for all $t \in [t_k, t_{k+1}]$, $k = 0, 1, \dots, n-1$

$$\begin{aligned} \mathbb{E}\|\tilde{X}_n^{RE}(t)\|^2 &\leq C_2 \left(\mathbb{E}\|X_n^{RE}(t_k)\|^2 + (t - t_k)^2 \cdot (1 + \mathbb{E}\|X_n^{RE}(t_k)\|^2) \right. \\ &\quad \left. + (1 + \mathbb{E}\|X_n^{RE}(t_k)\|^2) \cdot (t - t_k) + (1 + \mathbb{E}\|X_n^{RE}(t_k)\|^2) \cdot (t - t_k) \right) \\ &\leq C_3(1 + \mathbb{E}\|X_n^{RE}(t_k)\|^2), \end{aligned} \quad (9.32)$$

and by (9.26)

$$\sup_{0 \leq t \leq T} \mathbb{E}\|\tilde{X}_n^{RE}(t)\|^2 = \max_{0 \leq k \leq n-1} \sup_{t_k \leq t \leq t_{k+1}} \mathbb{E}\|\tilde{X}_n^{RE}(t)\|^2 \leq C_3(1 + \max_{0 \leq k \leq n-1} \mathbb{E}\|X_n^{RE}(t_k)\|^2) < +\infty. \quad (9.33)$$

Note that for now the bound in (9.33) depends on n . In the second part of the proof, we will show, with the help of Gronwall's lemma, that we can obtain the bound (9.25), with C that is independent of n .

We can write for all $t \in [0, T]$ that

$$\tilde{X}_n^{RE}(t) = \xi + \int_0^t \tilde{a}_n(s) ds + \int_0^t \tilde{b}_n(s) dW(s) + \int_0^t \tilde{c}_n(s) dN(s), \quad (9.34)$$

where

$$\tilde{a}_n(s) = \sum_{k=0}^{n-1} a(\theta_k, \tilde{X}_n^{RE}(t_k)) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \quad (9.35)$$

and for $f \in \{b, c\}$

$$\tilde{f}_n(s) = \sum_{k=0}^{n-1} f(U_{k,n}^{RE}) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s), \quad (9.36)$$

and recall that $U_{k,n}^{RE} = (t_k, X_n^{RE}(t_k)) = (t_k, \tilde{X}_n^{RE}(t_k))$. Note that $(\tilde{f}_n(t))_{t \in [0, T]} \in \mathbb{L}$ for $f \in \{a, b, c\}$. Furthermore, for $f \in \{b, c\}$ and all $t \in [0, T]$ it holds

$$\|\tilde{f}_n(t)\|^2 \leq \left(\sum_{k=0}^{n-1} \|f(U_{k,n}^{RE})\| \cdot \mathbf{1}_{(t_k, t_{k+1}]}(t) \right)^2 = \sum_{k=0}^{n-1} \|f(U_{k,n}^{RE})\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(t), \quad (9.37)$$

which, together with Fact 8.0.1, implies

$$\begin{aligned} \int_0^t \|\tilde{f}_n(s)\|^2 ds &\leq \int_0^t \sum_{k=0}^{n-1} \|f(U_{k,n}^{RE})\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &\leq K_1^2 \int_0^t \sum_{k=0}^{n-1} (1 + \|\tilde{X}_n^{RE}(t_k)\|^2) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &\leq 2K_1^2 \int_0^t \sum_{k=0}^{n-1} (1 + \|\tilde{X}_n^{RE}(t_k)\|^2) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &\leq 2K_1^2 \int_0^T \sum_{k=0}^{n-1} \mathbf{1}_{(t_k, t_{k+1}]}(s) ds + 2K_1^2 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &= 2K_1^2 T + 2K_1^2 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds. \end{aligned} \quad (9.38)$$

Analogously we get by (9.7)

$$\int_0^t \|\tilde{a}_n(s)\|^2 ds \leq 2\bar{D}^2 T + 2\bar{D}^2 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds. \quad (9.39)$$

By the fact that $N(t) = \tilde{N}(t) + m(t)$, we have

$$\begin{aligned} \mathbb{E} \|\tilde{X}_n^{RE}(t)\|^2 &\leq C_1 \left(\mathbb{E} \|\xi\|^2 + \mathbb{E} \left\| \int_0^t \tilde{a}_n(s) ds \right\|^2 + \mathbb{E} \left\| \int_0^t \tilde{b}_n(s) dW(s) \right\|^2 \right. \\ &\quad \left. + \mathbb{E} \left\| \int_0^t \tilde{c}_n(s) d\tilde{N}(s) \right\|^2 + \mathbb{E} \left\| \int_0^t \tilde{c}_n(s) dm(s) \right\|^2 \right). \end{aligned} \quad (9.40)$$

From the Hölder inequality, Itô isometry for stochastic integrals driven by martingales, and (9.39) we get for $t \in [0, T]$

$$\begin{aligned} \mathbb{E} \left\| \int_0^t \tilde{a}_n(s) ds \right\|^2 &\leq \mathbb{E} \left(\int_0^t \|\tilde{a}_n(s)\| ds \right)^2 \leq T \mathbb{E} \int_0^t \|\tilde{a}_n(s)\|^2 ds \\ &\leq K_2 + K_3 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds, \end{aligned} \quad (9.41)$$

$$\mathbb{E} \left\| \int_0^t \tilde{b}_n(s) dW(s) \right\|^2 = \mathbb{E} \int_0^t \|\tilde{b}_n(s)\|^2 ds \leq K_4 + K_5 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds, \quad (9.42)$$

$$\begin{aligned} \mathbb{E} \left\| \int_0^t \tilde{c}_n(s) d\tilde{N}(s) \right\|^2 &= \sum_{j=1}^{m_N} \mathbb{E} \int_0^t \|\tilde{c}_n^{(j)}(s)\|^2 \cdot \lambda_j(s) ds \leq \max_{1 \leq j \leq m_N} \|\lambda_j\|_\infty \cdot \mathbb{E} \int_0^t \|\tilde{c}_n(s)\|^2 ds \\ &\leq K_6 + K_7 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds, \end{aligned} \quad (9.43)$$

and

$$\begin{aligned} \mathbb{E} \left\| \int_0^t \tilde{c}_n(s) dm(s) \right\|^2 &= \sum_{i=1}^d \mathbb{E} \left| \sum_{j=1}^{m_N} \int_0^t \tilde{c}_{n,ij}(s) \cdot \lambda_j(s) ds \right|^2 \\ &\leq m_N \cdot \sum_{i=1}^d \sum_{j=1}^{m_N} \mathbb{E} \left| \int_0^t \tilde{c}_{n,ij}(s) \cdot \lambda_j(s) ds \right|^2 \\ &\leq m_N T \cdot \sum_{i=1}^d \sum_{j=1}^{m_N} \int_0^t \mathbb{E} |\tilde{c}_{n,ij}(s)|^2 \cdot \lambda_j^2(s) ds \\ &\leq m_N T \cdot \max_{1 \leq j \leq m_N} \|\lambda_j\|_\infty^2 \cdot \mathbb{E} \int_0^t \|\tilde{c}_n(s)\|^2 ds \\ &\leq K_8 + K_9 \int_0^t \sum_{k=0}^{n-1} \|\tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds. \end{aligned} \quad (9.44)$$

Hence

$$\begin{aligned}
\mathbb{E}\|\tilde{X}_n^{RE}(t)\|^2 &\leq K_{10}(1 + \mathbb{E}\|\xi\|^2) + K_{11} \int_0^t \sum_{k=0}^{n-1} \mathbb{E}\|\tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\
&\leq K_{10}(1 + \mathbb{E}\|\xi\|^2) + K_{11} \int_0^t \sup_{0 \leq u \leq z} \mathbb{E}\|\tilde{X}_n^{RE}(u)\|^2 dz
\end{aligned} \tag{9.45}$$

where K_{10}, K_{11} depend only on $K_1, T, m_N, \max_{1 \leq j \leq m_N} \|\lambda_j\|_\infty$. Therefore, for all $t \in [0, T]$

$$\sup_{0 \leq s \leq t} \mathbb{E}\|\tilde{X}_n^{RE}(s)\|^2 \leq K_{10}(1 + \mathbb{E}\|\xi\|^2) + K_{11} \int_0^t \sup_{0 \leq u \leq z} \mathbb{E}\|\tilde{X}_n^{RE}(u)\|^2 dz. \tag{9.46}$$

Note that the function

$$[0, T] \ni t \rightarrow \sup_{0 \leq s \leq t} \mathbb{E}\|\tilde{X}_n^{RE}(s)\|^2 \tag{9.47}$$

is Borel measurable (as a nondecreasing function) and bounded, which follows from (9.33). By using Lemma 13.0.11 we get the thesis. ■

Proof of Theorem 8.0.4. We have that for all $t \in [0, T]$

$$X(t) = \xi + \int_0^t \hat{a}(s) ds + \int_0^t \hat{b}(s) dW(s) + \int_0^t \hat{c}(s) dN(s), \tag{9.48}$$

where for $f \in \{a, b, c\}$

$$\hat{f}(s) = \sum_{k=0}^{n-1} f(s, X(s-)) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s). \tag{9.49}$$

(Note that for $f \in \{a, b, c\}$ the process $(\hat{f}(t))_{t \in [0, T]}$ is in $\mathbb{L}$ so all Lebesgue and stochastic integrals appearing in (9.48) are well-defined.) From the representations above we thus get

$$\mathbb{E}\|X(t) - \tilde{X}_n^{RE}(t)\|^2 \leq 3(A_{1,n}(t) + A_{2,n}(t) + A_{3,n}(t)) \tag{9.50}$$

where

$$\begin{aligned}
A_{1,n}(t) &= \mathbb{E}\left\|\int_0^t (\hat{a}(s) - \tilde{a}_n(s)) ds\right\|^2, \\
A_{2,n}(t) &= \mathbb{E}\left\|\int_0^t (\hat{b}(s) - \tilde{b}_n(s)) dW(s)\right\|^2, \\
A_{3,n}(t) &= \mathbb{E}\left\|\int_0^t (\hat{c}(s) - \tilde{c}_n(s)) dN(s)\right\|^2.
\end{aligned} \tag{9.51}$$

We have that

$$A_{1,n}(t) \leq 3(\mathbb{E}\|\tilde{A}_{1,n}^{RE}(t)\|^2 + \mathbb{E}\|\tilde{A}_{2,n}^{RE}(t)\|^2 + \mathbb{E}\|\tilde{A}_{3,n}^{RE}(t)\|^2), \tag{9.52}$$

where

$$\mathbb{E}\|\tilde{A}_{1,n}^{RE}(t)\|^2 = \mathbb{E}\left\|\int_0^t \sum_{k=0}^{n-1} \left(a(s, X(s-)) - a(s, X(t_k))\right) \mathbf{1}_{(t_k, t_{k+1}]}(s) ds\right\|^2, \quad (9.53)$$

$$\mathbb{E}\|\tilde{A}_{2,n}^{RE}(t)\|^2 = \mathbb{E}\left\|\int_0^t \sum_{k=0}^{n-1} \left(a(s, X(t_k)) - a(\theta_k, X(t_k))\right) \mathbf{1}_{(t_k, t_{k+1}]}(s) ds\right\|^2, \quad (9.54)$$

$$\mathbb{E}\|\tilde{A}_{3,n}^{RE}(t)\|^2 = \mathbb{E}\left\|\int_0^t \sum_{k=0}^{n-1} \left(a(\theta_k, X(t_k)) - a(\theta_k, \tilde{X}_n^{RE}(t_k))\right) \mathbf{1}_{(t_k, t_{k+1}]}(s) ds\right\|^2. \quad (9.55)$$

From Lemma 7.0.1 and the Hölder inequality we get for all $t \in [0, T]$ that

$$\begin{aligned} \mathbb{E}\|\tilde{A}_{1,n}^{RE}(t)\|^2 &\leq \mathbb{E}\left(\int_0^t \sum_{k=0}^{n-1} \|a(s, X(s-)) - a(s, X(t_k))\| \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds\right)^2 \\ &\leq tL^2 \mathbb{E} \int_0^t \left(\sum_{k=0}^{n-1} \|X(s-) - X(t_k)\| \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s)\right)^2 ds \\ &\leq TL^2 \mathbb{E} \int_0^T \sum_{k=0}^{n-1} \|X(s-) - X(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &= TL^2 \mathbb{E} \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} \|X(s-) - X(t_k)\|^2 ds \\ &= TL^2 \mathbb{E} \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} \|X(s) - X(t_k)\|^2 ds \\ &= TL^2 \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} \mathbb{E}\|X(s) - X(t_k)\|^2 ds \leq C_2(1 + \mathbb{E}\|\xi\|^2)n^{-1}. \end{aligned} \quad (9.56)$$

We now estimate $\sup_{t \in [0, T]} \mathbb{E}\|\tilde{A}_{2,n}^{RE}(t)\|^2$. To do that we use the following notation:

$$i(t) = \sup\{i = 0, 1, \dots, n \mid t_i \leq t\}, \quad (9.57)$$

$$\zeta(t) = t_{i(t)}, \quad (9.58)$$

for $t \in [0, T]$. Now we can write that

$$\mathbb{E}\|\tilde{A}_{2,n}^{RE}(t)\|^2 \leq 2\left(\mathbb{E}\|\tilde{A}_{21,n}^{RE}(t)\|^2 + \mathbb{E}\|\tilde{A}_{22,n}^{RE}(t)\|^2\right), \quad (9.59)$$

where

$$\mathbb{E}\|\tilde{A}_{21,n}^{RE}(t)\|^2 = \mathbb{E}\left\|\sum_{k=0}^{i(t)-1} \int_{t_k}^{t_{k+1}} \left(a(s, X(t_k)) - a(\theta_k, X(t_k))\right) ds\right\|^2, \quad (9.60)$$

$$\mathbb{E}\|\tilde{A}_{22,n}^{RE}(t)\|^2 = \mathbb{E}\left\|\int_{\zeta(t)}^t \left(a(s, X(\zeta(t))) - a(\theta_{i(t)}, X(\zeta(t)))\right) ds\right\|^2, \quad (9.61)$$

for all $t \in [0, T]$. Let for $k = 0, 1, \dots, n-1$

$$\tilde{Y}_k = \int_{t_k}^{t_{k+1}} a(s, X(t_k)) ds - a(\theta_k, X(t_k)) \cdot \Delta t_k = \int_{t_k}^{t_{k+1}} (a(s, X(t_k)) - a(\theta_k, X(t_k))) ds, \quad (9.62)$$

$$Z_j = \sum_{k=0}^j \tilde{Y}_k, \quad j = 0, 1, \dots, n-1, \quad (9.63)$$

and $Z_{-1} := 0$. Then we can write that for $t \in [0, T]$

$$\tilde{A}_{21,n}^{RE}(t) = Z_{i(t)-1} = \sum_{k=0}^{n-1} Z_{k-1} \cdot \mathbf{1}_{[t_k, t_{k+1})}(t) + Z_{n-1} \cdot \mathbf{1}_{\{T\}}(t). \quad (9.64)$$

Therefore, we have for all $t \in [0, T]$ that

$$\mathbb{E} \|\tilde{A}_{21,n}^{RE}(t)\|^2 \leq \mathbb{E} \left(\sup_{t \in [0, T]} \|\tilde{A}_{21,n}^{RE}(t)\|^2 \right) = \mathbb{E} \left(\max_{0 \leq j \leq n-1} \|Z_j\|^2 \right) = \mathbb{E} \left[\mathbb{E} \left(\max_{0 \leq j \leq n-1} \|Z_j\|^2 \mid \Sigma_\infty \right) \right]. \quad (9.65)$$

By Theorem 13.0.31 we get

$$\mathbb{E} \left(\max_{0 \leq j \leq n-1} \|Z_j\|^2 \mid \Sigma_\infty \right) = \mathbb{E} \left[\max_{0 \leq j \leq n-1} \left\| \sum_{k=0}^j \int_{t_k}^{t_{k+1}} (a(s, x_k) - a(\theta_k, x_k)) ds \right\|^2 \right]_{x_k = X(t_k), \quad k=0,1,\dots,n-1}. \quad (9.66)$$

Note that for all $x_0, x_1, \dots, x_{n-1} \in \mathbb{R}^d$ the process

$$\left(\sum_{k=0}^j \int_{t_k}^{t_{k+1}} (a(s, x_k) - a(\theta_k, x_k)) ds, \quad \sigma(\theta_0, \theta_1, \dots, \theta_j) \right)_{j=0,1,\dots,n-1} \quad (9.67)$$

is a square-integrable martingale. Hence, for all $x_0, x_1, \dots, x_{n-1} \in \mathbb{R}^d$ we have by Theorem 13.0.23

$$\begin{aligned} \mathbb{E} \left[\max_{0 \leq j \leq n-1} \left\| \sum_{k=0}^j \int_{t_k}^{t_{k+1}} (a(s, x_k) - a(\theta_k, x_k)) ds \right\|^2 \right] &\leq c_2^2 \sum_{k=0}^{n-1} \mathbb{E} \left\| \int_{t_k}^{t_{k+1}} (a(s, x_k) - a(\theta_k, x_k)) ds \right\|^2 \\ &\leq 2c_2^2 \cdot n \cdot \left(2\bar{D}(T/n)(1 + \max_{0 \leq k \leq n-1} \|x_k\|) \right)^2 = C_3 n^{-1} (1 + \max_{0 \leq k \leq n-1} \|x_k\|)^2, \end{aligned} \quad (9.68)$$

and therefore

$$\begin{aligned} \mathbb{E} \left[\max_{0 \leq j \leq n-1} \left\| \sum_{k=0}^j \int_{t_k}^{t_{k+1}} (a(s, x_k) - a(\theta_k, x_k)) ds \right\|^2 \right]_{x_k = X(t_k), \quad k=0,1,\dots,n-1} \\ \leq C_3 n^{-1} (1 + \max_{0 \leq k \leq n-1} \|X(t_k)\|)^2 \leq C_3 n^{-1} (1 + \sup_{0 \leq t \leq T} \|X(t)\|)^2. \end{aligned} \quad (9.69)$$

Combining the above estimates we obtain

$$\begin{aligned} \mathbb{E} \|\tilde{A}_{21,n}^{RE}(t)\|^2 &\leq \mathbb{E} \left[\mathbb{E} \left(\max_{0 \leq j \leq n-1} \|Z_j\|^2 \mid \Sigma_\infty \right) \right] \\ &\leq C_3 n^{-1} \cdot \mathbb{E} \left(1 + \sup_{t \in [0, T]} \|X(t)\| \right)^2 \leq C_4 (1 + \mathbb{E} \|\xi\|^2) n^{-1}. \end{aligned} \quad (9.70)$$

Using the Hölder inequality we get for all $t \in [0, T]$ that

$$\begin{aligned} \mathbb{E} \|\tilde{A}_{22,n}^{RE}(t)\|^2 &\leq (t - \zeta(t)) \cdot \int_{\zeta(t)}^t \mathbb{E} \|a(s, X(\zeta(t))) - a(\theta_{i(t)}, X(\zeta(t)))\|^2 ds \\ &\leq C_5(t - \zeta(t))^2 \cdot \left(1 + \sup_{t \in [0, T]} \mathbb{E} \|X(t)\|^2\right) \leq C_6(1 + \mathbb{E} \|\xi\|^2)n^{-2}. \end{aligned} \quad (9.71)$$

By (9.59), (9.70) and (9.71) we arrive at

$$\mathbb{E} \|\tilde{A}_{2,n}^{RE}(t)\|^2 \leq C_7(1 + \mathbb{E} \|\xi\|^2)n^{-1}, \quad (9.72)$$

for all $t \in [0, T]$. By the Hölder inequality we get for all $t \in [0, T]$ that

$$\begin{aligned} \mathbb{E} \|\tilde{A}_{3,n}^{RE}(t)\|^2 &\leq t \int_0^t \mathbb{E} \left(\sum_{i=0}^{n-1} \|a(\theta_i, X(t_i)) - a(\theta_i, \tilde{X}_n^{RE}(t_i))\| \cdot \mathbf{1}_{(t_i, t_{i+1}]}(s) \right)^2 ds \\ &\leq C_8 \int_0^t \sum_{i=0}^{n-1} \mathbb{E} \|X(t_i) - \tilde{X}_n^{RE}(t_i)\|^2 \cdot \mathbf{1}_{(t_i, t_{i+1}]}(s) ds. \end{aligned} \quad (9.73)$$

Combining (9.52) with (9.56), (9.72) and (9.73) we obtain for all $t \in [0, T]$ that

$$A_{1,n}(t) \leq C_8 \int_0^t \sum_{i=0}^{n-1} \mathbb{E} \|X(t_i) - \tilde{X}_n^{RE}(t_i)\|^2 \cdot \mathbf{1}_{(t_i, t_{i+1}]}(s) ds + C_9(1 + \mathbb{E} \|\xi\|^2)n^{-1}, \quad (9.74)$$

From the Itô isometry

$$A_{2,n}(t) = \mathbb{E} \int_0^t \|\hat{b}(s) - \tilde{b}_n(s)\|^2 ds. \quad (9.75)$$

Furthermore

$$A_{3,n}(t) \leq 2(B_{1,n}(t) + B_{2,n}(t)), \quad (9.76)$$

where

$$\begin{aligned} B_{1,n}(t) &= \mathbb{E} \left\| \int_0^t (\hat{c}(s) - \tilde{c}_n(s)) d\tilde{N}(s) \right\|^2, \\ B_{2,n}(t) &= \mathbb{E} \left\| \int_0^t (\hat{c}(s) - \tilde{c}_n(s)) dm(s) \right\|^2. \end{aligned} \quad (9.77)$$

From the isometry for martingale-driven stochastic integrals, we get

$$\begin{aligned} B_{1,n}(t) &= \sum_{j=1}^{m_N} \mathbb{E} \int_0^t \|\hat{c}^{(j)}(s) - \tilde{c}_n^{(j)}(s)\|^2 \cdot \lambda_j(s) ds \\ &\leq \max_{1 \leq j \leq m_N} \|\lambda_j\|_\infty \cdot \mathbb{E} \int_0^t \|\hat{c}(s) - \tilde{c}_n(s)\|^2 ds \end{aligned} \quad (9.78)$$

and by the Schwarz inequality

$$\begin{aligned}
B_{2,n}(t) &= \sum_{i=1}^d \mathbb{E} \left| \sum_{j=1}^{m_N} \int_0^t \left(\hat{c}_{ij}(s) - \tilde{c}_{n,ij}(s) \right) \cdot \lambda_j(s) ds \right|^2 \\
&\leq m_N \cdot \sum_{i=1}^d \sum_{j=1}^{m_N} \mathbb{E} \left| \int_0^t \left(\hat{c}_{ij}(s) - \tilde{c}_{n,ij}(s) \right) \cdot \lambda_j(s) ds \right|^2 \\
&\leq m_N T \cdot \sum_{i=1}^d \sum_{j=1}^{m_N} \int_0^t \mathbb{E} |\hat{c}_{ij}(s) - \tilde{c}_{n,ij}(s)|^2 \cdot \lambda_j^2(s) ds \\
&\leq m_N T \cdot \max_{1 \leq j \leq m_N} \|\lambda_j\|_\infty^2 \cdot \mathbb{E} \int_0^t \|\hat{c}(s) - \tilde{c}_n(s)\|^2 ds.
\end{aligned} \tag{9.79}$$

Hence, by (9.76), (9.78), and (9.79) we get

$$A_{3,n}(t) \leq C \cdot \mathbb{E} \int_0^t \|\hat{c}(s) - \tilde{c}_n(s)\|^2 ds. \tag{9.80}$$

Combining (9.50), (9.52), (9.75), and (9.80) we arrive at

$$\mathbb{E} \|X(t) - \tilde{X}_n^{RE}(t)\|^2 \leq C \left(A_{1,n}(t) + \sum_{f \in \{b,c\}} \mathbb{E} \int_0^t \|\hat{f}(s) - \tilde{f}_n(s)\|^2 ds \right). \tag{9.81}$$

Now, for $f \in \{b, c\}$ and $s \in [0, T]$ we get

$$\begin{aligned}
\|\hat{f}(s) - \tilde{f}_n(s)\| &= \left\| \sum_{k=0}^{n-1} \left(f(s, X(s-)) - f(U_{k,n}^E) \right) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \right\| \\
&\leq \sum_{k=0}^{n-1} \|f(s, X(s-)) - f(U_{k,n}^{RE})\| \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \\
&\leq K \sum_{k=0}^{n-1} \left((s - t_k) + \|X(s-) - \tilde{X}_n^{RE}(t_k)\| \right) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \\
&\leq K \max_{0 \leq k \leq n-1} \Delta t_k + K \sum_{k=0}^{n-1} \|X(s-) - X(t_k)\| \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \\
&\quad + K \sum_{k=0}^{n-1} \|X(t_k) - \tilde{X}_n^{RE}(t_k)\| \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s)
\end{aligned} \tag{9.82}$$

and hence

$$\begin{aligned}
\|\hat{f}(s) - \tilde{f}_n(s)\|^2 &\leq 3K^2 \max_{0 \leq k \leq n-1} (\Delta t_k)^2 + 3K^2 \sum_{k=0}^{n-1} \|X(s-) - X(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) \\
&\quad + 3K^2 \sum_{k=0}^{n-1} \|X(t_k) - \tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s).
\end{aligned} \tag{9.83}$$

Since any trajectory of X has only finite number of (jump) discontinuities in $[0, T]$, the functions

$$[0, T] \ni s \rightarrow \sum_{k=0}^{n-1} \|X(s-) - X(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s), \quad (9.84)$$

$$[0, T] \ni s \rightarrow \sum_{k=0}^{n-1} \|X(s) - X(t_k)\|^2 \cdot \mathbf{1}_{[t_k, t_{k+1})}(s) \quad (9.85)$$

differ only on a finite subset of $[0, T]$ (the first is a càglàd modification of the second one). This implies for $f \in \{a, b, c\}$ and $t \in [0, T]$ that

$$\begin{aligned} \int_0^t \|\hat{f}(s) - \tilde{f}_n(s)\|^2 ds &\leq 3K^2 T \max_{0 \leq k \leq n-1} (\Delta t_k)^2 + 3K^2 \int_0^t \sum_{k=0}^{n-1} \|X(s) - X(t_k)\|^2 \cdot \mathbf{1}_{[t_k, t_{k+1})}(s) ds \\ &\quad + 3K^2 \int_0^t \sum_{k=0}^{n-1} \|X(t_k) - \tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds. \end{aligned} \quad (9.86)$$

By Lemma 7.0.1 we have that

$$\begin{aligned} \int_0^t \sum_{k=0}^{n-1} \mathbb{E} \|X(s) - X(t_k)\|^2 \cdot \mathbf{1}_{[t_k, t_{k+1})}(s) ds &\leq C(1 + \mathbb{E}\|\xi\|^2) \int_0^T \sum_{k=0}^{n-1} (s - t_k) \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &\leq C_1(1 + \mathbb{E}\|\xi\|^2) \cdot \max_{0 \leq k \leq n-1} \Delta t_k. \end{aligned} \quad (9.87)$$

Since $\max_{0 \leq k \leq n-1} (\Delta t_k)^2 \leq T \max_{0 \leq k \leq n-1} \Delta t_k$, we get for all $t \in [0, T]$

$$\begin{aligned} \mathbb{E} \int_0^t \|\hat{f}(s) - \tilde{f}_n(s)\|^2 ds &\leq K_1 \cdot (1 + \mathbb{E}\|\xi\|^2) \cdot \max_{0 \leq k \leq n-1} \Delta t_k \\ &\quad + K_2 \int_0^t \sum_{k=0}^{n-1} \mathbb{E} \|X(t_k) - \tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds. \end{aligned} \quad (9.88)$$

and hence

$$\begin{aligned} \mathbb{E} \|X(t) - \tilde{X}_n^{RE}(t)\|^2 &\leq K_3 \cdot (1 + \mathbb{E}\|\xi\|^2) \cdot n^{-1} \\ &\quad + K_4 \int_0^t \sum_{k=0}^{n-1} \mathbb{E} \|X(t_k) - \tilde{X}_n^{RE}(t_k)\|^2 \cdot \mathbf{1}_{(t_k, t_{k+1}]}(s) ds \\ &\leq K_3 \cdot (1 + \mathbb{E}\|\xi\|^2) \cdot n^{-1} \\ &\quad + K_4 \int_0^t \sup_{0 \leq u \leq z} \mathbb{E} \|X(u) - \tilde{X}_n^{RE}(u)\|^2 dz. \end{aligned} \quad (9.89)$$

Therefore, for all $t \in [0, T]$

$$\begin{aligned} \sup_{0 \leq s \leq t} \mathbb{E} \|X(s) - \tilde{X}_n^{RE}(s)\|^2 &\leq K_3 \cdot (1 + \mathbb{E}\|\xi\|^2) \cdot n^{-1} \\ &\quad + K_4 \int_0^t \sup_{0 \leq u \leq z} \mathbb{E} \|X(u) - \tilde{X}_n^{RE}(u)\|^2 dz. \end{aligned} \quad (9.90)$$

Note that the function

$$[0, T] \ni t \rightarrow \sup_{0 \leq s \leq t} \mathbb{E} \|X(s) - \tilde{X}_n^{RE}(s)\|^2 \quad (9.91)$$

is bounded (due to Lemmas 7.0.1, 8.0.5) and Borel measurable (as a nondecreasing function). Hence, by Gronwall's lemma 13.0.11 we get

$$\sup_{0 \leq t \leq T} \mathbb{E} \|X(t) - \tilde{X}_n^{RE}(t)\|^2 \leq K_5 \cdot (1 + \mathbb{E} \|\xi\|^2) \cdot n^{-1}, \quad (9.92)$$

which completes the proof. ■

TODO: Dodac kod z implementacją oraz przykładowe rysunki...

9.0.1 Perturbed gradient descent method and Euler-Maruyama scheme

Let us consider again the minimalization problem (4.138). In order to handle plateaus we can add some random noise to the GD algorithm (9.93) and obtain

$$\begin{aligned} y_0 &= x_0, \\ y_{k+1} &= y_k - h \cdot \nabla f(y_k) + \sigma \Delta W_k + c \Delta N_k, \quad k = 0, 1, \dots \end{aligned} \quad (9.93)$$

where σ, c are additional hyperparameters, see [52] where the case of fractional Wiener noise case was considered. This is of course the Euler-Maruyama scheme when applied to the following SDE

$$dX(t) = -\nabla f(X(t))dt + \sigma dW(t) + c dN(t), t \geq 0, \quad (9.94)$$

and $X(0) = x_0$, which is the so-called stochastic Langevin dynamic connected with the minimization problem (4.138). For a more discussion on that topic, see [77].

Chapter 10

Introduction to optimal one-point approximation of solutions of SDEs

Let us consider the following SDE

$$\begin{cases} dX(t) = \kappa(\mu - X(t))dt + \sigma dW(t), & t \in [0, T], \\ X(0) = x_0, \end{cases} \quad (10.1)$$

with $T, \kappa, \sigma > 0$ and $\mu \in \mathbb{R}$. Its solution defines the well-known Ornstein-Uhlenbeck process which is given by

$$X(t) = e^{-\kappa t}x_0 + \mu(1 - e^{-\kappa t}) + \sigma e^{-\kappa t} \int_0^t e^{\kappa s} dW(s), \quad t \in [0, T]. \quad (10.2)$$

Using this process as a toy example, we will give an introduction to the broad theory of *optimal approximation* of solutions of SDEs. Namely, we assume that we have access only to the values of W at a finite number of time points, i.e., $W(t_1), W(t_2), \dots, W(t_n)$, and, based on this data, we are interested in optimal (in the $L^2(\Omega)$ sense) approximation of $X(T)$. Note that

$$\int_0^t e^{\kappa s} dW(s) = e^{\kappa t} W(t) - \kappa \int_0^t e^{\kappa u} W(u) du, \quad (10.3)$$

so

$$X(t) = e^{-\kappa t}x_0 + \mu(1 - e^{-\kappa t}) + \sigma W(t) - \kappa \sigma e^{-\kappa t} \int_0^t e^{\kappa u} W(u) du. \quad (10.4)$$

Hence, approximation of $X(T)$ is equivalent to the problem of approximating the weighted Lebesgue integral $\int_0^T e^{\kappa u} W(u) du$ having access only to discrete information about W . This is a more general case than (5.44), considered in Section 5.6. However, we can use a similar approach in order to define an optimal method that approximates $X(T)$.

Namely, an algorithm $\bar{X} = (\bar{X}_n)_{n \in \mathbb{N}}$ that approximates $X(T)$ is (a sequence) defined by a pair $(\bar{\Delta}, \bar{\varphi})$ of two sequences $\bar{\varphi} = (\varphi_n)_{n \in \mathbb{N}}$, $\bar{\Delta} = (\Delta_n)_{n \in \mathbb{N}}$, where each

$$\varphi_n : \mathbb{R}^n \rightarrow \mathbb{R} \quad (10.5)$$

is a Borel function and

$$\Delta_n = \{t_{0,n}, t_{1,n}, \dots, t_{n,n}\} \quad (10.6)$$

is a (possibly) non-uniform partition of $[0, T]$ such that

$$0 = t_{0,n} < t_{1,n} < \dots < t_{n,n} = T. \quad (10.7)$$

By

$$\bar{\mathcal{N}}(W) = (\mathcal{N}_n(W))_{n \in \mathbb{N}} \quad (10.8)$$

we mean a sequence of vectors $\mathcal{N}_n(W)$ which provides *standard information* with n evaluations about the Wiener process W at the discrete points from Δ_n , i.e.,

$$\mathcal{N}_n(W) = [W(t_{1,n}), W(t_{2,n}), \dots, W(t_{n,n})]. \quad (10.9)$$

(Recall that $W(0) = 0$ almost surely.) After computing the information $\mathcal{N}_n(W)$, we apply the mapping φ_n in order to obtain the n th approximation $\bar{X}_n$ in the following way

$$\bar{X}_n = \varphi_n(\mathcal{N}_n(W)). \quad (10.10)$$

We also assume that for all $n \in \mathbb{N}$

$$\mathbb{E}|\bar{X}_n|^2 < +\infty. \quad (10.11)$$

The informational cost of each $\bar{X}_n$ is equal to n evaluations of W . The set of all methods $\bar{X} = (\bar{X}_n)_{n \in \mathbb{N}}$, defined as above, is denoted by χ^{noneq} . We also consider the following subclass of χ^{noneq}

$$\chi^{\text{eq}} = \{\bar{X} \in \chi^{\text{noneq}} \mid \forall n \in \mathbb{N} \Delta_n = \Delta_n^{\text{eq}} := \{iT/n \mid i = 0, 1, \dots, n\}\}. \quad (10.12)$$

The n th error of a method $\bar{X}$ is defined as

$$e_n(\bar{X}) = \|X(T) - \bar{X}_n\|_{L^2(\Omega)} = \left(\mathbb{E}|X(T) - \bar{X}_n|^2 \right)^{1/2}. \quad (10.13)$$

Let us fix any method $\bar{X} \in \chi^{\text{noneq}}$ that is based on sequences of discretizations $(\Delta_n)_{n \in \mathbb{N}}$ and functions $(\varphi_n)_{n \in \mathbb{N}}$. By Theorem 13.0.29 we have for all $n \in \mathbb{N}$ that

$$(e_n(\bar{X}))^2 = \mathbb{E}|X(T) - \varphi_n(\mathcal{N}_n(W))|^2 \geq \mathbb{E}|X(T) - \mathbb{E}(X(T) \mid \mathcal{N}_n(W))|^2, \quad (10.14)$$

since $\mathbb{E}|X(T)|^2 < +\infty$ and $\sigma(\bar{X}_n) \subset \sigma(\mathcal{N}_n(W))$. Let us denote by

$$\bar{\mathcal{A}}_n(T) = \mathbb{E}(X(T) \mid \mathcal{N}_n(W)), \quad (10.15)$$

and, in particular, by

$$\bar{\mathcal{A}}_n^{\text{eq}}(T) = \mathbb{E}(X(T) \mid W(jT/n), j = 1, 2, \dots, n). \quad (10.16)$$

By (10.14) we see that the error of $\bar{X} \in \chi^{\text{noneq}}$, that is based on a sequence $(\Delta_n)_{n \in \mathbb{N}}$, cannot be smaller than the error of the algorithm $\bar{\mathcal{A}}(T) = (\bar{\mathcal{A}}_n(T))_{n \in \mathbb{N}}$ (which is also using $(\Delta_n)_{n \in \mathbb{N}}$). From the Doob-Dynkin representation theorem (see...) we get that for every $n \in \mathbb{N}$ there exists a Borel function $\bar{\varphi}_n : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$\bar{\mathcal{A}}_n(T) = \bar{\varphi}_n(\mathcal{N}_n(W)), \quad (10.17)$$

since $\sigma(\bar{\mathcal{A}}_n(T)) \subset \sigma(\mathcal{N}_n(W))$. Moreover, by the conditional Jensen inequality (see...), it holds that $\mathbb{E}|\bar{\mathcal{A}}_n(T)|^2 \leq \mathbb{E}|X(T)|^2 < +\infty$. Hence, $\bar{\mathcal{A}}(T) = (\bar{\mathcal{A}}_n(T))_{n \in \mathbb{N}} \in \chi^{\text{noneq}}$ for any sequence $\bar{\Delta}$ of the form (10.7) and $\bar{\mathcal{A}}^{\text{eq}}(T) \in \chi^{\text{eq}}$. In summary, for all $n \in \mathbb{N}$ we define

$$\begin{aligned} e^{\text{noneq}}(n) &:= \inf_{\bar{X} \in \chi^{\text{noneq}}} e_n(\bar{X}) = \inf_{\Delta_n} e_n(\bar{\mathcal{A}}(T)) \\ &= \inf_{0=t_{0,n} < t_{1,n} < \dots < t_{n,n}=T} \left(\mathbb{E}|X(T) - \mathbb{E}(X(T) \mid W(t_{1,n}), W(t_{2,n}), \dots, W(t_{n,n}))|^2 \right)^{1/2}, \end{aligned} \quad (10.18)$$

and

$$\begin{aligned} e^{\text{eq}}(n) &:= \inf_{\bar{X} \in \chi^{\text{eq}}} e_n(\bar{X}) = e_n(\bar{\mathcal{A}}^{\text{eq}}(T)) \\ &= \left(\mathbb{E} |X(T) - \mathbb{E}(X(T) \mid W(jT/n), j = 1, 2, \dots, n)|^2 \right)^{1/2}, \end{aligned} \quad (10.19)$$

where it holds

$$e^{\text{noneq}}(n) \leq e^{\text{eq}}(n), \quad (10.20)$$

since $\chi^{\text{eq}} \subset \chi^{\text{noneq}}$. The equations (10.18), (10.19) define the so-called *nth minimal errors* in the classes $\chi^{\text{noneq}}, \chi^{\text{eq}}$, respectively.

As we will see, in the case of the process X given by (10.4) we are able to compute the conditional expectation (10.15) (and therefore also the functions $\bar{\varphi}_n$) explicitly, see however, Remark 10.0.2. We also show the optimality of the algorithm $\bar{\mathcal{A}}(T)$.

From Theorem 13.0.30 we get

$$\mathbb{E}(X(T) \mid \mathcal{N}_n(W)) = e^{-\kappa T} x_0 + \mu(1 - e^{-\kappa T}) + \sigma W(T) - \kappa \sigma e^{-\kappa T} \int_0^T e^{\kappa u} \mathbb{E}(W(u) \mid \mathcal{N}_n(W)) du, \quad (10.21)$$

and from Lemma 5.3.1 we get the following explicit (and implementable) expression

$$\begin{aligned} \mathbb{E}(X(T) \mid \mathcal{N}_n(W)) &= e^{-\kappa T} x_0 + \mu(1 - e^{-\kappa T}) + \sigma W(T) - \kappa \sigma e^{-\kappa T} \sum_{j=0}^{n-1} \int_{t_j}^{t_{j+1}} e^{\kappa u} \mathbb{E}(W(u) \mid \mathcal{N}_n(W)) du \\ &= e^{-\kappa T} x_0 + \mu(1 - e^{-\kappa T}) + \sigma W(T) - \kappa \sigma e^{-\kappa T} \sum_{j=0}^{n-1} \frac{W(t_{j+1}) \cdot I_{1,j} + W(t_j) \cdot I_{2,j}}{\Delta t_j}, \end{aligned} \quad (10.22)$$

where

$$I_{1,j} = \int_{t_j}^{t_{j+1}} e^{\kappa u} \cdot (u - t_j) du = \frac{e^{\kappa t_j} + e^{\kappa t_{j+1}} (\kappa \Delta t_j - 1)}{\kappa^2}, \quad (10.23)$$

$$I_{2,j} = \int_{t_j}^{t_{j+1}} e^{\kappa u} \cdot (t_{j+1} - u) du = \frac{e^{\kappa t_{j+1}} - e^{\kappa t_j} (\kappa \Delta t_j + 1)}{\kappa^2}. \quad (10.24)$$

Hence, we have explicitly computed the conditional expectation $\mathbb{E}(X(T) \mid \mathcal{N}_n(W))$ and we see that it is of the form (10.17).

By (10.21) we have

$$X(T) - \mathbb{E}(X(T) \mid \mathcal{N}_n(W)) = -\kappa \sigma e^{-\kappa T} \int_0^T e^{\kappa u} \hat{W}_n(u) du, \quad (10.25)$$

where $\hat{W}_n(\cdot) = W(\cdot) - \mathbb{E}(W(\cdot) \mid \mathcal{N}_n(W))$ consists of independent Brownian bridges, see (5.16). Let us denote

by $S_j = \int_{t_j}^{t_{j+1}} e^{\kappa u} \hat{W}_n(u) du$. Then we get

$$\mathbb{E} |X(T) - \mathbb{E}(X(T) \mid \mathcal{N}_n(W))|^2 = (\kappa \sigma)^2 e^{-2\kappa T} \mathbb{E} \left| \sum_{j=0}^{n-1} S_j \right|^2, \quad (10.26)$$

and

$$\mathbb{E} \left| \sum_{j=0}^{n-1} S_j \right|^2 = \sum_{j=0}^{n-1} \mathbb{E}(S_j^2) + \sum_{i \neq j} \mathbb{E}(S_i S_j), \quad (10.27)$$

where for $i \neq j$ we get from (5.58) that

$$\mathbb{E}(S_i S_j) = \int_{t_i}^{t_{i+1}} \int_{t_j}^{t_{j+1}} e^{\kappa(t+s)} \mathbb{E}(\hat{W}_n(s) \hat{W}_n(t)) ds dt = 0, \quad (10.28)$$

and

$$\mathbb{E}(S_j^2) = \int_{t_j}^{t_{j+1}} \int_{t_j}^{t_{j+1}} e^{\kappa(t+s)} \mathbb{E}(\hat{W}_n(s) \hat{W}_n(t)) ds dt, \quad (10.29)$$

where, due to (5.56),

$$\mathbb{E}(\hat{W}_n(s) \hat{W}_n(t)) = \text{Cov}(\hat{W}_n(s) \hat{W}_n(t)) \geq 0, \quad (10.30)$$

for all $s, t \in [t_j, t_{j+1}]$. Since $\kappa > 0$, $t, s \in [0, T]$, we get the (rough) estimate

$$1 \leq 1 + \kappa(t + s) \leq e^{\kappa(t+s)} \leq e^{2\kappa T}. \quad (10.31)$$

Hence, by (10.29), (10.30), (10.31), (5.57)

$$\frac{1}{12} \sum_{j=0}^{n-1} (\Delta t_j)^3 \leq \sum_{j=0}^{n-1} \mathbb{E}(S_j^2) \leq \frac{e^{2\kappa T}}{12} \sum_{j=0}^{n-1} (\Delta t_j)^3, \quad (10.32)$$

which implies

$$(\kappa\sigma)^2 \cdot \frac{e^{-2\kappa T}}{12} \cdot \sum_{j=0}^{n-1} (\Delta t_j)^3 \leq \mathbb{E}|X(T) - \mathbb{E}(X(T) \mid \mathcal{N}_n(W))|^2 \leq (\kappa\sigma)^2 \cdot \frac{1}{12} \cdot \sum_{j=0}^{n-1} (\Delta t_j)^3. \quad (10.33)$$

For the upper error bound, we therefore get

$$e_n(\bar{\mathcal{A}}(T)) \leq \kappa\sigma \cdot \frac{T^{1/2}}{\sqrt{12}} \cdot \max_{0 \leq j \leq n-1} \Delta t_j, \quad (10.34)$$

and, in particular, for $\bar{\mathcal{A}}_n^{\text{eq}}(T) = \mathbb{E}(X(T) \mid W(jT/n), j = 1, 2, \dots, n)$, based on the equidistant discretization, we have

$$e_n(\bar{\mathcal{A}}^{\text{eq}}(T)) \leq \kappa\sigma \cdot \frac{T^{3/2}}{\sqrt{12}} \cdot n^{-1} \quad (10.35)$$

for all $n \in \mathbb{N}$. From the Jensen inequality, we get for any discretization (10.7) that

$$\sum_{j=0}^{n-1} (\Delta t_j)^3 \geq T^3 \cdot n^{-2}. \quad (10.36)$$

Combining (10.14), (10.33), and (10.36) we obtain for any algorithm $\bar{X} \in \chi^{\text{noneq}}$ and for all $n \in \mathbb{N}$ that

$$e_n(\bar{X}) \geq e_n(\bar{\mathcal{A}}(T)) \geq \kappa\sigma \cdot T^{3/2} \cdot \frac{e^{-\kappa T}}{\sqrt{12}} \cdot n^{-1}. \quad (10.37)$$

Since $\chi^{\text{eq}} \subset \chi^{\text{noneq}}$, we get by (10.35) and (10.37) that

$$\kappa\sigma \cdot \frac{T^{3/2}}{\sqrt{12}} \cdot e^{-\kappa T} \cdot n^{-1} \leq \inf_{\bar{X} \in \chi^{\text{noneq}}} e_n(\bar{X}) \leq \inf_{\bar{X} \in \chi^{\text{eq}}} e_n(\bar{X}) \leq \kappa\sigma \cdot \frac{T^{3/2}}{\sqrt{12}} \cdot n^{-1}, \quad (10.38)$$

for all $n \in \mathbb{N}$. Therefore, the n th minimal error in both classes is $O(n^{-1})$ and the algorithm $\bar{\mathcal{A}}^{\text{eq}}(T)$ is the optimal one. The bounds in (10.38) are sharp up to the constant $e^{-\kappa T} \in (0, 1]$ that might be small for large $\kappa T > 0$. This suggests that by using algorithms from χ^{noneq} we can reduce constants in the error bound but not the rate of convergence.

We now investigate how far we can (asymptotically) reduce the value of constants in error expressions when we use algorithms based on non-equidistant discretizations.

Firstly, note that for any $0 = t_0 < t_1 < \dots < t_n = T$ and all $t, s \in [t_j, t_{j+1}]$, $j = 0, 1, \dots, n-1$, we have

$$e^{2\kappa t_j} \leq e^{\kappa(t+s)} \leq e^{2\kappa t_{j+1}}, \quad (10.39)$$

and from (10.29) we get

$$\frac{1}{12} \sum_{j=0}^{n-1} e^{2\kappa t_j} (\Delta t_j)^3 \leq \sum_{j=0}^{n-1} \mathbb{E}(S_j^2) \leq \frac{1}{12} \sum_{j=0}^{n-1} e^{2\kappa t_{j+1}} (\Delta t_j)^3. \quad (10.40)$$

For any $\eta_j \in [t_j, t_{j+1}]$, $j = 0, 1, \dots, n-1$ we have, in the case of equidistant discretization $\Delta_n = \{t_j = jT/n : j = 0, 1, \dots, n\}$, that

$$\lim_{n \rightarrow +\infty} n^2 \cdot \sum_{j=0}^{n-1} e^{2\kappa \eta_j} (\Delta t_j)^3 = T^2 \cdot \lim_{n \rightarrow +\infty} \sum_{j=0}^{n-1} e^{2\kappa \eta_j} \cdot \frac{T}{n} = T^2 \cdot \int_0^T e^{2\kappa t} dt = \frac{T^2}{2\kappa} (e^{2\kappa T} - 1). \quad (10.41)$$

This and (10.40) imply

$$\lim_{n \rightarrow +\infty} n^2 \sum_{j=0}^{n-1} \mathbb{E}(S_j^2) = \frac{T^2}{12} \cdot \int_0^T e^{2\kappa t} dt, \quad (10.42)$$

and hence by (10.26)

$$\lim_{n \rightarrow +\infty} n \cdot e_n(\bar{\mathcal{A}}^{\text{eq}}(T)) = C^{\text{eq}}(\kappa, \sigma, T) \quad (10.43)$$

with

$$C^{\text{eq}}(\kappa, \sigma, T) = \kappa \sigma \cdot e^{-\kappa T} \cdot \frac{T}{\sqrt{12}} \cdot \left(\int_0^T e^{2\kappa t} dt \right)^{1/2} = \kappa^{1/2} \sigma \cdot \frac{T}{2\sqrt{6}} (1 - e^{-2\kappa T})^{1/2} > 0. \quad (10.44)$$

Hence, we have from (10.19), (10.43) what follows

$$\lim_{n \rightarrow +\infty} n \cdot \inf_{\bar{X} \in \chi^{\text{eq}}} e_n(\bar{X}) = \lim_{n \rightarrow +\infty} n \cdot e_n(\bar{\mathcal{A}}^{\text{eq}}(T)) = C^{\text{eq}}(\kappa, \sigma, T) > 0 \quad (10.45)$$

and the algorithm $\bar{\mathcal{A}}^{\text{eq}}(T)$ is asymptotically optimal in the class χ^{eq} .

We now turn to algorithms that use possibly non-equidistant discretizations of $[0, T]$. Recall that from (10.38) we know that the rate $O(n^{-1})$ cannot be improved. We investigate, however, already mentioned the possibility of reducing the asymptotic constant C^{eq} .

From (10.14), (10.26), (10.40) for any algorithm $\bar{X} \in \chi^{\text{noneq}}$ it holds

$$n^2 \cdot (e_n(\bar{X}))^2 \geq n^2 \cdot (e_n(\bar{\mathcal{A}}(T)))^2 \geq \frac{(\kappa \sigma)^2}{12} e^{-2\kappa T} \cdot n^2 \cdot \sum_{j=0}^{n-1} e^{2\kappa t_j} (\Delta t_j)^3, \quad (10.46)$$

whereas, according to the Jensen inequality,

$$\sum_{j=0}^{n-1} e^{2\kappa t_j} (\Delta t_j)^3 = \sum_{j=0}^{n-1} \left(e^{2\kappa t_j/3} \Delta t_j \right)^3 \geq n \cdot \left(\frac{\sum_{j=0}^{n-1} e^{2\kappa t_j/3} \Delta t_j}{n} \right)^3 = n^{-2} \cdot \left(\sum_{j=0}^{n-1} e^{2\kappa t_j/3} \Delta t_j \right)^3, \quad (10.47)$$

we obtain

$$n^2 \cdot (e_n(\bar{X}))^2 \geq \frac{(\kappa\sigma)^2}{12} e^{-2\kappa T} \cdot \left(\sum_{j=0}^{n-1} e^{2\kappa t_j/3} \Delta t_j \right)^3 \quad (10.48)$$

for all $n \in \mathbb{N}$. Since the sequence $(\Delta_n)_{n \in \mathbb{N}}$ (used by $\bar{X}$) might be arbitrary, the partial sum $\sum_{j=0}^{n-1} e^{2\kappa t_j/3} \Delta t_j$ does not have to be a partial Riemann sum and we only know that

$$\sum_{j=0}^{n-1} e^{2\kappa t_j/3} \Delta t_j \geq T. \quad (10.49)$$

For a meanwhile, let us therefore consider the following subclass of χ^{noneq}

$$\chi^{\text{noneq-qu}} = \{\bar{X} \in \chi^{\text{noneq}} \mid \lim_{n \rightarrow +\infty} \max_{0 \leq j \leq n-1} \Delta t_j = 0\}, \quad (10.50)$$

which is the class of all algorithms from χ^{noneq} that are based on normal sequences of discretizations (likewise in Riemann integral). We call such a discretization a *quasi-uniform* one. In this case, by (10.48), we get for any $\bar{X} \in \chi^{\text{noneq-qu}}$

$$\liminf_{n \rightarrow +\infty} n \cdot e_n(\bar{X}) \geq C^{\text{noneq}}(\kappa, \sigma, T), \quad (10.51)$$

where

$$C^{\text{noneq}}(\kappa, \sigma, T) = \frac{\kappa\sigma}{\sqrt{12}} e^{-\kappa T} \cdot \left(\int_0^T e^{2\kappa t/3} dt \right)^{3/2} = \frac{\sigma}{\sqrt{12\kappa}} \cdot e^{-\kappa T} \cdot \left[\frac{3}{2} (e^{2\kappa T/3} - 1) \right]^{3/2}. \quad (10.52)$$

The above considerations suggest the following technique of obtaining a lower bound in χ^{noneq} . Let us take any sequence $(m_n)_{n \in \mathbb{N}}$ of positive integers such that

$$\lim_{n \rightarrow +\infty} m_n = +\infty, \quad \lim_{n \rightarrow +\infty} \frac{m_n}{n} = 0, \quad (10.53)$$

and again let us consider arbitrary method $\bar{X}$ from χ^{noneq} that is based on sequences $(\varphi_n)_{n \in \mathbb{N}}$ and $(\Delta_n)_{n \in \mathbb{N}}$. By $\hat{\Delta} = \{\hat{\Delta}_n\}_{n \geq 1}$ we denote the sequence of discretizations given by $\{\hat{\Delta}_n\}_{n \geq 1} = \{\Delta_n \cup \Delta_n^{\text{eq}}\}_{n \geq 1}$, where $\Delta_n^{\text{eq}} = \{jT/m_n \mid j = 0, 1, \dots, m_n\}$. Hence, for all $n \geq 1$,

$$\hat{\Delta}_n = \left\{ 0 = \hat{t}_{0,n} < \hat{t}_{1,n} < \dots < \hat{t}_{k_n,n} = T \mid \hat{t}_{j,n} \in \Delta_n \cup \Delta_n^{\text{eq}}, j = 0, 1, \dots, k_n \right\}, \quad (10.54)$$

and, since $\{0, T\} \subseteq \Delta_n \cap \Delta_n^{\text{eq}}$ for all $n \geq 1$,

$$n \leq k_n \leq n + m_n - 1. \quad (10.55)$$

From (10.53) and (10.55) we have that

$$\lim_{n \rightarrow +\infty} \frac{n}{k_n} = 1. \quad (10.56)$$

Furthermore, since $\Delta_n^{\text{eq}} \subset \hat{\Delta}_n$, we have that

$$0 \leq \max_{0 \leq i \leq k_n-1} (\hat{t}_{i+1,n} - \hat{t}_{i,n}) \leq \frac{T}{m_n} \quad (10.57)$$

and by (10.53)

$$\lim_{n \rightarrow +\infty} \max_{0 \leq i \leq k_n-1} (\hat{t}_{i+1,n} - \hat{t}_{i,n}) = 0. \quad (10.58)$$

Hence, $(\hat{\Delta}_n)_{n \in \mathbb{N}}$ is normal. Let us denote by

$$\hat{\mathcal{N}}_n(W) = [W(\hat{t}_{1,n}), W(\hat{t}_{2,n}), \dots, W(\hat{t}_{k_n,n})]. \quad (10.59)$$

Then $\sigma(\bar{X}_n) \subset \sigma(\mathcal{N}_n(W)) \subset \sigma(\hat{\mathcal{N}}_n(W))$ for all $n \in \mathbb{N}$. Therefore, for all $n \in \mathbb{N}$

$$e_n(\bar{X}) \geq e_n(\hat{\mathcal{A}}(T)), \quad (10.60)$$

where the algorithm $\hat{\mathcal{A}}(T) = (\hat{\mathcal{A}}_n(T))_{n \in \mathbb{N}}$ is defined by

$$\hat{\mathcal{A}}_n(T) = \mathbb{E}(X(T) \mid \hat{\mathcal{N}}_n(W)). \quad (10.61)$$

By performing analogous computations as for $\bar{\mathcal{A}}(T)$ we have that for all $n \in \mathbb{N}$

$$\begin{aligned} n^2 \cdot (e_n(\bar{X}))^2 &\geq n^2 \cdot (e_n(\hat{\mathcal{A}}(T)))^2 \geq \frac{(\kappa\sigma)^2}{12} e^{-2\kappa T} \cdot n^2 \cdot \sum_{j=0}^{k_n-1} e^{2\kappa \hat{t}_j} (\Delta \hat{t}_j)^3 \\ &\geq \frac{(\kappa\sigma)^2}{12} e^{-2\kappa T} \cdot \left(\frac{n}{k_n}\right)^2 \cdot \left(\sum_{j=0}^{k_n-1} e^{2\kappa \hat{t}_j/3} \Delta \hat{t}_j\right)^3. \end{aligned} \quad (10.62)$$

Hence, from (10.56) and (10.58) we arrive at

$$\liminf_{n \rightarrow +\infty} n \cdot e_n(\bar{X}) \geq C^{\text{noneq}}(\kappa, \sigma, T). \quad (10.63)$$

Hence, from (10.51), (10.63) we see that we have the same asymptotic lower bounds in both classes χ^{noneq} , $\chi^{\text{noneq-qu}}$. Moreover, note that

$$C^{\text{noneq}}(\kappa, \sigma, T) < C^{\text{eq}}(\kappa, \sigma, T). \quad (10.64)$$

We now show a construction of (asymptotically) optimal algorithm in the class χ^{noneq} , for which the error can be reduced by a factor $C^{\text{noneq}}/C^{\text{eq}}$ when compared to $\bar{\mathcal{A}}^{\text{eq}}(T)$. We consider so-called *regular sequences of discretizations* generated by a probability density function h , see [80]. We will use the notation $\bar{\Delta}_h = \{\Delta_{h,n}\}_{n \geq 1}$ for a sequence of discretizations generated by a density $h : [0, T] \rightarrow (0, +\infty)$ that is assumed to be continuous and strictly positive. The knots

$$\Delta_{h,n} = \{0 = t_{0,n} < t_{1,n} < \dots < t_{n,n} = T\},$$

of the n th discretization is given by

$$\int_0^{t_{i,n}} h(s) ds = \frac{iT}{n}, \quad i = 0, 1, \dots, n. \quad (10.65)$$

Hence, by choosing such a density h one gets a whole sequence of discretizations $\bar{\Delta}_h$. One can obtain the sequence of equidistant discretizations by taking $h \equiv 1$. Note that algorithms based on the discretizations generated by h belong to the class χ^{noneq} . We will use the notation $\bar{\mathcal{A}}_h(T) = \{\bar{\mathcal{A}}_{h,n}(T)\}_{n \geq 1}$ for the algorithm $\bar{\mathcal{A}}(T)$ algorithm based on the sequence of discretizations $\bar{\Delta}_h$.

Note that by (10.65) and the mean value theorem we have that for all $n \in \mathbb{N}$, $i = 0, 1, \dots, n-1$ there exists $\xi_{j,n} \in [t_{j,n}, t_{j+1,n}]$ such that

$$\frac{T}{n} = \int_{t_{j,n}}^{t_{j+1,n}} h(s) ds = h(\xi_{j,n}) \cdot \Delta t_{j,n}. \quad (10.66)$$

This, gives, in particular,

$$Tn^{-1}\|h\|_{\infty} \leq \max_{0 \leq j \leq n-1} (t_{j+1,n} - t_{j,n}) \leq Tn^{-1}\|1/h\|_{\infty}, \quad (10.67)$$

since $\max\{\|h\|_{\infty}, \|1/h\|_{\infty}\} < +\infty$. From (10.40) we have that

$$n^2 \cdot \sum_{j=0}^{n-1} \mathbb{E}(S_j^2) \leq \frac{1}{12} \sum_{j=0}^{n-1} e^{2\kappa t_{j+1}} \cdot (n \cdot \Delta t_j)^2 \cdot \Delta t_j = \frac{T^2}{12} \sum_{j=0}^{n-1} \frac{e^{2\kappa t_{j+1}}}{h^2(\xi_j)} \cdot \Delta t_j, \quad (10.68)$$

and analogously

$$n^2 \cdot \sum_{j=0}^{n-1} \mathbb{E}(S_j^2) \geq \frac{1}{12} \sum_{j=0}^{n-1} e^{2\kappa t_j} \cdot (n \cdot \Delta t_j)^2 \cdot \Delta t_j = \frac{T^2}{12} \sum_{j=0}^{n-1} \frac{e^{2\kappa t_j}}{h^2(\xi_j)} \cdot \Delta t_j. \quad (10.69)$$

For each $n \in \mathbb{N}$ let us take any sequence of intermediate points $\eta_{j,n} \in [t_{j,n}, t_{j+1,n}]$, $j = 0, 1, \dots, n-1$. Then

$$\sum_{j=0}^{n-1} \frac{e^{2\kappa \eta_j}}{h^2(\xi_j)} \cdot \Delta t_j = \sum_{j=0}^{n-1} \frac{e^{2\kappa \xi_j}}{h^2(\xi_j)} \cdot \Delta t_j + \left(\sum_{j=0}^{n-1} \frac{e^{2\kappa \eta_j}}{h^2(\xi_j)} \cdot \Delta t_j - \sum_{j=0}^{n-1} \frac{e^{2\kappa \xi_j}}{h^2(\xi_j)} \cdot \Delta t_j \right). \quad (10.70)$$

Since for all $t, s \in [0, T]$

$$|e^{2\kappa t} - e^{2\kappa s}| \leq 2\kappa e^{2\kappa T} |t - s|, \quad (10.71)$$

we have by (10.67)

$$\begin{aligned} \left| \sum_{j=0}^{n-1} \frac{e^{2\kappa \eta_j}}{h^2(\xi_j)} \cdot \Delta t_j - \sum_{j=0}^{n-1} \frac{e^{2\kappa \xi_j}}{h^2(\xi_j)} \cdot \Delta t_j \right| &\leq \sum_{j=0}^{n-1} \frac{|e^{2\kappa \eta_j} - e^{2\kappa \xi_j}|}{h^2(\xi_j)} \cdot \Delta t_j \\ &\leq 2\kappa e^{2\kappa T} \|1/h\|_{\infty}^2 \sum_{j=0}^{n-1} |\eta_j - \xi_j| \cdot \Delta t_j \leq 2\kappa e^{2\kappa T} \|1/h\|_{\infty}^2 T \cdot \max_{0 \leq j \leq n-1} \Delta t_j \leq 2\kappa e^{2\kappa T} \|1/h\|_{\infty}^3 T^2 n^{-1}. \end{aligned} \quad (10.72)$$

Hence,

$$\lim_{n \rightarrow +\infty} \sum_{j=0}^{n-1} \frac{e^{2\kappa \eta_j}}{h^2(\xi_j)} \cdot \Delta t_j = \lim_{n \rightarrow +\infty} \sum_{j=0}^{n-1} \frac{e^{2\kappa \xi_j}}{h^2(\xi_j)} \cdot \Delta t_j = \int_0^T \frac{e^{2\kappa t}}{h^2(t)} dt, \quad (10.73)$$

and

$$\lim_{n \rightarrow +\infty} n \cdot e_n(\bar{\mathcal{A}}_h(T)) = \kappa \sigma e^{-\kappa T} \cdot \frac{T}{\sqrt{12}} \left(\int_0^T \frac{e^{2\kappa t}}{h^2(t)} dt \right)^{1/2}. \quad (10.74)$$

In order to find the optimal density h_0 we show that for fixed $g \in C([0, T], (0 + \infty))$ the function

$$h_0(t) = T \cdot \frac{(g(t))^{1/3}}{\int_0^T (g(t))^{1/3} dt} \quad (10.75)$$

minimizes the functional

$$F(h) = \int_0^T \frac{g(t)}{h^2(t)} dt \quad (10.76)$$

among all sampling densities $h \in C([0, T], (0, +\infty))$ such that $\int_0^T h(t)dt = T$. By the Hölder inequality for any

$h \in C([0, T], (0, +\infty))$ with $\int_0^T h(t)dt = T$ we have that

$$\begin{aligned} \int_0^T (g(t))^{1/3} dt &= \int_0^T \frac{(g(t))^{1/3}}{h^{2/3}(t)} \cdot h^{2/3}(t) dt \\ &\leq \left(\int_0^T \left(\frac{(g(t))^{1/3}}{h^{2/3}(t)} \right)^3 dt \right)^{1/3} \cdot \left(\int_0^T (h^{2/3}(t))^{3/2} dt \right)^{2/3} = T^{2/3} \cdot (F(h))^{1/3}, \end{aligned} \quad (10.77)$$

and

$$F(h_0) = \frac{1}{T^2} \left(\int_0^T (g(t))^{1/3} dt \right)^3. \quad (10.78)$$

Therefore

$$\left(\int_0^T (g(t))^{1/3} dt \right)^3 \leq T^2 \cdot \inf_{h \in C([0, T]), h > 0, \int_0^T h(t)dt = T} F(h) \leq T^2 F(h_0) = \left(\int_0^T (g(t))^{1/3} dt \right)^3, \quad (10.79)$$

and

$$\inf_{h \in C([0, T]), h > 0, \int_0^T h(t)dt = T} F(h) = F(h_0) = \frac{1}{T^2} \left(\int_0^T (g(t))^{1/3} dt \right)^3. \quad (10.80)$$

Hence, the optimal sampling density, that minimized the asymptotic constant in (10.74), is of the form

$$h_0(t) = T \cdot \frac{e^{2\kappa t/3}}{\int_0^T e^{2\kappa t/3} dt} = \frac{2\kappa T e^{2\kappa t/3}}{3(e^{2\kappa T/3} - 1)}. \quad (10.81)$$

Moreover, from the fact that equality in the Hölder inequality (10.77) holds only in the case when $C_1 g/h^2$ and $C_2 h$ are equal almost everywhere for some $C_1, C_2 \geq 0$ we get the uniqueness of h_0 .

We thus get

$$\lim_{n \rightarrow +\infty} n \cdot e_n(\bar{\mathcal{A}}_{h_0}(T)) = C^{\text{noneq}}(\kappa, \sigma, T). \quad (10.82)$$

Combining (10.63), (10.82) we obtain

$$\lim_{n \rightarrow +\infty} n \cdot \inf_{\bar{X} \in \chi^{\text{noneq}}} e_n(\bar{X}) = \lim_{n \rightarrow +\infty} n \cdot e_n(\bar{\mathcal{A}}_{h_0}(T)) = C^{\text{noneq}}(\kappa, \sigma, T) > 0, \quad (10.83)$$

and the algorithm $\bar{\mathcal{A}}_{h_0}(T)$ is asymptotically optimal in χ^{noneq} (and, in fact, also in the smaller class $\chi^{\text{noneq-qu}}$). Moreover, from (10.65), (10.81) we see that the optimal sampling points are of the form

$$t_{j,n} = \frac{3}{2\kappa} \ln \left[\frac{j}{n} (e^{2\kappa T/3} - 1) + 1 \right], \quad j = 0, 1, \dots, n. \quad (10.84)$$

Remark 10.0.1. *There is a vast, and still growing, literature on strong optimal approximation of solutions of SDEs in various computational settings and under different regularity assumptions, see, for example, [63], [65],...*

10.0.1 Some remarks on optimal reconstruction of trajectories of solutions of SDEs

We considered optimal one-point approximation of... However, the natural question is how to optimally approximate X not only at one fixed point but at the whole interval $[0, T]$.

Remark 10.0.2. *In the general case the conditional expectation (10.15) is hard, or even impossible, to compute explicitly. Hence, in the case of nonlinear SDEs other methods of construction of asymptotically optimal algorithms are used, see [63] for the pure Wiener case and [43] where jump-diffusion SDEs are considered.*

10.1 Exercises

1. Show (10.2).
2. Give a proof of the inequality (10.64).
3. Justify the inequality (10.71).
4. (*Optimal one-point approximation of Ornstein-Uhlenbeck process with jumps*) Using the approach and proof technique described in this chapter provide an optimal rate of convergence in the classes χ^{eq} , χ^{noneq} for the problem of approximating $X(T)$ where X is the unique solution of

$$\begin{cases} dX(t) = \kappa(\mu - X(t))dt + \sigma dW(t) + \gamma dN(t), & t \in [0, T], \\ X(0) = x_0, \end{cases} \quad (10.85)$$

under the assumption that we have access to values of W, N at finite number of discretization points, $\kappa > 0$, $\mu \in \mathbb{R}$, $\sigma, \gamma \geq 0$, $\sigma^2 + \gamma^2 \geq 0$, and with the positive intensity λ of the homogeneous Poisson process N .

5. Compute for $t \in [0, T]$ the conditional expectation

$$\bar{X}_n(t) = \mathbb{E}(X(t) \mid W(t_1), W(t_2), \dots, W(t_n)), \quad (10.86)$$

where X is

- (a) the geometric Brownian motion (7.11) with $c = 0$,
- (b) the Ornstein-Uhlenbeck process (10.4),

and $0 = t_0 < t_1 < \dots < t_n = T$ is an arbitrary fixed discretization of $[0, T]$. Note that it can be shown that $\bar{X}_n$ provides an optimal method (in a suitable sense) of reconstruction of whole trajectories of X based on the discrete data about W .

Hint. Use Lemma 5.3.1.

Chapter 11

Weak approximation of solutions of SDEs with application to option pricing

For the functions

$$a : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad (11.1)$$

$$b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m_W}, \quad (11.2)$$

$$c : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m_N}, \quad (11.3)$$

we assume that

(B) there exists $K \in (0, +\infty)$ such that for $f \in \{a, b, c\}$ and for all $s, t \in [0, T]$, $x, y \in \mathbb{R}^d$

$$\|f(t, x) - f(s, y)\| \leq K(|t - s| + \|x - y\|), \quad (11.4)$$

(C) the intensity functions $\lambda_j : [0, T] \rightarrow (0, +\infty)$, $j = 1, 2, \dots, m_N$, are Borel measurable and bounded.

where for $f = a$ the norm $\|\cdot\|$ is the euclidean norm $\|\cdot\|_2$, while for $f \in \{b, c\}$ the norm $\|\cdot\|$ is the Frobenius norm $\|\cdot\|_F$. From now we will denote both norms by $\|\cdot\|$ and the notion will be clear from the context. We assume that X is the unique strong solution of the following SDE

$$\begin{cases} dX(t) = a(t, X(t))dt + b(t, X(t))dW(t) + c(t, X(t-))dN(t), & t \in [0, T], \\ X(0) = \xi. \end{cases} \quad (11.5)$$

In many applications (for example, in option pricing) we need to compute

$$\mathbb{E}(F(X)) \quad (11.6)$$

where $F : D([0, T]) \rightarrow \mathbb{R}$ is a given measurable functional. For example, in the case of the European put option

$$F(X) = h(X(T)) = (K_0 - X(T))^+, \quad (11.7)$$

where $h(x) = (K_0 - x)^+$ for $x \geq 0$ and $h(x) = K_0$ for $x < 0$, while for Asian-type option

$$F(X) = \phi\left(\frac{1}{T} \int_0^T \psi(X(t))dt\right), \quad (11.8)$$

where ϕ, ψ are given functions and, this time,

$$F(w) = \phi\left(\frac{1}{T} \int_0^T \psi(w(t))dt\right), \quad w \in D([0, T]). \quad (11.9)$$

In general, the value of (11.6) cannot be computed explicitly. Hence, to have a good approximation of (11.6) we can use Monte Carlo methods.

Firstly, we consider European option pricing, i.e., we consider an approximation of

$$\mathbb{E}(h(X(T))), \quad (11.10)$$

where for $h : \mathbb{R}^d \rightarrow \mathbb{R}$ we assume that:

(i) there exists $D > 0$ such that for all $x \in \mathbb{R}^d$ it holds

$$|h(x)| \leq D, \quad (11.11)$$

(ii) there exists $L > 0$ such that for all $x, y \in \mathbb{R}^d$ it holds

$$|h(x) - h(y)| \leq L\|x - y\|. \quad (11.12)$$

Under the above assumptions, the following integral is finite

$$\mathbb{E}(h(X(T))) = \int_{\mathbb{R}^d} h(x) d\mathbb{P}^{X(T)}(x), \quad (11.13)$$

where $\mathbb{P}^{X(T)}$ is the law of random vector $X(T)$. Suppose that we know the law of $X(T)$, then we can sample M independent random vectors $\bar{X}_1(T), \dots, \bar{X}_M(T)$ from $\mathbb{P}^{X(T)}$ and take

$$\frac{1}{M} \sum_{l=1}^M h(\bar{X}_l(T)) \quad (11.14)$$

in order to approximate $\mathbb{E}(h(X(T)))$. However, the distribution of $X(T)$ is known only in some particular cases (for example, in the simplest Black-Scholes model). So in the case of general equation (11.5) we have to proceed in different way. Namely, we use the Euler-Maruyama approximation of X .

For the convenience of the reader, we recall here the classical Euler-Maruyama scheme. Let $n \in \mathbb{N}$ and take an arbitrary discretization $0 = t_0 < t_1 < \dots < t_n = T$ of the interval $[0, T]$. Set

$$X_n^E(0) = \xi, \quad (11.15)$$

and for $k = 0, 1, \dots, n-1$

$$X_n^E(t_{k+1}) = X_n^E(t_k) + a(U_{k,n}^E) \cdot \Delta t_k + b(U_{k,n}^E) \cdot \Delta W_k + c(U_{k,n}^E) \cdot \Delta N_k, \quad (11.16)$$

where

$$U_{k,n}^E = (t_k, X_n^E(t_k)), \quad (11.17)$$

$$\begin{aligned} \Delta t_k &= t_{k+1} - t_k, \\ \Delta W_k &= [\Delta W_k^1, \dots, \Delta W_k^{m_W}]^T, \\ \Delta N_k &= [\Delta N_k^1, \dots, \Delta N_k^{m_N}]^T, \end{aligned} \quad (11.18)$$

and

$$\Delta Z_k^j = Z^j(t_{k+1}) - Z^j(t_k), \quad Z \in \{N, W\}, \quad (11.19)$$

$$b(U_{k,n}^E) \cdot \Delta W_k = \left(\sum_{j=1}^{m_W} b_{ij}(U_{k,n}^E) \cdot \Delta W_k^j \right)_{i=1,2,\dots,d}, \quad (11.20)$$

$$c(U_{k,n}^E) \cdot \Delta N_k = \left(\sum_{j=1}^{m_N} c_{ij}(U_{k,n}^E) \cdot \Delta N_k^j \right)_{i=1,2,\dots,d}. \quad (11.21)$$

In addition, let us fix $M \in \mathbb{N}$ - the number of simulated trajectories. Simulation of M trajectories via Euler scheme X_n^E goes as follows. Let $\xi_1, \dots, \xi_M$ be independent random vectors with the same law as ξ . For $k = 0, 1, \dots, n-1$ and $l = 1, 2, \dots, M$ we set

$$\bar{X}_{n,l}^E(0) = \xi_l, \quad (11.22)$$

and for $k = 0, 1, \dots, n-1$

$$\bar{X}_{n,l}^E(t_{k+1}) = \bar{X}_{n,l}^E(t_k) + a(\bar{U}_{k,n,l}^E) \cdot \Delta t_k + (\Delta t_k)^{1/2} \cdot b(\bar{U}_{k,n,l}^E) \cdot Z_{k,l} + c(\bar{U}_{k,n,l}^E) \cdot U_{k,l}, \quad (11.23)$$

with $\bar{U}_{k,n,l}^E = (t_k, \bar{X}_{n,l}^E(t_k))$ and

$$Z_{k,l} = [Z_{k,l}^1, \dots, Z_{k,l}^{m_W}]^T, \quad (11.24)$$

$$U_{k,l} = [U_{k,l}^1, \dots, U_{k,l}^{m_N}]^T, \quad (11.25)$$

where

- $(Z_{k,l}^j)_{k=0,\dots,n-1,l=1,\dots,M,j=1,\dots,m_W}$ is an i.i.d sequence of random variables with $Z_{1,1}^1 \sim N(0, 1)$,
- $(U_{k,l}^j)_{k=0,\dots,n-1,l=1,\dots,M,j=1,\dots,m_N}$ is a sequence of independent random variables, where $U_{k,l}^j \sim \text{Pois}\left(\int_{t_k}^{t_{k+1}} \lambda_j(s) ds\right)$,
- the collection $\{\xi_1, \dots, \xi_M, (Z_{k,l}^j)_{k=0,\dots,n-1,l=1,\dots,M,j=1,\dots,m_W}, (U_{k,l}^j)_{k=0,\dots,n-1,l=1,\dots,M,j=1,\dots,m_N}\}$ consists of independent random variables.

Then, componentwise,

$$\bar{X}_{n,i,l}^E(0) = \xi_{i,l}, \quad (11.26)$$

$$\begin{aligned} \bar{X}_{n,i,l}^E(t_{k+1}) = \bar{X}_{n,i,l}^E(t_k) &+ a_i(\bar{U}_{k,n,l}^E) \cdot \Delta t_k \\ &+ (\Delta t_k)^{1/2} \cdot \sum_{j=1}^{m_W} b_{ij}(\bar{U}_{k,n,l}^E) \cdot Z_{k,l}^j \\ &+ \sum_{j=1}^{m_N} c_{ij}(\bar{U}_{k,n,l}^E) \cdot U_{k,l}^j, \end{aligned} \quad (11.27)$$

for $k = 0, 1, \dots, n-1$, $l = 1, 2, \dots, M$, $i = 1, 2, \dots, d$.

Note that

$$\bar{X}_{n,1}^E(T), \bar{X}_{n,2}^E(T), \dots, \bar{X}_{n,M}^E(T) \quad (11.28)$$

are independent random vectors with the same law as $X_n^E(T)$. Moreover, we can simulate them. Therefore, as an approximation of $\mathbb{E}(h(X(T)))$ we take

$$\frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)). \quad (11.29)$$

We now estimate the error

$$\left(\mathbb{E} \left| \mathbb{E}(h(X(T))) - \frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) \right|^2 \right)^{1/2}. \quad (11.30)$$

By SLLN we get for all $n \in \mathbb{N}$

$$\frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) \rightarrow \mathbb{E}(h(X_n^E(T))) = \int_{\mathbb{R}^d} h(x) d\mathbb{P}^{X_n^E(T)}(x) \text{ when } M \rightarrow +\infty \quad (11.31)$$

almost surely. By (3.149) we get for any $n, K \in \mathbb{N}$ that

$$\left(\mathbb{E} \left| \mathbb{E}(h(X_n^E(T))) - \frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) \right|^2 \right)^{1/2} = \frac{C_n(h)}{\sqrt{M}}, \quad (11.32)$$

where

$$0 \leq C_n^2(h) = \int_{\mathbb{R}^d} |h(x)|^2 d\mathbb{P}^{X_n^E(T)}(x) - \left(\int_{\mathbb{R}^d} h(x) d\mathbb{P}^{X_n^E(T)}(x) \right)^2 \leq D^2. \quad (11.33)$$

In addition, by (3.152)

$$\begin{aligned} & \mathbb{E} \left| \mathbb{E}(h(X(T))) - \frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) \right|^2 \\ &= \left| \mathbb{E}(h(X(T))) - \mathbb{E}(h(X_n^E(T))) \right|^2 + \mathbb{E} \left| \mathbb{E}(h(X_n^E(T))) - \frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) \right|^2 \\ &\leq \mathbb{E} |h(X(T)) - h(X_n^E(T))|^2 + D^2/M \\ &\leq L^2 \mathbb{E} \|X(T) - X_n^E(T)\|^2 + D^2/M \\ &\leq C_1 \max_{0 \leq k \leq n-1} \Delta t_k + D^2/M. \end{aligned} \quad (11.34)$$

So on the uniform mesh, we get

$$\left(\mathbb{E} \left| \mathbb{E}(h(X(T))) - \frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) \right|^2 \right)^{1/2} = O(n^{-1/2} + M^{-1/2}). \quad (11.35)$$

We now turn to the Asian options of the form (11.9), where we assume that the functions $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $\phi : \mathbb{R} \rightarrow \mathbb{R}_+$ are globally Lipschitz and ϕ is bounded. Then we have that there exists $L_1 \in (0, +\infty)$ such that for all $w_1, w_2 \in D([0, T])$

$$|F(w_1) - F(w_2)| \leq L_1 \int_0^T \|w_1(t) - w_2(t)\| dt. \quad (11.36)$$

Since ϕ is bounded, the following integral is finite

$$\mathbb{E}(F(X)) = \int_{D([0, T])} F(w) d\mathbb{P}^X(w), \quad (11.37)$$

where $\mathbb{P}^X$ is the law of random element $X : \Omega \rightarrow D([0, T])$. If we could sample M independent random elements $\bar{X}_1, \dots, \bar{X}_M$ from $\mathbb{P}^X$ then we could take

$$\frac{1}{M} \sum_{l=1}^M F(\bar{X}_l) \quad (11.38)$$

in order to approximate $\mathbb{E}(F(X))$. However, this is not possible even in the case $X = W$. Hence, again, we have to use the Euler-Maruyama scheme for the global approximation of X . In this case, we take

$$\hat{X}_n^E(t) = \sum_{k=0}^{n-1} X_n^E(t_k) \cdot \mathbf{1}_{[t_k, t_{k+1})}(t), \quad t \in [0, T), \quad (11.39)$$

and $\hat{X}_n^E(T) = X_n^E(T)$, where $\hat{X}_n^E \approx X$. As an approximation of M trajectories of X we take

$$\hat{X}_{n,l}^E(t) = \sum_{k=0}^{n-1} \bar{X}_{n,l}^E(t_k) \cdot \mathbf{1}_{[t_k, t_{k+1})}(t), \quad t \in [0, T], \quad (11.40)$$

and $\hat{X}_{n,l}^E(T) = \bar{X}_{n,l}^E(T)$ for $l = 1, 2, \dots, M$. Note that

$$\hat{X}_{n,1}^E, \hat{X}_{n,2}^E, \dots, \hat{X}_{n,M}^E \quad (11.41)$$

are independent $D([0, T])$ -valued random elements with the same law as $\hat{X}_n^E$. As an approximation of $\mathbb{E}(F(X))$ we take

$$\frac{1}{M} \sum_{l=1}^M F(\hat{X}_{n,l}^E), \quad (11.42)$$

where

$$F(\hat{X}_{n,l}^E) = \phi\left(\frac{1}{T} \int_0^T \psi(\hat{X}_{n,l}^E(t)) dt\right) = \phi\left(\frac{1}{T} \sum_{k=0}^n \psi(\bar{X}_{n,l}^E(t_k)) \Delta t_k\right). \quad (11.43)$$

We now estimate the mean square error

$$\left(\mathbb{E}\left|\mathbb{E}(F(X)) - \frac{1}{M} \sum_{l=1}^M F(\hat{X}_{n,l}^E)\right|^2\right)^{1/2}. \quad (11.44)$$

Again, by SLLN we have for all $n \in \mathbb{N}$

$$\frac{1}{M} \sum_{l=1}^M F(\hat{X}_{n,l}^E) \rightarrow \mathbb{E}(F(\hat{X}_n^E)) = \int_{D([0,T])} F(w) d\mathbb{P}^{\hat{X}_n^E}(w) \text{ as } M \rightarrow +\infty \quad (11.45)$$

almost surely. Furthermore, by (3.149) we get for any $M \in \mathbb{N}$ that

$$\left(\mathbb{E}\left|\mathbb{E}(F(\hat{X}_n^E)) - \frac{1}{M} \sum_{l=1}^M F(\hat{X}_{n,l}^E)\right|^2\right)^{1/2} = \frac{C_n(h)}{\sqrt{M}}, \quad (11.46)$$

where

$$0 \leq C_n^2(h) = \int_{D([0,T])} |F(w)|^2 d\mathbb{P}^{\hat{X}_n^E}(w) - \left(\int_{D([0,T])} F(w) d\mathbb{P}^{\hat{X}_n^E}(w)\right)^2 \leq D^2. \quad (11.47)$$

Hence, by Theorem 8.0.6

$$\begin{aligned} & \mathbb{E}\left|\mathbb{E}(F(X)) - \frac{1}{M} \sum_{l=1}^M F(\hat{X}_{n,l}^E)\right|^2 \\ &= \left|\mathbb{E}(F(X)) - \mathbb{E}(F(\hat{X}_n^E))\right|^2 + \mathbb{E}\left|\mathbb{E}(F(\hat{X}_n^E)) - \frac{1}{M} \sum_{l=1}^M F(\hat{X}_{n,l}^E)\right|^2 \\ &\leq \mathbb{E}|F(X) - F(\hat{X}_n^E)|^2 + D^2 M^{-1} \leq L_1^2 \mathbb{E}\left(\int_0^T \|X(t) - \hat{X}_n^E(t)\| dt\right)^2 + D^2 M^{-1} \\ &\leq L_1^2 T \mathbb{E} \int_0^T \|X(t) - \hat{X}_n^E(t)\|^2 dt + D^2 M^{-1} \leq C_1 \max_{0 \leq k \leq n-1} (\Delta t_k) + D^2 M^{-1}, \end{aligned} \quad (11.48)$$

Again on the uniform mesh we obtain

$$\left(\mathbb{E} \left| \mathbb{E}(F(X)) - \frac{1}{M} \sum_{l=1}^M F(\hat{X}_{n,l}^E) \right|^2 \right)^{1/2} = O(n^{-1/2} + M^{-1/2}). \quad (11.49)$$

Therefore, to have a good balance between discretization and Monte Carlo errors, it is reasonable to take $M = O(n)$.

11.1 Non-asymptotic (?) confidence intervals

By the Hoeffding inequality, we get for the European option price $\mathbb{E}(h(X(T)))$, with h satisfying (11.11), that

$$\mathbb{P} \left(\left| \mathbb{E}(h(X_n^E(T))) - \frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{\varepsilon^2 M}{2 \|h\|_\infty^2} \right). \quad (11.50)$$

Using analogous argumentation as for (3.38) and (3.39) we obtain for all $\delta \in (0, 1)$, $n, M \in \mathbb{N}$ that

$$\mathbb{P} \left(\left| \mathbb{E}(h(X_n^E(T))) - \frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) \right| > C(\delta) \cdot M^{-1/2} \right) \leq \delta, \quad (11.51)$$

where

$$C(\delta) = \sqrt{2 \ln(2/\delta)} \cdot D. \quad (11.52)$$

(Note that in the case of European pay-off h the bound D might be known explicitly.) Hence, we get that for all $\delta \in (0, 1)$, $n, M \in \mathbb{N}$

$$\mathbb{E}(h(X_n^E(T))) \in \left[\frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) - \frac{C(\delta)}{\sqrt{M}}, \frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) + \frac{C(\delta)}{\sqrt{M}} \right] \quad (11.53)$$

with the probability at least $1 - \delta$.

Recall that for non-negative random variables X, Y we have for all $\varepsilon > 0$ that

$$\mathbb{P}(X + Y > \varepsilon) \leq \mathbb{P}(X > \varepsilon/2) + \mathbb{P}(Y > \varepsilon/2). \quad (11.54)$$

Therefore, by the Hoeffding and Markov inequalities

$$\begin{aligned} & \mathbb{P} \left(\left| \mathbb{E}(h(X(T))) - \frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) \right| > \varepsilon \right) \\ & \leq \mathbb{P} \left(\left| \mathbb{E}(h(X(T))) - \frac{1}{M} \sum_{l=1}^M h(\bar{X}_l(T)) \right| > \varepsilon/2 \right) \\ & \quad + \mathbb{P} \left(\frac{L}{M} \sum_{l=1}^M \|\bar{X}_l(T) - \bar{X}_{n,l}^E(T)\| > \varepsilon/2 \right) \\ & \leq 2 \exp \left(-\frac{(\varepsilon/2)^2 M}{2 \|h\|_\infty^2} \right) + \frac{2L}{M\varepsilon} \sum_{l=1}^M \mathbb{E} \|\bar{X}_l(T) - \bar{X}_{n,l}^E(T)\| \\ & \leq 2 \exp \left(-\frac{(\varepsilon/2)^2 M}{2 \|h\|_\infty^2} \right) + \frac{2L}{\varepsilon} \left(\mathbb{E} \|X(T) - X_n^E(T)\|^2 \right)^{1/2} \\ & \leq 2 \exp \left(-\frac{(\varepsilon/2)^2 M}{2 \|h\|_\infty^2} \right) + \frac{2LC_1}{\varepsilon} \max_{0 \leq k \leq n-1} (\Delta t_k)^{1/2}. \end{aligned} \quad (11.55)$$

On the uniform mesh, we obtain

$$\begin{aligned} & \mathbb{P} \left(\left| \mathbb{E}(h(X(T))) - \frac{1}{M} \sum_{l=1}^M h(\bar{X}_{n,l}^E(T)) \right| > 2C(\delta) \cdot M^{-1/2} \right) \\ & \leq \delta + \frac{LC_1}{\sqrt{2D}} \frac{\sqrt{M}}{\sqrt{n \ln(2/\delta)}}. \end{aligned} \quad (11.56)$$

11.2 Implementation issues

Below we present the implementation of the Euler scheme for the Merton model, together with European option pricing via Monte Carlo simulations performed on multiple threads.

```
import numpy as np
import time
from multiprocessing import Pool
from multiprocessing import cpu_count

K = 80
T = 1
S_0 = 100
r = 0.08
sigma = 0.4
lam = 5.1012
c_0 = -0.12

def a(t, x):
    return (r - c_0 * lam) * x

def b(t, x):
    return sigma * x

def c(t, x):
    return c_0 * x

"""
European call
"""

def H(x, K):
    tmp = x - K
    tmp = np.where(tmp < 0, 0.0, tmp)

    return tmp

"""
European put
"""

def P(x, K):
    tmp = K - x
    tmp = np.where(tmp < 0, 0.0, tmp)

    return tmp

"""
Euler method on uniform grid
"""
```

```

def Euler_path(params):
    np.random.seed(seed=None)
    eta = params["eta"]
    lam = params["lam"]
    T = params["T"]
    n = params["n"]

    h = float(T / n)
    sq_h = np.sqrt(h)
    E = float(eta)
    t = 0.0
    lh = lam * h

    for i in range(1, n+1):
        Normals = np.random.normal(0, 1)
        Poiss = np.random.poisson(lh)
        E += a( t, E ) * h + b( t, E ) * sq_h * Normals + c( t, E ) * Poiss
        t = i * h

    return E

#####
if __name__ == '__main__':
    # M - number of simulated trajectories
    # K - option strike

    #M = 50001
    # n = 10231

    M = 3901
    n = 3702

    val_E = []
    val_EV = []

    num_proc = cpu_count()
    print("CPUs' number: ", num_proc)

    params = []
    for i in range(M):
        params.append({"eta": S_0,
                       "lam": lam,
                       "T": T,
                       "n": n})

    threads = num_proc - 1

    p = Pool(threads)
    print("Number of CPUs' used:", threads)

    start = time.time()

    results = p.map_async(Euler_path, params)
    E_T = np.asarray(results.get())

    end = time.time()
    print("Computation time : ", end - start, " s")

    val_E = H( E_T, K )

    mean_E = np.mean(val_E)
    price_E = np.exp(-1.0 * r * T) * mean_E

```

```

val_EV = np.power(val_E - mean_E, 2)
EV = M * np.mean(val_EV) / (M-1)
tmp_I = 2.0 * np.exp(-1.0 * r * T) * np.sqrt(EV / M)
I1 = price_E - tmp_I
I2 = price_E + tmp_I

print("European call option price = ", price_E)
print("Empirical variance:", EV)
print("[", I1, I2, "]")

```

Chapter 12

Parameter estimation and forecasting for SDEs-based models

Estimation of parameters for SDE-based models is an active area of research, driven by numerous applications in mathematical finance, biology, and engineering; see, for example, [32], [33], [81]. In this chapter, we begin with a brief introduction and a toy example, presenting the main ideas of parameter estimation in SDE-based models. We also discuss Monte Carlo forecasting and the construction of prediction intervals.

12.1 Introductory example - parameter estimation in the geometric Brownian motion model

In this section we assume that the price $(S(t))_{t \in [0, T]}$ of the risky asset satisfies the Black-Scholes equation

$$dS(t) = \mu S(t)dt + \sigma S(t)dW(t), \quad t \in [0, T], \quad S(0) \in (0, +\infty), \quad (12.1)$$

with $\mu \in \mathbb{R}$, $\sigma > 0$, $S(0) > 0$. Hence,

$$S(t) = S(0) \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W(t)\right). \quad (12.2)$$

12.1.1 Quadratic variation based estimation

Note that for the process $X(t) = \ln S(t)$ it holds for $t \in [0, T]$ that

$$d(X(t)) = \left(\mu - \frac{1}{2}\sigma^2\right)dt + \sigma dW(t), \quad (12.3)$$

hence

$$[X]_t = \sigma^2 t, \quad t \in [0, T], \quad (12.4)$$

and, by the definition of the quadratic variation, we obtain

$$\sigma^2 = \frac{1}{T}[X]_T \approx \frac{1}{T} \sum_{i=0}^{n-1} (X(t_{i+1}) - X(t_i))^2 = \frac{1}{T} \sum_{i=0}^{n-1} \left(\ln \frac{S(t_{i+1})}{S(t_i)}\right)^2. \quad (12.5)$$

Assuming knowledge of historical stock prices $(S(t_i))_{i=0,1,\dots,n}$ at the moments $0 = t_0 < t_1 < \dots < t_n = T$ we can take

$$\hat{\sigma}_n = \left(\frac{1}{T} \sum_{i=0}^{n-1} \left(\ln \frac{S(t_{i+1})}{S(t_i)}\right)^2\right)^{1/2}, \quad (12.6)$$

and

$$\hat{\mu}_n = \frac{1}{T} \sum_{i=0}^{n-1} \ln \frac{S(t_{i+1})}{S(t_i)} + \frac{1}{2}(\hat{\sigma}_n)^2 = \frac{1}{T} \ln \frac{S(T)}{S(0)} + \frac{1}{2}(\hat{\sigma}_n)^2 \quad (12.7)$$

as the simple estimators of μ and σ , respectively, under the assumed Black-Scholes model. We investigate the quality of these estimators.

We believe that the following results are known. However, we were unable to find suitable references in the literature. Hence, for the convenience of the reader, we present complete proofs.

The results below might be used in the context of high-frequency data, i.e. when $\lim_{n \rightarrow +\infty} \max_{0 \leq i \leq n-1} (t_{i+1} - t_i) = 0$. In this case by $\{t_0, t_1, \dots, t_n\}$ we mean $\{t_0^n, t_1^n, \dots, t_n^n\}$ and when considering the limit we use the sequence of discretizations $(\{t_0^n, t_1^n, \dots, t_n^n\})_{n \in \mathbb{N}}$. However, for the clarity of presentation, we omit the upper script.

Theorem 12.1.1. *There exists $C \in (0, +\infty)$ such that for all $n \in \mathbb{N}$ and any discretization $0 = t_0 < t_1 < \dots < t_n = T$ we have*

$$\left(\mathbb{E} |(\hat{\sigma}_n)^2 - \sigma^2|^2 \right)^{1/2} \leq C \max_{0 \leq i \leq n-1} (t_{i+1} - t_i)^{1/2}. \quad (12.8)$$

Moreover, $(\hat{\sigma}_n)^2$ is an asymptotically unbiased estimator of σ^2 , i.e.,

$$|\mathbb{E}(\hat{\sigma}_n)^2 - \sigma^2| = \frac{1}{T}(\mu - \frac{1}{2}\sigma^2)^2 \sum_{i=0}^{n-1} (t_{i+1} - t_i)^2 \rightarrow 0 \quad (12.9)$$

as $\lim_{n \rightarrow +\infty} \max_{0 \leq i \leq n-1} (t_{i+1} - t_i) = 0$. In addition, if

$$\sum_{n=1}^{+\infty} \max_{0 \leq i \leq n-1} (t_{i+1} - t_i) < +\infty, \quad (12.10)$$

then

$$\lim_{n \rightarrow +\infty} \hat{\sigma}_n = \sigma \quad (12.11)$$

almost surely.

Proof. Note that

$$\ln \frac{S(t_{i+1})}{S(t_i)} = \left(\mu - \frac{1}{2}\sigma^2 \right) \Delta t_i + \sigma \Delta W_i, \quad i = 0, 1, \dots, n-1, \quad (12.12)$$

where $\Delta t_i = t_{i+1} - t_i$ and $\Delta W_i = W(t_{i+1}) - W(t_i)$. Hence

$$(\hat{\sigma}_n)^2 = \frac{1}{T}(\mu - \frac{1}{2}\sigma^2)^2 \sum_{i=0}^{n-1} (\Delta t_i)^2 + \frac{2}{T}\sigma(\mu - \frac{1}{2}\sigma^2) \sum_{i=0}^{n-1} (\Delta t_i \cdot \Delta W_i) + \frac{\sigma^2}{T} \sum_{i=0}^{n-1} (\Delta W_i)^2, \quad (12.13)$$

and

$$\mathbb{E}|(\hat{\sigma}_n)^2 - \sigma^2|^2 = \frac{1}{T^2} \cdot \mathbb{E}|A_n + B_n + C_n|^2 = \frac{1}{T^2} \cdot \mathbb{E}(A_n^2 + B_n^2 + C_n^2 + 2A_n B_n + 2A_n C_n + 2B_n C_n) \quad (12.14)$$

where

$$\begin{aligned} A_n &= (\mu - \frac{1}{2}\sigma^2)^2 \cdot \sum_{i=0}^{n-1} (\Delta t_i)^2, \\ B_n &= 2\sigma(\mu - \frac{1}{2}\sigma^2) \cdot \sum_{i=0}^{n-1} (\Delta t_i \cdot \Delta W_i), \\ C_n &= \sigma^2 \cdot \left(\sum_{i=0}^{n-1} (\Delta W_i)^2 - T \right). \end{aligned} \quad (12.15)$$

Since $\mathbb{E}(\Delta W_i) = 0$ and $\mathbb{E}(\Delta W_i)^2 = \Delta t_i$, we have

$$\begin{aligned}\mathbb{E}(A_n B_n) &= A_n \cdot \mathbb{E}(B_n) = 0, \\ \mathbb{E}(A_n C_n) &= A_n \cdot \mathbb{E}(C_n) = 0.\end{aligned}\tag{12.16}$$

Moreover

$$\mathbb{E}(A_n)^2 = A_n^2 = \left(\mu - \frac{1}{2}\sigma^2\right)^4 \cdot \left(\sum_{i=0}^{n-1} (\Delta t_i)^2\right)^2,\tag{12.17}$$

and, since $\sigma(\Delta W_i)$ and $\sigma(\Delta W_j)$ are independent for $i \neq j$, we have

$$\begin{aligned}\mathbb{E}(B_n)^2 &= 4\sigma^2\left(\mu - \frac{1}{2}\sigma^2\right)^2 \cdot \left(\sum_{i=0}^{n-1} (\Delta t_i)^2 \cdot \mathbb{E}(\Delta W_i)^2 + \sum_{i \neq j} \Delta t_i \Delta t_j \mathbb{E}(\Delta W_i) \mathbb{E}(\Delta W_j)\right) \\ &= 4\sigma^2\left(\mu - \frac{1}{2}\sigma^2\right)^2 \cdot \sum_{i=0}^{n-1} (\Delta t_i)^3\end{aligned}\tag{12.18}$$

$$\begin{aligned}\mathbb{E}(C_n)^2 &= \sigma^4 \cdot \mathbb{E}\left(\sum_{i=0}^{n-1} \left((\Delta W_i)^2 - \Delta t_i\right)\right)^2 \\ &= \sigma^4 \cdot \left(\sum_{i=0}^{n-1} \mathbb{E}\left((\Delta W_i)^2 - \Delta t_i\right)^2 + \sum_{i \neq j} \mathbb{E}\left((\Delta W_i)^2 - \Delta t_i\right) \cdot \mathbb{E}\left((\Delta W_j)^2 - \Delta t_j\right)\right) \\ &= \sigma^4 \cdot \sum_{i=0}^{n-1} (\mathbb{E}(\Delta W_i)^4 - 2\Delta t_i \mathbb{E}(\Delta W_i)^2 + (\Delta t_i)^2) = 2\sigma^4 \cdot \sum_{i=0}^{n-1} (\Delta t_i)^2,\end{aligned}\tag{12.19}$$

and

$$\begin{aligned}\mathbb{E}(B_n C_n) &= 2\sigma^3\left(\mu - \frac{1}{2}\sigma^2\right) \cdot \left(\sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \Delta t_i \cdot \mathbb{E}(\Delta W_i \cdot (\Delta W_j)^2) - T \sum_{i=0}^{n-1} \Delta t_i \cdot \mathbb{E}(\Delta W_i)\right) \\ &= 2\sigma^3\left(\mu - \frac{1}{2}\sigma^2\right) \cdot \sum_{i=0}^{n-1} \Delta t_i \cdot \mathbb{E}(\Delta W_i)^3 = 0.\end{aligned}\tag{12.20}$$

Therefore

$$\begin{aligned}\mathbb{E}|\hat{\sigma}_n^2 - \sigma^2|^2 &= \frac{1}{T^2}\left(\mu - \frac{1}{2}\sigma^2\right)^4 \cdot \left(\sum_{i=0}^{n-1} (\Delta t_i)^2\right)^2 + \frac{4\sigma^2}{T^2}\left(\mu - \frac{1}{2}\sigma^2\right)^2 \cdot \sum_{i=0}^{n-1} (\Delta t_i)^3 \\ &\quad + \frac{2\sigma^4}{T^2} \cdot \sum_{i=0}^{n-1} (\Delta t_i)^2 \leq C(T, \mu, \sigma) \cdot \max_{0 \leq i \leq n-1} \Delta t_i,\end{aligned}\tag{12.21}$$

and this implies (12.8).

From (12.13) we get that

$$\mathbb{E}(\hat{\sigma}_n)^2 = \frac{1}{T}\left(\mu - \frac{1}{2}\sigma^2\right)^2 \cdot \sum_{i=0}^{n-1} (\Delta t_i)^2 + \sigma^2,\tag{12.22}$$

and the result (12.9) follows.

By (12.13) we have that

$$(\hat{\sigma}_n)^2 = \frac{1}{T}\left(\mu - \frac{1}{2}\sigma^2\right)^2 S_n^1 + \frac{2}{T}\sigma\left(\mu - \frac{1}{2}\sigma^2\right) S_n^2 + \frac{\sigma^2}{T} S_n^3\tag{12.23}$$

where

$$S_n^1 = \sum_{i=0}^{n-1} (\Delta t_i)^2, \quad (12.24)$$

$$S_n^2 = \sum_{i=0}^{n-1} (\Delta t_i \cdot \Delta W_i), \quad (12.25)$$

$$S_n^3 = \sum_{i=0}^{n-1} (\Delta W_i)^2. \quad (12.26)$$

Now, we have that

$$0 \leq S_n^1 \leq T \cdot \max_{0 \leq i \leq n-1} (t_{i+1} - t_i) \rightarrow 0, \quad (12.27)$$

as $n \rightarrow +\infty$. By (12.10) and Lemma 4.3, page 92 in [50] we get that

$$S_n^3 \rightarrow T \quad (12.28)$$

almost surely as $n \rightarrow +\infty$, and hence

$$|S_n^2| \leq \left(T \cdot \max_{0 \leq i \leq n-1} (t_{i+1} - t_i) \right)^{1/2} \cdot (S_n^3)^{1/2} \rightarrow 0 \quad (12.29)$$

almost surely as $n \rightarrow +\infty$. Combining (12.23), (12.27), (12.28), (12.29) we get (12.11). $\blacksquare$

Now we turn to the properties of $\hat{\mu}_n$. The first result is of positive nature.

Theorem 12.1.2. *$\hat{\mu}_n$ is an asymptotically unbiased estimator of μ , i.e.,*

$$|\mathbb{E}(\mu_n) - \mu| = \frac{1}{2T} (\mu - \frac{1}{2}\sigma^2)^2 \sum_{i=0}^{n-1} (t_{i+1} - t_i)^2 \rightarrow 0 \quad (12.30)$$

as $\lim_{n \rightarrow +\infty} \max_{0 \leq i \leq n-1} (t_{i+1} - t_i) = 0$.

Unfortunately, it turns out that $\hat{\mu}_n$ converge, in the mean-square sense, not to μ but to the random variable $\mu + \frac{\sigma}{T}W(T)$, which has the law $N(\mu, \sigma^2/T)$.

Theorem 12.1.3. *Let $\mu_T = \mu + \frac{\sigma}{T}W(T)$. We have that*

$$\left(\mathbb{E}|\hat{\mu}_n - \mu|^2 \right)^{1/2} \rightarrow \sigma/\sqrt{T}, \quad (12.31)$$

and

$$\mu_n \rightarrow \mu_T \text{ in } L^2(\Omega) \quad (12.32)$$

as $\lim_{n \rightarrow +\infty} \max_{0 \leq i \leq n-1} (t_{i+1} - t_i) = 0$, and

$$\mu_n \rightarrow \mu_T \text{ a.s.}, \quad (12.33)$$

if $\sum_{n=1}^{+\infty} \max_{0 \leq i \leq n-1} (t_{i+1} - t_i) < +\infty$. Moreover,

$$\lim_{T \rightarrow +\infty} \mu_T = \mu \text{ a.s.} \quad (12.34)$$

Proof. By (12.12) we obtain that

$$\hat{\mu}_n - \mu = \frac{1}{2} \left((\hat{\sigma}_n)^2 - \sigma^2 \right) + \frac{\sigma}{T} W(T), \quad (12.35)$$

and from the Minkowski inequality, and (12.8) we have

$$\frac{\sigma}{\sqrt{T}} - \frac{1}{2} C \max_{0 \leq i \leq n-1} (t_{i+1} - t_i)^{1/2} \leq \|\hat{\mu}_n - \mu\|_{L^2(\Omega)} \leq \frac{\sigma}{\sqrt{T}} + \frac{1}{2} C \max_{0 \leq i \leq n-1} (t_{i+1} - t_i)^{1/2}. \quad (12.36)$$

Taking both sides limits we get (12.31), (12.32), and (12.33). By the SLLN for the Wiener process, we get (12.34). ■

Other accurate estimators of μ were constructed in literature such as, for example, quasi-maximum likelihood estimators (QMLE), see [33]. However, in the case of geometric Brownian motion the simple estimators $\mu_n, (\sigma_n)^2$ might give very close results to that obtained from QML estimation, see page 93. in [33]. Moreover, μ_n should give quite good approximation to μ when T is large enough. Hence, estimation of μ is possible for long time horizons.

Remark 12.1.4. Note that we have

$$[S]_t = \sigma \int_0^t S^2(u) du, \quad t \in [0, T]. \quad (12.37)$$

Since

$$\frac{\sum_{i=0}^{n-1} (S(t_{i+1}) - S(t_i))^2}{\sum_{i=0}^{n-1} (S(t_i))^2 \Delta t_i} \rightarrow \frac{[S]_T}{\int_0^T (S(t))^2 dt} = \sigma^2 \quad (12.38)$$

in probability as $\lim_{n \rightarrow +\infty} \max_{0 \leq i \leq n-1} (t_{i+1} - t_i) = 0$, we obtained an alternative estimator of σ^2 .

12.1.2 Linear regression approach

It is an interesting fact that the problem of estimation of μ, σ in the Black-Scholes model (12.1) we can reduce to fitting parameters in a suitable linear regression model. Therefore, this can also be seen as an introductory example of the forthcoming maximum likelihood method for SDEs.

We assume that the real observations $(S(t_i))_{i=0,1,\dots,n}$ follow the Euler-Maruyama scheme for (12.1) (this assumption is justified for small stepsize h), i.e.,

$$S(t_{i+1}) = S(t_i) + \mu S(t_i) \Delta t_i + \sigma \sqrt{\Delta t_i} S(t_i) \xi_i, \quad (12.39)$$

for $i = 0, 1, \dots, n-1$, where $(\xi_i)_{i=0,1,\dots,n-1}$ is an iid sequence from $N(0, 1)$. Hence, we have each

$$Y_i := \frac{\Delta S_i}{S(t_i) \sqrt{\Delta t_i}} = \mu \sqrt{\Delta t_i} + \sigma \xi_i \quad (12.40)$$

has the law $N(\mu \sqrt{\Delta t_i}, \sigma^2)$ and all Y_i 's are independent. The likelihood function is the joint distribution of the random vector $(Y_0, \dots, Y_{n-1})$ and is equal to

$$f_{(Y_0, \dots, Y_{n-1})}(y_0, \dots, y_{n-1}) = \prod_{j=0}^{n-1} f_{Y_j}(y_j) = \left(\frac{1}{\sigma \sqrt{2\pi}} \right)^n \exp \left(-\frac{1}{2\sigma^2} \sum_{j=0}^{n-1} (y_j - \mu \sqrt{\Delta t_j})^2 \right). \quad (12.41)$$

Hence, the negative log-likelihood function is

$$\mathcal{L}(\mu, p) = -\ln f_{(Y_0, \dots, Y_{n-1})}(y_0, \dots, y_{n-1}) = \frac{n}{2} \ln(2\pi p) + \frac{1}{2p} \sum_{j=0}^{n-1} (y_j - \mu\sqrt{\Delta t_j})^2, \quad (12.42)$$

and we minimize $\mathcal{L}$ with respect to μ, p . We have that

$$\frac{\partial \mathcal{L}}{\partial \mu} = -\frac{1}{p} \sum_{j=0}^{n-1} (y_j - \mu\sqrt{\Delta t_j})\sqrt{\Delta t_j} = -\frac{1}{p} \left(\sum_{j=0}^{n-1} y_j \sqrt{\Delta t_j} - \mu T \right), \quad (12.43)$$

since $\sum_{j=0}^{n-1} \Delta t_j = T$, and

$$\frac{\partial \mathcal{L}}{\partial p} = \frac{n}{2p} - \frac{1}{2p^2} \sum_{j=0}^{n-1} (y_j - \mu\sqrt{\Delta t_j})^2. \quad (12.44)$$

By solving system of equations $\frac{\partial \mathcal{L}}{\partial \mu} = 0 = \frac{\partial \mathcal{L}}{\partial p}$ and using the fact that y_j 's are realizations of Y_j 's we get the following estimators

$$\hat{\mu}_n = \frac{1}{T} \sum_{j=0}^{n-1} \frac{\Delta S_j}{S(t_j)}, \quad (12.45)$$

and

$$\widehat{\sigma}_n^2 = \frac{1}{n} \sum_{j=0}^{n-1} \frac{1}{\Delta t_j} \left(\frac{\Delta S_j}{S(t_j)} - \hat{\mu}_n \cdot \Delta t_j \right)^2. \quad (12.46)$$

12.2 The maximum likelihood method (ML)

Given $x_0, x_1, \dots, x_n$, the realisations $X_\Theta(t_0), X_\Theta(t_1), \dots, X_\Theta(t_n)$ of a stochastic process X_Θ , a very popular used procedure to estimate unknown parameters $\Theta \in \mathbb{R}^s$ of the process is the maximum likelihood method (ML). In this method, the parameters are chosen to maximize the joint density $f_\Theta(x_0, x_1, \dots, x_n)$ of $(X_\Theta(t_0), X_\Theta(t_1), \dots, X_\Theta(t_n))$. For Markov processes (such as solutions of jump-diffusion SDEs), the joint density can be represented as a product of transitional densities as follows

$$f_\Theta(x_0, x_1, \dots, x_n) = f_{X(t_0)}(x_0) \cdot f_{X(t_1)|X(t_0)}(x_1|x_0) \cdot \dots \cdot f_{X(t_n)|X(t_{n-1})}(x_n|x_{n-1}), \quad (12.47)$$

where the transitional density $f_{X(t_{i+1})|X(t_i)}(\cdot|x_i)$ is a conditional density function of $X(t_{i+1})$ given $X(t_i) = x_i$. The aim is to minimize the negative log-likelihood defined as

$$\mathcal{L}(\Theta) = -\ln f_\Theta(x_0, x_1, \dots, x_n) = -\sum_{i=0}^{n-1} \ln f_{X(t_{i+1})|X(t_i)}(x_{i+1}|x_i), \quad (12.48)$$

see [81]. We now apply this idea to several SDEs for which the transition densities are known.

12.2.1 Geometric Brownian motion

First, we estimate the unknown parameters $\Theta = (\mu, \sigma)$ of the equation (12.3) or, equivalently, the equation (12.1). In what follows we assume that $t_i = ih$, $i = 0, 1, \dots, n$, $h = T/n$.

Let $s_0, s_1, \dots, s_n$ be discrete observations of S at $t_0, t_1, \dots, t_n$. We take $X(t) = \ln S(t)$ and let $x_k = \ln s_k$ be discrete realizations of X . By (12.3), for $s, t \in [0, T]$, $s < t$

$$X(t) = X(t) - X(s) + X(s) = \bar{\mu}(t-s) + X(s) + \sigma(W(t) - W(s)), \quad (12.49)$$

where $\bar{\mu} = \mu - \frac{1}{2}\sigma^2$. Since $X(s)$ and $W(t) - W(s)$ are independent, we obtain

$$f_{X(t_{k+1})|X(t_k)}(x|x_k) = \frac{1}{\sigma\sqrt{2\pi h}} \cdot \exp\left(-\frac{(x - x_k - \bar{\mu}h)^2}{2\sigma^2 h}\right). \quad (12.50)$$

By taking $q = \sigma^2$, the negative log-likelihood is as follows

$$\mathcal{L}(\mu, q) = \frac{1}{2}n \ln(2\pi qh) + \frac{1}{2h} \sum_{i=0}^{n-1} \left(\frac{\Delta x_k - (\mu - \frac{1}{2}q)h}{q^{1/2}} \right)^2, \quad (12.51)$$

with $\Delta x_k = x_{k+1} - x_k$. Note that

$$\sum_{k=0}^{n-1} \Delta x_k = \ln(s_n/s_0), \quad (12.52)$$

and

$$\frac{\partial \mathcal{L}}{\partial \mu} = -\frac{1}{q} \sum_{k=0}^{n-1} \left(\Delta x_k - (\mu - \frac{1}{2}q)h \right), \quad (12.53)$$

$$\frac{\partial \mathcal{L}}{\partial q} = \frac{N}{2q} + \frac{1}{hq^2} \sum_{k=0}^{n-1} \left(\Delta x_k - (\mu - \frac{1}{2}q)h \right) \left(\frac{qh}{2} - \frac{1}{2} \left(\Delta x_k - (\mu - \frac{1}{2}q)h \right) \right). \quad (12.54)$$

From $\frac{\partial \mathcal{L}}{\partial \mu} = 0$ we get

$$\mu - \frac{1}{2}q = \frac{1}{T} \sum_{k=0}^{n-1} \Delta x_k = \frac{1}{T} \ln(s_n/s_0), \quad (12.55)$$

while from $\frac{\partial \mathcal{L}}{\partial q} = 0$ we have

$$\begin{aligned} \frac{Tq}{2} + \frac{hq}{2}(x_n - x_0) - \frac{1}{2} \sum_{k=0}^{n-1} (\Delta x_k)^2 + \frac{1}{2}(\mu - \frac{1}{2}q)h(x_n - x_0) - (\mu - \frac{1}{2}q)\frac{Thq}{2} \\ + \frac{1}{2}(\mu - \frac{1}{2}q)h(x_n - x_0) - \frac{1}{2}(\mu - \frac{1}{2}q)^2 hT = 0. \end{aligned} \quad (12.56)$$

Inserting (12.55) to (12.56) we get

$$\frac{Tq}{2} + \frac{1}{2T} \left(\ln(s_n/s_0) \right)^2 h = \frac{1}{2} \sum_{k=0}^{n-1} \left(\ln(s_{k+1}/s_k) \right)^2, \quad (12.57)$$

and hence

$$\begin{aligned} q &= \frac{1}{T} \sum_{k=0}^{n-1} \left(\ln(s_{k+1}/s_k) \right)^2 - \frac{h}{T^2} \left(\ln(s_n/s_0) \right)^2 \\ &= \frac{1}{T} \sum_{k=0}^{n-1} \left(\ln(s_{k+1}/s_k) - \frac{h}{T} \ln(s_n/s_0) \right)^2. \end{aligned} \quad (12.58)$$

In summary we obtain the following ML-estimators of (μ, σ^2)

$$\widehat{\sigma}_n^2 = \frac{1}{T} \sum_{k=0}^{n-1} \left(\ln(s_{k+1}/s_k) - \frac{h}{T} \ln(s_n/s_0) \right)^2, \quad (12.59)$$

$$\widehat{\mu}_n = \frac{1}{T} \ln(s_n/s_0) + \frac{1}{2} \widehat{\sigma}_n^2. \quad (12.60)$$

12.2.2 Zero mean reverting OU-type process

We start with the following SDE

$$dX(t) = -aX(t)dt + \sigma dW(t), \quad t \in [0, T], \quad (12.61)$$

where $X(0) = x_0$, $a > 0$ is a constant mean reversion rate and $\sigma > 0$ is a deterministic volatility. Note that such a process is a base process for the Hull-White model and for the energy pricing models. Since for $t > s$

$$X(t) = e^{-a(t-s)}X(s) + e^{-at} \int_s^t e^{au} \sigma dW(u), \quad (12.62)$$

the random variables $X(s)$ and $\int_s^t e^{au} \sigma dW(u)$ are independent, and

$$\sigma^2 e^{-2at} \int_s^t e^{2au} du = \frac{\sigma^2}{2a} (1 - e^{-2a(t-s)}), \quad (12.63)$$

we get that

$$f_{X(t_{k+1})|X(t_k)}(x|x_k) = \frac{1}{\sqrt{2\pi} \left(\frac{\sigma^2}{2a} (1 - e^{-2ah}) \right)^{1/2}} \exp \left(-\frac{(x - e^{-ah}x_k)^2}{\frac{\sigma^2}{a} (1 - e^{-2ah})} \right). \quad (12.64)$$

The negative log-likelihood is as follows

$$\begin{aligned} \mathcal{L}(a, \sigma) &= \sum_{k=0}^{n-1} \left[\frac{1}{2} \ln \left(2\pi \cdot \frac{\sigma^2}{2a} (1 - e^{-2ah}) \right) \right. \\ &\quad \left. + \frac{a}{\sigma^2 (1 - e^{-2ah})} (x_{k+1} - e^{-ah}x_k)^2 \right]. \end{aligned} \quad (12.65)$$

We reparametrize it in terms of

$$A = e^{-ah}, \quad (12.66)$$

$$\Sigma = \frac{\sigma^2}{2a} (1 - e^{-2ah}), \quad (12.67)$$

and obtain

$$\mathcal{L}(A, \Sigma) = \frac{n}{2} \ln(2\pi\Sigma) + \frac{1}{2\Sigma} \sum_{k=0}^{n-1} (x_{k+1} - Ax_k)^2. \quad (12.68)$$

We have that

$$\frac{\partial \mathcal{L}}{\partial A} = \frac{1}{\Sigma} \sum_{k=0}^{n-1} (x_{k+1} - Ax_k) \cdot (-x_k), \quad (12.69)$$

and

$$\frac{\partial \mathcal{L}}{\partial \Sigma} = \frac{n}{2\Sigma} - \frac{1}{2\Sigma^2} \sum_{k=0}^{n-1} (x_{k+1} - Ax_k)^2. \quad (12.70)$$

From $\frac{\partial \mathcal{L}}{\partial A} = 0$ we have

$$A = \frac{\sum_{k=0}^{n-1} x_k x_{k+1}}{\sum_{k=0}^{n-1} x_k^2}, \quad (12.71)$$

while from $\frac{\partial \mathcal{L}}{\partial \Sigma} = 0$,

$$\Sigma = \frac{1}{n} \sum_{k=0}^{n-1} (x_{k+1} - Ax_k)^2. \quad (12.72)$$

Using (12.66) we arrive at the following MaxLike estimators of a and σ^2

$$\hat{a}_n = -\frac{1}{h} \ln \left(\frac{\sum_{k=0}^{n-1} x_k x_{k+1}}{\sum_{k=0}^{n-1} x_k^2} \right), \quad (12.73)$$

$$\hat{\sigma}_n^2 = \frac{1}{n} \cdot \left(\frac{2\hat{a}_n}{1 - e^{-2\hat{a}_n h}} \right) \cdot \sum_{k=0}^{n-1} \left(x_{k+1} - e^{-2\hat{a}_n h} x_k \right)^2. \quad (12.74)$$

Remark 12.2.1. *We have that*

$$\mathbb{E}(X(t)) = e^{-at} x_0 \rightarrow 0 \quad (12.75)$$

as $t \rightarrow +\infty$.

12.2.3 Mean reverting OU-type process

We now deal with the following version of the OU process

$$dX(t) = \kappa(\mu - X(t))dt + \sigma dW(t), \quad t \in [0, T], \quad (12.76)$$

which is also called the Vasicek process. Here $X(0) = x_0$ and $\kappa, \sigma > 0$. For $t > s$ we have

$$X(t) = e^{-\kappa(t-s)} X(s) + \mu(1 - e^{-\kappa(t-s)}) + \sigma e^{-\kappa t} \int_s^t e^{\kappa u} dW(u), \quad (12.77)$$

and hence

$$f_{X(t_{k+1})|X(t_k)}(x|x_k) = \frac{1}{\sqrt{2\pi} \left(\frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa h}) \right)^{1/2}} \exp \left(-\frac{(x - \mu(1 - e^{-\kappa h}) - e^{-\kappa h} x_k)^2}{\frac{\sigma^2}{\kappa} (1 - e^{-2\kappa h})} \right). \quad (12.78)$$

In this case, the negative log-likelihood is as follows

$$\begin{aligned} \mathcal{L}(\mu, \sigma, \kappa) &= \frac{n}{2} \ln \left(2\pi \cdot \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa h}) \right) \\ &\quad + \frac{\kappa}{\sigma^2 (1 - e^{-2\kappa h})} \sum_{k=0}^{n-1} (x_{k+1} - \mu - e^{-\kappa h} (x_k - \mu))^2, \end{aligned} \quad (12.79)$$

and we re-parametrize it in the following way

$$A = e^{-\kappa h}, \quad (12.80)$$

$$\Sigma = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa h}). \quad (12.81)$$

This gives

$$\mathcal{L}(\mu, \Sigma, A) = \frac{n}{2} \ln(2\pi\Sigma) + \frac{1}{2\Sigma} \sum_{k=0}^{n-1} (x_{k+1} - \mu - A(x_k - \mu))^2. \quad (12.82)$$

Minimalization of $\mathcal{L}$ gives the following ML estimators

$$\hat{\kappa}_n = -h^{-1} \ln(\hat{\beta}_{1,n}), \quad (12.83)$$

$$\hat{\mu}_n = \hat{\beta}_{2,n}, \quad (12.84)$$

$$\hat{\sigma}_n^2 = 2\hat{\kappa}_n \hat{\beta}_{3,n} (1 - \hat{\beta}_{n,1}^2)^{-1}, \quad (12.85)$$

where

$$\hat{\beta}_{1,n} = \frac{n^{-1} \sum_{k=0}^{n-1} x_{k+1} x_k - n^{-2} \sum_{k=0}^{n-1} x_{k+1} \sum_{k=0}^{n-1} x_k}{n^{-1} \sum_{k=0}^{n-1} x_k^2 - n^{-2} \left(\sum_{k=0}^{n-1} x_k \right)^2}, \quad (12.86)$$

$$\hat{\beta}_{2,n} = \frac{n^{-1} \sum_{k=0}^{n-1} (x_{k+1} - \hat{\beta}_{1,n} x_k)}{1 - \hat{\beta}_{1,n}}, \quad (12.87)$$

and

$$\hat{\beta}_{3,n} = n^{-1} \sum_{k=0}^{n-1} \left(x_{k+1} - \hat{\beta}_{1,n} x_k - \hat{\beta}_{2,n} (1 - \hat{\beta}_{1,n}) \right)^2, \quad (12.88)$$

see [11].

Remark 12.2.2. *We have that*

$$\mathbb{E}(X(t)) = e^{-\kappa t} x_0 + \mu(1 - e^{-\kappa t}) \rightarrow \mu \quad (12.89)$$

as $t \rightarrow +\infty$.

12.2.4 Geometric mean reverting OU-type process

Finally, we investigate the following OU-type process

$$dX(t) = \kappa X(t)(\mu - \ln(X(t)))dt + \sigma X(t)dW(t), \quad t \in [0, T], \quad (12.90)$$

with $X(0) = x_0 > 0$, which is called geometric Ornstein-Uhlenbeck process. It can be shown by the use of Itô formula that $0 < X(t) = \exp(U(t))$ where

$$dU(t) = \kappa(\bar{\mu} - U(t))dt + \sigma dW(t), \quad t \in [0, T], \quad (12.91)$$

and $\bar{\mu} = \mu - \frac{\sigma^2}{2\kappa}$. Having the discrete observations $x_0, \dots, x_n$ of X we take $u_k = \ln x_k$ as the observations of U . Now we can use (12.83)-(12.85) in order to obtain the estimators $\hat{\kappa}_n, \hat{\mu}_n, \hat{\sigma}_n^2$ that are based on the values $(u_k)_{k=0}^n$. The estimator of μ is defined by

$$\hat{\mu}_n = \hat{\mu}_n + \frac{\hat{\sigma}_n^2}{2\hat{\kappa}_n}. \quad (12.92)$$

12.3 CLS-conditional least square method

We now describe an alternative approach of parameter estimation, called CLS, which is especially useful in the case when the parameters of drift do not appear in diffusion term and vice versa - such models are considered in the literature, see, for example, Section 3.2.1 at page 122 in [32]. As we will see the CLS method allows us to get rid of the diffusion term and obtain efficient estimators for the drift parameters. However, the problem of estimation of parameters in the diffusion term has to be solved by means of other methods.

Basing on [47], we describe this procedure by constructing estimators for the parameters $(\alpha, \beta, \sigma) \in \mathbb{R} \times \mathbb{R} \times (0, +\infty)$ of the following SDE

$$\begin{cases} dX(t) = (\alpha + \beta f(X(t)))dt + \sigma dW(t), & t \in [0, T], \\ X(0) = x_0, \end{cases} \quad (12.93)$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is such that the SDE (12.93) has the unique strong solution. This is the case, for example, when f satisfies the global Lipschitz condition. On the other hand (12.93) has a unique strong solution also when f is only Borel and bounded or discontinuous and piecewise Lipschitz. Hence, (12.93) covers a wide range of interesting SDEs that appear in mathematical finance or in energy control/demand/storage modeling.

We have

$$\begin{aligned} \mathbb{E}(X(t_{k+1}) \mid \Sigma_{t_k}) &= X(t_k) + \int_{t_k}^{t_{k+1}} \mathbb{E}(\alpha + \beta f(X(s)) \mid \Sigma_{t_k}) ds \\ &\approx X(t_k) + (\alpha + \beta f(X(t_k)))h =: \hat{\mathbb{E}}(X(t_{k+1}) \mid \Sigma_{t_k}). \end{aligned} \quad (12.94)$$

We define

$$\varphi(\alpha, \beta) = \sum_{k=0}^{n-1} \left(X(t_{k+1}) - \hat{\mathbb{E}}(X(t_{k+1}) \mid \Sigma_{t_k}) \right)^2 = \sum_{k=0}^{n-1} \left(\Delta X_k - h\alpha - h\beta f(X(t_k)) \right)^2 \quad (12.95)$$

We have the following partial derivatives of φ

$$\frac{\partial \varphi}{\partial \alpha} = -2h \sum_{k=0}^{n-1} \left(\Delta X_k - h\alpha - h\beta f(X(t_k)) \right), \quad (12.96)$$

$$\frac{\partial \varphi}{\partial \beta} = -2h \sum_{k=0}^{n-1} \left(\Delta X_k - h\alpha - h\beta f(X(t_k)) \right) \cdot f(X(t_k)) \quad (12.97)$$

By solving the system of equations $\frac{\partial \varphi}{\partial \alpha} = 0 = \frac{\partial \varphi}{\partial \beta}$ we get

$$\hat{\alpha}_n = \frac{\sum_{k=0}^{n-1} \Delta x_k \sum_{k=0}^{n-1} (f(x_k))^2 - \sum_{k=0}^{n-1} f(x_k) \cdot \sum_{k=0}^{n-1} f(x_k) \Delta x_k}{T \cdot \sum_{k=0}^{n-1} (f(x_k))^2 - h \left(\sum_{k=0}^{n-1} f(x_k) \right)^2} \quad (12.98)$$

$$\hat{\beta}_n = \frac{\sum_{k=0}^{n-1} \Delta x_k - T \cdot \hat{\alpha}_n}{h \cdot \sum_{k=0}^{n-1} f(x_k)}. \quad (12.99)$$

We estimate σ by using an approximation of the quadratic variation, i.e.

$$\hat{\sigma}_n^2 = \frac{1}{T} \sum_{i=0}^{n-1} (X(t_{i+1}) - X(t_i))^2. \quad (12.100)$$

Remark 12.3.1. We apply the approach above to the so-called threshold diffusion

$$dX(t) = \left(\alpha + \beta \cdot \text{sgn}(X(t)) \right) dt + \sigma dW(t), \quad t \in [0, T], \quad (12.101)$$

with $X(0) = x_0$. We get

$$\hat{\alpha}_n = \frac{n(x_n - x_0) - \sum_{k=0}^{n-1} f(x_k) \cdot \sum_{k=0}^{n-1} f(x_k) \Delta x_k}{nT - h \left(\sum_{k=0}^{n-1} f(x_k) \right)^2} \quad (12.102)$$

since $(\text{sgn}(x))^2 = 1$.

Remark 12.3.2. In particular, for the OU process

$$\begin{cases} dX(t) = -aX(t)dt + \sigma dW(t), & t \in [0, T], \\ X(0) = x_0, \end{cases} \quad (12.103)$$

where $X(0) = x_0$, $a, \sigma > 0$, we obtain the following alternative estimator of a

$$\hat{a}_n = -\frac{\sum_{k=0}^{n-1} X(t_k) \Delta X_k}{h \sum_{k=0}^{n-1} X^2(t_k)}. \quad (12.104)$$

Remark 12.3.3. We can apply the estimators (12.98)-(12.99) to the problem (12.76), and we can define the following estimators for κ, μ

$$\hat{\mu}_n = -\frac{\hat{\alpha}_n}{\hat{\beta}_n}, \quad (12.105)$$

$$\hat{\kappa}_n = -\hat{\beta}_n. \quad (12.106)$$

12.4 Quasi maximum likelihood method (QML)

Let us consider the general problem of estimating unknown parameters of the following scalar SDE

$$dX(t) = a(t, X(t), \theta)dt + b(t, X(t), \theta)dW(t), \quad t \in [0, T] \quad (12.107)$$

where the solution X is sampled at the discrete points $t_i = iT/n$, $i = 0, 1, \dots, n$, and $x_0, x_1, \dots, x_n$ are the realisations of X at t_i 's. We also assume that $X(0) = x_0$ and $b(t, x, \theta) > 0$ for all $(t, x, \theta) \in [0, T] \times \mathbb{R} \times \mathbb{R}^s$. Moreover, $a, b : [0, T] \times \mathbb{R} \times \mathbb{R}^s \rightarrow \mathbb{R}$ are (at least) Borel measurable, $\theta \in \mathbb{R}^s$ and we assume that X is the unique strong solution of (12.107). For such a case, the transition density's explicit form is unknown. However, for the Euler scheme, defined as follows

$$X^E(t_0) = x_0, \quad (12.108)$$

$$X^E(t_{k+1}) = X^E(t_k) + a(t_k, X^E(t_k), \theta)h + b(t_k, X^E(t_k), \theta)\Delta W_k, \quad k = 0, 1, \dots, n-1, \quad (12.109)$$

we have the following conditional law

$$X^E(t_{k+1})|X^E(t_k) \sim N(X^E(t_k) + a(t_k, X^E(t_k), \theta)h, h|b(t_k, X^E(t_k), \theta)|^2). \quad (12.110)$$

Hence, we approximate the transition density with the following

$$f_{X^E(t_{k+1})|X^E(t_k)}(x|x_k) = \frac{1}{(2\pi b^2(t_k, x_k, \theta))^{1/2}} \exp\left(-\frac{(x - x_k - a(t_k, x_k, \theta)h)^2}{2hb^2(t_k, x_k, \theta)}\right). \quad (12.111)$$

Hence, the negative log-quasi-likelihood is defined by

$$\begin{aligned} \mathcal{L}(\theta) &= -\sum_{k=0}^{n-1} \ln f_{X^E(t_{k+1})|X^E(t_k)}(x_{k+1}|x_k) \\ &= \frac{1}{2} \sum_{k=0}^{n-1} \left[\ln(2\pi hb^2(t_k, x_k, \theta)) + \frac{(x_{k+1} - x_k - a(t_k, x_k, \theta)h)^2}{hb^2(t_k, x_k, \theta)} \right]. \end{aligned} \quad (12.112)$$

We obtain the quasi maximum likelihood estimator $\hat{\theta}_n$ of θ by minimizing $\mathcal{L}(\theta)$ with respect to $\theta \in \mathbb{R}^s$, i.e.

$$\hat{\theta}_n = \operatorname{argmin}_{\theta \in \mathbb{R}^s} \mathcal{L}(\theta). \quad (12.113)$$

Mostly, such minimalization has to be done numerically.

Remark 12.4.1. The approach described in this section can be extended to cover the multidimensional case.

Remark 12.4.2. We considered the case when the unknown estimated parameters θ do not depend on time. However, in more realistic models such parameters are often time-dependent. In such a case parameter estimation is much more involved and differs greatly from the time-independent case. See, for example, [45] where the application of the artificial neural network is discussed in the context of estimating time-dependent θ .

12.5 Monte Carlo forecasting in SDE-based models

Having calibrated the SDE-based model on $[0, T]$ to the real data $x_0, x_1, \dots, x_n$ we can make predictions for $t > T$. Namely, outside the interval $[0, T]$ we consider the following SDE

$$d\hat{X}(t) = a(t, \hat{X}(t), \hat{\theta}_n)dt + b(t, \hat{X}(t), \hat{\theta}_n)dW(t), \quad t \in [T, T + \Delta], \quad (12.114)$$

where $\hat{X}(T) = x_n$, $\Delta > 0$ is the length of the prediction horizon, $W = [W^1, \dots, W^m]^T$ is m -dimensional Wiener process, and $\hat{\theta}_n$ is the estimated value of the true parameter θ . We also assume that $a(t, x, \theta)$, $b(t, x, \theta)$ are defined for all $(t, x, \theta) \in [T, T + \Delta] \times \mathbb{R}^d \times \mathbb{R}^s$. We now describe (inductively) the forecasting procedure together with the construction of prediction intervals at confidence level $\alpha \in (0, 1)$ (which we take in practice close to 1). Since in general we do not know the analytical solution $\hat{X}$, the prediction procedure is based on the Euler-Maruyama scheme that approximates $\hat{X}$. Note that there are many trajectories that we can obtain as predictions of future behavior of $\hat{X}$. We can handle with this problem by computing a (suitable) single trajectory together with prediction intervals (at confidence level α) around it.

Let us take $h = \Delta/N$ for some $N \in \mathbb{N}$ and $t_k = T + kh$ for $k = 0, 1, \dots, N$.

12.5.1 Prediction intervals in the scalar case with $m = d = 1$

Approach 1: Assume that $\bar{x}_k$ is given prediction of the value of $\hat{X}(t_k)$ (for $k = 0$ we have $\bar{x}_0 := x_n$). The one step of Euler-Maruyama scheme from $\bar{x}_k$ at the time t_k to $t_{k+1} = t_k + h$ is defined as follows

$$\bar{X}_N^E(t_{k+1}) = \bar{x}_k + a(t_k, \bar{x}_k, \hat{\theta}_n) \cdot h + b(t_k, \bar{x}_k, \hat{\theta}_n) \cdot \Delta W_k, \quad (12.115)$$

where $\Delta W_k = W(t_{k+1}) - W(t_k)$. As in the Section 3.2.1, let q_α be the two-sided α -quantile of the normal distribution $N(0, 1)$. We define the prediction interval for (the unknown value) $\hat{X}(t_{k+1})$ as follows

$$\bar{\mathcal{P}}_{k+1}(\alpha) := \left[\bar{\mu}_{k+1} - q_\alpha \cdot \bar{\sigma}_{k+1}, \bar{\mu}_{k+1} + q_\alpha \cdot \bar{\sigma}_{k+1} \right], \quad (12.116)$$

with

$$\bar{\mu}_{k+1} := \bar{x}_k + a(t_k, \bar{x}_k, \hat{\theta}_n) \cdot h, \quad (12.117)$$

$$\bar{\sigma}_{k+1} := \sqrt{h} \cdot |b(t_k, \bar{x}_k, \hat{\theta}_n)|. \quad (12.118)$$

If $b(t_k, \bar{x}_k, \hat{\theta}_n) \neq 0$ then we have

$$\mathbb{P}(\bar{X}_N^E(t_{k+1}) \in \bar{\mathcal{P}}_{k+1}(\alpha)) = \mathbb{P}(|\bar{X}_N^E(t_{k+1}) - \bar{\mu}_{k+1}| \leq q_\alpha \cdot \bar{\sigma}_{k+1}) = \mathbb{P}\left(\frac{|\Delta W_k|}{\sqrt{h}} \leq q_\alpha\right) = \alpha, \quad (12.119)$$

while if $b(t_k, \bar{x}_k, \hat{\theta}_n) = 0$ then $\bar{X}_N^E(t_{k+1}) = \bar{\mu}_{k+1}$ with probability one. Since $\bar{X}_N^E(t_{k+1}) \approx \hat{X}(t_{k+1})$ and the law of $\bar{X}_N^E(t_{k+1})$ is $N(\bar{\mu}_{k+1}, \bar{\sigma}_{k+1}^2)$ (if $\bar{\sigma}_{k+1} > 0$), in the case when $\bar{\sigma}_{k+1} > 0$ the prediction $\bar{x}_{k+1}$ of the value $\hat{X}(t_{k+1})$ we take as a number drawn from the distribution $N(\bar{\mu}_{k+1}, \bar{\sigma}_{k+1}^2)$, while $\bar{x}_{k+1} := \bar{x}_k + a(t_k, \bar{x}_k, \hat{\theta}_n) \cdot h$ if $\bar{\sigma}_{k+1} = 0$.

The drawback of this approach is that with a small probability $1 - \alpha$ still, the drawn value $\bar{x}_{k+1}$ may lie outside the prediction interval $\bar{\mathcal{P}}_{k+1}(\alpha)$. We overcome this problem in the Approach 2.

Approach 2: Assume that we have given $\bar{x}_k$ and $\tilde{x}_k$, where $\bar{x}_k$ is the prediction of the value of $\hat{X}(t_k)$ and $\tilde{x}_k$ is equal to $\tilde{X}_N^E(t_k)$ - the simulated value of Euler-Maruyama scheme at t_k (for $k = 0$ we take $\tilde{x}_0 = \bar{x}_0 := x_n$). The one step of Euler-Maruyama scheme from t_k to $t_{k+1} = t_k + h$, starting from $\tilde{x}_k$ at t_k , is defined as follows

$$\tilde{X}_N^E(t_{k+1}) = \tilde{x}_k + a(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot h + b(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot \Delta W_k, \quad (12.120)$$

where $\Delta W_k = W(t_{k+1}) - W(t_k)$ is the increment of the Wiener process that is independent of Σ_{t_k} and, therefore, also of $\tilde{X}_N^E(t_k)$. Again, let q_α be the two-sided α -quantile of the normal distribution $N(0, 1)$. We define the prediction interval for (the unknown value) $\hat{X}(t_{k+1})$ as follows

$$\tilde{\mathcal{P}}_{k+1}(\alpha) := \left[\tilde{\mu}_{k+1} - q_\alpha \cdot \tilde{\sigma}_{k+1}, \tilde{\mu}_{k+1} + q_\alpha \cdot \tilde{\sigma}_{k+1} \right], \quad (12.121)$$

with

$$\tilde{\mu}_{k+1} := \tilde{x}_k + a(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot h, \quad (12.122)$$

$$\tilde{\sigma}_{k+1} := \sqrt{h} \cdot |b(t_k, \tilde{x}_k, \hat{\theta}_n)|. \quad (12.123)$$

Note that it holds $\tilde{\mu}_{k+1} = \mathbb{E}(\tilde{X}_N^E(t_{k+1}) \mid \Sigma_{t_k}) \approx \mathbb{E}(\hat{X}(t_{k+1}) \mid \Sigma_{t_k})$ and $\tilde{\sigma}_{k+1}^2 = \text{Var}(\tilde{X}_N^E(t_{k+1}) \mid \Sigma_{t_k}) \approx \text{Var}(\hat{X}(t_{k+1}) \mid \Sigma_{t_k})$. If $b(t_k, \tilde{x}_k, \hat{\theta}_n) \neq 0$, then we have

$$\begin{aligned} \mathbb{P}(\tilde{X}_N^E(t_{k+1}) \in \tilde{\mathcal{P}}_{k+1}(\alpha) \mid \Sigma_{t_k}) &= \mathbb{P}(|\tilde{X}_N^E(t_{k+1}) - \tilde{\mu}_{k+1}| \leq q_\alpha \cdot \tilde{\sigma}_{k+1} \mid \Sigma_{t_k}) \\ &= \mathbb{P}\left(\frac{|\Delta W_k|}{\sqrt{h}} \leq q_\alpha \mid \Sigma_{t_k}\right) = \mathbb{P}\left(\frac{|\Delta W_k|}{\sqrt{h}} \leq q_\alpha\right) = \alpha, \end{aligned} \quad (12.124)$$

and hence

$$\mathbb{P}(\tilde{X}_N^E(t_{k+1}) \in \tilde{\mathcal{P}}_{k+1}(\alpha)) = \alpha, \quad (12.125)$$

while if $b(t_k, \tilde{x}_k, \hat{\theta}_n) = 0$, then $\tilde{X}_N^E(t_{k+1}) = \tilde{\mu}_{k+1}$ with probability one. Now, the main difference, when comparing to Approach 1, is that in Approach 2 we take

$$\bar{x}_{k+1} := \tilde{\mu}_{k+1} \quad (12.126)$$

as the predicted value of $\hat{X}(t_{k+1})$, while for the next starting point for the one step of the Euler-Maruyama scheme we set

$$\tilde{x}_{k+1} := \tilde{X}_N^E(t_{k+1}) \quad (12.127)$$

if $\tilde{\sigma}_{k+1}^2 > 0$, and $\tilde{x}_{k+1} := \tilde{\mu}_{k+1}$ if $\tilde{\sigma}_{k+1} = 0$. (Hence, for fixed $\tilde{x}_k$, the value of $\tilde{x}_{k+1}$ might be drawn from the distribution $N(\tilde{\mu}_{k+1}, \tilde{\sigma}_{k+1}^2)$.) In this approach the predicted value $\bar{x}_{k+1}$ is the mid point of $\tilde{\mathcal{P}}_{k+1}(\alpha)$, while the drawn next starting point $\tilde{x}_{k+1}$ may lie outside the prediction interval $\tilde{\mathcal{P}}_{k+1}(\alpha)$ but with (small) probability $1 - \alpha$.

The procedure described in Approach 2 of Monte Carlo forecasting for SDEs might be implemented as follows:

- 1) draw independent random samples $Z_0, Z_1, \dots, Z_{N-1}$ form the distribution $N(0, 1)$,
- 2) let $\tilde{x}_0 := x_n$ and for $k = 0, 1, \dots, N - 1$ compute

$$\tilde{x}_{k+1} = \tilde{x}_k + a(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot h + \sqrt{h} \cdot b(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot Z_k, \quad (12.128)$$

- 3) having $\{\tilde{x}_k\}_{k=0,1,\dots,N}$, take $\bar{x}_0 := x_n$ and compute for $k = 1, 2, \dots, N$ the predictions $\bar{x}_k$ and the prediction intervals $\tilde{\mathcal{P}}_k(\alpha)$ as follows

$$\bar{x}_k = \tilde{\mu}_k = \tilde{x}_{k-1} + a(t_{k-1}, \tilde{x}_{k-1}, \hat{\theta}_n) \cdot h, \quad (12.129)$$

$$\tilde{\sigma}_k = \sqrt{h} \cdot |b(t_{k-1}, \tilde{x}_{k-1}, \hat{\theta}_n)|, \quad (12.130)$$

$$\tilde{\mathcal{P}}_k(\alpha) = [\tilde{\mu}_k - q_\alpha \cdot \tilde{\sigma}_k, \tilde{\mu}_k + q_\alpha \cdot \tilde{\sigma}_k]. \quad (12.131)$$

Remark 12.5.1. In the case when we have access to the real data (ground truth) at the time t_k we can take it as the starting point $\tilde{x}_k$ for the one step of the Euler-Maruyama algorithm (from t_k to t_{k+1}) in the Approach 2.

Remark 12.5.2. Note that if we take $\tilde{x}_k = \bar{x}_k = \tilde{\mu}_k$ for all k (no matter if $\tilde{\sigma}_k^2 = 0$ or not) in the Approach 2, then $(\tilde{x}_k)_{k=0,1,\dots,N}$ corresponds the Euler scheme when applied to the following ODE

$$\hat{x}'(t) = a(t, \hat{x}(t), \hat{\theta}_n), \quad t \in [T, T + \Delta], \quad \hat{x}(T) = x_n. \quad (12.132)$$

The drawback is that we lose in this case information about the noise modeled in (12.114) - the prediction of $\hat{X}$ is oversmoothed.

We define the following functions

$$\tilde{\mu}_{k+1}(x) = x + h \cdot a(t_k, x, \hat{\theta}_n), \quad (12.133)$$

$$\tilde{\sigma}_{k+1}^2(x) = h \cdot b^2(t_k, x, \hat{\theta}_n), \quad (12.134)$$

and let

$$\tilde{\mathcal{P}}_{k+1}(\alpha, x) = [\tilde{\mu}_{k+1}(x) - q_\alpha \cdot \tilde{\sigma}_{k+1}(x), \tilde{\mu}_{k+1}(x) + q_\alpha \cdot \tilde{\sigma}_{k+1}(x)]. \quad (12.135)$$

Then, for $k = 0, 1, \dots, N-1$

$$A_{k+1} := \{\tilde{X}_N^E(t_{k+1}) \in \tilde{\mathcal{P}}_{k+1}(\alpha, \tilde{X}_N^E(t_k))\} = \{|\Delta W_k| \leq q_\alpha \cdot \sqrt{h}\}, \quad (12.136)$$

and hence, the sets $(A_k)_{k=1,\dots,N}$ are independent. This implies that

$$\mathbb{P}\left(\bigcap_{k=0}^{N-1} A_{k+1}\right) = \prod_{k=0}^{N-1} \mathbb{P}(A_{k+1}) = \alpha^N, \quad (12.137)$$

which gives the probability for the *simultaneous prediction interval*

$$\bigtimes_{k=0}^{N-1} \tilde{\mathcal{P}}_{k+1}(\alpha, \tilde{X}_N^E(t_k)). \quad (12.138)$$

In order to have an accurate prediction for a relatively large number of steps (for example, $N = 20$) we should take $\alpha = 0,998$. This corresponds to $q_\alpha \approx 3.09$.

12.5.2 Ellipsoidal prediction regions in multidimensional case

Let us take, as in the one-dimensional case, that

$$\tilde{\mu}_{k+1}(x) := x + h \cdot a(t_k, x, \hat{\theta}_n), \quad x \in \mathbb{R}^d. \quad (12.139)$$

Assume that we have given $\bar{x}_k$ and $\tilde{x}_k$, where $\bar{x}_k$ is the prediction of the value of $\hat{X}(t_k)$ and $\tilde{x}_k$ is equal to $\tilde{X}_N^E(t_k)$ - the simulated value of Euler-Maruyama scheme at t_k (for $k = 0$ we take $\tilde{x}_0 = \bar{x}_0 := x_n$). Then for the one step of Euler-Maruyama scheme from t_k to $t_{k+1} = t_k + h$, starting from $\tilde{x}_k$ at t_k , we have

$$b^T(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot \left(\tilde{X}_N^E(t_{k+1}) - \tilde{\mu}_{k+1}(\tilde{x}_k) \right) = (b^T b)(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot \Delta W_k = \sqrt{h} \cdot (b^T b)(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot Z_k, \quad (12.140)$$

where $Z_k := \Delta W_k / \sqrt{h}$ has m -dimensional normal distribution $N(0, I_m)$. We impose the following assumption

(M) for all (t, x, z) the matrix $b(t, x, z) \in \mathbb{R}^{d \times m}$ has m linearly independent columns, i.e. $\text{rank}(b(t, x, z)) = m$.

Since for all (t, x, z) we have $\text{rank}(b(t, x, z)) \leq \min\{d, m\}$, the assumption (M) implies that

$$d \geq m. \quad (12.141)$$

Then, for all (t, x, z) , the matrix $(b^T b)(t, x, z) \in \mathbb{R}^{d \times d}$ is symmetric and positive definite. Hence,

$$h^{-1/2} \cdot (b^T b)^{-1}(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot b^T(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot \left(\tilde{X}_N^E(t_{k+1}) - \tilde{\mu}_{k+1}(\tilde{x}_k) \right) = Z_k = \Delta W_k / \sqrt{h} \quad (12.142)$$

has normal distribution $N(0, I_m)$. Therefore, for $B(0, R) = \{x \in \mathbb{R}^m \mid \|x\| \leq R\}$, $R \in [0, +\infty]$, we have

$$\begin{aligned} p(R) &= \mathbb{P} \left(h^{-1/2} \cdot (b^T b)^{-1}(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot b^T(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot \left(\tilde{X}_N^E(t_{k+1}) - \tilde{\mu}_{k+1}(\tilde{x}_k) \right) \in B(0, R) \right) \\ &= \int_{B(0, R)} \frac{1}{(2\pi)^{m/2}} e^{-\|x\|^2/2} dx, \end{aligned} \quad (12.143)$$

and we call $p(R)$ the *confidence level*. For all $y \in \mathbb{R}^d$ we define

$$\mathcal{P}_{k+1}(R, y) = \{z \in \mathbb{R}^d \mid h^{-1/2} \|b^+(t_k, y, \hat{\theta}_n) \cdot (z - \tilde{\mu}_{k+1}(y))\| \leq R\}, \quad (12.144)$$

where

$$b^+(t, y, z) := (b^T b)^{-1}(t, y, z) \cdot b^T(t, y, z) \in \mathbb{R}^{m \times d} \quad (12.145)$$

is the Moore-Penrose inverse of $b(t, y, z)$, since we have assumed that $\text{rank}(b(t, y, z)) = m$ and $d \geq m$, see [?]. Note that $b^+(t, y, z)$ is also the left-inverse of $b(t, y, z)$, since

$$b^+(t, y, z) \cdot b(t, y, z) = I_m. \quad (12.146)$$

Moreover, if we denote by

$$E_k(y) := (b^+(t_k, y, \hat{\theta}_n))^T \cdot b^+(t_k, y, \hat{\theta}_n), \quad (12.147)$$

then we can write

$$\mathcal{P}_{k+1}(R, y) = \{z \in \mathbb{R}^d \mid (z - \tilde{\mu}_{k+1}(y))^T \cdot E_k(y) \cdot (z - \tilde{\mu}_{k+1}(y)) \leq hR^2\}. \quad (12.148)$$

For all $y \in \mathbb{R}^d$ the matrix $E_k(y) \in \mathbb{R}^{d \times d}$ is symmetric and positive semi-definite, and for all $y, z \in \mathbb{R}^d$

$$(z - \tilde{\mu}_{k+1}(y))^T \cdot E_k(y) \cdot (z - \tilde{\mu}_{k+1}(y)) = \|b^+(t_k, y, \hat{\theta}_n) \cdot (z - \tilde{\mu}_{k+1}(y))\|^2 \geq 0. \quad (12.149)$$

We define the *prediction region* for $\hat{X}(t_{k+1})$ as

$$\tilde{\mathcal{P}}_{k+1}(R) := \mathcal{P}_{k+1}(R, \tilde{x}_k). \quad (12.150)$$

Then

$$\mathbb{P} \left(\tilde{X}_N^E(t_{k+1}) \in \tilde{\mathcal{P}}_{k+1}(R) \mid \Sigma_{t_k} \right) = \mathbb{P} \left(h^{-1/2} \|\Delta W_k\| \leq R \mid \Sigma_{t_k} \right) = \mathbb{P} \left(h^{-1/2} \|\Delta W_k\| \leq R \right) = p(R), \quad (12.151)$$

and therefore

$$\mathbb{P} \left(\tilde{X}_N^E(t_{k+1}) \in \tilde{\mathcal{P}}_{k+1}(R) \right) = p(R). \quad (12.152)$$

We take

$$\bar{x}_{k+1} := \tilde{\mu}_{k+1}(\tilde{x}_k) = \tilde{x}_k + h \cdot a(t_k, \tilde{x}_k, \hat{\theta}_n) \quad (12.153)$$

as the predicted value of $\hat{X}(t_{k+1})$, while for the next starting point for the one step of the Euler-Maruyama scheme we set

$$\tilde{x}_{k+1} := \tilde{X}_N^E(t_{k+1}). \quad (12.154)$$

Hence, for fixed $\tilde{x}_k$, the value of $\tilde{x}_{k+1}$ might be drawn from the distribution $N(\tilde{\mu}_{k+1}(\tilde{x}_k), \tilde{V}_{k+1}^2(\tilde{x}_k))$, where $\tilde{V}_{k+1}^2(y) := h \cdot (bb^T)(t_k, y, \hat{\theta}_n)$.

The procedure of Monte Carlo forecasting for SDEs might be described as follows:

1) draw independent random samples $Z_0, Z_1, \dots, Z_{N-1}$ form the distribution $N(0, I_m)$,

2) let $\tilde{x}_0 := x_n$ and for $k = 0, 1, \dots, N-1$ compute

$$\tilde{x}_{k+1} = \tilde{x}_k + a(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot h + \sqrt{h} \cdot b(t_k, \tilde{x}_k, \hat{\theta}_n) \cdot Z_k, \quad (12.155)$$

3) having $\{\tilde{x}_k\}_{k=0,1,\dots,N}$, take $\bar{x}_0 := x_n$, then, for $k = 1, 2, \dots, N$, compute the predictions $\bar{x}_k$ and set the prediction regions $\mathcal{P}_k(R)$ as follows

$$\bar{x}_k = \tilde{\mu}_k = \tilde{x}_{k-1} + a(t_{k-1}, \tilde{x}_{k-1}, \hat{\theta}_n) \cdot h, \quad (12.156)$$

$$\tilde{\mathcal{P}}_k(R) = \{z \in \mathbb{R}^d \mid h^{-1/2} \|b^+(t_{k-1}, \tilde{x}_{k-1}, \hat{\theta}_n) \cdot (z - \tilde{\mu}_k)\| \leq R\}. \quad (12.157)$$

For $k = 0, 1, \dots, N-1$

$$A_{k+1} := \{\tilde{X}_N^E(t_{k+1}) \in \mathcal{P}_{k+1}(R, \tilde{X}_N^E(t_k))\} = \{\|\Delta W_k\| \leq R \cdot \sqrt{h}\} = \{Z_k \in B(0, R)\}, \quad (12.158)$$

and hence, the sets $(A_k)_{k=1,\dots,N}$ are independent. This implies that

$$\mathbb{P}\left((\tilde{X}_N^E(t_1), \tilde{X}_N^E(t_2), \dots, \tilde{X}_N^E(t_N)) \in \bigtimes_{k=0}^{N-1} \mathcal{P}_{k+1}(R, \tilde{X}_N^E(t_k))\right) = \mathbb{P}\left(\bigcap_{k=0}^{N-1} A_{k+1}\right) = \prod_{k=0}^{N-1} \mathbb{P}(A_{k+1}) = (p(R))^N, \quad (12.159)$$

which gives the probability that the obtained N -step prediction $(\tilde{X}_N^E(t_1), \tilde{X}_N^E(t_2), \dots, \tilde{X}_N^E(t_N))$ belongs to the *simultaneous prediction region*

$$\bigtimes_{k=0}^{N-1} \mathcal{P}_{k+1}(R, \tilde{X}_N^E(t_k)). \quad (12.160)$$

In a special case, we can describe the geometry of $\mathcal{P}_{k+1}(R, y)$ in a more precise way. Namely, let us assume that for all (t, y, z)

$$\text{rank}(b^+(t, y, z)) = d. \quad (12.161)$$

Since $\text{rank}(b^+(t, y, z)) = d \leq \min\{d, m\} \leq m$, by (12.141) this assumption force the following

$$d = m, \quad (12.162)$$

and the matrix $b(t, y, z)$ has to be a square matrix in $\mathbb{R}^{d \times d}$. Moreover, it is of full rank (equal to d), and therefore it is also non-singular. In this case for all (t, y, z)

$$b^+(t, y, z) = b^{-1}(t, y, z), \quad (12.163)$$

and the symmetric matrix

$$E_k(y) = \left(b(t_k, y, \hat{\theta}_n) \cdot b^T(t_k, y, \hat{\theta}_n)\right)^{-1} \quad (12.164)$$

is positive definite (In particular, this also implies that $\tilde{V}_{k+1}^2(y)$ is positive definite, since $\tilde{V}_{k+1}^2(y) = h \cdot E_k^{-1}(y)$.) Moreover,

$$\begin{aligned} \mathcal{P}_{k+1}(R, y) &= \{z \in \mathbb{R}^d \mid h^{-1/2} \|b^{-1}(t_k, y, \hat{\theta}_n) \cdot (z - \tilde{\mu}_{k+1}(y))\| \leq R\} \\ &= \tilde{\mu}_{k+1}(y) + h^{1/2} b(t_k, y, \hat{\theta}_n) B(0, R), \end{aligned} \quad (12.165)$$

and, therefore, in the case when $E_k(y)$ is positive definite, the set $\mathcal{P}_{k+1}(R, y)$ is an ellipsoid centered at $\tilde{\mu}_{k+1}(y)$. The eigenvectors of $E_k(y)/(hR^2)$ are the principal axes of the ellipsoid, and length of each corresponding semi-axe is $1/(\lambda_j(y))^{1/2}$, where $\lambda_j(y)$'s are the (strictly positive) eigenvalues of $E_k(y)/(hR^2)$.

Remark 12.5.3. Note that we can write $\tilde{V}_{k+1}^2(y) = (\tilde{V}_{k+1}(y))^T \cdot \tilde{V}_{k+1}(y)$ where

$$\tilde{V}_{k+1}(y) = h^{1/2} \cdot b^T(t_k, y, \hat{\theta}_n), \quad (12.166)$$

and

$$\mathcal{P}_{k+1}(R, y) = \{z \in \mathbb{R}^d \mid \|[\tilde{V}_{k+1}(y)\tilde{V}_{k+1}^T(y)]^{-1}\tilde{V}_{k+1}(y)(z - \tilde{\mu}_{k+1}(y))\| \leq R\} \quad (12.167)$$

Remark 12.5.4. Of course in the case $m = d = 1$ we obtain the same prediction intervals as was described in the one-dimensional case, since

$$\mathcal{P}_{k+1}(q_\alpha, \tilde{x}_k) = \tilde{\mathcal{P}}_{k+1}(\alpha). \quad (12.168)$$

Remark 12.5.5. In this section, we showed how to construct ellipsoidal prediction regions for uncertainty characterization when time series under consideration are assumed to follow from SDEs-based models. See also [21] where the authors use ellipsoidal prediction regions in different econometric-fashioned approaches to time-series uncertainty characterization.

12.5.3 Concluding remarks - why the geometry of prediction regions matters

The procedure described in the previous section of obtaining prediction regions in the multidimensional case takes into account the fact that coordinates of $\hat{X}$ and $\tilde{X}_N^E$ are correlated. Because of that, it is possible to compute explicitly:

- the probability (12.151) that all of the coordinates $\tilde{X}_{N,j}^E$ are in $\tilde{\mathcal{P}}_{k+1}(R)$ simultaneously,
- the probability (12.159) that the whole predicted N -steps forward belong to the simultaneous prediction region (12.160).

One can however try an alternative (and, at first sight, more natural) approach in which one computes prediction intervals for each coordinate $\tilde{X}_j$ separately (by using an approach known from the one-dimensional case) and then takes the Cartesian product of them in order to obtain prediction region for ... (which is in fact a hypercube in). However, there is a pitfall in that approach, which we now describe in detail.

Note that for all $j = 1, 2, \dots, d$, $k = 0, 1, \dots, N - 1$

$$\tilde{X}_{N,j}^E(t_{k+1}) \mid \Sigma_{t_k} \sim N\left(\tilde{\mu}_{k+1,j}(\tilde{X}_N^E(t_k)), \tilde{\sigma}_{k+1,j}^2(\tilde{X}_N^E(t_k))\right), \quad (12.169)$$

where for all $x \in \mathbb{R}^d$

$$\tilde{\mu}_{k+1,j}(x) = x_j + h \cdot a_j(t_k, x, \hat{\theta}_n), \quad (12.170)$$

$$\tilde{\sigma}_{k+1,j}^2(x) = h \cdot \sum_{l=1}^m (b_l(t_k, x, \hat{\theta}_n))^2. \quad (12.171)$$

Let us take

$$\tilde{\mathcal{P}}_{k+1,j}(\alpha, x) = [\tilde{\mu}_{k+1,j}(x) - q_\alpha \cdot \tilde{\sigma}_{k+1,j}(x), \tilde{\mu}_{k+1,j}(x) + q_\alpha \cdot \tilde{\sigma}_{k+1,j}(x)], \quad (12.172)$$

and

$$B_{k+1,j} = \{\tilde{X}_{N,j}^E(t_{k+1}) \in \tilde{\mathcal{P}}_{k+1,j}(\alpha, \tilde{X}_N^E(t_k))\}, \quad (12.173)$$

then, by (12.169), we have for all $j = 1, 2, \dots, d$ that

$$\mathbb{P}\left(B_{k+1,j} \mid \Sigma_{t_k}\right) = \mathbb{P}\left(\frac{|\tilde{X}_{N,j}^E(t_{k+1}) - \tilde{\mu}_{k+1,j}(\tilde{X}_N^E(t_k))|}{\tilde{\sigma}_{k+1,j}(\tilde{X}_N^E(t_k))} \leq q_\alpha \mid \Sigma_{t_k}\right) = \alpha, \quad (12.174)$$

and hence for each coordinate of $\tilde{X}_N^E(t_{k+1})$ we have

$$\mathbb{P}(B_{k+1,j}) = \alpha. \quad (12.175)$$

Note that for fixed $k = 1, 2, \dots, d$ the sets

$$B_{k+1,1}, B_{k+1,2}, \dots, B_{k+1,d}$$

are not independent (even conditionally wrt Σ_{t_k}). Therefore, we do not have the exact value (or at least good estimates) of the probability

$$\mathbb{P}(B_{k+1}) = \mathbb{P}(\tilde{X}_N^E(t_{k+1}) \in \tilde{\mathcal{B}}_{k+1}(\alpha, \tilde{X}_N^E(t_k))), \quad (12.176)$$

where

$$B_{k+1} = \bigcap_{j=1}^d B_{k+1,j} \quad (12.177)$$

and

$$\tilde{\mathcal{B}}_{k+1}(\alpha, x) = \bigtimes_{j=1}^d \tilde{\mathcal{P}}_{k+1,j}(\alpha, x). \quad (12.178)$$

Moreover, the sets $\{B_{k+1}\}_{k=0,1,\dots,N-1}$ are also not independent. Hence, in general, it is not known how to compute the probability

$$\mathbb{P}\left((\tilde{X}_N^E(t_1), \tilde{X}_N^E(t_2), \dots, \tilde{X}_N^E(t_N)) \in \bigtimes_{k=0}^{N-1} \tilde{\mathcal{B}}_{k+1}(\alpha, \tilde{X}_N^E(t_k))\right) = \mathbb{P}\left(\bigcap_{k=0}^{N-1} B_{k+1}\right), \quad (12.179)$$

and this method of constructing prediction regions does not allow us to control the probability that the whole predicted sequence belongs to the simultaneous prediction region. This is an obvious drawback.

12.6 Exercises

1. Show for the equation (12.1) that the maximum likelihood estimators of (μ, σ^2) and quasi maximum likelihood estimators coincide.

Hint. Apply QML procedure to (12.3).

2. Give a proof of (12.165).
3. Show for $j = 1, 2, \dots, d$, $k = 0, 1, \dots, N-1$ that

$$\text{Var}\left(\sum_{l=1}^m b_{jl}(t_k, \tilde{X}_N^E(t_k), \hat{\theta}_n) \cdot \Delta W_k^l \mid \Sigma_{t_k}\right) = \tilde{\sigma}_{k+1,j}^2(\tilde{X}_N^E(t_k)). \quad (12.180)$$

4. Justify that $(X^E(t_k))_{k=0,1,\dots,n}$ is a discrete-time Markov process.
5. Show that if $\lim_{n \rightarrow +\infty} \max_{0 \leq j \leq n-1} (t_{j+1} - t_j) = 0$ then for the estimator (12.45) we have that

$$\lim_{n \rightarrow +\infty} \mathbb{E}(\hat{\mu}_n) = \mu. \quad (12.181)$$

Hint: show that

$$\sum_{j=0}^{n-1} (e^{\mu \Delta t_j} - 1) \rightarrow T\mu \quad (12.182)$$

as $\lim_{n \rightarrow +\infty} \max_{0 \leq j \leq n-1} (t_{j+1} - t_j) = 0$.

Chapter 13

Auxiliary results—part 1

In this lecture notes, we use many results from the theory of stochastic processes, including basics from measure and probability theory. We will now recall some of them. For a comprehensive exposition see, for example, [3], [12], [14], [17], [46], [61], [55], [72], [85].

By $\|\cdot\|_2$ we mean the Euclidean norm on $\mathbb{R}^d$.

Let X be an arbitrary set. A collection $\mathcal{A}$ of subsets of X is a σ -field on X if

- (i) $X \in \mathcal{A}$,
- (ii) for every $A \in \mathcal{A}$, the set $X \setminus A$ belongs to $\mathcal{A}$,
- (iii) for any infinite sequence $(A_n)_{n \in \mathbb{N}} \subset \mathcal{A}$ we have that $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{A}$,
- (iv) for any infinite sequence $(A_n)_{n \in \mathbb{N}} \subset \mathcal{A}$ we have that $\bigcap_{n \in \mathbb{N}} A_n \in \mathcal{A}$.

As we can see a σ -field on X is a subclass of 2^X that contains X and is closed with respect to complementation, countable unions, and intersections. Note also that, in the definition of σ -field we could have used only conditions (i), (ii), and (iii), or only conditions (i), (ii), and (iv), as it is sometimes in literature. Moreover, sometimes the authors use the term σ -algebra instead of σ -field.

A measurable space is a pair $(X, \mathcal{A})$ where X is a set and $\mathcal{A}$ is a σ -field on X .

Let X be a set and let $\mathcal{C}$ be an arbitrary class of subsets of X . By $\sigma(\mathcal{C})$ we denote the intersection of all those σ -fields that contain $\mathcal{C}$. It turns out that $\sigma(\mathcal{C})$ is the smallest σ -field that contains $\mathcal{C}$, and it is called the σ -field generated by $\mathcal{C}$. Such σ -field always exists, see, for example, Corollary 1.1.3 in [14]. If $\mathcal{C}$ is already a σ -field then $\sigma(\mathcal{C}) = \mathcal{C}$. Furthermore, the following holds (see, for example, [12]).

Proposition 13.0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be two families of subsets of X . Then:*

- (i) *If $\mathcal{C} \subset \mathcal{D}$ then $\sigma(\mathcal{C}) \subset \sigma(\mathcal{D})$*
- (ii) *If $\mathcal{C} \subset \sigma(\mathcal{D})$ then $\sigma(\mathcal{C}) \subset \sigma(\mathcal{D})$*
- (iii) *If $\mathcal{C} \subset \sigma(\mathcal{D})$ and $\mathcal{D} \subset \sigma(\mathcal{C})$, then $\sigma(\mathcal{C}) = \sigma(\mathcal{D})$*
- (iv) *If $\mathcal{C} \subset \mathcal{D} \subset \sigma(\mathcal{C})$, $\sigma(\mathcal{C}) = \sigma(\mathcal{D})$*

The intersection of an arbitrary (even uncountable) class of σ -fields on X is again a σ -field on X . This is not however true when considering the union of σ -fields. Namely, except in some particular cases, the union of two σ -fields $\mathcal{A}_1, \mathcal{A}_2$ on X is not a σ -field. The σ -field generated by $\mathcal{A}_1 \cup \mathcal{A}_2$ is denoted by $\mathcal{A}_1 \vee \mathcal{A}_2$,

$$\mathcal{A}_1 \vee \mathcal{A}_2 := \sigma(\mathcal{A}_1 \cup \mathcal{A}_2). \quad (13.1)$$

¹In fact, 2^X is the largest σ -field on X . The family $\{\emptyset, X\}$ is the smallest σ -field on X . For any σ -field on X the following inclusion holds $\{\emptyset, X\} \subset \mathcal{A} \subset 2^X$.

In general, if $(\mathcal{A}_i)_{i \in I}$ is a family of σ -fields on X , where the index set I might be even uncountable, then we define

$$\mathcal{A}_I := \sigma\left(\bigcup_{i \in I} \mathcal{A}_i\right), \quad (13.2)$$

the σ -field generated by $\bigcup_{i \in I} \mathcal{A}_i$. It is sometimes also denoted by $\bigvee_{i \in I} \mathcal{A}_i$. Note that the above definition of $\bigvee_{i \in I} \mathcal{A}_i$ is valid for a family $(\mathcal{A}_i)_{i \in I}$ of arbitrary classes of subsets of X (i.e., some of $\mathcal{A}_i$'s do not have to be σ -fields on X). This follows from the following fact.

Proposition 13.0.2. *Let $(\mathcal{A}_i)_{i \in I}$ be a family of classes of subsets of X . Then*

$$\sigma\left(\bigcup_{i \in I} \mathcal{A}_i\right) = \sigma\left(\bigcup_{i \in I} \sigma(\mathcal{A}_i)\right). \quad (13.3)$$

Proof. Since for all $i \in I$ we have $\mathcal{A}_i \subset \sigma(\mathcal{A}_i)$, we obtain that $\bigcup_{i \in I} \mathcal{A}_i \subset \bigcup_{i \in I} \sigma(\mathcal{A}_i)$ and hence

$$\sigma\left(\bigcup_{i \in I} \mathcal{A}_i\right) \subset \sigma\left(\bigcup_{i \in I} \sigma(\mathcal{A}_i)\right). \quad (13.4)$$

On the other hand, $\mathcal{A}_j \subset \bigcup_{i \in I} \mathcal{A}_i$ for all $j \in I$ and therefore $\sigma(\mathcal{A}_j) \subset \sigma\left(\bigcup_{i \in I} \mathcal{A}_i\right)$ for all $j \in I$. Hence,

$$\bigcup_{i \in I} \sigma(\mathcal{A}_i) \subset \sigma\left(\bigcup_{i \in I} \mathcal{A}_i\right) \text{ and} \quad \sigma\left(\bigcup_{i \in I} \sigma(\mathcal{A}_i)\right) \subset \sigma\left(\bigcup_{i \in I} \mathcal{A}_i\right). \quad (13.5)$$

Combining (13.4) and (13.5) we get the thesis. ■

If X is a topological space, then the σ -field generated by the collection $TopX$ of all open subsets of X is called the *Borel σ -field* on X and it is denoted by $\mathcal{B}(X)$ (i.e., $\mathcal{B}(X) = \sigma(TopX)$). Its elements are called *Borel sets*.

Let $(X_1, \mathcal{A}_1)$ and $(X_2, \mathcal{A}_2)$ be two measurable spaces. A function $f : X_1 \rightarrow X_2$ is called $\mathcal{A}_1$ -to- $\mathcal{A}_2$ measurable (or $\mathcal{A}_1/\mathcal{A}_2$ -measurable) if

$$f^{-1}(B) \in \mathcal{A}_1 \text{ for all } B \in \mathcal{A}_2, \quad (13.6)$$

where $f^{-1}(B) = \{x \in X_1 \mid f(x) \in B\}$. If $A \in \mathcal{A}_1$ then a function $f : A \rightarrow X_2$ is $\mathcal{A}_1/\mathcal{A}_2$ -measurable if $f^{-1}(B) \in \mathcal{A}_1$ for any $B \in \mathcal{A}_2$. Moreover, we define

$$\sigma(f) = \{f^{-1}(B) \mid B \in \mathcal{A}_2\}. \quad (13.7)$$

It turns out that $\sigma(f)$ is a σ -field on X_1 and it is the smallest σ -field on X_1 such that f is measurable relative to it and $\mathcal{A}_2$ (i.e., $\sigma(f)/\mathcal{A}_2$ -measurable). We call $\sigma(f)$ the σ -field generated by f . It is easy to check that f is $\mathcal{A}_1/\mathcal{A}_2$ -measurable iff $\sigma(f) \subset \mathcal{A}_1$. Moreover, if $(X_3, \mathcal{A}_3)$ is a measurable space then for any two functions $f_1 : X_1 \rightarrow X_2$ and $f_2 : X_2 \rightarrow X_3$, where f_2 is $\mathcal{A}_2/\mathcal{A}_3$ -measurable, we have $\sigma(f_2 \circ f_1) \subset \sigma(f_1)$. This implies the following fact.

Proposition 13.0.3. *Let $(X_i, \mathcal{A}_i)$, $i = 1, 2, 3$, be measurable spaces, let $f_1 : X_1 \rightarrow X_2$ be $\mathcal{A}_1/\mathcal{A}_2$ -measurable, while $f_2 : X_2 \rightarrow X_3$ is assumed to be $\mathcal{A}_2/\mathcal{A}_3$ -measurable. The the composition*

$$f_2 \circ f_1 : X_1 \rightarrow X_3 \quad (13.8)$$

is $\mathcal{A}_1/\mathcal{A}_3$ -measurable.

Proposition 13.0.4. *Let $(X, \mathcal{A})$ be a measurable space, and let $A \subset X$. Then the function*

$$X \ni x \rightarrow \mathbf{1}_A(x) \in \{0, 1\} \quad (13.9)$$

is $\mathcal{A}/\mathcal{B}(\mathbb{R})$ -measurable iff $A \in \mathcal{A}$.

In the case when X_2 is topological space, equipped with the Borel σ -field $\mathcal{A}_2 = \mathcal{B}(\text{Top}X_2)$, we refer to $\mathcal{A}_1/\mathcal{B}(X_2)$ -measurable function $f : X_1 \rightarrow X_2$ as *Borel function*.

Below we give important examples of Borel functions.

Proposition 13.0.5. *(i) If X_1 and X_2 are topological spaces and if $f : X_1 \rightarrow X_2$ is a continuous function then it is Borel measurable, that is $\mathcal{B}(X_1)/\mathcal{B}(X_2)$ -measurable.*

(ii) Every right (left)-continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ is Borel. Hence, every càdlàg (càglàd) function is Borel measurable.

(iii) If $f : [a, b] \rightarrow \mathbb{R}$ is a bounded variation function then f is Borel measurable.

(iv) Let I be a subinterval of $\mathbb{R}$, and let $f : I \rightarrow \mathbb{R}$ be a monotone function. Then f is Borel measurable.

(v) If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is a convex function then it is continuous and hence, Borel measurable.

Let us now assume that $(X_2, \|\cdot\|)$ is a real or complex Banach space and let $\mathcal{A}_2 = \mathcal{B}(X_2) = \sigma(\text{Top}X_2)$. Then we have another notion of measurability. A function $f : X_1 \rightarrow X_2$ is *strongly measurable* (or *Bochner measurable*) if it is Borel measurable and has a separable range (i.e., $f(X_1) \subset X_2$ is separable). If X_2 is separable, then every X_2 -valued Borel measurable function is strongly measurable. (It follows from the fact that a subset of separable metric space is itself separable.) Moreover, if $f : X_1 \rightarrow X_2$ is Borel measurable, then

$$X_1 \ni x \rightarrow \|f(x)\| \in [0, +\infty) \quad (13.10)$$

is $\mathcal{A}_1/\mathcal{B}(\mathbb{R})$ -measurable. Moreover, we have the following important result, see [14].

Theorem 13.0.6. *Let $(X_1, \mathcal{A}_1)$ be a measurable space, and let X_2 be a real or complex Banach space. Then the set of all strongly measurable functions from X_1 to X_2 is a vector space.*

We want to stress here that the set of all Borel measurable functions from X_1 to X_2 can fail to be a vector space, see page 403 in [14]. Therefore, the separability of the range in the definition of a strongly measurable function is crucial.

Let $(X, \mathcal{A}, \mu)$ be a measure space and let $p \in [1, +\infty)$. Then we define

$$L^p(X, \mathcal{A}, \mu; \mathbb{R}^d) = \left\{ f : X \rightarrow \mathbb{R}^d \mid f \text{ - Borel function, } \int_X \|f(x)\|_2^p \mu(dx) < +\infty \right\}. \quad (13.11)$$

If we identify functions which are equal μ -almost everywhere, then $L^p(X, \mathcal{A}, \mu; \mathbb{R}^d)$ is a Banach space with norm

$$\|f\|_{L^p(X; \mathbb{R}^d)} = \left(\int_X \|f(x)\|_2^p \mu(dx) \right)^{1/p}. \quad (13.12)$$

For the case $d = 1$ we write $L^p(X)$.

The proof of the following result can be found in [14].

Theorem 13.0.7. *(Change of measure in Lebesgue integral) Let $(X, \mathcal{A}, \mu)$ be a measure space, let $(Y, \mathcal{B})$ be a measurable space, and let $f : (X, \mathcal{A}) \rightarrow (Y, \mathcal{B})$ be measurable. Let g be an extended real-valued $\mathcal{B}$ -measurable function on Y . Then g is μf^{-1} -integrable if and only if $g \circ f$ is μ -integrable. If these functions are integrable, then*

$$\int_Y g d(\mu f^{-1}) = \int_X (g \circ f) d\mu. \quad (13.13)$$

For the proof of the mean value theorem we refer to, for example, page 167 in [61].

Theorem 13.0.8. (*Mean value theorem*) Let $E \subset \mathbb{R}^d$ be a connected set with $\lambda_d(E) < +\infty$. If $f : E \rightarrow \mathbb{R}$ is continuous and bounded, then there exists $c \in E$ such that

$$\int_E f(x) \lambda_d(dx) = f(c) \cdot \lambda_d(E). \quad (13.14)$$

We also recall fundamental integral inequalities, see, for example, [46].

Theorem 13.0.9. (*Hölder and Minkowski inequalities*) For any Borel measurable real-valued functions f and g on some measure space $(X, \mathcal{A}, \mu)$, we have

(i) for all $p, q, r > 0$ with $p^{-1} + q^{-1} = r^{-1}$ that it holds

$$\left(\int_X |f(x)g(x)|^r \mu(dx) \right)^{1/r} \leq \left(\int_X |f(x)|^p \mu(dx) \right)^{1/p} \cdot \left(\int_X |g(x)|^q \mu(dx) \right)^{1/q}, \quad (13.15)$$

(ii) for all $p > 0$

$$\left(\int_X |f(x) + g(x)|^p \mu(dx) \right)^{\min\{p, 1\}/p} \leq \left(\int_X |f(x)|^p \mu(dx) \right)^{\min\{p, 1\}/p} + \left(\int_X |g(x)|^p \mu(dx) \right)^{\min\{p, 1\}/p}. \quad (13.16)$$

Theorem 13.0.10. (*Jensen's inequality*) For any random vector $\xi : \Omega \rightarrow \mathbb{R}^d$, defined on a probability space $(\Omega, \Sigma, \mathbb{P})$, and convex function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, we have that

$$f(\mathbb{E}\xi) \leq \mathbb{E}(f(\xi)). \quad (13.17)$$

We also recall the very useful Gronwall's lemma for Lebesgue integrals, see, for example, [54] or Annex C in [69], where a more general version of the lemma can be found.

Lemma 13.0.11. (*continuous version of the Gronwall's lemma*) Let $-\infty < a < b < +\infty$ and let $u : [a, b] \rightarrow \mathbb{R}$ be a Borel measurable integrable function. If there exist $\alpha \in \mathbb{R}$, $\beta \in [0, +\infty)$ such that for all $t \in [a, b]$ the function u satisfies

$$u(t) \leq \alpha + \beta \int_a^t u(s) ds, \quad (13.18)$$

then for all $t \in [a, b]$ we have that

$$u(t) \leq \alpha e^{\beta(t-a)}. \quad (13.19)$$

Proof. Let us define the function $h : [a, b] \rightarrow \mathbb{R}$ as follows

$$h(t) = \alpha + \beta \int_a^t u(s) ds. \quad (13.20)$$

Then, by the assumption

$$u(t) \leq h(t). \quad (13.21)$$

for all $t \in [a, b]$. Moreover, h is absolutely continuous and

$$h'(t) = \beta \cdot u(t) \quad (13.22)$$

for almost all t in $[0, T]$. Since $\beta \geq 0$, we get from (13.21) and (13.22) that

$$h'(t) \leq \beta h(t) \quad (13.23)$$

for almost all t in $[0, T]$. Furthermore, by (13.23)

$$\left(e^{-\beta t} h(t)\right)' = -\beta e^{-\beta t} h(t) + e^{-\beta t} h'(t) \leq 0 \quad (13.24)$$

for almost all t in $[0, T]$. The functions $[a, b] \ni t \rightarrow e^{-\beta t}$ and h are absolutely continuous, hence so it is $[a, b] \ni t \rightarrow e^{-\beta t} h(t)$. Thereby

$$e^{-\beta t} h(t) - e^{-\beta a} h(a) = \int_a^t \left(e^{-\beta s} h(s)\right)' ds \leq 0 \quad (13.25)$$

for all $t \in [a, b]$. Hence,

$$u(t) \leq h(t) \leq e^{\beta(t-a)} h(a) = \alpha e^{\beta(t-a)}, \quad t \in [a, b]. \quad (13.26)$$

This completes the proof. $\blacksquare$

Below we also present a discrete version of the Gronwall inequality, which is extensively used, for example, in error estimation for various numerical schemes for ODEs, SDEs, etc.

Lemma 13.0.12. *Let us assume that for a nonnegative sequence of real numbers $(u_k)_{k=0}^n$ there exist $A, B \in [0, +\infty)$ such that*

$$u_{k+1} \leq A u_k + B, \quad k = 0, 1, \dots, n-1. \quad (13.27)$$

Then for $k = 0, 1, \dots, n$ it holds

$$u_k \leq A^k u_0 + C_k, \quad (13.28)$$

where

$$C_k = \begin{cases} B \frac{A^k - 1}{A - 1}, & A \neq 1, \\ k B, & A = 1. \end{cases} \quad (13.29)$$

Let $(X_i, \mathcal{A}_i)$, $i = 1, 2$, be measurable spaces. On the Cartesian product $X_1 \times X_2$ of the sets X_i , $i = 1, 2$, we consider the so called *product* σ -field $\mathcal{A}_1 \otimes \mathcal{A}_2$ defined by

$$\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma\left(\{A_1 \times A_2 \mid A_i \in \mathcal{A}_i, i = 1, 2\}\right). \quad (13.30)$$

Then $(X_1 \times X_2, \mathcal{A}_1 \otimes \mathcal{A}_2)$ is a measurable space. Note that we also have

$$\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma(\hat{\mathcal{A}}_1 \cup \hat{\mathcal{A}}_2), \quad (13.31)$$

where

$$\hat{\mathcal{A}}_1 = \{A \times X_2 \mid A \in \mathcal{A}_1\}, \quad (13.32)$$

$$\hat{\mathcal{A}}_2 = \{X_1 \times B \mid B \in \mathcal{A}_2\}, \quad (13.33)$$

see page. 5 in [12]. Now, let $(X_i, \mathcal{A}_i, \mu_i)$, $i = 1, 2$, be σ -finite measure spaces. Then there exists a unique measure $\mu_1 \times \mu_2$ on the σ -field $\mathcal{A}_1 \otimes \mathcal{A}_2$ such that

$$(\mu_1 \times \mu_2)(A_1 \times A_2) = \mu_1(A_1) \cdot \mu_2(A_2) \quad (13.34)$$

holds for any $A_i \in \mathcal{A}_i$, $i = 1, 2$. The measure $\mu_1 \times \mu_2$ is called the *product measure*. Of course $(X_1 \times X_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \mu_1 \times \mu_2)$ is a measure space.

Proposition 13.0.13. Let $(X, \mathcal{A})$, $(Y_i, \mathcal{C}_i)$ be measurable spaces and let $f_i : X \rightarrow Y_i$ for $i = 1, 2$. Define $H : X \rightarrow Y_1 \times Y_2$ by

$$H(x) = (f_1, f_2)(x) = (f_1(x), f_2(x)), \quad x \in X. \quad (13.35)$$

Then H is $\mathcal{A}/\mathcal{C}_1 \otimes \mathcal{C}_2$ -measurable iff $f_i : X \rightarrow Y_i$ is $\mathcal{A}/\mathcal{C}_i$ -measurable for $i = 1, 2$.

The proposition above has its multi-dimensional generalization, see, for example, Proposition 6.27, page 45 in [12] or Lemma 1.8 in [46].

Below we state the measurability of sections.

Proposition 13.0.14. Let $(X_i, \mathcal{A}_i)$, $(Y, \mathcal{C})$, $i = 1, 2$, be measurable spaces. Let us assume $f : X_1 \times X_2 \rightarrow Y$ to be $\mathcal{A}_1 \otimes \mathcal{A}_2/\mathcal{C}$ -measurable. For any $x \in X_1$ the section

$$X_2 \ni y \rightarrow f_x(y) = f(x, y) \in Y \quad (13.36)$$

is $\mathcal{A}_2/\mathcal{C}$ -measurable, while for any $y \in X_2$ the section

$$X_1 \ni x \rightarrow f^y(x) = f(x, y) \in Y \quad (13.37)$$

is $\mathcal{A}_1/\mathcal{C}$ -measurable.

Hence, product measurability implies the measurability of each section. We want to underline here that the converse is not true - without some additional requirements, the measurability of sections does not imply product measurability. We have however the following result, that is very useful from a point of view of stochastic processes (see, for example, page 46 in [12] or Remark 1.14 at page 5. in [60]).

Lemma 13.0.15. Let $(X, \mathcal{A})$ be a measurable space and let $f : X \times \mathbb{R} \rightarrow \mathbb{R}$ be such that the section $y \rightarrow f(x, y)$ is right-continuous (or left-continuous) for each $x \in X$ and that the section $x \rightarrow f(x, y)$ is $\mathcal{A}/\mathcal{B}(\mathbb{R})$ -measurable for each $y \in \mathbb{R}$. Then, f is $\mathcal{A} \otimes \mathcal{B}(\mathbb{R})/\mathcal{B}(\mathbb{R})$ -measurable.

Note that f may be a mapping with values in a separable metric space (such as the Euclidean space $\mathbb{R}^d$) and the thesis of Lemma 13.0.15 still holds, see Lemma 6.4.6 in [8].

Theorem 13.0.16. (Tonelli's Theorem) Let $(X_i, \mathcal{A}_i, \mu_i)$, $i = 1, 2$, be σ -finite measure spaces, and let $f : X_1 \times X_2 \rightarrow [0, +\infty]$ be $\mathcal{A}_1 \otimes \mathcal{A}_2/\mathcal{B}(\mathbb{R})$ -measurable. Then

(i) the function

$$X_1 \ni x_1 \rightarrow \int_{X_2} f(x_1, x_2) \mu_2(dx_2) \quad (13.38)$$

is $\mathcal{A}_1/\mathcal{B}(\mathbb{R})$ -measurable and the function

$$X_2 \ni x_2 \rightarrow \int_{X_1} f(x_1, x_2) \mu_1(dx_1) \quad (13.39)$$

is $\mathcal{A}_2/\mathcal{B}(\mathbb{R})$ -measurable, and

(ii) for the function f the following holds

$$\begin{aligned} \int_{X_1 \times X_2} f(x_1, x_2) \mu_1 \times \mu_2(dx_1, dx_2) &= \int_{X_1} \left(\int_{X_2} f(x_1, x_2) \mu_2(dx_2) \right) \mu_1(dx_1) \\ &= \int_{X_2} \left(\int_{X_1} f(x_1, x_2) \mu_1(dx_1) \right) \mu_2(dx_2). \end{aligned} \quad (13.40)$$

We stress that we can use (13.40) for every nonnegative and $\mathcal{A}_1 \otimes \mathcal{A}_2/\mathcal{B}(\bar{\mathbb{R}})$ -measurable function f , no matter if it is $\mu_1 \times \mu_2$ -integrable or not. Hence, for an arbitrary $\mathcal{A}_1 \otimes \mathcal{A}_2/\mathcal{B}(\bar{\mathbb{R}})$ -measurable function $f : X_1 \times X_2 \rightarrow \bar{\mathbb{R}}$ we can check if it is $\mu_1 \times \mu_2$ -integrable by applying (13.40) to $\int_{X_1 \times X_2} |f(x_1, x_2)| \mu_1 \times \mu_2(dx_1, dx_2)$.

Such a situation occurs many times in the theory of stochastic processes.

Theorem 13.0.17. (*Fubini's Theorem*) Let $(X_i, \mathcal{A}_i, \mu_i)$, $i = 1, 2$, be σ -finite measure spaces, and let $f : X_1 \times X_2 \rightarrow [-\infty, +\infty]$ be $\mathcal{A}_1 \otimes \mathcal{A}_2/\mathcal{B}(\bar{\mathbb{R}})$ -measurable and $\mu_1 \times \mu_2$ -integrable. Then

(i) for μ_1 -almost every $x_1 \in X_1$ the section f_{x_1} is μ_2 -integrable and for μ_2 -almost every $x_2 \in X_2$ the section f^{x_2} is μ_1 -integrable,

(ii) the functions I_f and J_f defined by

$$I_f(x_1) = \begin{cases} \int_{X_2} f_{x_1}(x_2) \mu_2(dx_2) & \text{if } f_{x_1} \text{ is } \mu_2\text{-integrable,} \\ 0 & \text{otherwise} \end{cases} \quad (13.41)$$

and

$$J_f(x_2) = \begin{cases} \int_{X_1} f^{x_2}(x_1) \mu_1(dx_1) & \text{if } f^{x_2} \text{ is } \mu_1\text{-integrable,} \\ 0 & \text{otherwise} \end{cases} \quad (13.42)$$

belong to $L^1(X_1, \mathcal{A}_1, \mu_1; \mathbb{R})$ and $L^1(X_2, \mathcal{A}_2, \mu_2; \mathbb{R})$, respectively, and

(iii) the relation

$$\int_{X_1 \times X_2} f(x_1, x_2) \mu_1 \times \mu_2(dx_1, dx_2) = \int_{X_1} I_f(x_1) \mu_1(dx_1) = \int_{X_2} J_f(x_2) \mu_2(dx_2) \quad (13.43)$$

holds.

Proofs of Theorems 13.0.16 and 13.0.17 (and their generalizations to a product of a finite number of σ -finite measure spaces) can be found in [14].

One of the corollaries from Tonelli's theorem is the following: Cavalieri's principle.

Lemma 13.0.18. Let $(X, \mathcal{A}, \mu)$ be a σ -finite measure space and let $g : X \rightarrow [0, +\infty]$ be a measurable function. Then

$$\int_X g(x) \mu(dx) = \int_0^{+\infty} \mu(\{x \in X \mid g(x) > t\}) dt, \quad (13.44)$$

and for all $\alpha \geq 0$

$$\int_X \min\{g(x), \alpha\} \mu(dx) = \int_0^\alpha \mu(\{x \in X \mid g(x) > t\}) dt. \quad (13.45)$$

Proof. For all $x \in X$ it holds

$$0 \leq g(x) = \lambda_1([0, g(x))) = \int_0^{+\infty} \mathbf{1}_{[0, g(x))}(t) dt = \int_0^{+\infty} \mathbf{1}_{\{g(x) > t\}}(x, t) dt. \quad (13.46)$$

Hence, by Tomelli's theorem 13.0.16 applied to the function $X \times [0, +\infty) \ni (x, t) \rightarrow \mathbf{1}_{\{g(x) > t\}}(x, t)$ we get

$$\begin{aligned} \int_X g(x) \mu(dx) &= \int_X \left(\int_0^{+\infty} \mathbf{1}_{\{g(x) > t\}}(x, t) dt \right) \mu(dx) = \int_0^{+\infty} \left(\int_X \mathbf{1}_{\{g(x) > t\}}(x, t) \mu(dx) \right) dt \\ &= \int_0^{+\infty} \mu(\{x \in X \mid g(x) > t\}) dt. \end{aligned} \quad (13.47)$$

The equality (13.45) follows from (13.44) when applied to the function $X \ni x \rightarrow \min\{g(x), \alpha\} \in [0, +\infty)$. $\square$

Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space, i.e., a measure space such that $\mathbb{P}(\Omega) = 1$. A $\Sigma/\mathcal{B}(\mathbb{R}^d)$ -measurable function $\xi : \Omega \rightarrow \mathbb{R}^d$ is called a random variable (or a random vector). If $(X, \mathcal{A})$ is a measurable space then we call a function $\xi : \Omega \rightarrow X$ a $(X$ -valued) *random element* (or a *random function*) provided that it is $\Sigma/\mathcal{A}$ -measurable. The law of ξ is given by

$$\mu(A) = \mathbb{P}(\{\xi \in A\}) = \mathbb{P}(\xi^{-1}(A)), \quad (13.48)$$

for all $A \in \mathcal{A}$. Hence, the random element ξ induces measure on $(X, \mathcal{A})$ and then the space $(X, \mathcal{A}, \mu)$ is a measure space equipped with the probability measure μ ($\mu(X) = 1$). The classical example is a Wiener space $(C([0, T]), \mathcal{B}(C([0, T])), w)$ equipped with the Wiener measure w . The measure w is a law of the random element $\hat{W} : \Omega \rightarrow C([0, T])$, generated by a Wiener process W (see Theorem 14.0.5).

Below we recall the Central Limit Theorem. For the proof see, for example, Theorem 10.3.16 in [14].

Theorem 13.0.19. (*Central Limit Theorem, CLT*) *Let $(X_j)_{j \in \mathbb{N}}$ be an i.i.d sequence of random variables with common mean μ and variance σ^2 . Let $S_n = \sum_{j=1}^n X_j, n \in \mathbb{N}$. Then the sequence*

$$\left(\frac{S_n - \mathbb{E}(S_n)}{\sqrt{\text{Var}(S_n)}} \right)_{n \in \mathbb{N}} \quad (13.49)$$

converges in distribution to a normal distribution with mean 0 and variance 1.

Note that under the assumptions of Theorem 13.0.19

$$\frac{S_n - \mathbb{E}(S_n)}{\sqrt{\text{Var}(S_n)}} = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \sqrt{n} \frac{\bar{S}_n - \mu}{\sigma}, \quad (13.50)$$

where $\bar{S}_n = S_n/n$.

The following result is a very useful consequence of the Borel-Cantelli lemma.

Theorem 13.0.20. ([48]) *Let $\alpha > 0$ and $K(p) \in [0, +\infty)$ for $p \geq 1$. In addition, let $Z_n, n \in \mathbb{N}$, be a sequence of random variables such that*

$$\left(\mathbb{E}|Z_n|^p \right)^{1/p} \leq K(p) \cdot n^{-\alpha} \quad (13.51)$$

for all $p \geq 1$ and all $n \in \mathbb{N}$. Then for all $\varepsilon \in (0, \alpha)$ there exists a random variable η_ε such that

$$|Z_n| \leq \eta_\varepsilon \cdot n^{-\alpha+\varepsilon} \text{ almost surely} \quad (13.52)$$

for all $n \in \mathbb{N}$. Moreover, $\mathbb{E}|\eta_\varepsilon|^p < +\infty$ for all $p \geq 1$.

Below we recall the famous Burkholder inequality, which is nowadays widely used when establishing error bounds for Monte Carlo algorithms (for references, see, [10], [57]). For the exposition of the inequality, we need a notion of a filtration and martingale in the discrete-time case.

A *discrete filtration* on a probability space $(\Omega, \Sigma, \mathbb{P})$ is an increasing family $(\Sigma_n)_{n \in \mathbb{N}_0}$ of sub- σ -fields of Σ . A sequence $M = (M_n)_{n \in \mathbb{N}_0}$ of $\mathbb{R}^d$ -valued random variables on $(\Omega, \Sigma, \mathbb{P})$ is said to be a *discrete-time martingale* with respect to $(\Sigma_n)_{n \in \mathbb{N}_0}$ provided that

- (i) $M = (M_n)_{n \in \mathbb{N}_0}$ is adapted to $(\Sigma_n)_{n \in \mathbb{N}_0}$, i.e., $\sigma(M_n) \subset \Sigma_n$ for all $n \in \mathbb{N}_0$,
- (ii) $\mathbb{E}\|M_n\|_2 < +\infty$ for all $n \in \mathbb{N}_0$,
- (iii) $\mathbb{E}(M_{n+1} \mid \Sigma_n) = M_n$ for all $n \in \mathbb{N}_0$.

We now recall two main inequalities for discrete-time martingales.

Theorem 13.0.21. (*Hoeffding inequality - original version, [30]*) Let $(X_i)_{i=1,2,\dots,n}$ be a sequence of independent random variables such that $a_i \leq X_i \leq b_i$ with probability one for $i = 1, 2, \dots, n$ and for some real numbers $a_1, \dots, a_n, b_1, \dots, b_n$. Let $S_n = \sum_{k=1}^n X_k$. Then for all $t > 0$

$$\mathbb{P}\left(|S_n - \mathbb{E}(S_n)| > t\right) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right). \quad (13.53)$$

Theorem 13.0.22. (*Hoeffding-Azuma inequality, [53]*) Let $(M_j)_{j=0,1,\dots,n}$ be a real-valued martingale such that $|M_j - M_{j-1}| \leq a_j$ a.s. for all $j = 1, 2, \dots, n$. Then, for all $r \geq 0$

$$\mathbb{P}\left(|M_n - M_0| \geq r\right) \leq 2 \exp\left(-\frac{r^2}{2 \sum_{j=1}^n a_j^2}\right). \quad (13.54)$$

Theorem 13.0.23. (*Burkholder's inequality, [10]*) For each $p \in (1, +\infty)$ there exist positive constants c_p and C_p such that for every $\mathbb{R}^d$ -valued discrete-time martingale $M = (M_n)_{n \in \mathbb{N}_0}$ and for every $n \in \mathbb{N}_0$ we have

$$c_p \| [M]_n^{1/2} \|_{L^p(\Omega)} \leq \left\| \max_{0 \leq j \leq n} \|M_j\|_2 \right\|_{L^p(\Omega)} \leq C_p \| [M]_n^{1/2} \|_{L^p(\Omega)}, \quad (13.55)$$

where $[M]_n = \|M_0\|_2^2 + \sum_{k=1}^n \|M_k - M_{k-1}\|_2^2$ is the quadratic variation of M .

In the case when $p \in [2, +\infty)$ we show some useful upper estimates for $\| [M]_n^{1/2} \|_{L^p(\Omega)}$. Firstly, let us denote by $\Delta M_k := M_k - M_{k-1}$ for $k = 1, 2, \dots, n$ and $\Delta M_0 := M_0$. Then, by the triangle inequality

$$\| [M]_n^{1/2} \|_{L^p(\Omega)}^2 = \| [M]_n \|_{L^{p/2}(\Omega)} = \left\| \sum_{k=0}^n \|\Delta M_k\|_2^2 \right\|_{L^{p/2}(\Omega)} \leq \sum_{k=0}^n \left\| \|\Delta M_k\|_2^2 \right\|_{L^{p/2}(\Omega)} = \sum_{k=0}^n \|\Delta M_k\|_{L^p(\Omega, \mathbb{R}^d)}^2,$$

hence

$$\left\| \max_{0 \leq j \leq n} \|M_j\|_2 \right\|_{L^p(\Omega)} \leq C_p \left(\sum_{k=0}^n \|\Delta M_k\|_{L^p(\Omega, \mathbb{R}^d)}^2 \right)^{1/2}. \quad (13.56)$$

Next, by the Hölder inequality

$$\sum_{k=0}^n \|\Delta M_k\|_{L^p(\Omega, \mathbb{R}^d)}^2 \leq \left(\sum_{k=0}^n \|\Delta M_k\|_{L^p(\Omega, \mathbb{R}^d)}^p \right)^{2/p} \cdot (n+1)^{1-\frac{2}{p}}. \quad (13.57)$$

Combining the estimates above we obtain

$$\left\| \max_{0 \leq j \leq n} \|M_j\|_2 \right\|_{L^p(\Omega)} \leq C_p \left(\sum_{k=0}^n \|\Delta M_k\|_{L^p(\Omega, \mathbb{R}^d)}^2 \right)^{1/2} \leq C_p \cdot (n+1)^{\frac{1}{2}-\frac{1}{p}} \cdot \left(\sum_{k=0}^n \|\Delta M_k\|_{L^p(\Omega, \mathbb{R}^d)}^p \right)^{1/p}. \quad (13.58)$$

We also recall the definition of a martingale and a local martingale in continuous time.

An $\mathbb{R}^d$ -valued stochastic process $M = (M(t))_{t \in [0, +\infty)}$ on $(\Omega, \Sigma, \mathbb{P})$ is said to be a *continuous-time martingale* with respect to the given filtration $(\Sigma_t)_{t \in [0, +\infty)}$ provided that

- (i) $M = (M(t))_{t \geq 0}$ is adapted to $(\Sigma_t)_{t \in [0, +\infty)}$, i.e., $\sigma(M(t)) \subset \Sigma_t$ for all $t \in [0, +\infty)$,
- (ii) $\mathbb{E}\|M(t)\|_2 < +\infty$ for all $t \in [0, +\infty)$,
- (iii) $\mathbb{E}(M(t) \mid \Sigma_s) = M(s)$ for all $0 \leq s < t$.

A random variable $\tau : \Omega \rightarrow [0, +\infty]$ is said to be *stopping time* with respect to the filtration $(\Sigma_t)_{t \in [0, +\infty)}$ if, for every $t \in [0, +\infty)$,

$$\{\tau \leq t\} \in \Sigma_t. \quad (13.59)$$

We also set

$$\Sigma_\tau = \{A \in \Sigma \mid \forall_{t \in [0, +\infty)} A \cap \{\tau \leq t\} \in \Sigma_t\}. \quad (13.60)$$

An $\mathbb{R}^d$ -valued stochastic process $M = (M(t))_{t \in [0, +\infty)}$ on $(\Omega, \Sigma, \mathbb{P})$ is said to be a *continuous-time local martingale* with respect to the given filtration $(\Sigma_t)_{t \in [0, +\infty)}$ if it is adapted to $(\Sigma_t)_{t \in [0, +\infty)}$ and if there exists an increasing sequence $(\tau_n)_{n \in \mathbb{N}}$ of $(\Sigma_t)_{t \in [0, +\infty)}$ -stopping times such that

- (i) $\mathbb{P}(\lim_{n \rightarrow +\infty} \tau_n = +\infty) = 1$,
- (ii) for every n the process $(M(t \wedge \tau_n), \Sigma_t)_{t \in [0, +\infty)}$ is a martingale.

Let $\{\mathcal{F}_1, \dots, \mathcal{F}_m\}$ be a family of sub- σ -fields of Σ . We call this collection a *family of independent σ -fields* or we say that the σ -fields $\mathcal{F}_1, \dots, \mathcal{F}_m$ are (*mutually*) *independent* if, for every $A_1 \in \mathcal{F}_1, \dots, A_m \in \mathcal{F}_m$,

$$\mathbb{P}(A_1 \cap \dots \cap A_m) = \mathbb{P}(A_1) \cdot \dots \cdot \mathbb{P}(A_m). \quad (13.61)$$

The independence of events from Σ can also be expressed in the term of independent σ -fields. Namely, the events $A_1, \dots, A_m \in \Sigma$ are independent iff the σ -fields $\sigma(\{A_1\}), \dots, \sigma(\{A_m\})$ are independent, where $\sigma(\{A_i\}) = \{\emptyset, \Omega, A_i, \Omega \setminus A_i\}$.

Random elements $\xi_j : \Omega \rightarrow X_j$, with values in arbitrary measurable spaces $(X_j, \mathcal{A}_j)$, $j = 1, 2, \dots, m$, are *independent* if and only if the σ -fields $\sigma(\xi_1), \dots, \sigma(\xi_m)$ are independent. Note that the independence of the events $A_1, \dots, A_m \in \Sigma$ is equivalent to the independence of the random variables $\mathbf{1}_{A_1}, \dots, \mathbf{1}_{A_m}$.

For a (possibly infinite) family $(\xi_i)_{i \in I}$ of random elements we say that they are *independent* if and only if the random elements $\xi_{i_1}, \dots, \xi_{i_m}$ are independent for every choice of a finite number $i_1, \dots, i_m$ of distinct indices in I . An analogous definition holds for an infinite family $(\mathcal{F}_i)_{i \in I}$ of sub- σ -fields of Σ .

Let $(\mathcal{A}_i)_{i \in I}$ be a family of (arbitrary) subsets of Σ . We say that $\mathcal{A}_i$'s are (*mutually*) *independent* iff for any finite subset J of I and for any $A_j \in \mathcal{A}_j$, $j \in J$, it holds

$$\mathbb{P}\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} \mathbb{P}(A_j). \quad (13.62)$$

Lemma 13.0.24. *Let $(\mathcal{A}_i)_{i \in I}$ be a family of π -systems and $\mathcal{A}_i \subset \Sigma$ for $i \in I$. Then $(\sigma(\mathcal{A}_i))_{i \in I}$ is a family of independent σ -fields iff $(\mathcal{A}_i)_{i \in I}$ are independent.*

We also say that the random element ξ is independent of the sub- σ -field $\mathcal{G} \subset \Sigma$ if the σ -fields $\sigma(\xi)$ and $\mathcal{G}$ are independent.

The following simple but useful lemma holds.

Lemma 13.0.25. *Let $\mathcal{A}_1, \dots, \mathcal{A}_m, \mathcal{B}_1, \dots, \mathcal{B}_m, \mathcal{C}_1, \dots, \mathcal{C}_m$ be a sub- σ -fields of Σ . Moreover, we assume that*

- (i) $\mathcal{A}_i, \mathcal{B}_i \subset \mathcal{C}_i$ for all $i = 1, \dots, m$,
- (ii) $(\mathcal{C}_i)_{i=1, \dots, m}$ is a family of independent σ -fields,
- (iii) for all $i = 1, \dots, m$ the σ -fields $\mathcal{A}_i$ and $\mathcal{B}_i$ are independent.

Then $\{\mathcal{A}_1, \dots, \mathcal{A}_m, \mathcal{B}_1, \dots, \mathcal{B}_m\}$ is a family of independent σ -fields.

Proof. Let us take arbitrary $A_i \in \mathcal{A}_i$, $B_i \in \mathcal{B}_i$, $i = 1, \dots, m$. By the assumption (i) we have $A_i \cap B_i \in \mathcal{C}_i$ for $i = 1, \dots, m$, and by the assumptions (ii), (iii) we get

$$\begin{aligned} \mathbb{P}(A_1 \cap \dots \cap A_m \cap B_1 \cap \dots \cap B_m) &= \mathbb{P}((A_1 \cap B_1) \cap \dots \cap (A_m \cap B_m)) \\ &= \prod_{j=1}^m \mathbb{P}(A_j \cap B_j) = \prod_{j=1}^m \mathbb{P}(A_j) \mathbb{P}(B_j). \end{aligned} \quad (13.63)$$

This ends the proof. $\blacksquare$

Pairwise independence is much weaker than being mutually independent. However, independence can be checked by repeated checks for pairwise independence, see, for example, [12] or [46].

Proposition 13.0.26. *The sub- σ -fields $\mathcal{F}_1, \mathcal{F}_2, \dots$ of Σ are independent if and only if $\bigvee_{i=1, \dots, n} \mathcal{F}_i$ and $\mathcal{F}_{n+1}$ are independent for all $n \in \mathbb{N}$.*

We also recall the grouping property, see, for example, [46].

Proposition 13.0.27. *Let $(\mathcal{F})_{i \in I}$ be a family of independent sub- σ -fields of Σ , and let $\mathcal{T}$ be a disjoint partition of I . Then the σ -fields*

$$\mathcal{F}_S = \bigvee_{s \in S} \mathcal{F}_s, \quad S \in \mathcal{T}, \quad (13.64)$$

are again independent.

Independence has its consequences for the product of random variables.

Lemma 13.0.28. *Let X, Y be independent and integrable random variables. Then*

$$\mathbb{E}(X \cdot Y) = \mathbb{E}(X) \cdot \mathbb{E}(Y). \quad (13.65)$$

The same holds true if X, Y are nonnegative and not necessarily integrable.

The proof of the following projection property of conditional expectation can be found, for example, in [12].

Theorem 13.0.29. *Let $\mathcal{G}$ be a sub- σ -field of Σ . For every $X \in L^2(\Omega, \Sigma, \mathbb{P})$ we have*

$$\mathbb{E}\left(X - \mathbb{E}(X \mid \mathcal{G})\right)^2 = \inf_{Y \in L^2(\Omega, \mathcal{G}, \mathbb{P})} \mathbb{E}(X - Y)^2. \quad (13.66)$$

Theorem 13.0.30. *(Conditional Fubini's Theorem, [9]) Suppose that $X \in L^1(\Omega \times [0, T], \Sigma \otimes \mathcal{B}([0, T]), \mathbb{P} \times \lambda_1)$ and let $\mathcal{G}$ be a sub- σ -field of Σ . Then*

$$\mathbb{E}\left(\int_0^T X(t) dt \mid \mathcal{G}\right) = \int_0^T \mathbb{E}(X(t) \mid \mathcal{G}) dt \text{ a.s.} \quad (13.67)$$

Theorem 13.0.31. *(The freezing lemma, [69]) Let X and Y be respectively a d -dimensional and a k -dimensional random vector, and let $\psi : \mathbb{R}^{d+k} \rightarrow \mathbb{R}$ be Borel measurable such that $\psi(X, Y) \geq 0$ a.s., or $\mathbb{E}|\psi(X, Y)| < +\infty$. Let $\mathcal{G}$ be a sub- σ -field of Σ such that X is $\mathcal{G}$ -measurable, and $\sigma(Y), \mathcal{G}$ are independent σ -fields. Then*

$$\mathbb{E}\left(\psi(X, Y) \mid \mathcal{G}\right) = \mathbb{E}\left(\psi(x, Y)\right) \Big|_{x=X} = \int_{\mathbb{R}^k} \psi(X, y) \mathbb{P}_Y(dy). \quad (13.68)$$

The following fact can be found for example in [2], page 238.

Lemma 13.0.32. *Let $W = (W(t))_{t \in [0, T]}$ be the Wiener process with respect to the filtration $(\Sigma_t)_{t \in [0, T]}$. If $f \in L^2([0, T])$ is a deterministic function, then for all $t \in [0, T]$ the σ -fields $\sigma\left(\int_t^T f(s)dW(s)\right)$ and Σ_t are independent.*

Chapter 14

Auxiliary results—part 2

In this section, we recall basic facts from the theory of continuous-time stochastic processes.

Let $(\Omega, \Sigma, \mathbb{P})$ be a *complete probability space*, that is all subsets of measure zero sets are also measurable (and hence, also of measure zero). We consider the product space $(\Omega \times [0, +\infty), \Sigma \otimes \mathcal{B}([0, +\infty)), \mathbb{P} \times \lambda_1)$. By a stochastic process X , with the base probability space $(\Omega, \Sigma, \mathbb{P})$ and with values in $\mathbb{R}^d$, we mean a mapping

$$X : \Omega \times [0, +\infty) \rightarrow \mathbb{R}^d, \quad (14.1)$$

such that for each $t \in [0, +\infty)$ the function

$$\Omega \ni \omega \rightarrow X(t, \omega) \in \mathbb{R}^d \quad (14.2)$$

is $\Sigma/\mathcal{B}(\mathbb{R}^d)$ -measurable, i.e., it is a random variable. We usually use the following notational convention $X = (X(t))_{t \in [0, +\infty)}$, which, in particular, underlines that a stochastic process might be seen as a collection of random variables. For a fixed $\omega \in \Omega$ the $\mathbb{R}^d$ -valued function

$$[0, +\infty) \ni t \rightarrow X(\omega, t) \in \mathbb{R}^d \quad (14.3)$$

is called a *trajectory* (or *realization* or *sample function*) corresponding to the elementary event ω . Note that, without any additional assumptions imposed on X , such a trajectory might not even be measurable. We say that the stochastic process X is *product measurable* if it is $\Sigma \otimes \mathcal{B}([0, +\infty))/\mathcal{B}(\mathbb{R}^d)$ -measurable. The following widely used fact is a direct consequence of product measurability and Fubini's theorem.

Proposition 14.0.1. *Let $X = (X(t))_{t \in [0, +\infty)}$ be a product measurable stochastic process. Then*

(i) *All trajectories of X are $\mathcal{B}([0, +\infty))/\mathcal{B}(\mathbb{R}^d)$ -measurable functions.*

(ii) *For any $t \in [0, +\infty)$ the function*

$$\Omega \ni \omega \rightarrow X(\omega, t) \in \mathbb{R}^d \quad (14.4)$$

is $\Sigma/\mathcal{B}(\mathbb{R}^d)$ -measurable, so it is $\mathbb{R}^d$ -valued random variable.

(iii) *if $\mathbb{E}(X(t))$ exists for all $t \in [0, +\infty)$ then*

$$[0, +\infty) \ni t \rightarrow \mathbb{E}(X(t)) \in \mathbb{R}^d \quad (14.5)$$

is $\mathcal{B}([0, +\infty))/\mathcal{B}(\mathbb{R}^d)$ -measurable.

(iv) *if $A \in \mathcal{B}([0, +\infty))$ and $\int_A \mathbb{E}|X(t)|dt < +\infty$, then*

$$\int_A |X(t)|dt < +\infty \text{ almost surely,} \quad (14.6)$$

and

$$\int_A \mathbb{E}(X(t))dt = \mathbb{E}\left(\int_A X(t)dt\right). \quad (14.7)$$

Let $(\Sigma_t)_{t \in [0, +\infty)}$ be a family of sub- σ -fields of Σ , such that

$$\Sigma_s \subset \Sigma_t \subset \Sigma, \text{ for all } 0 \leq s < t. \quad (14.8)$$

Then we call $(\Sigma_t)_{t \in [0, +\infty)}$ the *filtration* on $(\Omega, \Sigma, \mathbb{P})$. Moreover, we set

$$\Sigma_\infty = \sigma\left(\bigcup_{t \geq 0} \Sigma_t\right). \quad (14.9)$$

Of course $\Sigma_\infty \subset \Sigma$. We say that the filtration $(\Sigma_t)_{t \in [0, +\infty)}$ is *complete* if

$$\mathcal{N} \subset \Sigma_0, \quad (14.10)$$

where

$$\mathcal{N} = \{A \in \Sigma \mid \mathbb{P}(A) = 0\}. \quad (14.11)$$

If $(\Sigma_t)_{t \in [0, +\infty)}$ is not complete then we can always take

$$\bar{\Sigma}_t = \sigma(\Sigma_t \cup \mathcal{N}), \quad t \in [0, +\infty). \quad (14.12)$$

Of course the filtration $(\bar{\Sigma}_t)_{t \in [0, +\infty)}$ is complete and it is called *augmented* filtration. The filtration $(\Sigma_t)_{t \in [0, +\infty)}$ is called *right-continuous* if $\Sigma_t = \Sigma_{t+}$ for all $t \in [0, +\infty)$, where $\Sigma_{t+} = \bigcap_{s > t} \Sigma_s$. (We have that $\Sigma_t \subset \Sigma_{t+}$ for all $t \in [0, +\infty)$ and the family $(\Sigma_{t+})_{t \in [0, +\infty)}$ is already a right-continuous filtration.) If $(\Sigma_t)_{t \in [0, +\infty)}$ is complete and right-continuous then we say that it satisfies the *usual conditions*. Note that the filtration $(\tilde{\Sigma}_t)_{t \in [0, +\infty)}$, where

$$\tilde{\Sigma}_t = \bigcap_{s > t} \bar{\Sigma}_s (= \bar{\Sigma}_{t+}), \quad (14.13)$$

always satisfies the usual conditions, even if the initial filtration $(\Sigma_t)_{t \in [0, +\infty)}$ does not.

If for all $t \in [0, +\infty)$

$$\sigma(X(t)) \subset \Sigma_t, \quad (14.14)$$

then we say that the stochastic process $X = (X(t))_{t \in [0, +\infty)}$ is adapted to $(\Sigma_t)_{t \in [0, +\infty)}$. Every stochastic process X is adapted to $(\Sigma_t^X)_{t \in [0, +\infty)}$ where

$$\Sigma_t^X = \sigma(X(s) \mid s \in [0, t]) = \sigma\left(\bigcup_{s \in [0, t]} \sigma(X(s))\right). \quad (14.15)$$

We call $(\Sigma_t^X)_{t \in [0, +\infty)}$ the *natural filtration* of the process X . This is the smallest filtration to which X is adapted. We stress that the natural filtration of X neither has to be complete nor right-continuous, while $(\tilde{\Sigma}_t^X)_{t \in [0, +\infty)}$ satisfies the usual conditions. Analogously we also define the last element of $(\Sigma_t^X)_{t \in [0, +\infty)}$ by

$$\Sigma_\infty^X = \sigma\left(\bigcup_{t \geq 0} \Sigma_t^X\right). \quad (14.16)$$

Note that we have

$$\Sigma_\infty^X = \sigma\left(\bigcup_{t \geq 0} \sigma(X(t))\right). \quad (14.17)$$

14.0.1 What does it mean that two processes are equivalent?

Definition 14.0.2. Suppose that $X = (X(t))_{t \in [0, +\infty)}$ and $Y = (Y(t))_{t \in [0, +\infty)}$, are two $\mathbb{R}^d$ -valued stochastic processes defined on the same probability space $(\Omega, \Sigma, \mathbb{P})$.

(i) We say that X and Y are **stochastically equivalent** if

$$\forall_{t \in [0, +\infty)} \mathbb{P} \left(\{ \omega \in \Omega \mid X(\omega, t) = Y(\omega, t) \} \right) = 1. \quad (14.18)$$

We then say that Y is a **modification** (or a **version**) of X .

(ii) We say that X and Y are **indistinguishable** if

$$\mathbb{P} \left(\{ \omega \in \Omega \mid \forall_{t \in [0, +\infty)} X(\omega, t) = Y(\omega, t) \} \right) = 1. \quad (14.19)$$

Of course, if two processes are indistinguishable then they are stochastically equivalent, but the converse is in general not true. However, the following important result holds, see [17].

Theorem 14.0.3. *Suppose that $X = (X(t))_{t \in [0, +\infty)}$ and $Y = (Y(t))_{t \in [0, +\infty)}$ are stochastically equivalent, and that both processes are almost surely right-(or left-) continuous. Then they are indistinguishable.*

In summary, we have four different ways in which we can say that two stochastic processes are equal:

- (i) $X(\omega, t) = Y(\omega, t)$ for all $(\omega, t) \in \Omega \times [0, +\infty)$, but this situation occurs rarely,
- (ii) X and Y are stochastically equivalent,
- (iii) X and Y are indistinguishable,
- (iv) $X = Y$ $d\mathbb{P} \times d\lambda_1$ -almost everywhere, i.e.,

$$(\mathbb{P} \times \lambda_1) \left(\{ (\omega, t) \in \Omega \times [0, +\infty) \mid X(\omega, t) \neq Y(\omega, t) \} \right) = 0. \quad (14.20)$$

In order to use that notion of equivalence we have to assume that X and Y are product measurable processes.

14.0.2 Three notions of measurability of stochastic processes

Let $(\Omega, \Sigma, \mathbb{P})$ be a complete probability space with a filtration $(\Sigma_t)_{t \in [0, +\infty)}$ satisfying the usual conditions. We consider the following classes of stochastic processes

$$\mathbb{D} = \{ X : \Omega \times [0, +\infty) \rightarrow \mathbb{R}^d \mid X \text{ is adapted to } (\Sigma_t)_{t \in [0, +\infty)} \text{ and has càdlàg paths} \}, \quad (14.21)$$

and

$$\mathbb{L} = \{ X : \Omega \times [0, +\infty) \rightarrow \mathbb{R}^d \mid X \text{ is adapted to } (\Sigma_t)_{t \in [0, +\infty)} \text{ and has càglàd paths} \}. \quad (14.22)$$

Then, the *predictable* σ -field $\mathcal{P}$ on $\Omega \times [0, +\infty)$ is the smallest σ -field making all processes in $\mathbb{L}$ measurable, i.e.,

$$\mathcal{P} = \sigma \left(\bigcup_{X \in \mathbb{L}} \sigma(X) \right). \quad (14.23)$$

The *optional* σ -field $\mathcal{O}$ on $\Omega \times [0, +\infty)$ is the smallest σ -field making all processes in $\mathbb{D}$ measurable, i.e.,

$$\mathcal{O} = \sigma \left(\bigcup_{X \in \mathbb{D}} \sigma(X) \right). \quad (14.24)$$

A stochastic process $X = (X(t))_{t \in [0, +\infty)}$ is called *progressive* or *progressively measurable* if for each $t \in [0, +\infty)$ the mapping

$$\Omega \times [0, t] \ni (\omega, s) \rightarrow X(\omega, s) \in \mathbb{R}^d \quad (14.25)$$

is $\Sigma_t \otimes \mathcal{B}([0, t]) / \mathcal{B}(\mathbb{R}^d)$ -measurable. The *progressive σ -field* $\mathcal{A}$ on $\Omega \times [0, +\infty)$ is the smallest σ -field that makes all progressive processes measurable. We have that

$$\mathcal{P} \subset \mathcal{O} \subset \mathcal{A} \subset \Sigma \otimes \mathcal{B}([0, +\infty)), \quad (14.26)$$

see [72]. For the proof of the fact below we refer to [55].

Proposition 14.0.4. *Let $f : [0, +\infty) \rightarrow \mathbb{R}^d$ be a Borel function, and let us consider the (deterministic process) $X(\omega, t) = f(t)$ for all $(\omega, t) \in \Omega \times [0, +\infty)$. Then $X = (X(t))_{t \in [0, +\infty)}$ is predictable.*

14.0.3 Stochastic process as a random element with values in functional space and vice versa

Sometimes it is convenient to switch between two possible ways of looking at stochastic processes. The first possibility is to consider a stochastic process X as a product measurable function (14.3) with values in $\mathbb{R}^d$. However, if (almost) all trajectories of X belong to some functional space E equipped with σ -field $\mathcal{E}$, then we can consider a mapping $\hat{X} : \Omega \rightarrow E$ defined as

$$\hat{X}(\omega) = X(\omega, \cdot), \quad \omega \in \Omega. \quad (14.27)$$

If $\hat{X}$ is $\Sigma/\mathcal{E}$ -measurable then we say that the process X generates the random element $\hat{X}$ in $(E, \mathcal{E})$. Moreover, the law μ of $\hat{X}$ is a probabilistic measure induced by $\hat{X}$ on the measure space $(E, \mathcal{E})$, see (13.48).

We listed below the three most common cases. For the proofs see, for example, [24].

Theorem 14.0.5. (i) *Let $E = C([0, T])$, equipped with supremum norm $\|\cdot\|_\infty$, and $\mathcal{E} = \mathcal{B}(E)$. If X is a stochastic process with continuous trajectories, then it generates a random element in $C([0, T])$.*

(ii) *Let $E = D([0, T])$ be the Skorokhod space, equipped with the Skorokhod metric d , and $\mathcal{E} = \mathcal{B}(E)$. If X is a stochastic process with càdlàg trajectories, then it generates a random element in $D([0, T])$.*

(iii) *Let $E = L^2([0, T])$, equipped with the norm $\|\cdot\|_{L^2([0, T])}$, and $\mathcal{E} = \mathcal{B}(E)$. If $X \in L^2(\Omega \times [0, T], \Sigma \otimes \mathcal{B}([0, T]), \mathbb{P} \times \lambda_1)$, then it generates a random element in $L^2([0, T])$.*

Having a random element $\hat{X}$ in some functional space E it is natural to ask if there is a product measurable stochastic process X satisfying (14.27). In the case when $E = L^2([0, T])$ the situation is more subtle since $L^2([0, T])$ consists of equivalence classes.

Theorem 14.0.6.

(i) *Let $\hat{X}$ be a random element in $C([0, T])$. Then the process X defined by*

$$X(\omega, t) = (\hat{X}(\omega))(t), \quad (\omega, t) \in \Omega \times [0, T], \quad (14.28)$$

is product measurable and has continuous trajectories.

(ii) *Let $\hat{X}$ be a random element in $D([0, T])$. Then the process X defined by (14.28) is product measurable and has càdlàg trajectories.*

(iii) *Let $\hat{X}$ be a random element in $L^2([0, T])$. Then there exists a product measurable process X such that for almost all ω , the equality $X(\omega, t) = (\hat{X}(\omega))(t)$ holds almost everywhere on $[0, T]$.*

For the proof of (iii) see Proposition 2, page 741. in [61]. In order to show (i) and (ii) we recall after [68] what follows.

Lemma 14.0.7. (a) The coordinate mappings $\delta_t : C([0, T]) \rightarrow \mathbb{R}$ defined by

$$\delta_t(f) = f(t), \quad f \in C([0, T]) \quad (14.29)$$

are $\mathcal{B}(C([0, T]))/\mathcal{B}(\mathbb{R})$ -measurable for all $t \in [0, T]$.

(b) The coordinate mappings $\delta_t : D([0, T]) \rightarrow \mathbb{R}$ defined by

$$\delta_t(f) = f(t), \quad f \in D([0, T]) \quad (14.30)$$

are $\mathcal{B}(D([0, T]))/\mathcal{B}(\mathbb{R})$ -measurable for all $t \in [0, T]$.

Proof of Theorem 14.0.6 (i). For any $t \in [0, T]$ we can write (14.28) as

$$X(\omega, t) = \delta_t(\hat{X}(\omega)), \quad \omega \in \Omega. \quad (14.31)$$

By Lemma 14.0.7 (a) and from the fact that the random element $\hat{X}$ is $\Sigma/\mathcal{B}(C([0, T]))$ -measurable, we get for any $B \in \mathcal{B}(\mathbb{R})$ that

$$(X(t))^{-1}(B) = \left(\delta_t(\hat{X})\right)^{-1}(B) = \hat{X}^{-1}(\delta_t^{-1}(B)) \in \Sigma, \quad (14.32)$$

so for any $t \in [0, T]$ the mapping

$$\Omega \ni \omega \rightarrow X(\omega, t) \in \mathbb{R} \quad (14.33)$$

is $\Sigma/\mathcal{B}(\mathbb{R})$ -measurable. For any $\omega \in \Omega$ we have that $\hat{X}(\omega) \in C([0, T])$. Hence, by (14.28) we get that for any $\omega \in \Omega$ the function

$$[0, T] \ni t \rightarrow X(\omega, t) \in \mathbb{R} \quad (14.34)$$

is continuous. By Lemma 13.0.15 we obtain that X is $\Sigma \otimes \mathcal{B}([0, T])/\mathcal{B}(\mathbb{R})$ -measurable. ■

The proof of Theorem 14.0.6 (ii) is analogous and therefore it is left as an exercise.

Remark 14.0.8. The spaces $C([0, T])$, $L^2([0, T])$ considered in Theorem 14.0.5 (i), (iii) are separable Banach spaces. Hence, the random elements in Theorem 14.0.5 (i), (iii) are Bochner measurable. Since the space $D([0, T])$, under the Skorokhod metric d , is a complete separable metric space, the random element in Theorem 14.0.5 (ii) is Borel measurable. (Recall also that the vector space $D([0, T])$ under the supremum norm is Banach but not separable.)

14.1 Exercises

1. Show that $\mathcal{N}$ in (14.11) is a π -system. Moreover, give a proof that

$$\mathcal{C} = \{A \in \Sigma \mid \mathbb{P}(A) \in \{0, 1\}\} \quad (14.35)$$

is a σ -field on Ω and $\sigma(\mathcal{N}) = \mathcal{C}$.

2. Let $\mathcal{G}$ be a sub- σ -field of Σ . Show that the σ -fields $\sigma(\mathcal{N})$ and $\mathcal{G}$ are independent.
3. Let $(\Sigma_t)_{t \in [0, +\infty)}$ be a complete filtration of a complete probability space $(\Omega, \Sigma, \mathbb{P})$. Show that for each $t \in [0, +\infty)$ the probability space $(\Omega, \Sigma_t, \mathbb{P})$ is complete.
4. Give a proof of (14.17).
Hint: Use Proposition 13.0.2.
5. Let $f : [0, +\infty) \times \mathbb{R}^d \rightarrow \mathbb{R}^m$ be a Borel function and let $X = (X(t))_{t \in [0, +\infty)}$ be a $\mathbb{R}^d$ -valued predictable (optional, progressive) process. Show that the process $(f(t, X(t)))_{t \in [0, +\infty)}$ is predictable (optional, progressive).

6. Let $f : [0, +\infty) \times \mathbb{R}^d \rightarrow \mathbb{R}^m$ be a Borel function and let $X = (X(t))_{t \in [0, +\infty)}$ be a $\mathbb{R}^d$ -valued adapted and product measurable process. Show that the process $(f(t, X(t)))_{t \in [0, +\infty)}$ is adapted and product measurable.
7. Let us assume that $(\Omega, \Sigma, (\Sigma_t)_{t \in [0, +\infty)}, \mathbb{P})$ is a complete probability space with a complete filtration $(\Sigma_t)_{t \in [0, +\infty)}$. Suppose that $X = (X(t))_{t \in [0, +\infty)}$ and $Y = (Y(t))_{t \in [0, +\infty)}$ are two stochastically equivalent processes, where X is adapted to $(\Sigma_t)_{t \in [0, +\infty)}$. Show that Y is also adapted.
8. For any two product measurable stochastic processes X, Y prove that the following are equivalent:
 - (i) $(\mathbb{P} \times \lambda_1) \left(\{(\omega, t) \in \Omega \times [0, +\infty) \mid X(\omega, t) \neq Y(\omega, t)\} \right) = 0,$
 - (ii) $\mathbb{P} \left(\{\omega \in \Omega \mid X(\omega, t) \neq Y(\omega, t)\} \right) = 0$ for $d\lambda_1$ -almost all $t \in [0, +\infty),$
 - (iii) $\lambda_1 \left(\{t \in [0, +\infty) \mid X(\omega, t) \neq Y(\omega, t)\} \right) = 0$ for $d\mathbb{P}$ -almost all $\omega \in \Omega.$
9. Show that for any $t \in [0, T]$ the coordinate mapping (14.29) is continuous.
10. Give a proof of Theorem 14.0.6 (ii).

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